MILITARY ENGINEER SERVICES
(EASTERN COMMAND)
SHILLONG ZONE

PROVISION OF OTM ACCOMMODATION AT HAYULIANG UNDER GE (P) LOHITPUR

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Encls : Drawings _____Sheets

_____
SIGNATURE OF CONTRACTOR

DY DIR/AAD (CONTRACTS)
FOR ACCEPTING OFFICER

Dated : _____

e-Tender System

Tele :72-6733(Mil)/0364-2534846
Fax :0364-2537640
E-Mail: ceshiz-ec-army@nic.in

मुख्यालय मुख्य अभियन्ता
HQ Chief Engineer
शिलाँग आंचल
Shillong Zone
एस. ई. फाल्स
Spread Eagle Falls
शिलाँग-७९३०११
Shillong-793011

89770 /HAY/ 15 /E8 **19 Sep 2024**

M/S_____
_____
_____
_____

NAME OF WORK : PROVISION OF OTM ACCOMMODATION AT HAYULIANG UNDER GE (P) LOHITPUR

Dear Sir(s),

1. Tender documents in respect of above work are uploaded on the site **https://defproc.gov.in**. The tender is on single stage two cover e-tendering system. The contents of Cover I & Cover II are specified in **NOTICE OF TENDER**.

2. Bids will be received online by **ACCEPTING OFFICER** upto the date and time mentioned in the **NOTICE INVITING TENDER (NIT).** No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as non bonafide tender/bid.

3. Bid will be opened on due date and time fixed for opening in the presence of tenderers/bidders or their authorised representatives, who have uploaded their quotation bid and who wish to be present at the time of opening the bids.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents and/or depute your technical representative for discussion on tender/ drawings and to clarify doubts, if any, on or before as mention web site . You are requested not to write piece meal points and forward your points duly consolidated before due date as mentioned in web site.

5. Un-enlisted contractors are required to submit the scanned copies (in pdf file) of documents required as per eligibility criteria mentioned in instructions for filling the tender documents and Appendix 'A' to NIT alongwith **EARNEST MONEY DEPOSIT (EMD)** and tender fee on e-procurement portal and submit the physical documents in the office of **HQ Chief Engineer Shillong Zone, SE Falls, Shillong-793011** within time limit specified in NIT. Inadequacy/deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.

6. (a) Contractor having not executed standing security bond and standing security deposit in any MES formation shall upload scanned copy of **EARNEST MONEY DEPOSIT (EMD)** mentioned in Notice of Tender and shall ensure receipt of hard copy of EMD in the office of tender issuing authority before date & time fixed for this purpose. In case of failure to abide by any of these two requirements. The finance bid will not be opened.

   (b) Please refer condition 19 of IAFW-2249 (As amended under amendment No 47 to IAFW- 2249).

7. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appx ‘A' to NIT on e- procurement portal and submit physical documents in the office of **HQ Chief Engineer Shillong Zone, SE Falls, Shillong-793011** before date & time fixed for this purpose .

8. The contractor must ensure that the tender/bid on the proper form is uploaded in time as the Accepting Officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand/ through post from tenderer/bidder even if they are received in time.

9. In view of delays due to system failure or other communication related failures, it is suggested that the tender/bid be uploaded, if necessary, sufficiently in advance of the last due date and time fixed.

10. General Conditions of Contracts (IAFW-2249) (1989 Print), Schedule of minimum fair wages and MES SSR (Part -1 and Part -II) are not enclosed with these documents. These are available for perusal in the Office of GE concerned and this office.

11. **ANY TENDER / BID, WHICH PROPOSES ALTERATIONS TO ANY OF THE CONDITIONS, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR WHICH PROPOSES ANY OTHER CONDITION OR PRESCRIPTION, WHATSOEVER, IS LIABLE TO BE REJECTED.**

THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY OTHER BID

Yours faithfully,

Encls: As above

SD/-

SIGNATURE OF CONTRACTOR

DY DIR/AAD (CONTRACTS)
FOR ACCEPTING OFFICER

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S)

1. **EARNEST MONEY DEPOSIT (EMD):-**Contractor(s) who are not enlisted with MES/who are enlisted but have not executed the Standing Security Bond shall submit Earnest Money Deposit as detailed in Notice of Tender in one of the following forms, alongwith their tender/bid :-

(a) Deposit at Call Receipt from a Scheduled Bank in favour of respective **GE** as mentioned in the NIT.

(b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of respective **GE** as mentioned in the NIT.

It is advisable that Earnest Money is deposited in the form of deposit call receipt from an approved Schedule Bank for easy refund. In case the tenderer/bidder wants to lodge 'EARNEST MONEY DEPOSIT' in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of concerned GE.

NOTES: Earnest Money Deposit (EMD) in the form of cheque/Bank Guarantee etc will not be accepted. **NON-SUBMISSION OF EARNEST MONEY DEPOSIT (EMD)** (scanned copy alongwith Technical Bid & hard copy before the date & time fixed for opening of BOQ) WILL RENDER **THE BID DISQUALIFIED FOR OPENING OF COVER II (FINANCE BID)**.

2. **PERFORMANCE SECURITY DEPOSIT (APPLICABLE FOR BOTH UNENLISTED AS WELL AS ENLISTED CONTRACTORS):** In case, the tender submitted by contractor is accepted, the contractor will be required to lodge "**Performance Security Deposit**" for an amount equivalent to 5% of the Contractor Sum/Lump Sum in favour of the Accepting officer within 28 (twenty Eight) days of the receipt by the contractor of notification of acceptance of his tender/bid in the form of "Bank Guarantee" or FDR or insurance surety bonds.. Failure of the contractor to comply with the requirements of Performance Security shall constitute sufficient grounds for cancellation of the award of work.

3 **GENERAL INSTRUCTIONS FOR COMPLIANCE**

3.1 The bids received only in the electronic form will be considered. All bids shall be submitted on **'http://defproc.gov.in'** portal. Documents should be scanned and forwarded in 'pdf form and 'xls' form as indicated.

3.2 Bids shall be uploaded on **'http://defproc.gov.in'** portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like email/ fax/by hand/ through post will be considered.

3.3 Bid should be **DIGITALLY** signed using **valid DSC**. All pages of tender documents, corrections/ alterations shall be signed/ initialed by the lowest bidder after acceptance.

3.4 Drawings, if issued in physical form, must be returned duly initialed by the tenderer/bidder in separate envelope indicating his name and address.

3.5 The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialed. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

3.6 In the technical bid, a scanned copy of **Power of Attorney** in favour of the person uploading the bid using his/her DSC shall be uploaded. In case the digital signatory himself is the sole proprietor, scanned copy of an affidavit on stamp paper of appropriate value to this effect stating that he has authority to bind the firm in all matters pertaining to contract including the Arbitration Clause, shall be attached in 'pdf' form. In case of partnership concern or a limited company, digital signatory of the bid/tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Articles of Association of the Company) in all the matters pertaining to the contracts with Union of India including arbitration clause. A scanned copy of the documents confirming of such authority shall be attached with the tender/bid in 'pdf form, if not submitted earlier. The person uploading the bid on behalf of another partner(s) or on behalf of a firm or company using his **DSC** shall upload with the tender/bid a scanned copy (in 'pdf form) of Power of Attorney duly executed in his favour by such other or all of the Partner(s) or in accordance with constitution of the company in case of company, stating that he has authority to bind such other person of the firm or the Company, as the case may be, in all matters pertaining to the contract including the Arbitration Clause. Even in case of Firms or Companies which have already given Power of Attorney to an individual authorizing him to sign tender in pursuance of which bids are being uploaded by such person as a routine, fresh Power of Attorney duly executed in his favour stating specifically that the said person has authority to bind

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (Contd…..)

such partners of the Firm, or the Company as the case may be, including the condition relating to Arbitration Clause, should be uploaded in 'pdf' form with the tender/bid; unless such authority has already been given to him by the Firm or the Company. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Article of Association.

3.7 Hard copies of all above documents should be sent by the contractor to the Tender issuing authority well in advance to be received before the date & time fixed for the same.

3.8 Bid (Cover 1 & 2) shall be uploaded online well in time.

3.9 The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to special condition 3 referred hereinafter and also conditions 24 & 25 of IAFW 2249 (General conditions of contract).

3.10 Tenderers/bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tenders/bids, may do so at the appointed time.

4 BOQ :-

4.1 The tenderer shall quote his rates on the BOQ EXCEL FILE only as per guide line of e-procure web portal. No alteration to the format will be accepted and the bid will be disqualified.

4.2 In case the tenderer wish to revise/modify the rates quoted in the BOQ file, he can do so only in the BOQ files before uploading the tender through **https://defproc.gov.in** site only before closing date & time.

4.3 After the uploading of tender, Department may upload the errata/ amendment through corrigendum. The tenderer/ bidder should submit their offer considering the errata/amendment carried out through corrigendum issued from time to time.

4.4 While uploading the quotation, the tenderers/ bidders should specifically check whether any **revised BOQ** has been uploaded by department through corrigendum prior to **Bid submission start date**. Tenderers/ bidders attention is specifically drawn to the fact that they should submit their offer on **revised BOQ only**. In case any tenderer submits offer on pre-revised BOQ in lieu of Revised BOQ, **it will be considered as a willful negligence by the tenderer/bidder and quotation shall be consider as a non-bonafide**.

5 **REVOCATION/REVISION OF OFFER UPWARD/ OFFERING VOLUNTARY REDUCTION, AFTER CLOSING OF BID SUBMISSION DATE AND TIME:-**In the event of any tenderer/bidder revoking his offer or revising his rates upward/ offering voluntary reduction, after closing of bid submission date & time, his offer will be treated as revoked and the Earnest Money deposited by him shall be forfeited. In case of MES enlisted Contractors, the amount equal to the Earnest Money stipulated in the Notice of tender, shall be notified to the tenderer/bidder for depositing the amount through MRO. Bids of such Contractors/bidders shall not be opened till the aforesaid amount equal to the earnest money is deposited by him in Govt Treasury. In addition, bids of such tenderer/bidder and his related firm shall not be opened in second call or subsequent calls. Reduction offered by the tenderer/bidder on the freak high rates referred for review shall not be treated as voluntary reduction.

6 CPM (Critical Path Method):-

6.1 The project planning for work covered in the scope of tender is based on CPM.

6.2 The tenderer/bidder is expected to be fully conversant with the CPM technique and employ technical staff who can use the technique in sufficient details. Sufficient books and other literature on the subject are widely available in the market which the tenderer,/bidder may make use of.

6.3 The tenderer's/bidder's attention is drawn to special condition of the tender regarding preparation of the detailed network analysis and time schedule for the work and his liability for employing sufficient resources to adhere to this schedule. Any inability on the part of the tenderer/bidder in

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (Contd…..)

using the technique will be taken as his technical inefficiency and will affect his class of enlistment and future prospect/invitation to tenders for future works.

6.4 The tenderer shall make/ update the CPM Chart using software's like MS Project or PRIME VERA or any other suitable software after taking prior approval from GE.

7. Department may issue amendments/errata in form of **CORRIGENDUM** to tender /revised BOQ to the tender documents. The tenderer/bidder is requested to read the tender documents in conjunction with all the errata/ amendments/corrigendum, if any, issued by the department.

8. These instructions shall form part of the tender documents.

SIGNATURE OF CONTRACTOR
DATED:

DY DIR/AAD (CONTRACTS)
FOR ACCEPTING OFFICER

MILITARY ENGINEER SERVICES
NOTICE OF TENDER IAFW-2162
APPLICABLE FOR LUMP SUM/MEASUREMENT CONTRACTS (Contd…..)

1 A tender is invited for the work as mentioned in Appendix 'A' to this **NOTICE INVITING TENDER (NIT)**

2 The work is estimated to cost as indicated in aforesaid Appendix 'A'. This estimate, however, is not a guarantee and is merely given as a rough guide and if the work costs more or less, a tenderer/bidder will have no claim on that account. The tender shall be based on as mentioned in aforesaid Appendix 'A'.

3 The work is to be completed within the period as indicated in aforesaid Appendix 'A' in accordance with the phasing, if any, indicated in the tender from the date of handing over site mentioned in Work Order issued by GE after submission of Performance Security Deposit.

4.1 Normally contractors whose names are on the MES approved list for the area in which the work lies, and within whose financial category the estimated amount would fall, may tender/bid but in case of term contracts, contractors of class SS to E may tender/bid.

4.2 In case, the tender submitted by enlisted as well as un-enlisted contractor is accepted, the contractor will be required to lodge "**Performance Security Deposit**" for an amount equivalent to 5% of the Contract Sum/Lump Sum in favour of the Accepting Officer within 28 (Twenty Eight) days of the receipt by the contractor of notification of acceptance of his tender/bid in the form of "Bank Guarantee" or FDR or insurance surety bonds. Failure of the contractor to comply with the requirements of Performance Security shall constitute sufficient grounds for cancellation of the award of work and forfeiture of the Earnest Money. In case of MES enlisted contractor, amount equal to the Earnest Money stipulated in the Appendix 'A', shall be notified to the tenderer for depositing the amount through MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury {See Condition 19 of General Conditions of Contract (IAFW-2249) (As amended under amendment No. 47 to IAFW-2249)}.

4.3 Not more than one tender/bid shall be submitted/uploaded by one contractor or one firm of contractors. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

5 The **CHIEF ENGINEER, SHILLONG ZONE, SPREAD EAGLE FALLS, SHILLONG – 793011** will be the Accepting Officer here in after referred to as such for purpose of the contract.

6 The Technical Bid and Financial Bid (Cover-1 and Cover-2) shall be uploaded by the tenderer/bidder on or before the date & time mentioned in NIT. A scanned copy of DD with enlistment details/documents shall be uploaded as packet 1/cover-1 ('T' bid) of the tender/bid on e-tendering portal. DD is refundable in case 'T' bid is not accepted resulting in non-opening of 'Q' bid. The applicant contractor shall bear the cost of bank charges for procuring and encashing the DD and shall not have any claim from Government whatsoever on this account.

6.1 **INTEGRITY PACT (IP)** duly signed by Accepting Officer / Officer designated by Accepting Officer is uploaded as part of NIT as Annexure – I of Appendix 'A' to NIT and the same shall form part of tender documents. The scanned copy of **INTEGRITY PACT (IP)** duly signed on each page by the bidder(s) shall be uploaded as part of technical bid (Cover – I) and original **INTEGRITY PACT (IP)** duly signed on each page shall be forwarded to this office by post alongwith demand draft and all other documents as required and mentioned in Appendix 'A' to NIT. Bidders who do not upload scanned copy of **INTEGRITY PACT (IP)** duly signed, shall be intimated of the same alongwith intimation of other such deficient documents, through option of 'Short Fall Documents' in e-tendering portal before 'T' Bid evaluation. Any bidder who fails to forward the copy of **INTEGRITY PACT (IP)** duly signed even after this communication, shall be disqualified in the Technical Bid (Cover – I).

MILITARY ENGINEER SERVICES
NOTICE OF TENDER IAFW-2162
APPLICABLE FOR LUMP SUM/MEASUREMENT CONTRACTS (Contd…..)

6.2 Tender form and conditions of contract and other necessary documents shall be available on **https://defproc.gov.in** web site for download and shall form part of contract agreement in case the tender/bid is accepted.

6.3 In Case of contractor who has not executed the Standing Security Bond, the Cover-1 shall be accompanied with Earnest Money of such amount as mentioned in Appendix 'A' in the form of deposit at call receipt in favour of concerned GE/ GE(I)/ AGE(I) (see Appendix 'A') by a scheduled Bank or in receipted treasury Challan the amount being credited to the revenue deposit of the concerned GE/ GE(I)/ AGE(I) (see Appendix 'A').

6.4 The GE/ GE(I)/ AGE(I) will return the Earnest Money wherever applicable to all unsuccessful tenderers/bidders by endorsing an authority on the deposit-at-call receipt for its refund, on production by the tenderer, bidder a certificate of the Accepting Officer that a bonafide tender/bid was received and all documents were returned.

6.5 The GE/ GE(I)/ AGE(I) will either return the Earnest Money to the successful tenderer/bidder by endorsing an authority on the deposit-at-call Receipt for its refund on receipt of an appropriate amount of Security Deposit or will retain the same in part or full on account of security deposit if such a transaction is feasible.

6.6 Copies of the drawings and other document pertaining to the work signed for the purpose of identification by the Accepting Officer or his accredited representative, sample of materials and stores to be supplied by the contractor will also be available for inspection by the tenderer/bidder at the office of Accepting Officer and concerned GE/ GE(I)/ AGE(I) during working hours.

7 The tenderers/bidders are advised to visit the site of work by making prior appointment with GE/ GE (I)/ AGE(I) who is also the Executing Agency of the work (see appendix 'A'). The tenderers/bidders are deemed to have full knowledge of all relevant documents, samples, site etc., whether they have inspected them or not.

8 Any tender/bid which proposes any alteration to any of the conditions laid down or which proposes any other condition or prescription whatsoever, is liable to be rejected.

9 The uploading of bid implies that bidder has read this notice and the Conditions of Contract and has made himself aware of the scope and specification of work to be done and of the conditions and rates at which stores, tools and plants etc will be issued to him and local conditions and other factors having bearing on the execution of the work.

10 Tenderers/bidders must be in possession of a copy of the MES Standard Schedule of Rates (see appendix 'A') including amendments and errata thereto.

11 Invitation for e-tender does not constitute any guarantee for validation of 'T' bid and subsequent opening of finance bid of any applicant/bidder, even of enlisted contractors of appropriate class, merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the 'T' bid and not open the finance bid of any applicant/bidder. 'T' bid validation shall be decided by the Accepting Officer based on, inter alia, capability of the firm as per criteria given in Appx 'A' to this **NIT.** The applicant contractor/bidder will be informed regarding non- validation of his 'T' bid assigning reasons thereof through the **https://defproc.gov.in** website. The applicant contractor/bidder if he so desires may appeal to the next higher Engineer authority viz CE Eastern Command on email id **dircontceengrkl-mes@nic.in** with copy to the Accepting Officer on email before the scheduled date of opening of Finance Bid. The decision of the Next Higher Engineer Authority (NHEA) shall be final and binding. The contractor/bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

12 The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking,

MILITARY ENGINEER SERVICES
NOTICE OF TENDER IAFW-2162
APPLICABLE FOR LUMP SUM/MEASUREMENT CONTRACTS (Contd…..)

giving a price preference over other Tender(s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible from such tenderer/bidder whose tender/bid is rejected.

13 Accepting Officer does not bind himself to accept the lowest or any tender/bid or to give any reason for not doing so.

14 This **Notice Inviting Tender (NIT)** including Appendix 'A' shall form part of the contract.

SIGNATURE OF CONTRACTOR **DY DIR/AAD (CONTRACTS)**
DATED: **FOR ACCEPTING OFFICER**

SINGLE STAGE TWO COVER SYSTEM FOR ISSUE OF TENDER ONLINE

APPENDIX 'A' TO (NIT)

1.	Name of Work	**PROVISION OF OTM ACCOMMODATION AT HAYULIANG UNDER GE (P) LOHITPUR**
2.	Estimated Cost	**10625.47** Lakhs (At Par Market)
3.	Period of Completion	**1080 days**
4.	Tender Cost	**Rs. 3000/-** in the shape of DD/Bankers Cheque from any Scheduled Bank in favour of **GE (P) Lohitpur** and payable at **TEZU**. (Note: In case of retendering, the contractor who had quoted in the previous call is not required to submit the cost of tender.)
5.	Website/portal address	https://defproc.gov.in
6.	Type of Contract	The tender shall be based on drawings and specifications (IAFW-2159) and General Conditions of Contract (IAFW-2249) with Schedule 'A' (List of items of work) to be pre-priced by MES. The tenderer is required to work out detailed quantity for each building/structure listed in Schedule 'A' Part-I(A), apply his own rate and quote the lumpsum amount that he will charge for the complete Schedule 'A' Part-I(A). The tenderers are required to quote their lumpsum amounts for pre-priced parts (except Part-I(A) i.e. buildings/structures) of Schedule 'A' and quote rates against each item of other parts (except pre-priced parts) of Schedule 'A'.
7.	Timeline details:	
	(a) Bid submission start date	Refer critical dates as shown in website
	(b) Bid submission end date	
	(c) Date of bid opening	
8	Eligibility Criteria	
	(a) For MES enlisted contractrs.	Contractor shall satisfy the following: - (i) Enlistment in Class 'SS' and category a(i) as per MES enlistment rules and satisfying Prequalification criteria (PQC) as in Note No.1 here-in-after. (ii) Shall not carry adverse remarks in Work Load Return or any similar report circulated by the competent engineer authority.
	(b) For contractors not enlisted with MES.	(i) Contractor shall meet the enlistment criteria of 'SS' Class and category a(i) and should satisfactorily completed requisite value of works as per Prequalification criteria (PQC) as in Note No.1 here-in-after. With Central/State Government/Central/State PSUs/AWHO/AFNHB/CGEWHO/DGMAP, annual turnover, bank solvency, working capital and other requirements given in Para1.4 & 1.5 of Section 1 of MES Manual of Contracts 2020 as available in all MES formations as well as MES website (www.mes.gov.in). (ii) Not carrying adverse remarks in Work Load Report (WLR) or any other similar report circulated by any competent authority, if already working in MES. (iii) Not suspended/debarred/ blacklisted(either permanently or temporarily) from participating in any bid or for business dealings by any Central/State Government Department or any Central/State Government PSU or any Autonomous Body under Central/State Government or any Local Body as on the bid submission end date. (iv) Details of works completed and under progress in MES to be submitted in the following format:-

Srl No	CA NO. and Name of work	Value of CA	Date of Commencement	Date of completion	Extended date of completion

(v) Unlisted contractor who have secured two work in MES should get themselves registered in the appropriate designated Class with any registering authority, else the firm will not be eligible for participation in the tender unless until the firm in enlisted with the MES.

APPENDIX 'A' TO (NIT) CONT'D....

	Tender issuing and Accepting Officer	Chief Engineer Shillong Zone, Military Engineer Services, Spread Eagle Falls, Shillong, Meghalaya-793011 Tele : 72-6733(Mil)/0364-2534846 FAX : 0364-2537640 Email :ceshiz-ec-army@nic.in
10	Executing Agency	**GE (P) Lohitpur**
11	Earnest Money	**Rs.15,00,000.00** In favour of **GE (P) Lohitpur** payable at **TEZU** in the form of deposit at call receipt, FDR not acceptable.

NOTES:

1. **PRE-QUALIFICATION CRITERIA (PQC):**

(a) Average Annual financial turnover during the last three years, ending 31st March of the previous financial year, should be at least 30% of the estimated cost. (This average annual turnover of 30% is for a work with period of completion as three years and where the period of completion is less than three years, the requirement of average turnover shall be proportionately specified).

(b) Experience of having successfully completed similar works during last seven years, ending last day of month previous to the month of bid submission start date, should be either of the following:

Three similar completed works costing not less than the amount of Rs 42,50,19,000.00

OR

Two similar completed works costing not less than the amount of Rs 53,12,74,000.00

OR

One similar completed work costing not less than the amount of Rs 85,00,38,000.00

(c) Similar work means works involving RCC frame building work alongwith all services i.e. external electric supply (with HT), water supply, road, site development, fire fighting etc.

(d) For the purpose of value of completed works, the value of previously completed works shall be enhanced @ 5% per year to bring them at par with present cost.

2. After opening of Cover 1, if the number of MES enlisted contractors of 'SS Class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT are less than 7(Seven), applications in respect of contractors two classes below the eligible class shall also be considered subject to fulfilment of PQC as per Note 1 above and other eligibility criteria given in the NIT. Therefore, MES contractors one/two class below may also bid for this tender. However, contractors of one/two classes below the eligible class shall not be considered in case their present residual work in hand is more than five their present tendering limit. Such bidders shall upload in their Cover-1 bid details of works in hand showing names of work, names of Accepting Officers, Contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contractors are considered for evaluation.

3. In case after opening of Cover 1, the number of MES enlisted contractors of 'SS' Class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT, are 7 (Seven) or more, applications of only those one class below the eligible class bidders shall be considered who fulfil the PQC as per Note 1 above and are meeting the criteria of upgradation in 'SS' Class with respect of past experience of completed works (individual work experience and average annual turnover as applicable) and financial soundness (solvency/financial soundness and working capital) as per details given in Manual on Contracts. Therefore, such contractors shall upload the requisite information/documents in Cover-1.

4. Un-enlisted contractor shall be considered provided their firm meets the eligibility criteria. Foreign firms shall not be eligible for this tender. However, Indian Firms having foreign national / Indian nationals staying abroad / Indian national having taken foreign citizenship, as director(s) shall be considered subject to security clearance from the concerned authorities of Government of India.

5. Contractors enlisted with MES will upload following documents in Cover 1 for checking eligibility: -

(a) Application for tender on Firm's letter head.

(b) Enlistment letter issued by the Registering Authority duly renewed for the cycle period in vogue.

(c) Scanned copy of DD/Bankers Cheque toward cost of tender and EMD instrument in case SSD bond is not signed at the time of registration, if applicable.

(d) Documents required in respect of PQC stipulations.

(e) Works experience and financial soundness and other necessary documents as applicable as per Para 2 of Notes above for contractors one/two class below the eligible class.

APPENDIX 'A' TO (NIT) CONT'D....

6. Contractors not enlisted with MES will be required to submit the following documents in Cover1 for checking eligibility: -

(a) Application for tender on Firm's letter head.

(b) Scanned copy of DD/Bankers cheque toward cost of tender and EMD instrument.

(c) Copy of Police Verification Report/Police Clearance Certificate/ Character Certificate from the Police Authority of the area where the registered office of the firm is located/notarized copy of valid passport of Proprietor/each Partner/each Director.

(d) Documents required for enlistment in MES for SS class as per Para 1.5 of Section 1 MES Manual on Contracts 2020.

(e) Documents required in respect of PQC stipulations.

(f) Details of works being executed, which fulfils PQC, with address of concerned department.

(g) Any other document required as described in this Appendix.

7. Tenders not accompanied by scanned copies of requisite DD/Bankers Cheque towards cost of tender and earnest money (as applicable) in Cover 1 shall not be considered for validation of 'T' bid and their Financial bids will not be opened.

8. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instruments (as applicable) reach the office of Accepting Officer within 10 days of bid submission end date failing which following action shall be taken:-

(a) In case of tenders from an enlisted contractor of MES, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will be opened. However non-submission of physical copies of cost of tender shall be considered as wilful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding for a period of six months commencing from the date of opening of Financial Bid(Cover2).

(b) In case of tenders from un-enlisted contractor, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will not be opened. Name of such contractors along with complete address shall be circulated for not opening of their bids for a period of six months commencing from the date of opening of Financial bid (Cover 2).

(c) In case of tenders from enlisted and un-enlisted contractors, where scanned copies of instruments for Earnest Money Deposit (as applicable) have been uploaded in Cover 1 but the same are not received in physical form within stipulated period, such tenders shall not qualify for opening of financial bid(Cover2).

9. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on their behalf. However, a contractor can execute the work through power of attorney to sons/daughters/spouse of Proprietor/Partner/Director and firm's own employees, director, project manager provided they are not having a separate enlisted firm in MES in their name as Proprietor/Partner/Director.

10. After opening of Cover1 and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, a communication in the form of e-mail/ SMS/ Speed Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication, failing which their financial bid(Cover2) shall not be opened and contractor shall not have any claim on the same.

11. Invitation for e-tender does not constitute any guarantee for validation of Technical bid and subsequent opening of financial bid of any applicant/bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in this Appendix. Tenderer/bidder will be informed regarding non-validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher Engineer Authority (NHEA)viz CE, Eastern Command on email id jtdgcontcengrkl_mes@nic.in with copy to the Accepting Officer on email before the scheduled date of opening of Cover 2. NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of the NHEA shall be final and binding. The tenderer/bidder shall not be entitled for any compensation whatsoever for rejection of their bid.

APPENDIX 'A' TO (NIT) CONT'D....

12. In case the BOQ is revised through the corrigendum and the bidder has failed to quote on revised BOQ (i.e. the firm has quoted on pre revised BOQ), such bid shall be treated as wilful negligence by the bidder and their quotation shall be considered non-bonafide. In such cases the lowest tender shall be determined from amongst the valid/bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.

13. Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after closing of bid submission date & time shall be considered as a wilful default. For this default a penalty of an amount equal to Earnest Money shall be levied. In case of an un-enlisted tenderer, Earnest Money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the Standing Security Bond, an amount equal to the earnest money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the afore mentioned amount is deposited in the Government Treasury. No other disciplinary/administrative action shall be taken against such tenderers. In such a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for this tender in second call or subsequent calls.

14. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/one or more Partners/Directors are common. Decision of Accepting Officer on issue/deny the tender to any one of the related firms shall be final and binding.

15. Contractor shall upload copy of Goods and Services Tax (GST) Registration Certificate of their company. In case of failure to upload and submit the document, the firm shall be disqualified in Technical Bid Evaluation.

16. The INTEGRITY PACT duly signed by Accepting Officer/Officer specifically designated by Accepting Officer is enclosed with NIT (NOTICE OF TENDER). The scanned copy of Integrity Pact duly signed on each page by the bidder(s) shall be uploaded as part of technical bid and original Integrity Pact duly signed on each page shall be forwarded by post alongwith DD/Bankers Cheque towards tender cost.

17. JOINT VENTURE: The provisions for Joint Venture is at Annexure 'II' and format for Joint Venture is given at Annexure 'III'.

18. Even though the tenderers meet the above criteria, they are liable to be disqualified if they have made misleading or false information in bidding documents submitted.

19. The Bidders shall be responsible for all of the costs associated with the preparation of their Bids and their participation in the Bidding Process. The Employer will not be responsible or in any way liable for such costs, regardless of the conduct or outcome of the Bidding Process.

20. Bidders are encouraged to submit their respective bids after visiting the Project site and ascertaining for themselves the site conditions, soil condition, location, surroundings, climate, availability of power, water & other utilities for construction, access to site, handling and storage of materials, weather data, applicable laws and regulations, and any other matter considered relevant by them by carrying out necessary survey/investigation. Bidders are advised to visit the site and familiarise themselves with the Project within the stipulated time of submission of the Bid. No extension of time is likely to be considered for submission of Bids.

21. Performance and other requirement: Bidder shall fulfil the following requirement: -

(a) There should not be poor/slow progress in running work. (If yes, submit details and reasons of delay to check that these are not attributable to him or are beyond his control).

(b) There are no serious defect observed in works which stand un-rectified (If yes, submit details and reasons).

(c) There are no cancelled/abandoned contracts in which govt unrealized recoveries exist (If yes, submit details and reasons).

(d) He/They have not been black listed by any Govt Deptt. If yes, submit details and reasons).

(e) There is no Government dues outstanding against the firm (If yes, submit details and reasons).

(f) Proprietor/ partners/directors of firm are not involved in ant-national/anti-social activities and have neither been convicted nor any proceeding are pending in court of such activities.

(g) Firm should have been considered capable of taking more load in work load return circulated by MES. In case the firm is not considered capable of taking more work load due to unsatisfactory performance in the running works, in the prevailing report circulated by DGBR their technical bid shall be rejected.

APPENDIX 'A' TO (NIT) CONT'D....

22. If bidders/ tenderers desire that any condition or stipulation given in the tender documents is to be modified/ amended or deleted, they may submit their comments/ suggestions before last working date of clarification as shown in critical date sheet in subject tender ID alongwith supporting documents, as applicable, for consideration by the Deptt for issue of corrigendum/ amendments to tender documents. If Deptt considers comments/suggestions to be appropriate, corrigendum/amendments to tender documents shall be issued and also upload on E-tendering Portal. If department does not consider comments/ suggestions suitable, corrigendum/ amendments to tender documents shall not be issued/uploaded on E-tendering Portal and tenderers shall quote strictly complying with the various provisions given in the tender documents.

23. All disputes arising out of or in respect of this tender before acceptance shall be decided by Court of appropriate jurisdiction at Shillong only. After acceptance of tender, Condition 72 (Jurisdiction of Courts) of IAFW – 2249 shall be applicable.

24. **Performance Security:** Please refer condition 19 of IAFW-2249 (As amended under amendment No. 47 to IAFW- 2249).

Sd/-
(Ajay Kumar Agarwal)
AE (QS&C)
AAD (Contracts)
For Accepting Officer

89770/HAY/ 14 /E-8दिनांक/Dated **19 Sep 2024**

HQ Chief Engineer
Shillong Zone
Spread Eagle Falls
Shillong - 793011

Annexure to
Appendix 'A' to NIT

INTEGRITY PACT

Tele :72-6733(Mil)/0364-2534846
Fax :0364-2537640
E-Mail: ceshiz-ec-army@nic.in

मुख्यालय मुख्य अभियन्ता
HQ Chief Engineer
शिलाँग आंचल
Shillong Zone
एस. ई. फाल्स
Spread Eagle Falls
शिलाँग-७९३०११
Shillong-793011

89770/HAY/ 14 /E8 **19 Sep 2024**

To
_____
_____
_____
_____

SUB: TENDER ID NO. …… FOR THE WORK : PROVISION OF OTM ACCOMMODATION AT HAYULIANG UNDER GE (P) LOHITPUR (CA NO: CESZ/HAY/ OF 2024-2025)

Dear Sir(s),

1. It is hereby declared that MES is committed to follow the principle of transparency, equity and competitiveness in public procurement.
2. The subject Notice Inviting Tender (NIT) is an invitation to offer made on the condition that the Bidder will sign the Integrity Pact, which is an integral part of tender / bid documents, failing which the tenderer / bidder will stand disqualified from the tendering process and the bid of the bidder would be summarily rejected.
3. This declaration shall form part and parcel of the Integrity Pact and signing of the same shall be deemed as acceptance and signing of the Integrity Pact on behalf of the MES.

Yours faithfully,

Sd/-

Chief Engineer / Designated Officer
(Nominated by Chief Engineer)

Annexure to
Appendix 'A' to NIT (Contd...)

INTEGRITY PACT

To,

Chief Engineer
HQ Chief Engineer Shillong Zone
Spread Eagle Falls
Shillong – 793 011

SUB: TENDER ID NO. …… FOR THE WORK : PROVISION OF OTM ACCOMMODATION AT HAYULIANG UNDER GE (P) LOHITPUR (CA NO: CESZ/HAY/ OF 2024-2025)

Dear Sir,

1. I/We acknowledge that MES is committed to follow the Principles thereof as enumerated in the Integrity Pact enclosed with the tender/bid document.

2. I/We agree that the Notice Inviting Tender (NIT) is an invitation to offer made on the condition that I/We will sign the Integrity Pact, which is an integral part of tender documents, failing which I/We will stand disqualified from the tendering process. I/We acknowledge that THE MAKING OF THE BID SHALL BE REGARDED AS AN UNCONDITIONAL AND ABSOLUTE ACCEPTANCE of the conditions of the NIT.

3. I/We confirm acceptance and compliance with the Integrity Pact in letter and sprit and further agree that execution of the said Integrity Pact shall be separate and distinct from the main contract, which will come into existence when tender/bid is finally accepted by MES. I/We acknowledge and accept the validity of the Integrity Pact, which shall be in line with Para 15 of the enclosed Integrity Pact.

4. I/We acknowledge that in the event of my/our failure to sign and accept the Integrity Pact, while submitting the tender/bid, MES shall have unqualified, absolute and unfettered right to disqualify the tenderer/bidder and reject the tender/bid in accordance with terms and conditions of the tender/bid.

Yours faithfully

(Duly authorized signatory of the Bidder)

INTEGRITY PACT

GENERAL

Whereas the President of India, represented by Chief Engineer, HQ Chief Engineer Shillong Zone, Spread Eagle Falls, Shillong – 793 011hereinafter referred to as **'Principal / Owner'** and the first part, has floated the Tender **(CA NO: CESZ/HAY/ OF 2024-2025)** and intends to award, under laid down organizational procedure, contract for **PROVISION OF OTM ACCOMMODATION AT HAYULIANG UNDER GE (P) LOHITPUR (CA NO: CESZ/HAY/ OF 2024-2025)** hereinafter referee to as works / Services and M/s..... represented by, ...(which term unless expressly indicated by the contract, shall be deemed to include its successors and its assignees), hereinafter referred to as the Bidder / Contractor and the second part is willing to carryout the works / services.

1. Whereas the Bidder is a Proprietorship Concern / Partnership Firm / Limited Liability Firm / Privet Limited Company / Limited Company / Joint Venture constituted in accordance with the relevant law in the matter and the Principal / Owner is Chief Engineer (CE) performing its functions on behalf of the President of India.

OBJECTIVES

2. Now, therefore, the Principal / Owner and the Bidder agree to enter into this pre-contract agreement, referred to as **INTEGRITY PACT(IP),** to avoid all forms of corruption by following a system that is fair, transparent and free from any influence / prejudiced dealings prior to, during and subsequent to conclusion of the contract to be entered into with a view to:-

2.1. Enabling the Principal / Owner to get the desired works / services at a competitive price in conformity with the defined specifications of the services by avoiding high cost and the distortionary impact of corruption on public procurement.

2.2. Enabling Bidders to abstain from bribing or any corrupt practice in order to secure the contract by providing assurance to them that their competitors will also refrain from bribing and other corrupt practices and the Principal / Owner will commit to prevent corruption, in any form, by their officials by following transparent procedures.

COMMITMENTS OF THE PRINCIPAL / OWNER

3. The Principal / Owner commits itself to the following:-

3.1. The Principal / Owner undertakes that, no official of the Principal / Owner, connected directly or indirectly with the contract will demand, take a promise for or accept, directly or through intermediaries, any bribe, consideration, gift, reward, favor or any material or for any person, organization or third party related to the contract; in exchange for an advantage; in the bidding process, bid evaluation, contracting or implementation process related to the Contract.

3.2. The Principal / Owner will, during the pre-contract stage, treat all Bidders alike and will provide to all Bidders the same information and will not provide any such information to any particular Bidder which could afford an advantage to that particular Bidder in comparison to other Bidders.

3.3. All the officials of the Principal / Owner will report to the appropriate Government office any attempted or completed breach(s) of the above commitments as well as any substantial suspicion of such a breach.

4. In case of any such preceding misconduct on the part of such official(s) is reported by the Bidder to the Principal / Owner willful and verifiable facts and the same is prima facie found to be corrected by the Principal / Owner, necessary disciplinary proceedings, or any other action as deemed fit, including criminal proceedings may be initiated by the Principal / Owner and such a person shall be debarred from further dealing related to the tender / contract process. In such a case while an Inquiry is being conducted by the Principal / Owner the tender process / proceedings under the contract would not be stalled.

COMMITMENTS OF BIDDERS

5. The Bidder commits himself to take all measures necessary to prevent corrupt practices, unfair means and illegal activities during any stage of his bid or during any pre-contract or post-contract stage in order to secure the contract or in furtherance to secure it and in particular commits himself to the following-:

5.1. Bidder will not offer, directly or through intermediaries, any bribe, gift, consideration, reward, favor any material or non-material benefit or other advantage, commission, fee, brokerage or inducement to any official of the Principal / Owner, connected directly or indirectly with the bidding process, or to any person, organization or third party related to the contract in exchange for any advantage in the bidding, evaluation, contracting and implementation of the Contract.

5.2. The bidder further undertakes that he has not given, offered or promised to give, directly or indirectly any bribe, gift, consideration, reward, favour any material or non-material benefits or other advantage, commission, fees, brokerage or inducement to any official of the Principal / Owner or otherwise in procuring the Contract or forbearing to do or having done any act in relation to the obtaining or execution of the contract or any other Contract with the Government for showing or forbearing to show favor or disfavor to any person in relation to the Contract or any other Contract with the Government.

5.3. The bidder will not collude with other parties interested in the contract to impair the transparency, fairness and progress of the bidding process, bid evaluation, contracting and implementation of the contract.

5.4. The bidder will not accept any advantage in exchange for any corrupt practice, unfair mean and illegal activities.

5.5. The bidder would not enter into conditional contract with any Agent(s), broker(s) or any other intermediaries wherein payment is made or penalty is levied, directly or indirectly, on success or failure of the award of the contract.

5.6. The Bidder commits to refrain from giving any complaint directly or through any other manner without supporting it with full and verifiable facts. Complaint will be processed as per **Guidelines for Handling of Complaints** in vogue. In case the complaint is found to be vexatious, frivolous or malicious in nature, it would be construed as a violation of Integrity Pact.

6. **PREVIOUS TRANSGRESSION**

6.1. The Bidder declares that no previous transgression occurred in the last three years immediately before signing of this Integrity Pacts with any other company in respect of any corrupt practices envisaged hereunder or with any Public Sector Enterprise in India or any Government Department in India.

6.2. If the Bidder makes incorrect statement on this subject, Bidder can be disqualified from tender process or the contract and if already awarded, same can be terminated for such reason.

7. **COMPANY CODE OF CONDUCT**

7.1. Bidder are advised to have a company code of conduct (clearly rejecting the use of bribes and other unethical behavior) and a compliance program for the implementation of the code of conduct throughout the country.

8. **SANCTION FOR VIOLATION**

8.1. Any breach of the aforesaid provisions by the Bidder or any one employed by him or acting on his behalf (whether with or without the knowledge of the Bidder) or the commission of any offence by the Bidder or any one employed by him or acting on his behalf, as defined in Chapter IX of the Indian Penal Code, 1860 or the Prevention of Corruption Act 1988 or any other act enacted for the prevention of corruption shall entitle the Principal / Owner to take all or any one of the following actions, wherever required:-

(i) Technical bid of the Bidder will not be opened. Bidder will not be entitled to or give any compensation. However, the proceedings with the other Bidder(s) would continue.

(ii) Financial bid of the Bidder will not be opened. Bidder will not be entitled to or give any compensation. However, the proceedings with the other Bidder(s) would continue.

(iii) The Earnest Money Deposit shall stand forfeited either fully or partially, as decided by the Principal / Owner, in case contract is not awarded to the Bidder and the Principal / Owner shall not be required to assign any reason therefore. For enlisted contractors an amount less than or equal to Earnest Money Deposit as decided by the Principal / Owner shall be deducted from any amount held with the Department / any payment due.

(iv) To immediately cancel the contract, if already concluded / awarded without any compensation to the Bidder.

(v) To encash the Performance Security furnished by the Bidder.

(vi) To cancel all or any other Contract(s) with the Bidder.

(vii) To temporarily suspend or temporarily debar / permanently debar the bidder as per the extant policy.

(viii) If adequate amount is not available in the present tender / contract, the deficient amount can be recovered from any outstanding payment due to the Bidder from the Principal / Owner in connection with any other contract for any other works / services.

(ix) If the Bidder or any employee of the Bidder or any person acting on behalf of the Bidder, either directly or indirectly, is closely related to any of the officers of the Principal / Owner, or alternatively if any close relative of an officer of the Principal / Owner has financial interest / stake in the Bidder's firm, the same shall be disclosed by the Bidder at the time of submission of tender. Any failure to disclose the interest involved shall entitle the Principal / Owner to debar the Bidder from the bid process or rescind the contract without payment of any compensation to the Bidder. The term 'close relative' for this purpose would mean spouse whether residing with the Government servant or not, but does not include a spouse separated from the Government servant by a decree or order of a competed Court; son or daughter or step son or step daughter and wholly dependent upon Government servant, but does not include a child or a step child who is no longer in any way dependent upon the Government servant or of whose custody the Government servant has been deprived of by or under any law; any other person related, whether by blood or marriage, to the Govt servant or to the Government servant's wife or husband and wholly dependent upon Government servant.

(x) The Bidder shall not lend to or borrow any money from or enter into any monetary dealings or transactions, directly or indirectly, with any employee of the Principal / Owner and if he does so the Principal / Owner shall be entitle forthwith to cancel the contract and all other contracts with the Bidder.

8.2. The decision of the Principal / Owner to the effect that a breach of the provisions of this Integrity Pact has been committed by the Bidder shall be final and binding on the Bidder. **However, the Bidder can approach the Independent External Monitor(s) (IEMs) appointed for the purposes of this Pact.**

9. **INDEPENDENT EXTERNAL MONITORS (IEMs)**

9.1. MoD has appointed the following Independent External Monitors for this pact in consultation with the Central Vigilance Commission:-

Sl No	Name of IEM	Contact details with e-mail ID
(i)	Shri Narayan Murthy Ganpathy, IFoS (Retd), IEM	3/268, Vipul Khand, Gomtinagar, Lucknow – 226010, Mo. No 9415147459, E-mail gana_narayan@yahoo.com
(ii)	Shri Lalatendu Mohanti, IPS (Retd), IEM	Flat No. 9367 (Ground Floor & First Floor), Pocket 9, Sector-C, Vasant Kunj, New Delhi-110070, Mo. No. 9717095659, E-mail-L.mohanti@gmail.com

9.2. Details of Nodal officer nominated by E-in-C's Branch are as follows:-

Name : Shri Bhavesh Gupta, Director (Contracts), E-in-C's Branch
E-Mail ID : dircont1einc-mec@nic.in
Mobile No : 9868145022

9.3. In case of any complaint with regard to violation of Integrity Pact, either party can approach IEMs with copy to the Nodal Officer and the other party. If any such complaint from bidder is received by the Principal / Owner, the Principal / Owner shall refer the complaint to the Independent External Monitors for their recommendations / inquiry report.

9.4. If the IEMs need to peruse the relevant records of the Principal / Owner and /or of the Bidder / Contractor in connection with the complaint sent to them, the Principal / Owner and/or the Bidder/ Contractor shall make arrangement for such perusal of records by the IEMs as demanded by them including unrestricted and unconditional access to the project documentation and minutes of meeting. If records / documents of Sub-Contractor(s) are also required to be perused by the IEMs, the Bidder shall make arrangement for such perusal of records by the IEMs as demanded by them. IEMs are under obligation to treat the information and documents of the Principal / Owner and Bidder / Contractor / Sub-Contractors with confidentiality.

9.5. The task of the IEMs, is to review independently and objectively, any complaint received with regard to violation Integrity Pact and offer recommendations or carry out inquiry as deemed fit. The IEMs are not subject to any instructions by the representatives of the parties and shall perform their functions neutrally and independently. The report of inquiry, if any, made by the IEMs shall be submitted to either of the following for a final and appropriate decision in the matter keeping in view the provision of this Pact:-

(a) Engineer-in-Chief in normal cases

(b) CVO (MES & BRO) / MoD in cases involving vigilance angle

10. **EXAMINATION OF BOOKS OF ACCOUNTS**

In case of any allegation of violation of any provisions of this Integrity Pact or payment of commission, the Principal / Owner or its agencies shall be entitled to examine the Books of Account of the Bidder and the Bidder shall provide necessary information of the relevant financial documents in English and shall extend all possible help for the purpose of such examination.

11. **LAW AND PLACE OF JURISDICTION**

This Pact is subject to Indian Law. The place of performance and jurisdiction is the seat of the Principal / Owner.

12. **OTHER LEGAL ACTIONS**

The actions stipulated in this Integrity Pact are without prejudice to any other legal action that may follow in accordance with the provisions of the extant law in force relating to any civil or criminal proceedings.

13. **SIGNING OF INTEGRITY PACT ON BEHALF OF BIDDER**

(a) **Proprietorship Concern** - The Integrity Pact must be signed by the proprietor or by an authorised signatory holding power of attorney signed by the proprietor.

(b) **Partnership firm -** The Integrity Pact must be signed by all partners or by one or more partner holding power of attorney signed by all partners.

(c) **Limited Liability firm -** The Integrity Pact must be signed by all partners or by one or more partner holding power of attorney signed by all partners.

(d) **Private Limited/Limited Company** - The Integrity Pact must be signed by a representative duly authorised by Board resolution.

(e) **Joint Venture** - The Integrity Pact must be signed by all partners and members to Joint Venture or by one or more partner holding power of attorney signed by all partners and members to the Joint Venture.

14. **Validity**

14.1. The validity of this Integrity Pact shall be from date of its signing. It expires for the Contractor after the final payment under the contract has been made or till the continuation of Defect Liability Period, whichever is later and for all other bidders, till the Contract has been awarded.

14.2. Should one or several provisions of this Pact turn out to be invalid; the remainder of this Pact remains valid. In this case, the parties will strive to come to an agreement to their original intentions.

……

(For and on behalf of Principal/Owner)

……

(For and on behalf of Bidder/Contractor)

WITNESSES

1. **……..**

……..

……..

(Signature, Name and Address)

2. **……**

……

……

(Signature, Name and Address)

Place:

Date:

Annexure – II to
Appendix 'A' to NIT (Contd...)

PROVISIONS OF JOINT VENTURE

1. Joint venture (JV) shall not comprise more than two firms (called parties of JV). The lead party shall have at least 60% share of interest in the JV. The other than lead party shall have at least 30% share of interest in the JV. The format of agreement is at Annexure-III. The JV shall be considered as un-enlisted contractor. The JV shall submit Earnest Money Deposit for all tenders mandatorily. Any bid received from JV firm without EMD shall be considered as non-bonafide tender.

2. A valid agreement shall exist between the parties of JV defining clearly the role, responsibility and scope of work of each party, percentage share of each party along with nomination of leader (Lead Party) for the purpose of this work and a confirmation that the parties of the JV are jointly and severally responsible.

3. a) Foreign Companies shall not be permitted to participate in JV.

b) Indian Companies having Director(s) of foreign origin and Indian Companies having Director(s) of Indian origin but residing abroad / having foreign citizenship shall be permitted to participate in JV. However, security clearance in such cases shall be obtained by following procedure laid down by Ministry of Home Affairs.

c) Case for security clearance shall be processed to E-in-C's Branch on PRIORITY after opening 'T' bid (Cover-1). Further, processing of tender to open Finance Bid (Cover-2) shall not be held up awaiting receipt of security clearance. However, if the JV requiring security clearance of Director(s) becomes L1, the tender shall be accepted only on receipt of security clearance. For this, the Accepting Officer will pursue the security clearance vigorously.

4. JV as a single unit or each party of the JV shall have Permanent Account Number (PAN) and GSTIN. However, if the contract is awarded to the JV, then PAN & GSTIN shall be obtained by the JV as a single unit.

5. Parties to the JV shall meet the following laid down criteria for the purpose of forming JV:-

Criteria	Lead Party	Other than lead Party	Overall as a JV
Average Annual turnover as per NIT/Tender documents	Min 60 %	Min 30 %	100 %
Technical Capacity as per NIT/Tender documents	Min 30 % of Pre-qualification Criteria	Min 30 % of Pre-qualification Criteria	-

(i) The JV shall collectively meet the working capital criteria as contained in the NIT/Tender documents.

(ii) The JV shall collectively meet the Bank solvency/Financial soundness for engagement criteria as contained in the NIT/Tender documents.

6. Both the parties to the JV either individually (single entity of the JV) or collectively (lead and other than lead) shall fulfill the 100% of the Technical qualifying criteria as stipulated in the NIT/Tender documents.

7. Both the Parties of JV shall jointly possess the required T&P, machinery and engineering/supervisory staff. T&P can be either on ownership basis or leasehold as stipulated in NIT/tender documents and documentary proof of the same shall be submitted.

Annexure – II to
Appendix 'A' to NIT (Contd...)

8. JV concluded up to the date of bid submission are permitted to apply. Copy of JV should be uploaded alongwith Technical Bid (Cover-I). The Department reserves the right to verify the particulars furnished by the applicant independently. If any information furnished by the applicant JV is found incorrect and/or misleading and / or false representation and/or deliberately suppressed information, at a later stage, the JV and both the parties shall be liable to be debarred from tendering/taking up of any work in MES.

9. Party/ parties will not be allowed to bid for the same tender in their independent capacity as well as under JV. Also no party will be allowed to bid for the same tender under multiple JVs. In case of violation (party/parties bidding independently as well as under JV for the same tender, party/parties bidding for the same tender under multiple JVs), the bid of the party/parties concerned as well as the bid(s) of the related JV(s) shall not be opened (i.e. shall not be qualified in 'T' bid Cover '1 ').

10. The lead party shall attend all progress review meetings and shall be answerable to all issues relating to the project.

11. For any of the defaults as under of the JV, administrative action shall be taken against both the parties of JV **:-**

a) In case of non-submission of physical original documents of cost of tender, EMD - Barring from bidding for six months.

b) Due to default in performance of contract etc - Administrative actions as per existing instructions.

12. Any unrealised recovery from JV shall be recovered proportionately from the parties in proportion to their percentage share in the JV. If it is not possible to recover proportionate share (partly/ fully) from one party, it shall be recovered from other party.

13. No JV shall be allowed to participate if either or both the parties are found banned / adversely remarked in WLR of MES or debarred from tendering by any authority.

SIGNATURE OF CONTRACTOR Dy DIR/AAD (CONTRACTS)

DATED : _____ FOR ACCEPTING OFFICER

Annexure – III

FORMAT FOR JOINT BIDDING AGREEMENT FOR JOINT VENTURE
(to be executed on stamp paper of appropriate value)

THIS JOINT BIDDING AGREEMENT is entered into on this the .....day of .....

AMONGST

.....having its **registered** office at ..... (here-in-after referred to as the **'**First Part**'** which expression shall, unless repugnant to the context include its successors and permitted assigns)
AND

1. .....having its registered office at ..... (here-in-after referred to as the **'**Second Part**'** which expression shall, unless repugnant to the context include its successors and permitted assigns)

The above mentioned parties of the FIRST and SECOND PART are collectively referred to as the **"Parties"** and each is individually referred to as a **"Party"**

WHEREAS,

A. The Military Engineer Services, represented by HQ CHIEF ENGINEER SHILLONG ZONE and having its office at SPREAD EAGLE FALLS, SHILLONG – 793 011 (here-in-after referred to as the **"Authority"**) which expression shall, unless repugnant to the context or meaning thereof, include its administrators, successors and assigns) has invited bid by its Tender ID No. 2022_MES_ _1 for "PROVISION OF OTM ACCOMMODATION AT HAYULIANG UNDER GE (P) LOHITPUR".

B. The Parties are interested in jointly bidding for the tender as member of a Joint Venture and in accordance with the terms and conditions of the tender document in respect of the work, and

C. It is necessary condition under the Pre-qualification Criteria (PQC) that the parties of the Joint Venture shall enter into a Joint Bidding Agreement and furnish a copy thereof with the Application.

NOW IT IS HEREBY AGREED as follows :-

1. Definitions and Interpretations

In this Agreement, the capitalised terms shall, unless the context otherwise requires, have the meaning ascribed thereto under the PQC in the Notice Inviting Tender.

2. Joint Venture

a) The parties do hereby irrevocably constitute a Joint Venture for the purposes of jointly participating in the Bidding Process for the project.

b) The Parties hereby undertake to participate in the Bidding Process only through this Joint Venture and not individually and/or through any other Joint Venture constituted for this Project, either directly or indirectly.

3. Covenants

The parties hereby undertake that in the event the Joint Venture is declared the selected Bidder and awarded the contract, it shall perform all its obligations as the Contractor in terms of the Contract.

4. Role of the Parties

The parties hereby undertake to perform the roles and responsibilities as described below **:-**

a) Party of the First Part shall be the Lead member of the Joint Venture and shall have the power of attorney from the other Party for conducting all business for and on behalf of the Joint Venture during the Bidding Process and execution process as well as post execution process.

b) Party of the Second Part shall be the Member of the Joint Venture.

5. Joint and Several Liability

The parties do hereby undertake to be jointly and severally responsible for all obligations and liabilities relating to the Project and in accordance with the terms and conditions given in the NIT and Tender Documents.

6. Field of Expertise

The parties do hereby declare that the field of expertise of the parties are as under**:-**

First Party **:**

Second Party **:**

7. Share of Work in the Project

The parties agree that the proportion of the Contract to be allocated among the parties shall be as follows **:-**

First Party **:**

Second Party **:**

8. Representation of the Parties

Each Party represents to the other Party as of the date of this Agreement that **:-**

a) Such Party is duly organised, validly existing all in good standing under the laws of its incorporation and has all requisite power and authority to enter into this Agreement;

b) The execution, delivery and performance by such Party of this Agreement has been authorised by all necessary and appropriate corporate or governmental action and a copy of the extract of the charter documents and board resolution /power of attorney in favour of the person executing this Agreement for the delegation of power and authority to execute this Agreement on behalf of the Joint Venture party is annexed to this Agreement, and will not, to the best of its knowledge **:-**

(i) Require any consent or approval not already obtained;

(ii) Violate any applicable law presently in effect and having applicability to it;

(iii) Violate the memorandum and articles of association, by-laws or other applicable organisational documents thereof;

(iv) Violate any clearance, permit, concession, grant, license or other governmental authorisation, approval, judgment, order or decree or any mortgage agreement, indenture or any other instrument to which such Party is a Party or by which such Party or any of its properties or assets are bound or that is otherwise applicable to such Party; or

(v) Create or impose any liens, mortgages, pledges, claims, security interest, charges or encumbrances or obligations to create a lien, charge, pledge, security, interest, encumbrances or, mortgage in or on the property of such Party, except for encumbrances that would not, individually or in the aggregate, have a material adverse effect on the financial condition or prospects or business of such Party so as to prevent such Party from fulfilling its obligations under this Agreement.

c) This Agreement is the legal and binding obligation of such Party, enforceable in accordance with its terms against it; and

d) There is no litigation pending or, to the best of such Party's knowledge, threatened to which it or any of its Affiliates is a Party that presently affects or which would have a material adverse effect on the financial condition or prospects or business of such Party in the fulfilment of its obligations under this Agreement.

9. Termination

This Agreement shall be effective from the date hereof and shall continue in full force and effect until completion of Defect Liability Period under and in accordance with the Contract, in case the Project is awarded to the Joint Venture. However, in case the Joint Venture does not pre-qualify for the Project or does not get selected for award of the Project, the Agreement will stand terminated in case the Applicant does not pre-qualify or upon return of the Bid Security/Earnest Money Deposit by the Authority to the Bidder, as the case, may be.

10. Miscellaneous

a) This Joint Bidding Agreement shall be governed by Laws of India.
b) The Parties acknowledge and accept that this Agreement shall not be amended by the Parties without the prior written consent of the Authority.

IN WITNESS WHEREOF THE PARTIES ABOVE NAMED HAVE EXECUTED AND DELIVERED THIS AGREEMENT AS OF THE DATE FIRST ABOVE WRITTEN

SINGED, SEALED AND DELIVERED	SINGED, SEALED AND DELIVERED
For and on behalf of	
LEAD member by **:**	SECOND PART
(Signature) (Name) (Designation) (Address)	(Signature) (Name) (Designation) (Address)

WITNESS

In the presence of **:**

1. 2.

SCHEDULE 'A'(BOQ) : NOTES
(LIST OF ITEMS OF WORKS)

1. GENERAL:-

1.1. This SCHEDULE is divided into **16 PARTS** as mentioned here-in-below **:-**

(i)	SCHEDULE 'A' PART–I(A)	BUILDING WORKS
(ii)	SCHEDULE 'A' PART–I(B)	TRAVERSE RETAINING WALL ESH 1, 2, 3 & 4
(iii)	SCHEDULE 'A' PART–II	SITE CLEARANCE, SITE DEVELOPMENTS & EARTH WORKS
(iv)	SCHEDULE 'A' PART–III	INTERNAL WATER SUPPLY
(v)	SCHEDULE 'A' PART–IV	INTERNAL ELECTRIFICATION
(vi)	SCHEDULE 'A' PART–V	ROAD, PATH, CULVERT & HARDSTANDING
(vii)	SCHEDULE 'A' PART–VI	EXTERNAL WATER SUPPLY
(viii)	SCHEDULE 'A' PART–VII	DG SET
(ix)	SCHEDULE 'A' PART–VIII	EXTERNAL ELECTRIC SUPPLY
(x)	SCHEDULE 'A' PART–IX	AREA DRAINAGE
(xi)	SCHEDULE 'A' PART–X	COMPOUND WALL, FENCING & GATE
(xii)	SCHEDULE 'A' PART–XI	RETAINING WALL
(xiii)	SCHEDULE 'A' PART–XII	FIRE ALARM SYSTEM
(xiv)	SCHEDULE 'A' PART–XIII	LIGHTNING PROTECTION
(xv)	SCHEDULE 'A' PART–XIV	FIRE FIGHTING SYSTEM
(xvi)	SCHEDULE 'A' PART–XV	MISC B/R, E/M SERVICES (TO BE QUOTED BY CONTRACTOR)

1.2. Description of Building/Structures and Services/Works given in various Parts of SCHEDULE 'A' are in brief. These are deemed to be amplified and read in conjunction with Special Conditions, Particular Specifications, Specification for Materials and Workmanship and Conditions in relevant Trade Section of the MES STANDARD SCHEDULE OF RATES PART-I (2009), PART-II (2020) and Contract Drawings.

1.3. **ALL PRE-PRICED SCHEDULE :-**

1.3.1. Pre-priced Rates inserted under Column 4 of SCHEDULE 'A' PART-II to SCHEDULE 'A' PART-XIV are deemed to be at par with the Rates contained in the MES STANDARD SCHEDULE OF RATES PART-II (2020) or ANALOGOUS RATES thereto. Contractor's attention is invited to Condition 6A(B) of IAFW-2249, where under the LUMP SUM PRICE shall be worked out by him independently of the Prices or Rates inserted by MES in the Tender and irrespective of any Errors or inaccuracies therein.

1.3.2. The tenderer to work out detailed quantity for each building / structure listed in Schedule "A" Part-I(A) apply his own rate including all tax as specified and quote the lumpsum amount that he will charge for the complete Schedule "A" Part-I(A). The tenderers to quote their lump sum amounts including all tax for all pre-priced parts of Schedule 'A'. It is an express Condition of the Contract that the tendered amount shall be deemed to include for the full and entire completion of the items or building/ structure of work and the contractor shall have **no claims on account of any error in the unit rates/prices inserted by MES**.

1.3.3. The Lump Sum Amount shall be Quoted / entered in Figures at **green Space on the BOQ (GENERAL SUMMARY).**

1.4. In SCHEDULE 'A' PART-XV the UNIT RATES for various Items shall be entered by the Tenderer / Bidder in FIGURES under Green Column of BOQ and quoted amount of each items shall be deemed to include for the full and entire completion of the items of work.

1.5. The under mentioned remarks shall be deemed to have been against each item of BOQ:-

(a) Refer list of drawing

(b) Refer note No : All applicable Schedule 'A' Notes

SCHEDULE 'A'(BOQ) : NOTES
(LIST OF ITEMS OF WORKS)

1.6. The under mentioned remarks shall be deemed to have been inserted in respective columns against each item of Schedule 'A' parts prepriced in tender by MES:-

(a)	Under Column 3 (Drg No.)	:	Refer List of Drawings
(b)	Under Column 7 (Period of Completion of Individual Item from the Date of handing over Site)	:	Refer Note No. 1.23 of SCHEDULE 'A' NOTES here-in-after
(c)	Under Column 8 (REMARKS)	:	Refer SCHEDULE 'A' NOTES for all Items

1.7. Probable distributions of various Items of Internal / External Services are indicated on Drawings. These are tentative and may be varied as per Site Requirement, where necessary, at the discretion of the Engineer-in-Charge. The effect of such Changes/Variations in quantities due to realignment or re-sitting shall be deemed to be included in the QUOTED RATES and no Price Adjustment on this account shall be allowed.

1.8. In case, details in respect of Items shown on Main Drawings are not given in the Drawings referred to in the Main Drawings then the same shall be followed from any Other Drawings included in the LIST OF DRAWINGS. Any Drawings mentioned in any of the Contract Drawings if inadvertently not included in the LIST OF DRAWINGS shall also be deemed to form Part of the Contract. The Tenderer shall see such Drawings in the office of ACCEPTING OFFICER/CWE/GE.

1.9. The LUMP SUM AMOUNT quoted against SCHEDULE 'A' PART-I(A), SCHEDULE 'A' PART-I(B), SCHEDULE 'A' PART-II to SCHEDULE 'A' PART-XIV and Unit Rate(s) Quoted against Items catered for under SCHEDULE 'A' PART-XV shall include all TAXES/DUTIES/CESS including GST, etc.

1.10. Unless otherwise specifically mentioned, unit rate of Each Item of Work is inclusive of all 'Material & Labour' or 'Supply & Fixing' or 'Supply & Laying' including Tools and Plants, Installing, Commissioning and Testing complete as required for entire Completion of Works.

1.11. The Lump Sum Amount quoted for Works under SCHEDULE 'A' PART-I(A), SCHEDULE 'A' PART-I(B), SCHEDULE 'A' PART-II to SCHEDULE 'A' PART-XIV & Unit Rate(s) quoted against Items catered for under SCHEDULE 'A' PART – XV shall be deemed to include for all minor extra constructional details which have not been specifically shown on drawings or given in particular specification but are essential for the execution of works and services in the workmen like manner, sound construction and established engineering practice. In all the above and similar cases the details indicated elsewhere in the drawings that are similar or near similar to the missed out items of works shall be followed. In the absence of any other similar or near similar details, minimum essential requirement for completion of the work from the structural and utility point of view shall be deemed to be included in the quoted lump sum. The details of such items not specifically shown on drawings/specified shall be furnished during execution of works.

1.12. All Reinforcement shall be of TMT Bars (Fe 500D) (except Mild Steel Round Bars where specifically shown in Drawings/mentioned in Schedule 'A' Items) irrespective of whatever shown on Drawings or Specified elsewhere in these Tender documents. However, wherever 6mm dia Mild Steel Round Bars is indicated in the Drawing as reinforcement, the same shall be amended to read as 8mm dia TMT Bar (Fe 500D).

1.13. The description of the Items / Works mentioned are in brief and be read as "Complete all as specified and shown on Drawings" whether specifically written at the end of respective Item in particular or not.

1.14. Special Conditions and Preambles to Items given in MES Schedule Part-I and MES Schedule Part-II under respective Trades shall be applicable. If any Provision in the description of Items of Schedule 'A' and/or in Particular Specifications is at variance with the Provisions in Special Conditions and Preambles to MES Schedule Items, the Provisions in description of Items of Schedule 'A' and Provisions in Particular Specifications shall take precedence thereover.

1.15. Government shall not Pay anything Extra if the Contractor has to Pay Royalty to the State Government for the Earth brought to Site by him from Outside of the MD Land.

1.16. In case any Royalty is required to be Paid by the Contractor for bringing Approved Earth from Outside of the MD Land, the Rate quoted shall be deemed to include for the same and nothing Extra shall be Paid to the Contractor by the Department.

SCHEDULE 'A'(BOQ) : NOTES
(LIST OF ITEMS OF WORKS)

1.17. The Rate quoted for Rough Excavation is independent of the Actual Method of Excavation permitted by the Private Land Owners and nothing Extra shall be Paid by the Department.

1.18. Blasting in the Work site is not permitted.

1.19. Excavation and earthwork required for the execution of items covered under Schedule 'A' Part-I(B) and Schedule 'A' Part- V to Part- XV shall be measured and paid under Schedule 'A' Part-II except where excavation and earthwork specifically included in the description of particular items of respective Schedule 'A'.

1.20. Cost of Testing including Transportation of Materials or Equipments or Items, provision of all facilities for Testing in accordance with Specifications and/or IS shall be borne by the Contractor unless otherwise mentioned.

1.21. **VISIT TO SITE**

The Tenderer are advised to visit the Work Site by taking prior appointment with GE either in person before uploading of BOQ. The Tenderer shall have no Claim what-so-ever on this account at a Later Stage whether he has actually inspected the Work Site or not.

1.22. **LAYOUT**

1.22.1. Layout of Building/Structures and Works Services indicated in the SITE PLAN are Tentative and no Adjustment in Price shall be done on account of Final Approved Layout.

1.23. **PERIOD OF COMPLETION :-** The entire Work under the Contract including its Connected Services shall be completed within **1080 days** from the Date of Commencement as per WORK ORDER No. 1 to be issued by the GE. Site for all Works i.e Building Works and other Services shall be handed over simultaneously on Date of Commencement indicated in WORK ORDER No. 1 and their Completion Period shall run concurrently.

1.24. **QUOTED RATES/PERCENTAGES/AMOUNT**

1.24.1. The LUMP SUM quoted by the Tenderer shall be deemed to include for All Minor Details of construction which are not specifically shown on Drawings or specified in Particular Specification but which are essential for execution of WORKS/ SERVICES in Workman like manner and sound Construction. In case of differences in opinion of the Contractor and the GE as to whether or not a certain Item of Work constitutes Minor Details of Construction, the Decision of ACCEPTING OFFICER shall be final, conclusive and binding.

1.25. **TAXES, DUTIES & LEVIES, ETC.:-**

1.25.1. The rate quoted by the tenderer shall be inclusive of all types of taxes, levies including GST, labour welfare tax etc. as prevailing on the date of submission of tender.

1.25.2. Certain taxes such as central excise duty, service tax, additional custom duty, state level value added tax, octori and other levies which were applicable on interstate transportation of goods are subsumed by GST, thus special condition for **"Reimbursement/refund on variation in prices in taxes directly related to contract value"** as specified here-in-after in tender shall be deemed to be amended incorporating GST in lieu of taxes mentioned in the condition but subsumed by GST.

1.25.3. Cost of Testing including Transportation of Materials or Equipments or Items, provision of all facilities for Testing in accordance with Specifications and/or IS shall be borne by the Contractor unless otherwise mentioned.

1.26. **SAFETY MEASURES, PRECAUTION'S, RISKS ETC.,**

1.26.1. The Work shall be carried out with utmost care to ensure that no damage to Existing/Adjoining Work is done failing which the damage, if any done shall be rectified by the Contractor to match with the Existing/ Adjoining Work to the entire satisfaction of Engineer-in-Charge under Contractor's own arrangement and at his own expenses.

1.26.2. Suitable Tools, Plants, Equipments, Mechanism, etc., as considered necessary shall be adopted during execution of the Work. The Contractor shall take all Precautions, Safety Measures, etc., to avoid any Damage, Miss-happening, Accident, etc., to the Workmen engaged by him to carryout

the Work. The UNIT RATES quoted by Contractor shall be deemed to have included the Element of adopting Safety Measures, Precautions and also the Risks, etc. involved in Work and nothing EXTRA shall be admissible on this account.

1.26.3. **The Contractor has to follow all the directives issued by MES / Station Authority / Concerned State Authority / GOI to avoid the spread of Corona Virus (COVID-19). For this account nothing extra shall be admissible.**

1.27. **METHOD OF MEASUREMENTS**

1.27.1. The method of Measurements to be adopted in the Contract shall be as laid down in MES STANDARD SCHEDULE OF RATES PART-I & II, **IS-1200 (latest)** and relevant BIS standard. The General Rules, Special Conditions, Preambles and Other Provisions in MES STANDARD SCHEDULE OF RATES PART-I & II, **IS-1200** and relevant BIS standard shall be deemed to apply to the Work under this Contract unless otherwise mentioned in the Tender documents. However, where the UNITS of Work given in Column 4 against Items of SCHEDULE 'A' are at variance with that mentioned against respective Items in MES STANDARD SCHEDULE OF RATES , the Measurements of Work against such Items of SCHEDULE 'A' shall be done as per the UNITS given in SCHEDULE 'A'.

2. FOR SCHEDULE 'A' PART-I(A)

2.1. **GENERAL**

2.1.1. Buildings and/or Item of Works as indicated in SCHEDULE 'A' PART-I(A) includes Verandah, Steps, Built-in-Furniture, Platform (as applicable and/or as shown on Drawing) etc. likewise (even if not specifically mentioned in description of Items under SCHEDULE 'A' PART-I(A) all as shown in Key Plans and other Drawings and as specified in Particular Specifications are included in the scope of Work of this Tender unless otherwise specifically stated to be excluded in SCHEDULE 'A' or in Particular Specifications & Contractor's quoted LUMP SUM against SCHEDULE 'A' PART-I(A) shall be deemed to include for the above Provisions.

2.1.2. The LUMP SUM quoted against SCHEDULE 'A' PART-I(A) Item, shall deemed to include for all Items of Works complete as specified in Particular Specifications and shown on DRAWINGS, NOTES thereon for completion of Work and all minor details though not specifically shown on DRAWINGS or specified in Particular Specifications except Works covered under ALL REMAINING SCHEDULE as specified here-in-before or unless specifically stated elsewhere in the Tender documents.

2.1.3. Cutting and Forming Chases in Masonry, Concrete Work, etc. including Cutting and or Leaving Holes/ Recesses, Sinking, etc. where required and as directed by the Engineer-in-Charge shall be done as far as possible while the Work is in Progress. The Cost of Materials and Labour for Cutting and/or Forming Chases, Cutting or Leaving Holes/Recesses, Sinking and Making Good in Cement Mortar 1:3 for filling up to 20mm and in PCC 1:2:4 (12.5mm Graded Crushed Stone Aggregate) for filling more than 20mm shall deemed to be included in the LUMP SUM quoted by the Contractor. In addition cutting for Forming Chases, Cutting or Leaving Holes/Recesses, Sinking and Making Good in Cement Mortar 1:3/PCC 1:2:4 (12.5mm Graded Crushed Stone Aggregate) in connection with the Work required in Plumbing, Internal Water Supply, Internal Electrification, etc. shall deemed to be included in the UNIT RATES of respective Building Works in SCHEDULE 'A' PART-I(A) & I(B). However no Price Adjustment shall be made for any variation in the Items/ Quantities of Chases/Holes /Recesses, etc. consequent upon Changes in Quantities of Plumbing, Internal Water Supply, Internal Electrification Works, etc.

2.1.4. In all the above and other similar cases, the details indicated elsewhere in the drawings that are similar or near similar to the missed out items of work shall be followed. In the absence of any other similar or near similar details, minimum essential requirement for completion of the work from structural and utility point of view shall be deemed to be included in the lump sum /rates quoted by contractor.

2.1.5. In case of varying provisions between Schedule of Finishes drawings and other drawings with regard to finishes except cooking platform, bar counter, service counter, reception counter, step of stair case, other steps, Lab bench, shelve, wall cladding & sloping roof finishing the Schedule of Finishes drawing(s) shall take precedence over other drawings and cooking platform, bar counter, service counter, reception counter, step of stair case, other steps, Lab bench, shelve finishes, wall cladding & sloping roof finishing shall be provided as per architectural drawing. However, in case there are varying provisions with regard to height of skirting/dado in Schedule of Finishes drawings and other drawings, provisions in other drawings shall be followed.

SCHEDULE 'A'(BOQ) : NOTES
(LIST OF ITEMS OF WORKS)

2.2. **LUMP SUM QUOTED** by the Contractor for the Building/Works under SCHEDULE 'A' PART-I(A) shall be deemed to include for the following **:-**

2.2.1. **EXCAVATION AND EARTHWORK**

2.2.1.1. Surface Dressing and Leveling of the Premises around 3.0 METRE Perimeter of the Building from External Wall.

2.2.1.2. The existing Areas/ground where the Buildings / Structures covered in SCHEDULE 'A' PART-I(A) are to be provided, are assumed not having much undulations. In case large undulations are existing at site and site clearance/development works are required to achieve the modified ground level, the payment for such site clearance/development works shall be paid separately. At the initial stage(without disturb existing ground levels), the levels of the existing ground shall be recorded as per methods as specified in relevant clause in particular specification and final proposed modified ground levels shall be approved by CWE **under intimation to AO.**

2.2.2. **ANTI-TERMITE TREATMENT**

2.2.2.1. Pre-construction anti-termite treatment to the building listed under particular specification.

2.2.2.2. Pre-construction Anti-Termite Treatment shall be carried out as per IS, to the Building of SCHEDULE 'A' PART- I(A) & PART-I(B) as mentioned as above and its Cost shall be deemed to have been included in the Cost of Building in SCHEDULE 'A' PART- I(A) & PART-I(B).

2.2.2.3. Irrespective of whatever Shown/Specified in Tender document the Contractor has to give the Guarantee for Anti-Termite Treatment for TEN YEARS.

2.2.3. Damp Proof Course (DPC) as specified.

2.2.4. Seismic Strengthening Measures as specified.

2.2.5. **WATER PROOFING ROOF TREATMENT:** Heavy water proofing roof treatment to the building listed under particular specification. Irrespective of whatever Shown/Specified in Tender document the Contractor has to give the Guarantee for Water Proofing Treatment over RCC Roof Slabs for TEN YEARS.

2.2.6. PCC Plinth Protection, PCC Plinth Protection with PCC Surface Drain as indicated/shown in Drawings or specified in tender document.

2.2.7. RCC lintels band shall be provided as per mentioned structural drawing.

2.2.8. RCC lintels beam/continuous lintel beam shall be provided over all doors, windows or any openings if required as per site condition and as well as structural purpose.

2.2.9. Masonry work under external plinth beam shall be provided as mentioned main structural drawing/ CESZ/STD/26

2.2.10. Builders hardware for doors/windows etc. wherever not indicated on drawings but essential for functional requirement shall be provided and deemed to be included in quoted rate.

2.2.11. Stair Cases and Railings, etc. as indicated/shown in Drawings.

2.2.12. Steps, Ramp, Platforms, etc. as indicated/shown in Drawings.

2.2.13. Crumple Joints as indicated/shown in Drawings.

2.2.14. All structural details & structural note shall be provide as shown/ mention in main drawing & TD drawing No CESZ/STD/26.

2.2.15. Fitting and Fixture, Fitments, Built-in-Cupboard, Built-in-locker, Cabinets below Platforms, Other Furniture Articles as indicated/shown in Drawings.

2.2.16. PVC (HDPE) water tank over 75mm thick cement concrete 1:2:4 (using 20mm graded crushed stone aggregate) platform over roof slab including PVC valve with PVC float of size to suit inlet pipe for each service tank. Over flow pipe and wash out pipe (galvanised iron 'medium grade' of dia same as inlet pipe) shall be projected suitably beyond the external face of tank wall. Both the pipes than shall be connected with each other and one common pipe shall be brought down along the wall upto and beyond plinth protection to drain out the water in drain. Location of PVC tank if not shown on drawings shall be provided as directed by GE and capacity of tank shall be as given in schematic diagram for water supply (incase the capacity of tank is not mentioned in drawing or also in tender documents then the same shall be considered as 500 liters capacity).

2.2.17. Flushing Cistern (dual flushing type), Water Closet, Urinal, Readymade PVC Connecting Pipes to Wash Hand Basin, Flushing Cistern, stainless steel sink etc, Wash Hand Basin with bottle trap including Mirror. Readymade PVC Connecting Pipes shall be of 15mm Outer Diameter of Low Density Coloured Polythene Pipe, 450mm long conforming to IS-3076 with and including Brass Unions with Nuts on Both Sides complete.

2.2.18. **Local Sunk** shall be provided at upper RCC floors of the buildings and shall be complete with all as shown on TD drawing No. CESZ/2015/TD/011 and as specified.

SCHEDULE 'A'(BOQ) : NOTES
(LIST OF ITEMS OF WORKS)

2.2.19. Nahani trap with stainless steel grating, gully trap with grating, soil, waste and vent pipe, HDPE drain pipe upto and including first manhole and gully trap where bath/WC/toilet are occurring. The average distance of Gully Trap if not shown on Drawings shall be considered 1.50 METRE from the outer face of External Wall of the Buildings. The average distance of First Manhole if not shown on Drawings shall be considered as 3000mm from the Outer Face of External Wall of the Buildings.

2.2.20. Provision for hanging arrangement for ceiling fan and fixing arrangement for exhaust fan, AC, etc. (if shown on Drawings) shall be deemed to be included against SCHEDULE 'A' PART- I(A).

2.2.21. The foundation depth wherever marked on drawings shall be taken as depth from natural ground level and in no case any foundation shall be laid on filled earth.

2.2.22. Wherever door/window/ventilator is marked but type of door/window/ventilator is not shown in drawings, the same shall be provided with similar type of door/window/ventilator as provided for other rooms in the same building with complete all as specified in particular specifications and shown on TD drawings, except toilets where PVC doors and frames shall be provided if nothing is marked on drawings.

2.2.23. Provision of required type of "CB-3 (cup board) for living room" & "CB-4 (cup board) for other than living room i.e store room, cook house, etc" all as specified in particular specifications and shown on TD drawings wherever cup board (CB) is marked but type of cup board (CB) is not shown in main drawings.

2.2.24. Provision of required type of RCC shelves / RCC loft / working platform shall be all as specified in particular specifications and shown on drawings.

2.2.25. The water proofing compound to be used in this work shall be liquid water proofing compound of approved make and shall be used @ 200 ml per 50 kg bag of cement or as per manufacturer instructions whichever is more. The water proofing compound shall be mixed in water used for concrete/mortar.

2.2.26. Provision of 3 coat of white wash on internal walls, 2 coats of exterior acrylic emulsion paint over a coat of primer on external walls and 3 coat of white wash on soffits of RCC slab / ceiling surfaces / chajjah after preparation of surfaces, wherever Schedule of Finishes are missing for walls and ceilings respectively.

2.2.27. Provision of 40 mm thick PCC M-15(Nominal Mix) (using 20 mm graded crushed stone aggregate) finished even and smooth using extra cement in flooring all as specified in particular specifications wherever floor finishes are missing in schedule of finishes.

2.2.28. Provision of 20 mm thick kota stone on RCC counters/platforms/cooking platforms/shelve/stair case step in one piece (Max 1.00 m length) all as specified in particular specifications wherever finishes over counters/platforms/cooking platforms/shelve/ reception counter / Lab bench/stair case step are missing in drawings.

2.2.29. In addition to list of drawings attached with tender documents, if any other drawing(s) are referred to in drawings or in the tender documents, but not listed in the list of drawings, these drawing shall also form part of the tender documents. The contractor shall be deemed to have taken the details etc, shown in these missing drawing into consideration before quoting the lump sum amount, nothing extra shall be paid on this account.

2.2.30. In case where type, size and reinforcement details of footing, pile, pile cap, beam, continuous lintel beam/chajja, slab, columns, fins, truss & purlin, etc are not indicate on structural drawing these shall be provided as directed by accepting officer as per details of similar footing, pile, pile cap, beam, continuous lintel beam/chajja, slab, columns, truss & purlin etc and cost of same and shall be deemed to be included in the lump sum. The decision of the accepting officer as to the similar of footing, pile, pile cap, beam, continuous lintel beam/chajja, slab, columns, truss & purlin etc shall be final binding.

2.2.31. **Comaflauge painting with exterior emulsion paint over a coat of primer including preparation of surfaces(disruptive pattern) shall be provided in external surfaces of building mentioned in Sl No 1 to 5 of Schedule 'A' part-1(A).**

2.2.32. **RAISING OF EXISTING GROUND LEVELS :**

2.2.32.1. Irrespective of whatever is shown on architectural and structural main drawings, after surface dressing on existing ground level, the raising of existing ground level up to made up ground level (MGL) as per filling height mentioned in drawing of respective buildings & structure is to be carried out by filling with approved earth.

2.2.32.2. For filling from above original/ natural ground level **(OGL/NGL)** to plinth height/finished floor level**(FFL)** of respective buildings/structure as shown on main drawings/ as mentioned here-in-

after, the lump sum quoted rate shall be inclusive of the cost of approved earth obtained from outside MOD land. The soil /earth obtained from excavation shall be used for returning filling after separating it from grass, vegetations and other foreign materials. Surplus earth/soil available after returning filling of respective building shall be re-used for filling in original ground level (OGL) to plinth height/finished floor level (FFL). The lead for this purpose shall be considered as not exceeding 50 m for pricing deviation if any. However to calculate the quantity of earth to be brought from outside MOD land of individual building/structure, same shall be calculated after deduction of surplus earth. The earth shall be brought from outside MOD land with any lead, any royalty including spreading, leveling watering, rolling and compaction complete and nothing extra shall be paid on this account.

2.2.32.3. Irrespective of what is shown on architectural and structural main drawings, the lumpsum price inserted in respect of various buildings/structures mentioned in Sch 'A' part-I(A) caters for height of entire structure from OGL to MGL including filling as mentioned in drawing. Any change during execution as per site condition etc. shall be regularised through DO. The bidder/tenderer should quote their offer accordingly.

2.2.32.4. The foundation depth(DF) for Magazine shed wherever marked on drawings shall be taken as depth from lowest point of natural ground level and in no case any foundation shall be laid on filled earth.

2.2.32.5. **Fire Symbols:-** All magazine sheds shall be provided with two tablets, one on each side, of size 1200 mm x 1200 mm x 10 mm made out of cement plaster fully painted with fluorescent paints and inscribed with digit there on. The overall size of digit shall be length 500 mm x width 350 mm x thickness of line 100 mm. The fire symbol tablets shall be of square, octagonal and cross shape and painted with white with black colour line border/orange and the digits shall be painted with black colour and all as per relevant IS and as directed at site.

2.2.32.6. **Explosive Content Boards** :-Two Nos of cemented explosive content boards per magazine shed shall be provided one on each inner side long of each magazine shed. The boards shall be made out of cement plaster fully painted with fluorescent paints and as directed.

2.2.33. Cost of furniture i.e. in-built furniture and locker as marked in architectural drawings shall be deemed to be included in quoted rate.

2.3. **SBC (SAFE BEARING CAPACITY) for bldgs/structure listed under Schedule 'A' Part-I(A) & I(B):-** The SBC based on which foundation has been designed has been indicated in the structural drawings. While quoting the offer, the tenderer should consider the SBC given in the main structural drawings. However, in case GE observes soil of unusual nature at site during the execution, which may result in lower SBC. GE shall get tested the SBC of soil and if actual SBC found at site is at variance with reference to SBC mentioned in main drawings, the case shall be referred to Accepting Officers for approval and the change shall be regularized through deviation order (DO) and the cost for such testing of SBC of soil shall be borne by the department.

3. FOR SCHEDULE 'A' PART-I(B) & SCHEDULE 'A' PART-II TO SCHEDULE 'A' PART XIV

3.1. All Quantities shown under Col. 5 are PROVISIONAL. Any variation in Quantities if considered necessary due to Site Requirement and/or any Other Technical Reasons as decided by the GE may be executed to the extent as specified in Condition 7 of General Conditions of Contract (IAFW-2249).

3.2. Unless otherwise specified, the UNIT RATES of SCHEDULE 'A' PART-I(B) & SCHEDULE 'A' PART-II to SCHEDULE 'A' PART-XIV under Col. 4/5 of Each Item of Work inserted by MES and/or Rate quoted by the Tenderer shall be deemed to includes 'Materials and Labours' i.e. 'M&L' or 'Supplying and Fixing' i.e. 'S&F' or 'Supplying and Laying' complete unless otherwise specifically mentioned.

3.3. Special Conditions, Conditions in MES STANDARD SCHEDULE PART-I and the PREAMBLES to the Items given in MES STANDARD SCHEDULE OF RATES PART-II under respective Trades shall be applicable, if any provision in the description of Items of SCHEDULE 'A' and PARTICULAR SPECIFICATIONS are at variance with the provision laid in the Conditions of MES SCHEDULE and PREAMBLES to MES SCHEDULE Item, the provision in description of Items of SCHEDULE 'A' and provision in PARTICULAR SPECIFICATIONS shall take precedence.

SCHEDULE 'A'(BOQ) : NOTES
(LIST OF ITEMS OF WORKS)

4. APPLICABLE FOR SCHEDULE 'A' PART- XV

4.1. All Quantities shown under Col. 6 are PROVISIONAL. Any variation in Quantities if considered necessary due to Site Requirement and/or any Other Technical Reasons as decided by the GE may be executed to the extent as specified in Condition 7 of General Conditions of Contract (IAFW-2249).

4.2. Special Conditions, Conditions in MES STANDARD SCHEDULE PART-I and the PREAMBLES to the Items given in MES STANDARD SCHEDULE OF RATES PART-II under respective Trades shall be applicable, if any provision in the description of Items of SCHEDULE 'A' and PARTICULAR SPECIFICATIONS are at variance with the provision laid in the Conditions of MES SCHEDULE and PREAMBLES to MES SCHEDULE Item, the provision in description of Items of SCHEDULE 'A' and provision in PARTICULAR SPECIFICATIONS shall take precedence.

5. FOR ALL SCHEDULE 'A' PARTS (EXCEPT SCHEDULE 'A' PART-XV)

5.1. VALUATION OF DEVIATIONS APPLICABLE FOR LUMP SUM CONTRACTS BASED ON DRAWINGS & SPECIFICATIONS.

5.2. Refer Condition 62 (B) & (C) of the General Conditions of Contracts (IAFW-2249).

5.3. In case of any DEVIATION involved in respect of SCHEDULE 'A' PART-I(A), SCHEDULE 'A' PART-I(B) & Schedule 'A' Part-II to Part-XIV, the Pricing of the Deviation shall be done at applicable RATES of MES SCHEDULE PART-II (2020) enhanced/reduced by Contractor's derived percentages i.e Represent (%) Percentage of BOQ.

5.4. Items of Work involved in Deviation Order for which there are no Rates in MES Schedule be priced in accordance with the Conditions 62(G) of IAFW-2249.

5.5. **APPROVAL OF SPECIAL (STAR) RATE**

5.5.1. In case of any deviation, mode of pricing shall be decided by Accepting Officer in terms of Condition 62 of IAFW-2249.

5.5.2. In the event of a deviation order involving fixation of Special (Star) Rate, Draft Rate shall be prepared by GE (within a maximum period of 30 days) while initiating the proposal for deviation seeking approval of Accepting Officer and notified to contractor. While notifying the Draft Rate, it will be clearly stipulated that the same is merely an estimated rate and firm rate shall be fixed based on actual and receipt of supporting documents from contractor such as vouchers/literature of product/test certificates etc. (as applicable) on completion of the work involving Star Rate. Any objection to the method of fixing Star Rate will be dealt as per conduction 7 of IAFW-2249.

5.5.3. Draft Star Rate shall be made based on market enquiry through telephonic enquiry / quotations / email / rate lists / internet based sources, material & labour constants available in various civil engg books and record available in respect of Star Rates approved in the past for similar item of work etc. Contractor may also assist GEs office in preparation of draft Star Rate.

5.5.4. The Draft Star Rate shall be purely a draft rate and shall not be used for claiming final payment during execution of work. However GE shall allow part payment to the tune of 80% during execution to avoid any financial hardship to contractor.

5.5.5. After completion of the item of work involving Star Rate, contractor shall submit the vouchers / literature of product / test certificates (as applicable, decision of GE being final in case of any disagreement) for finalisation of Star Rate. The Star Rate shall be technically checked by DCWE (C) / Director (C) depending upon the financial effect and approved by competent authority within a period of one month from submission of the relevant documents by contractor as mentioned above.

5.5.6. The Star Rate as approved by competent authority after technical check by DCWE (C) / Director (C) depending upon the financial effect shall be referred as "the rate decided by GE" under Para 62(G) of IAFW-2249.

6. EXECUTION OF ELECTRICAL WORKS / ELECTRICAL COMPONENT OF WORKS

6.1. For execution of electrical works / electrical component of works requiring Electrical License in accordance with Rule 29 under Chapter III of Central Electricity Authority (Measures Relating to Safety and Electric Supply) Regulation, 2010, the contractor shall have valid Electrical License issued by the concerned State / Union Territory in the name of Firm or get the work executed through an agency having valid Electrical License issued by the concerned Sate / Union Territory where the work is being executed.

SCHEDULE 'A'(BOQ) : NOTES
(LIST OF ITEMS OF WORKS)

6.2. Prior to commencement of electrical works / electrical component of works, copy of valid Electrical License in the name of the contractor or copy of agreement with an agency having valid Electrical License alongwith agency's Electrical License shall be submitted by the contractor to the Accepting Officer through GE with copy to CWE. Incase contractor does not possess valid Electrical License and he intends to get such work executed through agency having valid Electrical License and with whom he has agreement, prior approval of such agency in all contracts shall be got done from competent authority (i.e. concern CWE) (concerned CE Zone incase there is no CWE). Copy of License shall be kept on record by GE as well as CE Zone.

6.3. GE and concerned AGE / Engineer-in-Charge shall ensure that electrical work / electrical component of works required to be executed in accordance with Rule 29 mentioned above are executed by contractor / agency having valid Electrical License, as applicable. In addition, Supervisor for execution of electrical works / electrical component of works employed by the contractor shall possess Supervisor Certificate of Competency issued by concerned State Government / Union Territory and the worker / tradesmen for execution of electrical works / electrical component of works shall hold necessary permit issued by concerned State Government / Union Territory.

6.4. Reference of Electrical License and details of Supervisor's Certificate of Competency and Permit of tradesmen employed for execution of electrical works / electrical component of works shall be duly incorporated in the Works Diary and relevant site documents respectively for the whole period during which such work is executed.

7. **INTEGRITY PACT (IP) :- Integrity Pact (IP)** is an integral part of contract and both parties are bound by its provisions.

8. **INSPECTION OF THE WORKS (DEFECTS LIABILITY PERIOD) :** {Refer Condition 46 of IAFW – 2249 (General Conditions of Contracts) (As amended under amendment No. 48 to IAFW- 2249)}:-

"The **Defects liability period for the scope of work(s) covered under the contract is TWENTY FOUR CALENDAR MONTHS**. Any defects in the work noticed during this period shall be rectified by the contractor forthwith, on demand in writing from the GE specifying the area affected not withstanding that the work executed might have been inadvertently passed, certified and paid. The contractor shall carry out the rectification of the defects notified at his own expense during the defects liability period and in event of his failure to do within a period to be specified by the GE in his demand, the GE shall undertake such rectification work at the risk and expense of the contractor**"**.

9. **YARD STICK :** - The Yard Stick against various Items of Schedule 'A' Part-I(A) has not been included in the tender documents. The same shall be prepared by JE(B/R) & AGE(B/R) of concerned sub division and shall be technical checked first by AGE(Contracts) of GE office and thereafter by DCWE/ACWE(Contracts) of CWE office and shall be approved by CWE. Copy of approved yardstick shall be forwarded to the Accepting Officer for his record and comments.

10. **HORIZONTAL AND VERTICAL SCREEN**:

The contractor shall be provide horizontal and vertical camouflage net matching with local terrain (forest area) to camouflage the site of work during the execution of work as approved by Engr-in Charge. The quoted rates are deemed to be inclusive of this cost. Nothing extra will be paid on this account.

11. **WEIGH BRIDGE:**

Weight bridge of minimum capacity of 40 tonne shall be installed at site for the measurement of steel, cement and other material etc. The quoted rates are deemed to be inclusive of this cost. Nothing extra will be paid on this account.

12. **PERFORMANCE SECURITY** :- Condition 19 of IAFW-2249 (amended vide Amendment No 47).

13. Works under this contract lies in **'UN-RESTRICTED'** Area.

SCHEDULE 'A'(BOQ) : NOTES
(LIST OF ITEMS OF WORKS)

14. All T & P shall preferably be new and in no case shall be older than 03 Years. Any T & P which remain unserviceable for more than 30 days shall be replaced by new one immediately.

15. The Contractors shall quote same rates for the identical items which may inadvertently appear in more than one place. If different rates are quoted by the tenderers for such identical items, the same shall be rationalized by considering the lowest quoted rate for such items permanent.

16. Wherever any reference to any Indian Standards occurs in the documents relating to this contract, the same shall be inclusive of all amendments issued thereto or revisions thereof, if any, up to the date of receipt of tenders.

17. The Contractor shall establish, maintain and assume responsibility for grades, lines, levels and benchmarks. He shall report any errors or inconsistencies regarding grades, lines, levels, dimensions etc. to the Engineer -in-Charge before commencing work. Commencement of work shall be regarded as the Contractor's acceptance of such grades, lines, levels, and dimensions. If at any time, any error appears due to grades, lines, levels and benchmarks during the progress of the work, the Contractor shall, at his own expense rectify such error, if so required, to the satisfaction of the Engineer -in-Charge. Nothing extra shall be payable on this account for any errors found at a later date.

18. The Contractor shall protect and maintain temporary/ permanent benchmarks at the site of work throughout the execution of work. These benchmarks shall be got checked by the Engineer-in-Charge or his authorized representatives. The work at different stages shall be checked with reference to bench marks maintained for the said purpose. Nothing extra shall be payable on this account.

19. The approval by the Engineer-in-Charge, of the setting out by the Contractor, shall not relieve the Contractor of any of his responsibilities and obligation to rectify the errors/ defects, if any, which may be found at any stage during the progress of the work or after the completion of the work.

20. The Contractor shall be entirely and exclusively responsible for the horizontal, vertical and other alignments, the level and correctness of every part of the work and shall rectify effectively any errors or imperfections therein. Such rectifications shall be carried out by the Contractor at his own cost to the entire satisfaction of the Engineer- in-Charge.

21. Any legal or financial implications resulting out of disposal of earth shall be sole responsibility of the contractor. Nothing extra over the schedule shall be paid on this account.

22. The Contractor shall take all precautions to abide by the environmental related restrictions imposed by any statutory body having jurisdiction in the area as well as prevent any pollution of streams, ravines, river bed and waterways. All waste or superfluous materials shall be transported by the Contractor, and disposed of at designated places only. No claim on account of site constraints mentioned above or any other site constraints such as lack of public transport, inadequate availability of skilled, semi-skilled or unskilled workers in the near vicinity, non-availability of construction machinery spare parts etc. or any other constraints not specifically stated here shall be entertained from the Contractor. Therefore, the tenderers are advised to visit site and get first-hand information of site constraints. Accordingly, they should quote their tenders. Nothing extra shall be payable on this account. Any hindrances claimed by the contractor on this account shall not be considered while giving extension of time under clause 5 of General condition of contract amended upto date till last date or extended last date of submission of tender.

23. The Contractor(s) shall take all precautions to avoid accidents by exhibiting necessary caution boards day and night. The contractor shall ensure entire necessary precaution during the entire period of work and site related activities to ensure full safety to workers and avoid any kind of accident. In case of any accident of labour's/ contractual staffs or any other human being the entire responsibility will rest on the part of the contractor both legally and financially and any compensation under such circumstances, if becomes payable, shall be entirely borne by the contractor. Existing drains, pipes, cables, over-head wires, sewer lines, water lines and similar services encountered in the course of the execution of work shall be protected against the damage by the contractor, at his own expense, for which nothing is payable. The contractor shall not store materials or otherwise occupy any part of the site in a manner likely to hinder the operation of such services.

24. Unless otherwise provided in the Schedule of quantities/BOQ, the rates tendered by the contractor shall be all inclusive and shall apply to all heights, lifts, leads and depths of the building and nothing extra shall be payable to him on this account.

SCHEDULE 'A'(BOQ) : NOTES
(LIST OF ITEMS OF WORKS)

25. The Contractor(s) shall take instructions from the Engineer-in-Charge regarding collection and stacking of materials at any place. No excavated earth or building rubbish shall be stacked on areas where other buildings, roads, services and compound walls are to be constructed. The stacking shall take place as per stacking plan however, if any change is required, the same shall be done with the approval of Engineer-in-Charge.

26. Any cement slurry added over base surface (or) for continuation of concreting for better bond is deemed to have been built in the items and nothing extra shall be payable or extra cement considered in consumption on this account.

27. The contractor shall make arrangement for Two (2 nos.) Inspection 4x4 totally covered vehicles (minimum 2000 CC engine with 185 mm ground clearance or equivalent. Model of vehicles shall not be older than 2017), from start to completion of entire work and upto 6 months after completion or till final bill is passed , whichever is earlier, at disposal of Engineer in charge to facilitate work inspection, quality control, coordination with multiple agencies and better liaisoning. The average mileage of each inspection vehicle shall be approximately 2500 Km/month. The inspection vehicle shall be made available for 12 hours per day on daily basis including holidays as per the direction of Engineer in charge. All expenses of this inspection vehicle including running and maintenance, fuel charges, driver's salary, toll tax, parking charges etc. shall be borne by the contractor. Recovery @ Rs. 2000/- per day per Inspection vehicles shall be made, if contractor fails to provide Inspection vehicles within 15 days of letter of award/acceptance. Recovery @ Rs. 2000/- per day per Inspection vehicles shall be made, in case of occurrence of a default i.e. non-availability of vehicle, breakdown of vehicle etc. If department decides for not availing the services of inspection vehicles than deduction @ Rs. 1500/- per day per vehicle shall be affected in running account bill or any amount payable to contractor. The quoted rates are deemed to be inclusive of this cost. Nothing extra will be paid on this account.

28. The contractor shall arrange one Additional inspection vehicle other than as mentioned above to enable Employer or his authorized representative(s) to attend site and inspect the execution of the work beyond normal working hours or on authorized public holidays, if required by Engineer in charge. Nothing extra will be paid on this account.

29. The contractor shall make arrangement for Helmets and leather shoes (meant of construction work at sites) for all field staff of the department during the entire period of construction for safety reasons. One helmet and two pairs of shoes per staff member (maximum ten members) of the departments per year shall be arranged by the contractor.

30. The contractor shall construct suitable godowns, yard at the site of work for storing all other materials so as to be safe against damage by sun, rain, damages, fire, theft etc. at his own cost and also employ necessary watch and ward establishment for the purpose at his cost.

31. The rates quoted by the Contractor are deemed to be inclusive of site clearance, setting out work, profile, establishment of reference bench mark(s), taking spot levels, construction of all safety and protection devices, barriers, preparatory works, working during monsoon, working at all depths, height, lead, lift and location etc. unless clearly specified otherwise in the description of item in schedule of quantities of work and any other incidental works required to complete this work. Total station survey instruments only shall be used for layout, fixing boundaries, centre lines and getting initial level and final level etc. Nothing extra shall be payable on this account.

32. The Contractor shall make all necessary arrangements to protect the work already executed from rain and for carrying out the further work, during monsoon including providing and fixing temporary shelters, protections etc. Nothing extra shall be payable on this account. Also, no claims for hindrance shall be entertained on this account.

33. In case of flooding of site on account of rain or any other cause and any consequent damage, whatsoever, no claim financially or otherwise shall be entertained not withstanding any other provisions elsewhere in the contract agreement. Also, the Contractor shall make good, at his own cost, the damages caused, if any. Also, no claims for hindrance shall be entertained on this account.

34. The contractor will take reasonable precautions to prevent his workman and employees from removing and damaging any flora (tree/plant/vegetation) from the project area.

35. The Contractor shall maintain all the work in good condition till the completion of entire work. The Contractor shall be responsible for and shall make good, all damages and repairs, rendered necessary due to fire, rain, traffic, floods or any other causes. The Engineer-in-Charge shall not be responsible for any claims for injuries to person/workmen or for structural damage to property

SCHEDULE 'A'(BOQ) : NOTES
(LIST OF ITEMS OF WORKS)

happening from any neglect, default, want of proper care or misconduct on the part of the Contractor or of any other of his representatives, in his employment during the execution of the work. The compensation, if any, shall be paid directly to the Department / authority / persons concerned, by the Contractor at his own cost. Nothing extra shall be paid on this account.

(SIGNATURE OF CONTRACTOR) **DY DIR/AAD (CONTRACTS)**
DATED: **FOR ACCEPTING OFFICER**

SCHEDULE 'A' PART I(A)

(LIST OF ITEMS OF WORKS & PRICES) (BUILDING & STRUCTURES WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Construction of a **AMN STORAGE ACCN URP HD 1.1, CHAMBER 5 X 2MT. (NEC) (URP NO.1, 2 & 4 WITH TRAVERSE** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings. Note: Items of traverse shall be measured and paid under seperate schedule i.e Schedule I(B).		30,615,800.00 **Each Block**	3.00	91,847,400.00		
2	Construction of a **AMN STORAGE ACCN URP HD 1.1, CHAMBER 6 X 2MT. (NEC) (URP NO. 3 WITH TRAVERSE** with all fittings, fixtures complete all as specified and shown on drawings. Note: Items of traverse shall be measured and paid under seperate schedule i.e Schedule I(B).		36,664,000.00 **Each Block**	1.00	36,664,000.00		
3	Construction of a **AMN STORAGE ACCN (W P SHED) HD 1.2, W/O TRAVERSE(ESH-5)** with all fittings, fixtures complete all as specified and shown on drawings.		2,846,600.00 **Each Block**	1.00	2,846,600.00		
4	Construction of a **AMN STORAGE ACCN (STORAGE) HD 1.3, W/O TRAVERSE (ESH-6)** with all fittings, fixtures complete all as specified and shown on drawings.		11,360,400.00 **Each Block**	1.00	11,360,400.00		
5	Construction of a **AMN STORAGE ACCN (STORAGE) HD 1.4, W/O TRAVERSE (ESH-7)** with all fittings, fixtures complete all as specified and shown on drawings.		5,898,700.00 **Each Block**	1.00	5,898,700.00		
6	Construction of a **RETURNED AMMUNITION AND EMPTIES AREA ESH NO.8** with all fittings, fixtures complete all as specified and shown on drawings.		4,833,400.00 **Each Block**	1.00	4,833,400.00		
7	Construction of a **RECEIPT AND ISSUE SHED/COUNTER** with all fittings, fixtures complete all as specified and shown on drawings.		1,719,300.00 **Each Block**	2.00	3,438,600.00		
8	Construction of a **AMN POINT MAIN OFFICE** with all fittings, fixtures complete all as specified and shown on drawings.		1,769,633.92 **Each Block**	1.00	1,769,633.92		

SCHEDULE 'A' PART I(A)

(LIST OF ITEMS OF WORKS & PRICES) (BUILDING & STRUCTURES WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
9	Construction of a **GARRAGES FOR 'B' VEH** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		1,078,000.00 **Each Block**	1.00	1,078,000.00		
10	Construction of a **FIRE STATION** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		4,359,700.00 **Each Block**	1.00	4,359,700.00		
11	Construction of a **FUEL STORE** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		908,700.00 **Each Block**	1.00	908,700.00		
12	Construction of a **GUARD HOUSE** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		2,541,370.17 **Each Block**	1.00	2,541,370.17		
13	Construction of a **MAIN GATE WITH INFRASTRUCTURE AT IN & OUT GATE** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		2,863,196.62 **Each Block**	2.00	5,726,393.24		
14	Construction of a **PUMP HOUSE CUM CHLORINATION PLANT** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		1,284,600.00 **Each Block**	1.00	1,284,600.00		
15	Construction of a **PUMP HOUSE** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		1,178,139.72 **Each Block**	1.00	1,178,139.72		
16	Construction of a **SENTRY ROOM** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		252,500.00 **Each Block**	3.00	757,500.00		
17	Construction of a **GARAGE FOR PARKING OF 33 CYCLES AND 02 SCOOTERS** with all fittings, fixtures items including complete all as specified and shown on drawings.		637,800.00 **Each Block**	1.00	637,800.00		

SCHEDULE 'A' PART I(A)

(LIST OF ITEMS OF WORKS & PRICES) (BUILDING & STRUCTURES WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
18	Construction of a **SHED FOR GEN SET** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		518,800.00 **Each Block**	2.00	1,037,600.00		
19	Construction of a **STATIC WATER TANK 225 KL** with all fittings, fixtures items including complete all as specified and shown on drawings.		2,512,800.00 **Each Block**	5.00	12,564,000.00		
20	Construction of a **SUB STATION BLDG** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		983,300.00 **Each Block**	1.00	983,300.00		
21	Construction of a **RCC UG SUMP 225 KL CAPACITY** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		2,485,100.00 **Each Block**	2.00	4,970,200.00		
22	Construction of a **GARAGE FOR 2 X TFF** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		2,480,500.00 **Each Block**	1.00	2,480,500.00		
23	Construction of a **UNIT STORAGE ACCN** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		2,313,997.03 **Each Block**	1.00	2,313,997.03		
24	Construction of a **GUARD ROOM WITH WATCH TOWER** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		2,708,041.83 **Each Block**	3.00	8,124,125.49		
25	Construction of a **WATCH TOWER** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		1,647,500.00 **Each Block**	3.00	4,942,500.00		

SCHEDULE 'A' PART I(A)

(LIST OF ITEMS OF WORKS & PRICES) (BUILDING & STRUCTURES WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
26	Construction of a **INSPECTION PIT** with all fittings, fixtures items including complete all as specified and shown on drawings.		119,300.00 **Each Block**	2.00	238,600.00		
27	Construction of a **ALARM POST** with all fittings, fixtures including complete all as specified and shown on drawings.		16,300.00 **Each Block**	4.00	65,200.00		
28	Construction of a **FIRE POINT** with all fittings, fixtures items including complete all as specified and shown on drawings.		29,500.00 **Each Block**	25.00	737,500.00		
29	Construction of a **INCINRETOR** with all fittings, fixtures items including complete all as specified and shown on drawings.		95,800.00 **Each Block**	1.00	95,800.00		
30	Construction of a **FIRE SYMBOL BOARD** with all fittings, fixtures items including complete all as specified and shown on drawings.		15,300.00 **Each Block**	10.00	153,000.00		
31	Construction of a **SHED NO DISPLAY BOARD** with all fittings, fixtures items including complete all as specified and shown on drawings.		15,300.00 **Each Block**	10.00	153,000.00		
32	Construction of a **COOK HOUSE WITH DINING HALL** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		6,247,616.28 **Each Block**	1.00	6,247,616.28		
33	Construction of a **GUARD ROOM** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		1,596,392.68 **Each Block**	3.00	4,789,178.04		

SCHEDULE 'A' PART I(A)

(LIST OF ITEMS OF WORKS & PRICES) (BUILDING & STRUCTURES WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
34	Construction of a **DRYING AND IRONING ROOM** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		668,300.00 **Each Block**	1.00	668,300.00		
35	Construction of a **SINGLE LIVING ACCN FOR JCO'S (JCO-01, DSC-01)** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		2,384,012.58 **Each Block**	1.00	2,384,012.58		
36	Construction of a **LIVING ACCN FOR 2 HAV AND 28 OR'S** with all fittings, fixtures and built in furniture items including complete all as specified and shown on drawings.		13,967,058.39 **Each Block**	1.00	13,967,058.39		
37	Construction of a **SCOOTER SHED FOR 2 HAV AND 28OR'S** with all fittings, fixtures items including complete all as specified and shown on drawings.		2,433,530.70 **Each Block**	1.00	2,433,530.70		
38	Construction of a **DHOBI GHAT** with all fittings, fixtures items including complete all as specified and shown on drawings.		134,300.00 **Each Block**	1.00	134,300.00		
39	Construction of a **GARBAGE BIN** with all fittings, fixtures items including complete all as specified and shown on drawings.		420,806.00 **Each Block**	1.00	420,806.00		
40	Construction of a **SEPTIC TANK AND SOAK PIT FOR 10 USERS** with all fittings, fixtures items including complete all as specified and shown on drawings.		276,400.00 **Each Block**	7.00	1,934,800.00		

SCHEDULE 'A' PART I(A)

(LIST OF ITEMS OF WORKS & PRICES) (BUILDING & STRUCTURES WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
41	Construction of a **SEPTIC TANK AND SOAK PIT FOR 50 USERS** with all fittings, fixtures items including complete all as specified and shown on drawings.		638,400.00 **Each Block**	1.00	638,400.00		

Total of Schedule 'A' Part-I(A) carried over to BOQ (General Summary) **249,608,261.56**

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART I(B)

(LIST OF ITEMS OF WORKS & PRICES) (TRAVERSE RETAINING WALL ESH 1, 2, 3 & 4)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Material & Labour for Anti Termite Treatment treating the soil under plinth protection at the rate of 5 litres of emulsion per square metre of surface area complete all as specified and as directed.		66.80 **Sqm**	1001.70	66,913.56		
2	Material & Labour for Anti Termite Treatment treating the soil along with external perimeter of building at the rate of 2.25 litre of emulsion per metre of the perimeter wall complete all as specified and as directed.		33.20 **Sqm**	1335.60	44,341.92		
3	Plain Cement Concrete 1:4:8 type D2 using 40 mm graded stone aggregate as in foundation filling and mass concrete complete all as specified and directed		5,483.10 **Cum**	524.58	2,876,324.60		
4	Reinforced Cement Concrete M-25 (Design mix) using 20 mm graded stone aggregate as in Walls, retaining walls, basement walls, ballast walls and the like, any thickness; above top of footings; including attached pilasters and buttresses complete all as specified and as directed.		8,028.60 **Cum**	3093.72	24,838,240.39		
5	High strength steel deformed TMT bars (Fe 500D) 10mm dia and over, cut to length, bent to shape required, including cranking, bending spirally for hooping for columns, hooking ends and binding with and including mild steel wire (annealed) not less than 0.9mm dia or securing with clips complete all as specified and directed.		82.60 **Kg**	463806.00	38,310,375.60		
6	Precast cement concrete block masonry with PCC 1:3:6 type C1 solid blocks, exceeding 10 cm in width and setting in cement mortar 1:6 including necessary formwork and moulds complete all as specified.		8,439.60 **Cum**	183.12	1,545,459.55		
7	Use and Waste of formwork to sides of concrete foundations, foundation beams, footing, bases of column, raft and raft beams, sides and soffits of foundation beams, etc. flat rough finished surfaces of concrete complete all as specified.		316.72 **Sqm**	1486.38	470,766.27		

SCHEDULE 'A' PART I(B)

(LIST OF ITEMS OF WORKS & PRICES) (TRAVERSE RETAINING WALL ESH 1, 2, 3 & 4)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
8	Use and waste of formwork to faces of walls, retaining walls, sides of drain wall, abutments, parapets and staircase railings and similar work including attached pillasters, buttresses, vertical or to batter in rough finished flat surface of concrete complete all as specified and as directed.		493.55 **Sqm**	8446.20	4,168,622.01		
9	Providing precast cement concrete 1:3:6 type C1 using 20mm graded stone aggregate in window or door cills, lintels, bressumers, steps, quoin blocks (for use with masonry), string courses, copings. kneelers, apex stones, pier caps, columns, posts, boundary pillars, bed plates, pintle blocks, anchorage blocks, bonders (for stone walling). corbels, brackets, etc. (of any section indicated), with plain fair faces, including chamfers, weatherings, throating etc. and setting in cement mortar 1:4 complete with joints to match complete all as specified.		10,400.90 **Cum**	10.08	104,841.07		
10	Rendering 5 mm thick in Cement Mortar 1:3 on fair faces of brick work or concrete surfaces, fair finished complete all as specified.		145.24 **Sqm**	2489.76	361,612.74		
11	Rendering 15 mm thick in Cement Mortar 1:6 on other than fair faces of brick work or concrete surfaces finished even and fair complete all as specified and as directed.		281.25 **Sqm**	201.60	56,700.00		
	Total of Schedule 'A' Part-I(B) carried over to BOQ (General Summary)				**72,844,197.72**		

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART II

(LIST OF ITEMS OF WORKS & PRICES) (SITE CLEARANCE, SITE DEVELOPMENTS & EARTH WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Earth work in all kind of soil excavation by mechanical means (Hydraulic excavator) means over areas (exceeding 30 cm and not exceeding 1.5m in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5m, complete all as specified and directed.		166.62 **Cum**	48838.30	8,137,437.55		
2	Earth work in all kind of soil excavation by mechanical means (Hydraulic excavator) means over areas (exceeding 1.5m and not exceeding 3m in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5m, complete all as specified and directed.		199.93 **Cum**	18423.08	3,683,326.38		
3	Earth work in all kind of soil excavation by mechanical means (Hydraulic excavator) means over areas (exceeding 3m and not exceeding 4.5m in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5m, complete all as specified and directed.		233.24 **Cum**	8792.75	2,050,821.01		
4	Earth work in all kind of soil excavation by mechanical means (Hydraulic excavator) means over areas (exceeding 4.5m and not exceeding 6m in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5m, complete all as specified and directed.		266.55 **Cum**	5656.81	1,507,822.71		
5	Earth work in all kind of soil excavation by mechanical means (Hydraulic excavator) means over areas (exceeding 6m and not exceeding 7.5m in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5m, complete all as specified and directed.		299.86 **Cum**	4008.04	1,201,850.87		
6	Earth work in all kind of soil excavation by mechanical means (Hydraulic excavator) means over areas (exceeding 7.5m and not exceeding 9m in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5m, complete all as specified and directed.		333.17 **Cum**	2982.36	993,632.88		

SCHEDULE 'A' PART II

(LIST OF ITEMS OF WORKS & PRICES) (SITE CLEARANCE, SITE DEVELOPMENTS & EARTH WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
7	Earth work in all kind of soil excavation by mechanical means (Hydraulic excavator) means over areas (exceeding 9m and not exceeding 10.5m in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5m, complete all as specified and directed.		366.48 **Cum**	2168.48	794,704.55		
8	Earth work in all kind of soil excavation by mechanical means (Hydraulic excavator) means over areas (exceeding 10.5m and not exceeding 12m in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5m, complete all as specified and directed.		399.79 **Cum**	1258.03	502,947.81		
9	Earth work in all kind of soil excavation by mechanical means (Hydraulic excavator) means over areas (exceeding 10.5m and not exceeding 12m in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5m, complete all as specified and directed.		433.10 **Cum**	162.13	70,218.50		
10	Forming embankments including raising (or lowering) earth, spreading in layers not exceeding 30 cm thick; watering. ramming/ rolling and finishing to required size, shape, etc., not exceeding 1.5 m high from base complete all as specified and as directed.		168.10 **Cum**	36052.21	6,060,376.50		
11	Forming embankments including raising (or lowering) earth, spreading in layers not exceeding 30 cm thick; watering. ramming/ rolling and finishing to required size, shape, etc., exceeding 1.5m but not exceeding 3 m high from base complete all as specified and as directed.		333.40 **Cum**	2116.80	705,741.12		
12	Forming embankments including raising (or lowering) earth, spreading in layers not exceeding 30 cm thick; watering. ramming/ rolling and finishing to required size, shape, etc., exceeding 3m but not exceeding 4.5 m high from base complete all as specified and as directed.		498.70 **Cum**	1381.38	688,894.21		

SCHEDULE 'A' PART II

(LIST OF ITEMS OF WORKS & PRICES) (SITE CLEARANCE, SITE DEVELOPMENTS & EARTH WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
13	Surface dressing not exceeding 15 cm deep in Hard dense soil complete all as specified and directed.		33.50 **Sqm**	6452.70	216,165.45		
14	Rough excavation not exceeding 1.5 m deep and getting out in soft/loose soil complete all as specified and as directed.		179.10 **Cum**	184.00	32,954.40		
15	Excavation in trenches not exceeding 1.50 m wide for foundations or for shafts, wells, cesspits, manholes, pier holes etc. not exceeding 10 Sqm on plan and not exceeding 1.50 m in depth and getting out in soft/loose soil.		383.70 **Cum**	3627.00	1,391,679.90		
16	Excavation in trenches not exceeding 1.50 m wide for foundations or for shafts, wells, cesspits, manholes, pier holes etc. not exceeding 10 Sqm on plan and exceeding 1.5m and not exceeding 3.0 m in depth and getting out in soft/loose soil.		542.40 **Cum**	10.00	5,424.00		
17	Excavation in trenches not exceeding 1.50 m wide for foundations or for shafts, wells, cesspits, manholes, pier holes etc. not exceeding 10 Sqm on plan and not exceeding 1.50 m in depth and getting out in hard dense soil.		675.00 **Cum**	8425.45	5,687,178.75		
18	Excavation in trenches not exceeding 1.50 m wide for foundations or for shafts, wells, cesspits, manholes, pier holes etc. not exceeding 10 Sqm on plan and exceeding 1.5m but not exceeding 3 m in depth and getting out in hard dense soil.		833.70 **Cum**	894.90	746,078.13		
19	Earth work in excavation by mechanical means (Hydraulic excavator)/ in foundation trenches or drains (not exceeding 1.5 m in width or 10 sqm on plan), including dressing of sides and ramming of bottoms, lift upto 1.5 m, including getting out the excavated any type of soil and disposal of surplus excavated soil as directed, within a lead of 50 m complete all as specified and directed.		228.55 **Cum**	315.00	71,993.25		

SCHEDULE 'A' PART II

(LIST OF ITEMS OF WORKS & PRICES) (SITE CLEARANCE, SITE DEVELOPMENTS & EARTH WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
20	Excavation over area not exceeding 1.5 m deep and getting out in hard dense soil complete all as specified and as directed.		450.00 **Cum**	5129.53	2,308,288.50		
21	Returning, filling in, including spreading, levelling, watering and well ramming in layers not exceeding 25 cm as in soft/loose soil complete all as specified and directed.		139.10 **Cum**	9132.40	1,270,316.84		
22	Surface excavation not exceeding 30 cm deep and averaging 15 cm deep and getting out in hard dense soil		93.20 **Sqm**	29117.50	2,713,751.00		
23	Surface excavation not exceeding 30 cm deep and averaging 15 cm deep and getting out in soft/loose soil		53.10 **Sqm**	2950.00	156,645.00		
24	Add to (or deduct from) **Sr item No.22 above** for every 3 cm or part thereof above or below 15 cm average depth Surface excavation and getting out in hard dense soil		9.30 **Sqm**	29117.50	270,792.75		
25	Removing excavated material (soil) not exceeding 50 m and depositing where directed at a level n. exc. 1.50 m above the starting point complete all as specified and as directed.		320.40 **Cum**	15293.00	4,899,877.20		
26	Removing excavated material (soil) exceeding 50m but not exceeding 100 m and depositing where directed at a level not exceeding 1.50 m above the starting point complete all as specified and as directed.		364.10 **Cum**	38028.69	13,846,246.03		
27	Removing excavated material (soil) exceeding 250m but not exceeding 500 m and depositing where directed at a level not exceeding 1.50 m above the starting point complete all as specified and as directed.		582.60 **Cum**	7193.91	4,191,171.97		

SCHEDULE 'A' PART II

(LIST OF ITEMS OF WORKS & PRICES) (SITE CLEARANCE, SITE DEVELOPMENTS & EARTH WORKS)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
28	Excavation in post holes (or similar holes) each not exceeding 0.5 cum and getting out and returning filling in, ramming earth or broken rubble mixed with earth around, in layers not exceeding 25 cm thick, watering and removing surplus soil to a distance not exceeding 50 m and making good surface in soft/loose soil.		259.10 **Each**	1758.00	455,497.80		
	Total of Schedule 'A' Part-II carried over to BOQ (General Summary)				**64,661,835.06**		

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART III

(LIST OF ITEMS OF WORKS & PRICES) (INTERNAL WATER SUPPLY)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Chlorinated Polyvinyl Chloride (CPVC) pipes 25 mm outside dia, SDR 11 (IS 15778) & Plain CPVC SDR 11 fittings concealed on walls / ceiling or laying in floors complete all as specified.		353.40 **Rm**	558.00	197,197.20		
2	Chlorinated Polyvinyl Chloride (CPVC) pipes 20 mm outside dia, SDR 11 (IS 15778) & Plain CPVC SDR 11 fittings concealed on walls / ceiling or laying in floors complete all as specified.		244.60 **Rm**	434.00	106,156.40		
3	Supply and fixing of 3- layer PP-R (Polypropylene random copolymer) concealed pipes 32 mm outer dia ,SDR 7.4 (PN-16) pipes UV stabilized and antimicrobial fusion welded having thermal stability for hot & cold water supply all PP-R plain & brass threaded polypropylene random fittings laid in trenches complete all as specified and directed.		261.40 **Rm**	255.00	66,657.00		
4	Bib taps, chromium plated cast copper alloy, with crutch or butterfly handle, screwed down, screwed for iron pipe or for brass ferrule and fixed for 15 mm bore pipe		465.70 **Each**	30.00	13,971.00		
5	Pillar taps, fancy type, cast copper alloy, with capstan heads, chromium plated, screwed down high pressure, with or without lettered 'Hot' or 'Cold' with long screwed shanks and fly-nuts, screwed for iron pipe and fixed for 15 mm bore pipe complete all as specified and directed.		676.44 **Each**	44.00	29,763.36		
6	Stop valves/Angle valves fancy type concealed, chromium plated, cast copper alloy, screwed down, high pressure with crutch or butterfly handle, screwed both ends for iron pipe or for unions and of size 25 mm bore complete all as directed and specified		801.76 **Each**	21.00	16,836.96		
7	Stop valves/Angle valves fancy type, chromium plated, cast copper alloy, screwed down, high pressure with crutch or butterfly handle, screwed both ends for iron pipe or for unions and of size 20 mm bore complete all as directed and specified		716.70 **Each**	21.00	15,050.70		
8	Stop valves/Angle valves fancy type, chromium plated, cast copper alloy, screwed down, high pressure with crutch or butterfly handle, screwed both ends for iron pipe or for unions and of size 15 mm bore complete all as directed and specified		682.03 **Each**	33.00	22,506.99		

SCHEDULE 'A' PART III

(LIST OF ITEMS OF WORKS & PRICES) (INTERNAL WATER SUPPLY)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
9	3 layered fusion welded PP-R (polypropylene Random Co -Polymer) gate valve (brass insert as per requirement of the site with handle, screwed both ends for PP-R pipes or for unions fixed of grade SDR - 7.4 (PN-16) and size 32 mm complete all as specified and directed.		604.51 **Each**	19.00	11,485.69		
10	Brass chromium plated shower rose 125mm size suitable for CPVC SDR-11 pipe of outer dia 20mm complete all as specified and directed.		409.25 **Each**	19.00	7,775.75		
11	Brass chromium plated basin mixer for mixing hot & cold water screwed down for 15mm bore dia of pipe		1,743.41 **Each**	11.00	19,177.51		
	Total of Schedule 'A' Part-III carried over to BOQ (General Summary)				**506,578.56**		

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART IV

(LIST OF ITEMS OF WORKS & PRICES) (INTERNAL ELECTRIFICATION)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Material and Labour for concealed point wiring with PVC insulated unsheathed multistranded FRLS copper conductor in PVC conduit of 20 mm dia(min) with cable 1.5 Sqmm size 1100 volts grade conforming to IS:694 and terminate to flush box of required size suitable mounting modular switches and sockets including single core multi stranded FRLS copper conductor 1.5 Sqmm PVC insulated cable as earth wire connected to earth and common earth dolly drawn in same concealed conduit for the One light/ceiling fan/ exhaust fan/building security light point controlled by one No., one way 6 Amp switch complete all as specified and directed.		992.75 **Per Point**	521.00	517,222.75		
2	Material and Labour for concealed point wiring with PVC insulated unsheathed multistranded FRLS copper conductor in PVC conduit of 20 mm dia(min) with cable 1.5 Sqmm size 1100 volts grade conforming to IS:694 and terminate to flush box of required size suitable mounting modular switches and sockets including single core multi stranded FRLS copper conductor 1.5 Sqmm PVC insulated cable as earth wire connected to earth and common earth dolly drawn in same concealed conduit for the One light point controlled by two way 6 Amp switches complete all as specified and directed.		1,397.96 **Per Point**	7.00	9,785.72		
3	Material and Labour for concealed point wiring with PVC insulated unsheathed multistranded FRLS copper conductor in PVC conduit of 20 mm dia(min) with cable 1.5 Sqmm size 1100 volts grade conforming to IS:694 and terminate to flush box of required size suitable mounting modular switches and sockets including single core multi stranded FRLS copper conductor 1.5 Sqmm PVC insulated cable as earth wire connected to earth and common earth dolly drawn in same concealed conduit for the One three pin 6 Amp socket outlet point controlled by one no, one way switch on same board complete all as specified and directed.		284.10 **Per Point**	18.00	5,113.80		

SCHEDULE 'A' PART IV

(LIST OF ITEMS OF WORKS & PRICES) (INTERNAL ELECTRIFICATION)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
4	Material and Labour for concealed point wiring with PVC insulated unsheathed multistranded FRLS copper conductor in PVC conduit of 20 mm dia(min) with cable 1.5 Sqmm size 1100 volts grade conforming to IS:694 and terminate to flush box of required size suitable mounting modular switches and sockets including single core multi stranded FRLS copper conductor 1.5 Sqmm PVC insulated cable as earth wire connected to earth and common earth dolly drawn in same concealed conduit for the One three pin 6 Amp socket outlet point controlled by one no, one way switch on independent board complete all as specified and directed.		1,070.40 **Per Point**	99.00	105,969.60		
5	Material and Labour for concealed point wiring with PVC insulated unsheathed multistranded FRLS copper conductor in PVC conduit with cable 2.5 Sqmm size 1100 volts grade conforming to IS:694 in new metal flush box of 1.6 mm thick including one coat of syenthetic enamelled black paint with new modular white cover plate with frame of suitable size of required colour fixed with brass screws and cup washers including single core multi stranded FRLS copper conductor 2.5 Sqmm PVC insulated cable as earth wire to common earth and connected to earth dolly and earth terminal box drawn in same concealed conduit for One 3 Pin socket 15/16 A Indepentent board complete all specified and directed.		1,131.39 **Per Point**	90.00	101,825.10		
6	Material and Labour for concealed point wiring with PVC insulated unsheathed multistranded FRLS copper conductor in PVC conduit with cable 4.0 Sqmm size 1100 volts grade conforming to IS:694 in new metal flush box of 1.6 mm thick including one coat of syenthetic enamelled black paint with new modular white cover plate with frame of suitable size of required colour fixed with brass screws and cup washers including single core multi stranded FRLS copper conductor 4.0 Sqmm PVC insulated cable as earth wire to common earth and connected to earth dolly and earth terminal box drawn in same concealed conduit for Modular Box Metal box, power socket Indepentent board for Geysers complete all specified and directed.		1,930.50 **Per Point**	10.00	19,305.00		

SCHEDULE 'A' PART IV

(LIST OF ITEMS OF WORKS & PRICES) (INTERNAL ELECTRIFICATION)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
7	Supply & Fix switch 6 Amps 230 Volt 50 Hz modular type, 1 Module with red indication, single pole one way, ISI marked complete all as specified.		260.50 **Each**	638.00	166,199.00		
8	Supply & Fix bell push, modular type,1 Module 6 amps complete all as specified and directed.		307.30 **Each**	64.00	19,667.20		
9	Supply & Fix switch 6 Amps 230 Volt 50 Hz modular type, 1 Module with red indication, two way 5/6 amps complete all as specified and directed.		307.30 **Each**	14.00	4,302.20		
10	Supply & Fix modular socket 6 Amps 230 Volt 50 Hz, 5 pin combined modular type 2 Module, shuttered type, ISI marked complete all as specified.		340.54 **Each**	117.00	39,843.18		
11	Supply & Fix modular socket 6/16 Amps 230 Volt 50 Hz, 6 pin combined modular type 2 Module, shuttered type , ISI marked complete all as specified.		528.40 **Each**	100.00	52,840.00		
12	Supply & Fix switch 16 Amp 230 Volt 50 Hz rating Modular type, 1 Module, one way switch complete all as specified and directed.		343.70 **Each**	100.00	34,370.00		
13	Supply & Fix Modular cover plate with frame & flush box of 8 modules, complete all as specified and as directed.		665.50 **Each**	18.00	11,979.00		
14	Supply & Fix Modular cover plate with frame & flush box of 6 modules, complete all as specified and as directed.		593.70 **Each**	14.00	8,311.80		
15	Supply & Fix Modular cover plate with frame & flush box of 4 modules, complete all as specified and as directed.		460.50 **Each**	14.00	6,447.00		
16	Supply & Fix Modular cover plate with frame & flush box of 3 modules, complete all as specified and as directed.		397.50 **Each**	103.00	40,942.50		
17	Supply & Fix Modular cover plate with frame & flush box of 2 modules, complete all as specified and as directed.		374.90 **Each**	10.00	3,749.00		

SCHEDULE 'A' PART IV

(LIST OF ITEMS OF WORKS & PRICES) (INTERNAL ELECTRIFICATION)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
18	Supply & Fix Modular cover plate with frame & flush box of 1 modules, complete all as specified and as directed.		371.80 **Each**	16.00	5,948.80		
19	Supply and fixing ceiling fan regulator Modular type 2 Module 120 watt, 230 Volt 50 Hz all as specified and directed.		1,017.95 **Each**	60.00	61,077.00		
20	Ceiling rose 3 phase/ PVC/Polycarbonate, isolated body complete all as specified and as directed.		65.70 **Each**	461.00	30,287.70		
21	Supply & Fix of Hot water storage (Geyser) 25 Ltrs capacity, BEE 5 star rated,vertical type, digital temperature display with suitable size of fastner and incl provision and connecting up of ss braided pipe with adopter for inlet and outlet water connection, copper element corrosion resistance antiscaling magnesium anode, having adjustable knob for temperature control and inner tank material of superior glassline coating / percelain Enamel Coating / Titanium Enamel Coating / Nano Blue Diamond Glass Coating / Coating with Single Weld line tank and outer body material ABS with high pressure with standing capacity upto 8 bar presuure, Multiple safety systems for protection against dry heating, overheating and over pressure, IPX4 waterproof technology for longer product life having heating element 2 KW and minimum two year warranty on product, minimum 3 years warranty on heating element 5 years on tank complete all as specified and directed.		10,029.54 **Each**	13.00	130,384.02		
22	Supply & Fix Exhaust fan 300 mm sweep, copper winding,metal body, four metal blades BEE approved/ISI marked, air delivery shall not be less than 1600 CMH at 1400 RPM, mounted in suitable opening arrangement with louvers shutters, suitable for operation on 240 volt, single phase, 50 cycle per second, AC supply and connecting up using flexible cord wire 3 core using PVC flexible conduit from ceiling rose or connector to fan to cover PVC cable fixed in position as recommended by manufacturer complete all as specified..		3,548.59 **Each**	34.00	120,652.06		
23	Supply and fix MCB 4 pole, TPN, 16 to 32 Amps, 10KA breaking capacity 415 Volts complete all as specified and directed complete all as specified and directed.		1,721.04 **Each**	3.00	5,163.12		

SCHEDULE 'A' PART IV

(LIST OF ITEMS OF WORKS & PRICES) (INTERNAL ELECTRIFICATION)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
24	Supply and fix RCBO with earth leakage, over load and short circuit proptection, DP, 240 Volts, 32 Amps, 30 mA senstivity, 4 Module complete all as specified and directed		3,452.32 **Each**	22.00	75,951.04		
25	Supply and fixing 240V 10KA 'C' curve miniature circuit breaker (MCB) of Single pole and current carrying capacity 6 Amps to 32 Amps in the DB complete with connection testing and commissioning all as specified and directed.		294.00 **Each**	150.00	44,100.00		
26	S & F energy saving BLDC Ceiling fan 1200mm sweep BEE 5 star rated suitable for regulator and suspended from fan hook and electrically connecting with twisted three core PVC insulated unsheated flexisble cord of 0.70 Sqmm copper conductor cable from ceiling rose to fan.		2,649.47 **Each**	60.00	158,968.20		
27	S&F Triple pole and earth enclosure with a three pin and earth plug and socket complete with one triple pole MCB 20 Amps capacity 415 V complete all as specified and as directed.		3,348.90 **Each**	1.00	3,348.90		
28	Supply and fix RCBO with earth leakage, over load and short circuit proptection, DP, 240 Volts, 32 Amps, 30 mA senstivity, 4 Module complete all as specified and directed		4,414.52 **Each**	14.00	61,803.28		
29	Earth continuity conductor or main earthing lead fixed to wall on batten or recess or chases or buried in ground or drawn in conduit/pipe or fixed to poles or any other indicated situation for loop earthing etc. as required 4mm dia galvanised iron or steel wire complete all as specified and directed.		60.20 **RM**	50.00	3,010.00		

SCHEDULE 'A' PART IV

(LIST OF ITEMS OF WORKS & PRICES) (INTERNAL ELECTRIFICATION)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
30	Supply and fixing Earth continuity conductor or main earthing lead fixed to wall on batten or recess or chases or buried in ground or drawn in conduit/pipe or fixed to wall or any other indicated situation for loop earthing etc. as required with copper srtip of size 20 X3 mm complete all as specified and directed. (For provide main earthing lead for anti static flooring all arround the explosive bldg)		426.80 **RM**	1140.00	486,552.00		

Total of Schedule 'A' Part-IV carried over to BOQ (General Summary) **2,335,118.97**

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART V

(LIST OF ITEMS OF WORKS & PRICES)(ROAD, PATH, CULVERT & HARDSTANDING)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Laying and compacting 150 mm thick (compacted thickness) Granular Sub Base Course (GSB) grading as per Grade I as specified Table 400-1 of MaRTH specifications for Road and Bridge works, 2013, in a single layer over compacted subgrade with natura sand, crushed gravel, crushed stone, crushed slag or combination thereof as per the grading required The materials are mechanically mixed, spread and leveling on the subgrade by using motor grader of adequate capacity, and rolling with vibratory roller of capacity 80-1 00 KN static weight for achieving required compaction density of atleast 98% of the Maximum Dry Density. Consolidating to required gradient and camber complete all as specified and directed with finished surface tolerances as per Table 900-1 of MaRTH specifications for Road and Bridge works, 2013.		3,936.71 / **10 Sqm**	2890.25	11,378,076.08		
2	Laying and compacting 225 mm thick (compacted thickness) Wet Mix Macadam (WMM) laid in three layers of 75mm each grading as per table 400-13 of MoRTH specifications for Road and Bridge works 2013 manufactured in WMM plant material comprising of clean, crushed, graded aggregate and granular material premixed with water to a dense mass, spreading with sensor paver and rolling with vibratory roller of capacity 80-100 KN static weight and consolidating to required gradient and camber complete all as specified and directed. The finished surface tolerances to be as per Table 900-1 of MoRTH specifications for Road and Bridge works 2013 complete all as specified and as directed.		6,304.01 / **10 Sqm**	3105.25	19,575,527.05		
3	Preparation of unsurfaced WMM surfaces by brushing with wire brushes for removing caked mud etc., sweeping with brooms and finally fanning the cleaned surfaces with gunny bags to remove all loose dirt etc. complete all as specified and as directed.		360.60 / **10 Sqm**	2890.25	1,042,224.15		
4	Applying evenly a priming/tack coat with bituminous primer @ 10 Kg per 10 Square Metre on road surfaces using paving bitumen VG-10 with mechanical pressure sprayer complete all as specified and as directed.		512.50 / **10 Sqm**	2890.25	1,481,253.13		

SCHEDULE 'A' PART V

(LIST OF ITEMS OF WORKS & PRICES)(ROAD, PATH, CULVERT & HARDSTANDING)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
5	Providing premix bituminous, macadam laid/rolled and compacted to required gradient and camber (consolidated thickness not exc. 75 mm) using paving bitumen VG-30 @ 4 % by weight of total mix complete all as specified and directed. Note: If actual design mix comes less than 4% binder content, then the rate of recovery of bitumen shall be as per rate in purchase voucher per Kg for the purpose of adjustment. However, no adjustment shall be made if the binder content is more than 4%.		8,915.70 **Cum**	1445.13	12,884,300.96		
6	Applying evenly a priming/tack coat with bituminous primer @ 5 Kg per 10 Square Metre on road surfaces using paving bitumen VG-10 with mechanical pressure sprayer complete all as specified and as directed.		263.00 **10 Sqm**	2890.25	760,135.75		
7	Bituminous premix semi dense asphaltic concrete of 40mm consolidated thickness with 5.5% binder content by weight of total mix, rolled and compacted to required camber and gradient complete (using paving bitumen VG-30), rolled with 8 to 12 tonne power roller to required camber and gradient complete all as specified and as directed.		3,828.50 **10 Sqm**	2890.25	11,065,322.13		
8	Material & Labour lime or sand stone Kerb roughly square stone of size 20cm x 30 cm all surface hammer dressed so that the maximum deression from a straight edge held against the dressed surface shall not exceed 20mm jointed in cement mortar 1:6 and struck flush jointed in Cement Mortar 1:6 and stuck flush complete all as secified and directed.		634.37 **Running metre**	10510.00	6,667,228.70		
9	Material & Labour for Moorum/ Red bajri 100 mm thickness rolled and consolidated to required camber and gradient complete all as specified and directed.		799.30 **10 Sqm**	1313.75	1,050,080.38		
10	Material & Labour for morrum /red bajri Filling of low lying portion with approved earth including ramming, levelling etc complete all as specified and directed.		1,240.10 **Cum**	225.00	279,022.50		
11	Rolling and consolidating formation surfaces in cutting with 8 to 12 tonne power roller complete all as specified and as directed.		26.00 **10 Sqm**	510.00	13,260.00		

SCHEDULE 'A' PART V

(LIST OF ITEMS OF WORKS & PRICES)(ROAD, PATH, CULVERT & HARDSTANDING)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
12	Hard core gauge not exceeding 63mm in size of broken stone or boulders, deposited spread and levelled in layers not exceeding 15cm thickness watered and rammed to a true surfaces complete all as specified & as directed.		2,070.80 **Cum**	295.00	610,886.00		
13	Material and labour for sand filling under floors or in foundation including watering and consolidation complete all as specified and as directed.		1,974.30 **Cum**	37.50	74,036.25		
14	60 mm thickness Machine pressed precast concrete interlocking paver block M-35 Grade fixed with grey cement and pigment of any shape and size confirming to IS 15658-2006 complete all as specified and as directed.		774.40 **Sqm**	750.00	580,800.00		
15	Plain Cement Concrete 1:4:8, type D2 using 40 mm graded stone aggregate as in foundation filling and mass concrete complete all as specified and as directed.		5,483.10 **Cum**	182.10	998,472.51		
16	Soling (or sub-base) 200 mm thick (spread thickness) with broken boulders or quarried stone, interstices filled, surface formed and rolled and consolidated to required Gradient and Camber with 8 to 12 Tonne Power Roller, complete all as specified and as directed.		3,694.10 **10 Sqm**	215.00	794,231.50		
17	Cement concrete work in pavement/hard standing of flexural strength 40 kg/per square centimeter in field using 400 kg of cement per cubic meter thickness of concretee 25 cm or less complete all as specified and directed.		7,107.10 **Cum**	430.00	3,056,053.00		
18	Forming 12 mm thick expansion joint filled with pre-formed bitumenous filler and top 30mm preimed and sealed with sealing compound Grade'A' thickness of concrete slab 200mm complete all as specified and directed.		1,101.60 **10 RM**	73.10	80,526.96		

SCHEDULE 'A' PART V

(LIST OF ITEMS OF WORKS & PRICES)(ROAD, PATH, CULVERT & HARDSTANDING)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
19	Forming construction joint 30mm deep x 10mm wide filled with sealing compound Grade 'A' complete all as specified and directed.		335.90 **10 RM**	33.50	11,252.65		
20	100mm thck Plain cement concrete Type B-1, 1:2:4 (using 20 mm graded aggregate) as in floor complete all as specified and as directed.		633.20 **Sqm**	2200.00	1,393,040.00		
21	Plain Cement concrete 1:3:6, type C2 using 40 mm graded stone aggregate in foundation filling and mass concrete complete all as specified and as directed.		5,862.00 **Cum**	220.00	1,289,640.00		
22	Reinforced Cement Concrete M-25 (Design mix) (using 20 mm graded stone aggregate) as in slabs supported on walls, beams and Columns in floors, roofs, landings, balconies, canopies, deck slabs and in shelves and the like complete all as specified and as directed.		7,388.40 **Cum**	214.94	1,588,062.70		
23	Reinforced Cement Concrete M-25 (Design mix) using 20 mm graded stone aggregate as in Walls, retaining walls, basement walls, ballast walls and the like, any thickness; above top of footings; including attached pilasters and buttresses complete all as specified and as directed.		8,028.60 **Cum**	429.88	3,451,334.57		
24	Reinforced Cement Concrete M-25 (Design mix) using 20 mm graded stone aggregate as in foundations, including rafts, footings, foundation beams; plinth beams; bases for columns and mass concrete complete all as specified and as directed.		7,640.60 **Cum**	224.74	1,717,148.44		
25	High strength steel deformed TMT bars (Fe 500D) 10mm dia and over, cut to length, bent to shape required, including cranking, bending spirally for hooping for columns, hooking ends and binding with and including mild steel wire (annealed) not less than 0.9mm dia or securing with clips complete all as specified and directed.		82.60 **Kg**	99433.41	8,213,199.67		

SCHEDULE 'A' PART V

(LIST OF ITEMS OF WORKS & PRICES)(ROAD, PATH, CULVERT & HARDSTANDING)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
26	Use and waste of formwork to soffits of suspended slabs such as roof slab, floor slabs, landings not exceeding 200 mm thick (horizontal or sloping) for flat rough finished surfaces of concrete complete all as specified and as directed.		495.26 **Sqm**	806.80	399,575.77		
27	Use and waste of formwork to faces of walls, retaining walls, sides of drain wall, abutments, parapets and staircase railings and similar work including attached pillasters, buttresses, vertical or to batter in rough finished flat surface of concrete complete all as specified and as directed.		493.55 **Sqm**	3227.20	1,592,784.56		
28	Rendering 10 mm thick in Cement Mortar 1:6 on fair faces of brick work or concrete surfaces finished even and fair complete all as specified and as directed.		189.35 **Sqm**	1288.00	243,882.80		
	Total of Schedule 'A' Part-V carried over to BOQ (General Summary)				**92,291,358.19**		

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART VI

(LIST OF ITEMS OF WORKS & PRICES)(EXTERNAL WATER SUPPLY)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Gun-metal, gate valves suitable for 40mm bore pipe, with iron wheel head, screwed both ends for iron pipe and fixed complete all as specified and as directed.		784.97 **Each**	6.00	4,709.82		
2	Gun-metal, gate valves suitable for 32mm bore pipe, with iron wheel head, screwed both ends for iron pipe and fixed complete all as specified and as directed.		670.26 **Each**	15.00	10,053.90		
3	Gun-metal, gate valves suitable for 25mm bore GI pipe, with iron wheel head, screwed both ends for iron pipe and fixed complete all as specified and as directed.		587.24 **Each**	8.00	4,697.92		
4	Galvanised steel water tubing 80 mm bore, medium grade laid in trenches including flat iron clamps of suitable size, with nuts, bolts, etc. complete all as specified and as directed.		760.50 **Rm**	2600.00	1,977,300.00		
5	Galvanised steel water tubing 50 mm bore, medium grade laid in trenches including all fittings, complete all as specified and as directed.		421.20 **Rm**	400.00	168,480.00		
6	Galvanised steel water tubing 40 mm bore, medium grade laid in trenches including all fittings, complete all as specified and as directed.		396.50 **Rm**	150.00	59,475.00		
7	Galvanised steel water tubing 32 mm bore, medium grade laid in trenches including all fittings, complete all as specified and as directed.		344.50 **Rm**	510.00	175,695.00		
8	Galvanised steel water tubing 25 mm bore, medium grade laid in trenches including all fittings, complete all as specified and as directed.		184.60 **Rm**	200.00	36,920.00		
	Total of Schedule 'A' Part-VI carried over to BOQ (General Summary)				**2,437,331.64**		

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART VII
(LIST OF ITEMS OF WORKS & PRICES)(DG SET)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Supply and fixing Earth continuity conductor or main earthing lead fixed to wall on batten or recess or chases or buried in ground or drawn in conduit/pipe or fixed to poles or any other indicated situation for loop earthing etc. as required with Galvanized iron strip 32 x 6mm complete all as specified and directed		230.80 **Rm**	60.00	13,848.00		
2	LT XLPE insulated, armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 95 Sqmm x 3.5 Core.		526.72 **Rm**	350.00	184,352.00		
3	LT XLPE insulated, armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 35 Sqmm x 3.5 Core.		269.75 **Rm**	500.00	134,875.00		
4	LT XLPE insulated, armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 50 Sqmm x 3.5 Core.		337.52 **Rm**	700.00	236,264.00		
	Total of Schedule 'A' Part-VII carried over to BOQ (General Summary)				**569,339.00**		

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART VIII

(LIST OF ITEMS OF WORKS & PRICES)(EXTERNAL ELECTRIC SUPPLY)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Supply, laying, jointing HT XLPE cable insulated, screened, PVC bedded, galvanised steel strip or wire armoured, electric power cables (heavyduty) with aluminium conductor, 11000 volts Earthed, grade cross sectional area 95 sq. mm 3 core laid in pipe/ ducts/ trenches/ under floor/ under road including testing complete all as specified and directed.		1,067.97 **Rm**	70.00	74,757.90		
2	Cable jointing kit for 11 KV (Earthed) grade cable for outdoor termination heat shrink type joint complete with jointing material and accessories suitable for 3 core XLPE armoured aluminium conductor cable of size 50 sqmm to 95 sqmm complete all as specified and directed.		6,150.70 **Each**	2.00	12,301.40		
3	Cable jointing kit for 11 KV (Earthed) grade cable for straight through heat shrink type joint complete with jointing material and accessories suitable for 3 core XLPE armoured aluminium conductor cable of size 95 sqmm complete all as specified and as directed		16,042.80 **Each**	2.00	32,085.60		
4	LT XLPE insulated, PVC bedded galvanised steel strip or wire armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 10 Sqmm x 2 Core.		143.61 **Rm**	750.00	107,707.50		
5	LT XLPE insulated, armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 10 Sqmm x 4 Core.		174.68 **Rm**	5000.00	873,400.00		
6	LT XLPE insulated, armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 16 Sqmm x 4 Core.		193.50 **Rm**	600.00	116,100.00		
7	LT XLPE insulated, armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 35 Sqmm x 3.5 Core.		269.75 **Rm**	500.00	134,875.00		
8	LT XLPE insulated, armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 50 Sqmm x 3.5 Core.		337.52 **Rm**	700.00	236,264.00		

SCHEDULE 'A' PART VIII

(LIST OF ITEMS OF WORKS & PRICES)(EXTERNAL ELECTRIC SUPPLY)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
9	LT XLPE insulated, armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 95 Sqmm x 3.5 Core.		526.72 **Rm**	200.00	105,344.00		
10	LT XLPE insulated, armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 120 Sqmm x 3.5 Core.		648.15 **Rm**	30.00	19,444.50		
11	LT XLPE insulated, PVC bedded galvanised steel strip or wire armoured heavy duty electric cable 1100 volt grade ISI marked and specification as per IS : 7098 (Part-I) with alumimium conductor multi strand cross sectional area 150 Sqmm x 3.5 Core.		759.22 **Rm**	70.00	53,145.40		
12	Supply & fix Steel Tubular Swaged Poles Type 410-SP-44 (10.00 Metre long) complete with cast iron base plate, finial taper plug, cap, mild steel bolts, screws, etc. including applying two coats of black bituminous paint internally and externally upto planting depth and applying two coats of aluminium paint over a coat of aluminium primer (externally from top of PCC copping) including writting MES Pole No. with synthetic enamel paint (first grade quality) complete all as specified. Note:- (a) Earthwork will be measured and paid under appropriate items in Schedule 'A' Part-II and PCC will be measured and paid under relevant items in this Schedule. (b) Volume occupied by Steel Pole shall be deducted to derive the quantity of PCC.		13,746.80 **Each**	6.00	82,480.80		
13	Lightning arresters expulsion type complete with all fittings, transmission class, discharge capacity 65 kilo amp suitable for working pressure 11000 Volts complete all as specified and directed.		10,694.90 **Per set**	1.00	10,694.90		
14	Material & Labour for cross arms, bracings, supports, clamps and back plates fabricated from structural steel sections including nuts, bolts, washers, welding, bending (cold or hot), drilling holes, for bolts any shape or size as indicated or directed and inclusive of one primer coat of red oxide and two coats of aluminium paint complete all as specified.		11,931.40 **Quintal**	1.00	11,931.40		

SCHEDULE 'A' PART VIII

(LIST OF ITEMS OF WORKS & PRICES)(EXTERNAL ELECTRIC SUPPLY)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
15	Aluminium Conductor, Steel Reinforced of any size and type on HT, LT Poles including straining,binding with insulators with copper binding wire with insulators as in OH conductors, providing PVC Sleeves for jumper complete all as specified and as directed.		17,563.10 **Quintal**	0.30	5,268.93		
16	Anti climbing device; spikes type fabricated out of M.S. bars (round, square and flat) including nuts, bolts, washers and welding in two halves made to shape to suit the cross section of the pole complete all as specified and as directed.		12,242.60 **Quintal**	1.00	12,242.60		
17	Plain Cement Concrete 1:4:8, type D2 using 40 mm graded stone aggregate as in foundation filling and mass concrete complete all as specified and as directed.		5,483.10 **Cum**	7.00	38,381.70		
18	Plain Cement concrete 1:3:6, type C2 using 40 mm graded stone aggregate in foundation filling and mass concrete complete all as specified and as directed.		5,862.00 **Cum**	94.00	551,028.00		
19	Sand cushioning to the full width of trenches for protection of cables including punning down complete all as specified. Note :- Thickness of sand cushioning shall be the thickness after punning down.		1,934.60 **Cum**	583.00	1,127,871.80		
20	Supply and laying unreinforced precasted concrete cable cover class HV type 1 with peak size 300mm x 180mm x 40mm in trenches for protection of UG cable with width equal to length of PCC cover, laid dry and flat complete all as specified and as directed.		57.40 **Each**	1056.00	60,614.40		
21	Supply and laying unreinforced precasted concrete cable cover class LV type 1 with peak size 250mm x 150mm x 40mm in trenches for protection of UG cable with width equal to length of PCC cover, laid dry and flat complete all as specified and as directed.		44.20 **Each**	53267.00	2,354,401.40		

SCHEDULE 'A' PART VIII

(LIST OF ITEMS OF WORKS & PRICES)(EXTERNAL ELECTRIC SUPPLY)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
22	Supply and laying unreinforced precasted concrete cable cover class HVP type 1 with peak size 300mm x 180mm x 40mm in trenches for protection of UG cable with width equal to length of PCC cover, laid dry and flat complete all as specified and as directed.		85.40 **Each**	335.00	28,609.00		
23	Supply and fixing Earth continuity conductor or main earthing lead fixed to wall on batten or recess or chases or buried in ground or drawn in conduit/pipe or fixed to poles or any other indicated situation for loop earthing etc. as required with Galvanized iron strip 32 x 6mm complete all as specified and directed		230.80 **Rm**	120.00	27,696.00		
24	Earth continuity conductor or main earthing lead fixed to wall on batten or recess or chases or buried in ground or drawn in conduit/pipe or fixed to poles or any other indicated situation for loop earthing etc. as required 4mm dia galvanised iron or steel wire complete all as specified and directed.		60.20 **RM**	80.00	4,816.00		
25	Supply & install Voltage stablizer(servo controlled) for automatic operation three phase naturally oil cooled indoor type & designed for input variation between 340 V to 460 V with output voltage stabilized at 415 with accuracy +/_ 2.5% and speed of correction 8 volts per second and of capacity 500 KVA complete all as specified and directed.		1,033,863.81 **Each**	1.00	1,033,863.81		

Total of Schedule 'A' Part-VIII carried over to BOQ (General Summary) **7,115,326.04**

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART IX

(LIST OF ITEMS OF WORKS & PRICES) (AREA DRAINAGE)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Plain Cement Concrete 1:4:8, type D2 using 40 mm graded stone aggregate as in foundation filling and mass concrete complete all as specified and as directed.		5,483.10 **Cum**	606.87	3,327,528.90		
2	Reinforced Cement Concrete M-25 (Design mix) (using 20 mm graded stone aggregate) as in slabs supported on walls, beams and Columns in floors, roofs, landings, balconies, canopies, deck slabs and in shelves and the like complete all as specified and as directed.		7,388.40 **Cum**	618.14	4,567,065.58		
3	Reinforced Cement Concrete M-25 (Design mix) using 20 mm graded stone aggregate as in Walls, retaining walls, basement walls, ballast walls and the like, any thickness; above top of footings; including attached pilasters and buttresses complete all as specified and as directed.		8,028.60 **Cum**	1374.76	11,037,398.14		
4	Reinforced Cement Concrete M-25 (Design mix) using 20 mm graded stone aggregate as in foundations, including rafts, footings, foundation beams; plinth beams; bases for columns and mass concrete complete all as specified and as directed.		7,640.60 **Cum**	634.66	4,849,183.20		
5	High strength steel deformed TMT bars (Fe 500D) 10mm dia and over, cut to length, bent to shape required, including cranking, bending spirally for hooping for columns, hooking ends and binding with and including mild steel wire (annealed) not less than 0.9mm dia or securing with clips complete all as specified and directed.		82.60 **Kg**	100523.19	8,303,215.49		
6	High strength steel deformed TMT bars (Fe 500D) 6mm to 8mm dia, cut to length, bent to shape required, including cranking, bending spirally for hooping for columns, hooking ends and binding with and including mild steel wire (annealed) not less than 0.9mm dia or securing with clips complete all as specified and directed.		84.55 **Kg**	118615.97	10,028,980.26		
7	Use and Waste of formwork to sides of concrete foundations, foundation beams, footing, bases of column, raft and raft beams, sides and soffits of foundation and plinth beams, etc. flat rough finished surfaces of concrete complete all as specified and directed.		316.72 **Sqm**	6181.38	1,957,766.67		

SCHEDULE 'A' PART IX

(LIST OF ITEMS OF WORKS & PRICES) (AREA DRAINAGE)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
8	Use and waste of formwork to faces of walls, retaining walls, sides of drain wall, abutments, parapets and staircase railings and similar work including attached pillasters, buttresses, vertical or to batter in rough finished flat surface of concrete complete all as specified and as directed.		493.55 **Sqm**	12163.70	6,003,394.14		
9	Rendering 10 mm thick in Cement Mortar 1:6 on fair faces of brick work or concrete surfaces finished even and fair complete all as specified and as directed.		189.35 **Sqm**	9416.40	1,782,995.34		
	Total of Schedule 'A' Part-IX carried over to BOQ (General Summary)				**51,857,527.71**		

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART X

(LIST OF ITEMS OF WORKS & PRICES) (COMPOUND WALL, FENCING & GATE)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Plain Cement Concrete 1:4:8, type D2 using 40 mm graded stone aggregate as in foundation filling and mass concrete complete all as specified and as directed.		5,483.10 **Cum**	253.68	1,390,952.81		
2	Providing precast cement concrete 1:2:4 type C1 using 20mm graded stone aggregate in window or door cills, lintels, bressumers, steps, quoin blocks (for use with masonry), string courses, copings. kneelers, apex stones, pier caps, columns, posts, boundary pillars, bed plates, pintle blocks, anchorage blocks, bonders (for stone walling). corbels, brackets, etc. (of any section indicated), with plain fair faces, including chamfers, weatherings, throating etc. and setting in cement mortar 1:6 complete with joints to match complete all as specified.		11,036.80 **Cum**	581.76	6,420,768.77		
3	Reinforced Cement Concrete M-25 (Design mix) using 20 mm graded stone aggregate as in foundations, including rafts, footings, foundation beams; plinth beams; bases for columns and mass concrete complete all as specified and as directed.		7,640.60 **Cum**	602.02	4,599,794.01		
4	Reinforced Cement Concrete M-25 (Design mix) using 20 mm graded stone aggregate as in Columns, pillars, piers, post and struts, etc. complete all as directed.		8,650.20 **Cum**	380.37	3,290,276.57		
5	High strength steel deformed TMT bars (Fe 500D) 10mm dia and over, cut to length, bent to shape required, including cranking, bending spirally for hooping for columns, hooking ends and binding with and including mild steel wire (annealed) not less than 0.9mm dia or securing with clips complete all as specified and directed.		82.60 **Kg**	94877.63	7,836,892.24		
6	High strength steel deformed TMT bars (Fe 500D) 6mm to 8mm dia, cut to length, bent to shape required, including cranking, bending spirally for hooping for columns, hooking ends and binding with and including mild steel wire (annealed) not less than 0.9mm dia or securing with clips complete all as specified and directed.		84.55 **Kg**	50096.00	4,235,616.80		

SCHEDULE 'A' PART X

(LIST OF ITEMS OF WORKS & PRICES) (COMPOUND WALL, FENCING & GATE)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
7	High strength steel deformed TMT bars (Fe 500D) 5mm dia and over upto and including 10mm dia in stirrups, spacers and binders cut to length, bent to shape required including cranking, bending spirally and binding with and including mild steel (annealed) wire not less than 0.90 mm dia or securing with clips complete all as specified and directed.		85.18 **Kg**	84007.45	7,155,754.59		
8	Use and Waste of formwork to sides of concrete foundations, foundation beams, footing, bases of column, raft and raft beams, sides and soffits of foundation and plinth beams, etc. flat rough finished surfaces of concrete complete all as specified and directed.		316.72 **Sqm**	5436.32	1,721,791.27		
9	Use and waste of formwork to sides of columns, pillars, piers, post and struts, etc, square, rectangular or polygonal, rough finished surfaces of concrete complete all as directed.		598.08 **Sqm**	3941.28	2,357,200.74		
10	Use and waste of formwork to faces of walls, retaining walls, sides of drain wall, abutments, parapets and staircase railings and similar work including attached pillasters, buttresses, vertical or to batter in rough finished flat surface of concrete complete all as specified and as directed.		493.55 **Sqm**	11635.20	5,742,552.96		
11	Mild steel framed work such as grills, gratings, etc.. with ends of bars shouldered and/or riveted, or forged into spikes; framed guard bars; barred iron doors; ladders; framed balusters; walk ways; railings; framework of water tanks and similar work conforming to Fe.290.Gde-E-165 complete all as specified and directed.		164.86 **Kg**	5232.81	862,681.06		
12	Painting with two coats of synthetic enamel paint on steel and iron surfaces of any description over 10 cm in width or girth including preparation of surfaces and applying one coat of red oxide zinc chrome primer complete all as specified and as directed.		141.30 **Sqm**	2371.07	335,032.19		
13	Preparation of new plastered or unplastered of wall surfaces and applying one coat of alkali resistant primer and then applying two coats of oil bound distemper complete all as specified and directed.		101.50 **Sqm**	11635.42	1,180,995.13		

SCHEDULE 'A' PART X

(LIST OF ITEMS OF WORKS & PRICES) (COMPOUND WALL, FENCING & GATE)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
14	Plain Cement concrete 1:3:6, type C2 using 40 mm graded stone aggregate in foundation filling and mass concrete complete all as specified and as directed.		5,862.00 **Cum**	263.64	1,545,457.68		
15	Mild Steel Hand rails, half round or other steel section or core for wooden hand-rail straight portion, drilled and countersunk for balusters running bars, for doors, balusters, nosing to steps including embeding in concrete and screwing etc. wind ties for roof including securing bolts, angle posts and struts in fencing including cutting tops to shape as directed, drilling holes, notching for wire, etc.; flat iron fillets(for securing wire mesh etc.) including all necessary drilling, bolting or riveting, etc., and similar work conforming to Fe 290 (Gde E-165) complete all as specified and directed.		103.18 **Kg**	45449.78	4,689,508.30		
16	Supply only Chain Link MS galvanized fencing of mesh 50 x 50 mm and of any type of shape (minimum dia 4mm and weight of mesh not exceeding 4 Kg/Sqm) conforming to IS 2721 complete all as specified and as directed.		76.26 **Kg**	26234.21	2,000,620.85		
17	Fixing only chain link fencing including line wires, any mesh size and type of standards, rails, straining bolts, etc. including provision of binding wire.		6.35 **Sqm**	8517.60	54,086.76		
18	Supply and fixed of mild steel screw bolts 12 mm diameter of shank (machine made) with hexagonal or square heads at one end and screwed the other, or screwed both ends, each screwed end to be fitted with washers and one hexagonal or square nut, length of bolt not exceeding 5 cm complete all as specified and directed.		197.46 **Kg**	1583.87	312,750.97		
19	Galvanised, 2 strand, steel barbed wire 2.24mm dia barbed with 2mm dia barbs at 75mm spacings and straining and fixing to any type of standard rails straining bolts, including, securing or tying at crossings with and provision of galvanised mild steel wire, staples or steel pins, etc. as directed (each line wire to be measured) complete all as specified and directed.		1,342.01 **100 Rm**	234.77	315,063.69		

SCHEDULE 'A' PART X

(LIST OF ITEMS OF WORKS & PRICES) (COMPOUND WALL, FENCING & GATE)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
20	Painting with two coats of synthetic enamel paint on steel and iron surfaces of any description not exceeding 10 cm in width or girth including preparation of surfaces and applying one coat of red oxide zinc chrome primer complete all as specified and as directed.		144.10 **10 Rm**	554.52	79,906.33		
21	Rolled mild steel of grade Fe-410-W (Gde E-250) (quality A) in framed work as in doors or gates of angle or other section with gusset plates, rails, braces, wheels etc., complete, drilled for fixing of steel sheeting or other covering. Doors, etc. to be prepared for hanging or sliding with and including either hooks and hinges or steel hanging door fittings (exclusive of steel sheeting or other covering, running rails and guides) and hanging; also fastening and fixing complete.		117.70 **Kg**	4886.56	575,148.11		
22	High strength steel round bars (Fe 500D) 10 mm dia and over, cut to length, bent to shape required including cranking, bending spirally for hooping for columns, hooking ends and binding with and including mild steel wire (annealed) not less than 0.90 mm dia complete all as specified and directed.		81.87 **Kg**	12679.60	1,038,078.85		
	Total of Schedule 'A' Part-X carried over to BOQ (General Summary)				**57,740,930.69**		

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART XI

(LIST OF ITEMS OF WORKS & PRICES) (RETAINING WALL)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Plain Cement Concrete 1:4:8, type D2 using 40 mm graded stone aggregate as in foundation filling and mass concrete complete all as specified and as directed.		5,483.10 **Cum**	544.68	2,986,534.91		
2	Reinforced Cement Concrete M-25 (Design mix) using 20 mm graded stone aggregate as in foundations, including rafts, footings, foundation beams; plinth beams; bases for columns and mass concrete complete all as specified and as directed.		7,640.60 **Cum**	2126.16	16,245,138.10		
3	Reinforced Cement Concrete M-25 (Design mix) using 20 mm graded stone aggregate as in Walls, retaining walls, basement walls, ballast walls and the like, any thickness; above top of footings; including attached pilasters and buttresses complete all as specified and as directed.		8,028.60 **Cum**	4009.49	32,190,591.41		
4	High strength steel deformed TMT bars (Fe 500D) 10mm dia and over, cut to length, bent to shape required, including cranking, bending spirally for hooping for columns, hooking ends and binding with and including mild steel wire (annealed) not less than 0.9mm dia or securing with clips complete all as specified and directed.		82.60 **Kg**	487871.49	40,298,185.07		
5	Use and Waste of formwork to sides of concrete foundations, foundation beams, footing, bases of column, raft and raft beams, sides and soffits of foundation and plinth beams, etc. flat rough finished surfaces of concrete complete all as specified and directed.		316.72 **Sqm**	2302.65	729,295.31		
6	Use and waste of formwork to faces of walls, retaining walls, sides of drain wall, abutments, parapets and staircase railings and similar work including attached pillasters, buttresses, vertical or to batter in rough finished flat surface of concrete complete all as specified and as directed.		493.55 **Sqm**	25444.90	12,558,330.40		
7	Rendering 15 mm thick in cement mortar 1:4 on fair faces of brick work or concrete surfaces in two layers with first layer 10 mm thick and second layer 5mm thick, finished even and smooth without using extra cement, finishing coat shall be mixed with water proofing compound (WPC) as per manufacturers instruction complete all as specified and as directed. Note : WPC shall be measured and paid separately.		304.24 **Sqm**	9826.00	2,989,462.24		

SCHEDULE 'A' PART XI

(LIST OF ITEMS OF WORKS & PRICES) (RETAINING WALL)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
8	Supply only integral water proofing compound.		46.10 **Kg**	740.00	34,114.00		
9	Loose filling with aggregate of gauge 40mm to 80mm of broken stone or boulder including excavation in weep holes/ HDPE pipes etc complete all as specified and directed		1,924.00 **Cum**	3543.20	6,817,116.80		
	Total of Schedule 'A' Part-XI carried over to BOQ (General Summary)				**114,848,768.24**		

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART XII

(LIST OF ITEMS OF WORKS & PRICES) (FIRE ALARM SYSTEM)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	supply and fix Fire alarm control panel complying with BS EN 54 Single loop panel of capacity from 99 detectors & 99 modules upto 200 detector and modules all as specified in SSR Part-I. Note : (i)The panel shall be fixed outside the building. (ii) Earthing wire 4mm dia for Fire Alarm control panel shall be measured and paid in relevant items of Schedule of Internal Electrification Schedule i.e Sch 'A' Part-IV.		64,641.00 **Each**	8.00	517,128.00		
2	Annunciating devices Addressable control module complete all as specified and directed.		3,290.00 **Each**	29.00	95,410.00		
3	Annunciating devices Addressable manual call point resettable type complete all as specified and directed.		3,132.00 **Each**	29.00	90,828.00		
4	Smoke Detectors flame proof complying with and type certified to EN-45 Addressable type smoke detector Photoelectric / Optical with base with self built intelligence as per IS: 11360 complete all as specified and directed.		2,060.00 **Each**	56.00	115,360.00		
5	Fire Alarm sounders / Sirens electronic type horizontal double mounting single stage continuous normal operation for at least 50 minutes with sound level 65 DB to 120 DB measured at 1.5 meter with sound output variation of + 5 DB for not less than 3500 meter range (diameter) with DOL starter with addlitional auxillary control (NO+NC) complete. (to be fixed outside the building)		8,918.00 **Each**	29.00	258,622.00		

Total of Schedule 'A' Part-XII carried over to BOQ (General Summary) **1,077,348.00**

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART XIII

(LIST OF ITEMS OF WORKS & PRICES)(LIGHTNING PROTECTION)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Test point terminal block made of gun metal or phosphorus bronze size 75x75x25mm drilled and screwed including 3 nos. 8mm dia 25mm long hex. head screw complete all as specified and directed.		689.70 **Each**	25.00	17,242.50		
2	Aluminium strip 25 x 3.15mm with saddles and fixing screws etc. for roof conductors and down conductors above ground complete all as specified and directed.		125.10 **Rm**	150.00	18,765.00		
3	Earth continuity conductor or main earthing lead fixed to wall on batten or recess or chases or buried in ground or drawn in conduit/pipe or fixed to poles or any other indicated situation for loop earthing etc. as required 4mm dia galvanised iron or steel wire complete all as specified and directed.		60.20 **RM**	1200.00	72,240.00		
4	Supply and fixing Earth continuity conductor or main earthing lead fixed to wall on batten or recess or chases or buried in ground or drawn in conduit/pipe or fixed to poles or any other indicated situation for loop earthing etc. as required with Galvanized iron strip 32 x 6mm complete all as specified and directed		230.80 **Rm**	250.00	57,700.00		

Total of Schedule 'A' Part-XIII carried over to BOQ (General Summary) **165,947.50**

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'A' PART XIV

(LIST OF ITEMS OF WORKS & PRICES) (FIRE FIGHTING SYSTEM)

Ser No	Description of items/works	Drg Nos (See list of drgs)	Rate per unit (Rs)	No of unit required	Amount (Rs.)	Period of completion of individual items from the date of handing over the site	Remarks
1	2	3	4	5	6	7	8
1	Hose box made of 18 guage MS Sheet 800x600x250mm with double door 4mm thick glasses and door frame made of 16 guage MS sheet painted fire red outside and brilliant white inside yard complete all as specified and as directed.		3,359.00 **Each**	19.00	63,821.00		
2	300mm dia heavy, 'c' class MS ERW pipes (ERW quality conforming to IS 1239) with ISI marked with screwed or welded joints, including all fitting, supported by galvanized hangers (Size and Spacing as per drawing) including anchor fasteners for fixing in ceiling testing to 15 kg/cm2 pressure, painting with a coat of Zinc Dichromate primer and two coat of approved enamel paint, making direction arrows and labelling etc.complete (Pipes 50mm and above diameter will be welded and pipes 50mm and below will be screwed jointed complete all as specified and directed.		4,221.00 **Rm**	1500.00	6,331,500.00		
3	150mm dia heavy, 'c' class MS ERW pipes (ERW quality conforming to IS 1239) with ISI marked with screwed or welded joints, including all fitting, supported by galvanized hangers (Size and Spacing as per drawing) including anchor fasteners for fixing in ceiling testing to 15 kg/cm2 pressure, painting with a coat of Zinc Dichromate primer and two coat of approved enamel paint, making direction arrows and labelling etc.complete (Pipes 50mm and above diameter will be welded and pipes 50mm and below will be screwed jointed complete all as specified and directed.		1,260.00 **Rm**	200.00	252,000.00		

Total of Schedule 'A' Part-XIV carried over to BOQ (General Summary) **6,647,321.00**

Signature of Contractor
Date :

Jt/Dy/Asstt/Addl. Asstt. Dir (Contracts)
For Accepting Officer

SCHEDULE 'B'
LIST OF MATERIALS TO BE ISSUED TO THE CONTRACTOR
(REFER CONDITION 10 OF GENERAL CONDITIONS OF CONTRACTS - 1AFW-2249)

Sl No.	Particulars	Rate at which Stores, etc. will be issued to the Contractor		Place of issue (by name)	Remarks
		UNIT	RATE		
1	2	3	4	5	6

---------- **NIL** ----------

(SIGNATURE OF CONTRACTOR) **Dy DIR/AAD (CONTRACTS)**
DATED: **FOR ACCEPTING OFFICER**

SCHEDULE 'C'
ISSUE OF TOOLS AND PLANTS (OTHER THAN TRANSPORT) WHICH WILL BE HIRED BY THE CONTRACTOR
(REFER CONDITION 15, 34 AND 35 OF IAFW-2249)

Ser. No.	Quantity	Particulars	Details of MES Crew supplied	Hire charges per unit per working day	Stand by charges per unit per day	Place of Issue	Remarks
1	2	3	4	5	6	7	8

---------- NIL – ----------

(SIGNATURE OF CONTRACTOR) **Dy DIR/AAD (CONTRACTS)**
DATED: **FOR ACCEPTING OFFICER**

SCHEDULE 'D'
TRANSPORT TO BE HIRED BY THE CONTRACTOR
(REFER CONDITION 16, AND 35 OF IAFW-2249)

Sr. No.	Quantity	Particulars	Rate per unit per working day	Stand by charges per unit per day	Place of Issue	Remarks
1	2	3	4	5	6	7

---------- NIL – ----------

(SIGNATURE OF CONTRACTOR) **Dy DIR/AAD (CONTRACTS)**
DATED: **FOR ACCEPTING OFFICER**

TENDER

To

The President of India

Having examined and perused the following documents **: -**

1. Specifications signed by Joint Director (Contracts)/ Dy Director (Contracts /Asst Director (Contracts)/ Addl Asst Director (Contracts)

2. Drawings details in the specifications.

3. Schedule 'A', 'B', 'C', 'D' attached hereto.

4. MES Standard Schedule of Rates Part-I (2009) Specifications and Part-II (2020) Rates inclusive of amendment No. 1 to 122 to SSR Part-II(Rates) including Section No. 30 for "Runway & Pavement" containing item Nos. 30001 to 30053 in SSR Part-II(Rates) (here-in-before) and in IAFW-2249 referred to as the MES schedule) together with Errata/amendment nos 1(a) to 3 for Part-I as applicable.

5. General Conditions of Contracts, IAFW-2249 (1989 Print) together with **errata No 1 to 20** and **amendments 1 to 49**.

6. **WATER :** Water will not be supplied by MES.

Should this tender be accepted, I/We agree

(a) *That the sum **of ₹**_____ forwarded as Earnest money, shall either be retained as part of Performance Security Deposit or be refunded by the Government on receipt of the appropriate amount of Performance Security Deposit, all as per Condition 19 of IAFW-2249.

(b) To execute all the works referred to in the said documents upon the terms and condition contained or referred to therein and as detailed in General Summary here-in-after and to carry out such deviations as may be ordered vide condition 7 of IAFW-2249 upto a maximum of **10 % (ten percent)** and further agree to refer all disputes (in case disagreement with the decision of such **DRB or conciliator** as applicable as per condition 71 of IAFW 2249), as required by Condition 70 of IAFW-2249 to the sole arbitrator of a serving Officer having degree in Engineering or equivalent or having passed final/direct final Examination of Sub-division II of Institution of Surveyors (India) recognised by the Government of India to be appointed by the **ENGINEER-IN-CHIEF** or in his absence the Officer Officiating as **ENGINEER-IN-CHIEF** or **DIRECTOR GENERAL OF WORKS** if specifically delegated in writing by the **ENGINEER-IN-CHIEF ARMY HEADQUARTERS NEW DELHI** whose decision shall be final, conclusive and binding.

*Delete if not applicable.

Brought forwarded **₹.....(Rupees.....**

.....)

Signature_____ (*Name of Signatory*_____) in the Capacity of _____ duly authorised to sign the TENDER and on behalf of M/S_____ DATED _____

(IN BLOCK CAPITALS)

Postal Address .....

.....

…….....

……..

Telephone No. .....

Mobile No **1.** .....

(Alternate) **2.** .....

E-Mail ID **1**. ……..……@……

2. ……..……@……

Fax No .....

Witness.....

Address.....

.....

…….. .....

ACCEPTANCE

.....alterations have been made in these documents as evidence that these alterations were made before the execution of the contract Agreement, they have been initialed by the contractor and Shri .....

The said officer(s) is/are hereby authorized to sign and initial on my behalf of the documents forming part of this contract.

The above tender is/was accepted by me on behalf of the President of India for the lump sum of ₹.....……

(Rupees.....……

……)

Signature _____

Appointment **:** Chief Engineer Shillong Zone

(For and on behalf of the President of India)

Dated this _____ day of _____ 2024

CA NO : CESZ/HAY/ OF 2024-2025 **SERIAL PAGE NOS 127 to 160**

GENERAL CONDITIONS OF CONTRACTS (IAFW-2249 : PRINT – 1989)

FOR

ITEM RATE CONTRACT (IAFW-1779A (REVISED 1955))/ LUMP SUM CONTRACT IAFW-2159

A copy of the GENERAL CONDITIONS OF CONTRACTS (IAFW-2249-1989 print) has been supplied to me/us and is in my/our possession. I/We have read and understood the provisions contained in the aforesaid GENERAL CONDITIONS OF CONTRACTS before submission of this tender and I/We agree that I/We shall abide by the terms and conditions therefore, as modified, if any, else where in this tender documents.

It is hereby further agreed and declared by me/us, that the GENERAL CONDITIONS OF CONTRACTS (IAFW-2249-1989 Print) including condition 70 thereof pertaining to settlement of dispute by Arbitration, containing 33 pages (Ser Page No. **127 to 160**) with **errata 1 to 20** and **amendment Nos 1 to 49** form part of this tender documents.

It is also agreed by me/us that in case of any discrepancy in the interpretation of the Contents between ENGLISH and HINDI VERSION, ENGLISH VERSION shall take precedence over HINDI VERSION.

_____
SIGNATURE OF CONTRACTOR
DATED :

Jt/Dy DIRECTOR/AAD (CONTRACTS)
FOR ACCEPTING OFFICER

Note :- In case the tenderer is not in possession of any of the documents referred above he should obtain a copy from the office of the HQ Chief Engineer Shillong Zone, Spread Eagle Falls, Shillong - 793011 before submission of his tender.

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MILITARY ENGINEER SERVICES
ERRATA TO GENERAL CONDITIONS OF
CONTRACT IAFW - 2249 (1989 PRINT)

ERRATA

Errata No	Page No	Particulars
1	2	3
1.	4	Condition 3, Para 3, Line 1: For “An’ Read “ and”
2.	7	Condition 7, last Para, line 9 ; For “Objection” Read “objection”
3.	7	Condition 7, last Para, line 11 : For “thetein” Read “therein”
4.	8	Condition 8, line 7 “Delete” after the word “necessary”
5.	10	Condition 15, Para 2, line 3, For “actory” Read “ Factory”
6.	11	Condition 15, Para 2, line 6 For “escept” Read “except”
7.	11	Condition 15, Para 2, line 6 For “authorise Read “ authorised”
8.	11	Condition 15, Para 5, line 1, For “Maintence” Read “ Maintenance”
9.	16	Condition 32, Para 1, line 4 For “sitning” Read “siting”
10.	16	Condition 32, Para 2, line 6 For "reinstead" Read"reinstated"
11.	16	Condition 34, line 6 For "revese " Read "revest"
12.	16.17	Condition 36, para 2, line 2 For "conbusitible" Read"combustible"
13.	17	Condition 37, line 3 For "whese " Read "whose"
14	17	Condition 43,line 1 For "shal" Read " Shall"
15.	17	Condition 44, para 2, line 4 For"visit this " Read "visit his"
16.	18	Condition 48, para (a), line 5 For "airising" Read "arising"
17	20	Condition 54, line 1 and 2 illigible word is "Contractor"
18.	20	Condition 54, para below (a), Insert`b' in the exsiting bracket "( )"
19.	30	ANNEXURE `B', Clause 6, line 2 Delete the word "after"
20.	30	ANNEXURE `B', Clause 6 but line 3 Insert fullstop"(.)" after the word "thereof" and For "laddor" Read "Ladder"

SIGNATURE OF CONTRACTOR
DATED :

Jt/Dy/DIRECTOR/AAD (CONTRACTS)
FOR ACCEPTING OFFICER

MILITARY ENGINEER SERVICES
AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)

Amendment No	Page No	Particulars
1	2	3

1. 28 & 29

Annexure `A' Military Engineer Services Contractor's labour Regulations

(a) Para 2 (a) line 3

For : Not exceeding Rs.400 per month

Read: Not exceeding Rs.500 per month"

(b) Para 11, line 5 Add the following after the words labourers concerned. "The Garrison Engineer shall payments to the labour concerned within 45 days from the receipt of the report from the labour welfare officer or other person authorised as aforesaid, as the case may be."

2 9

Condition 10(R), Line 2 to 6.

For : Materials which_____ as aforesaid.

Read: "Materials which Govt shall supply are shown in Schedule `B' which also stipulates place of issue and rate(s) to be charged in respect thereof".

3. 19

Condition 50, Sub para 4(b) (iii) Delete the existing description against sub para 4(b) (iii) and insert as under:-

(iii) Where the contract sum exceeds Rs.20lakhs but does not exceed Rs. 150 lakhs.	Seven and half percent of the Contract value of the item or group of items of work for which a separate period of completion is given or Rupees seven and a half lakhs whichever is less.
(iv) Where contract exceeds Rs.150 lakhs but does not exceed Rs. 300 Lakhs.	Five percent of the contract value of the item or group of the items of work for which a separate period of completion is given or Rupees twelve and a half lakhs whichever is less.
(v) Where the contract sum exceeds Rs. 300 lakhs.	Four percent of the contract value of the item or group of items of work for which a separate period of completion is given subject to a maximum of Rupees twenty five lakhs."

4. 27

CONDITION 70 TO SUB PARA-9

Add the following at the end of the sub para : "If the value of the claims or counter claims in an arbitration reference exceeds Rs. 1 Lakh the arbitrator shall give reasons for the award."

5. 14 &15

CONDITION 25

Delete para 2 and 3 of the above condition and substitute the following :- "Where the contractor is not a qualified Engineer or even if he is so qualified, he cannot, in the opinion of the GE give his full personal attention to the works, he shall at his own expense employ a person/person(s) possessing the following qualifications, and/or experience as his accredited agent(s)to supervise the works and to receive instructions from the Engineer-in-Charge:-

MILITARY ENGINEER SERVICES
AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)

Amendment No	Page No	Particulars
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(a) For works costing over Rs. 300 lakhs other than those covered in(g) below.	Two Degree holders in Engineering from Government recognised Institution or equivalent with atleast 5 years practical experience of works. and Adequate numbers of Diploma holders in Engineering from a government recognized Institution with at least 8 years practical experience of works.

(Note :- The above provision shall be applicable irrespective of the fact whether contractor himself is a qualified Engineer or not).

(b) For works costing between Rs.100 lakhs and Rs.300 lakhs. other than those covered in (g) below	A degree holder in Engineering from a Government recognised Institution or equivalent with at least 4 years practical experience of works OR Two diploma holders in Engineering from a Government recognised Institution with at least 6 years practical experience of works
(c) For works costing between Rs.60 lakhs and Rs.100 lakhs other than those covered in (g) below	A degree holder in Engineering from a Government recognised Institution or equivalent with at least 3 years practical experience of works. OR A diploma holder in Engineering from a Government recognised Institution with at least 5 years practical experience of works.
(d) For works costing between Rs.7.5 lakhs and Rs.60 lakhs other than those covered in (g) below	A degree holder in Engineering from a Government recognised Institution or equivalent with at least 2 years practical experience of works. OR A diploma holder in Engineering from a Government recognised Institution with at least 4 years practical experience of works.
(e) For works costing between Rs.2 lakhs and Rs.7.5 lakhs other than those covered in (g) below	A degree or diploma holder in Engineering from a Government recognised Institution with adequate practical experience of works.
(g) For works of any value for repairs to buildings and roads, periodical Services to building, term contract, grass cutting and fencing	A competent person with adequate practical experience of works. The employment of Agent(s) as aforesaid shall be to the approval of GE who may verify his / their qualifications and experience by referring to original degree/diploma/testimonials which shall be made available to him by the contractor or by the individual employed or proposed to be employed.

MILITARY ENGINEER SERVICES
AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)

Amendment No	Page No	Particulars
1	2	3
6	27	Delete the description added at the end of sub para 9 vide amendment No.4 (1989 Print) and insert as under :- "The arbitrator shall give reasons for the award in each and every case irrespective of the value of claims or counter-claims".
7	24	CONDITION 64 Delete para 4 viz `Provided the amount_____ shall not exceed Rs. 1.5 lakhs in toto.
8.	26	CONDITION 70 ARBITRATION PARA 1, LINE 4 Amend the words "Engineer Officer" to Read" Serving Officer having degree in Engineering or equivalent or having passed final/direct final Examination of of Sub-Division II of Institution of Surveyor (India) recognised by the Govt. Of India."
9.	2	Add the following new condition under chapter IV:- "71. Jurisdiction of Courts".
	27	Add the following new condition:- "71. Jurisdiction of Courts-Irrespective of the place of issue of tenders, the place of acceptance of tenders, the place of execution of contract or the place of payment under the contract, the contract shall be deemed to have been made at the place from where the acceptance of tenders has been issued and the work is executed/executable, The courts of the place from where the acceptance of the tender has been issued or the place where the work is executed/under execution shall alone have jurisdiction to decide any dispute arising out of or in respect of the contract."
10.	14 & 15	CONDITION 25 AS AMENDED VIDE AMENDMENT No.5 FOR " Degree holders in Engineering from a Government recognised Institution or equivalent" wherever occuring. READ "Degree holders in Engineering from a Government recognised Institution or equivalent/final or direct final pass of sub division II of the Institution of Surveyors (India)."
11.	27	CONDITION 70 : ARBITRATION PARA 8 FOR : "The Arbitrator may, from time..... Publishing the award." READ: "The arbitrator may, from time to time with the consent of the parties, enlarge the time for making and publishing the award."
12.	24	Add the following Para 4 after Para 3 (c) :- "The amount so retained from the contractor shall be called retention money and shall be released to the contractor alongwith the final bill. However, in case the amount of this retention money is more than 1.50 lakhs, then after retaining an amount of Rs. 1.50 lakhs or 1% of the contract amount as executed whichever is more, the balance amount of retention money will be refunded to the contractor at satisfactory completion of works as certified by the GE."
13.	23	Condition 62 (G) valuation of deviation para 1 line 3 For : "Plus 10 %" Read : "Plus 15 %"

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AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)

Amendment No	Page No	Particulars
1	2	3
14.	5	(a) Condition 4A(a), Line 4 FOR: "Rs. 2,500/-" READ :"Rs.5,000/-" (b) Condition 4A(d), Line 2 & 3 FOR : "Rs. 5,000/-" READ: 10,000/-
15.	13	Condition 22, Security Deposit, sub para 3, Line 3 and 4 For : "Controller of Defence Accounts concerned" Read : "Accepting Officer"
16.	24	Condition 64 Sub para 7, lines 4 & 5 For:"Controller of Defence Accounts" Read:"Accepting Officer"
17.	19	(a) Condition 50, Sub Para 2, Line 2a For : " Deviation orders on the contractor" Read : "Deviation order, escalation amount" (b) Condition 50, Sub Para 4 (b) (i) to 4 (b) (v) Delete existing sub para 4 (b) (i) to 4 (b) (v) in toto and insert as under ; "Ten percent of the contract value of the item or group of items of work for which a separate period of Completion is given." (c) Condition 50, Sub Para para 5 Add Sub para 5 as under "Amount of retention money plus compensation during currency of contract shall not exceed ten percent of contract value. Contractor shall furnish BGB or Fixed Deposit Receipt in lieu of retention money and if compensation amount is not fully met from retention money, the difference if any shall be recovered from the payments due to the contractor."
18.	3 and 4	(a) Condition 1(a), line 4 For : " the M.E.S. Schedule, the Specifications" Read : "the M.E.S. Schedule, the Special Conditions, the Specifications" (b) Condition 1(b), line 2 For : " these conditions, and the Specifications" Read : "these conditions, the Special Conditions and the Specifications" (c) Condition 1(b), line 3 For : " loaned" Read : "sold" (d) Condition 1(g), line 3 For : " Garrison Engineer" who administers" Read : "Garrison Engineer" (or Assistant Garrison Engineer (Independent) if applicable) who administers" (e) Condition 1(g), line 3 In the end, add the following:- " Where the office of the CWE does not exist and the GE is directly under the Chief Engineer, the authority of the CWE stipulated in these conditions shall vest in the concerned Chief Engineer." (f) Condition 1(h), line 3

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AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)

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		For : "the Superintendent Grade I" Read: " the Junior Engineer (JE)"
18(contd.)		(g) Condition 1(i), line 1 For : "Indian Standards Instiution." Read: " Bureau of Indian Standards."
		(h) Condition 1(p), line 4 For : "air craft and acts of God" Read: "air craft and natural calamities"
		(j) Condition 1(p), line 5 For : "floods and tornado." Read: "floods, tornado and Tsunami."
19.	5	(a) Condition 4A, line 7 For : "Rs. 60,000/-" Read: "Rs. 1,50,000/-"
		(b) Condition 4A(a), line 4 For : "Rs. 2,500/-" Read: "Rs. 10,000/-"
		(c) Condition 4A(c), line 4 For : "Rs. 2,500/-" Read: "Rs. 10,000/-"
		(d) Condition 4A(d), lines 2 and 3 For : "Rs. 5,000/-" Read: "Rs. 20,000/-"
		(e) Condition 4A(f), line 1 For : "Rs. 60,000/-" Read: "Rs. 1,50,000/-"
20.	5	(a) Condition 6, subpara 1 The existing contents shall be substituted as under:- "**6. Provisional Items**- The amount pertaining to provisional items need not be deducted from the Contract Sum. The Engineer-in-Charge and Contractor shall set out the works covered under provisional items and provisional lump sum based on the description of items, drawings forming part of contract agreement and considering the ground conditions as encountered at site of works. The statement of variation in quantities, new items or deletion of items from the contract agreement as required will be worked out for approval of Engineer-in-Charge who will convey approval of such changes through site order book. For the purpose of payment the variation in value of work executed under these items shall be ascertained by measurement or valuation as for deviation. The variations shall be regularized as per condition 7."
21.	6 and 7	(a) Condition 6A, line 6 Add following after the words 'preference to scale.':- "However the provisions of any drawing of a later date shall take precedence over that of a drawing of previous date further subject to the condition that working drawing will take precedence over a Typical Detail Drawing." (b) Condition 6 A(a), line 1 For: "Quantities."

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Read: "Quantities, Schedule 'B', 'C', 'D',"

(c) Condition 6 A(a), subpara 3, line 1
For: "one document forming"
Read: "one document as defined hereinabove forming"

(d) Condition 6 A(A)
(i) Existing subpara (b) to (f) shall be enumerated as subpara (c) to (g)
(ii) Add new subpara (b) after existing subpara (a) as under:-

"(b). Items and their quantities for which no rates or prices have been quoted shall be deemed to have been covered by the rates and prices quoted for the other items of Schedule 'A' or bill of quantities."

(e) Condition 6 A(C) (iii)
The existing contents shall be substituted as under:-

(iii)	Contract sum exceeding Rs. 10 lakh but not exceeding Rs. 100 lakh	: Rs. 20,000 plus 2% of Contract Sum exceeding Rs. 10 lakh subject to a maximum of Rs. 1,00,000/-
(iv)	Contract sum exceeding Rs. 100 lakh	: 1% of Contract Sum subject to a maximum of Rs. 5 lakh

22. 8 (a) Condition 9 (c), line 19
For: "labour at Site"
Read: "labour and tools and plants at Site"

(b) Condition 9(c), line 19
Add following at the end of para (c) viz line 21:-
"The contractor shall give the details of idle labour, workmen, employees, tools and plants and such other details as asked by the GE, within seven days of issue of order of suspension and the GE shall verify the necessity and correctness thereof. Proper record of these details shall be maintained duly signed by the GE and the contractor."

23. 8, 9 and 10 (a) Condition 10 A, subpara 2

Add in the end after the word "comply" following:-

"However if the cost of particular item of material in a contract exceed Rs. 1 lakh, these materials shall be procured only from the manufactures or from their authorised dealers/ stockist (except in case of material of local origin) and the contractor shall furnish proof thereof to the satisfaction of the GE that the material so comply."

(b) Condition 10 A, subpara 2
Add in the end after the word "in the Contract." following:-

"The approved samples of materials which loose their identity after incorporation in the work shall be preserved with the GE till completion of work. Thereafter the same shall be removed in 'as is where is' condition by the contractor with prior permission of the GE without any extra cost to the Government. However, the approved samples of material which do not loose their identity after incorporation and which can be incorporated in the work as decided by the Engineer-in-Charge shall be allowed to be used in the sample quarter/ block/ work."

(c) Condition 10 B, line 1 of last subpara of page 9
For: "cost of loading,"
Read: "cost of carriage, loading,"

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AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)

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		(d) Condition 10 B, line 3 of subpara 3 of page 10 For: "the Engineer-in-Charge may," Read: "the GE may," (e) Condition 10 B, line 5 of subpara 3 of page 10 For: "as decided by the GE." Read: "on the date of notifying to the contractor by GE (after technical check of final bill in CWE's Office) as decided by the GE or the stock book rate as fixed by the department, whichever is higher."
23.(Contd)	8, 9 and 10 (Contd)	(f) Condition 10 B, line 6 of subpara 3 of page 10 For: "the market rate," Read: "the recovery rate,"
24.	10 and 11	(a) Condition 11(A), subpara (i) to (vii) Existing contents of subpara (i) to (vii) shall be replaced as under with sub para (i) to (iv):- "(i) by reason of civil commotion, local combination of workmen, strike or lockout, affecting any of the trades employed on the work, or (ii) by reason of delay on part of nominated sub contractors, or nominated suppliers which the Contractor has , in the opinion of G.E., taken all practicable steps to avoid, or reduce, or (iii) by reason of delay on the part of Contractors or tradesmen engaged by Government in executing works not forming part of the contract, or (iv) by reason of any other cause (except force majeure) which in the absolute discretion of the Accepting Officer is beyond the Contractor's control;"
24 (Contd.)		(b) Condition 11(A), last subpara, line 2 For: "G.E." Read: "Accepting Officer" (c) Condition 11(B), last but 3rd line from end For: "G.E." Read: "Accepting Officer" (d) Condition 11(C) The existing contents shall be substituted as under:- "(C) Extension of time if due shall be granted within 45 days of receipt of request from the contractor along with supporting documents, but before expiry of original/ extended period of completion. (D) No claim in respect of compensation or otherwise, for idle labour and/ or idle machinery etc. and/ or business loss or any such loss, howsoever arising, as a result of extensions granted under Conditions (A) and (B) above shall be admissible. The decision on reason and quantum of extension shall be final binding. (E) DELAY ON ACCOUNT OF FORCE MAJEURE – Should any force majeure circumstances arise, each of the contracting party will be excused for the non fulfilment or for the delayed fulfilment of any of its contractual obligations, if the affected party within 15 days of its occurrence informs the other party in writing. Force majeure shall mean fires, floods, natural calamities such as earth quakes, lightening or other acts such as war, turmoils, strikes (otherwise than

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contractor's employees), invasion, act of foreign enemies, hostilities, civil war, rebellion, revolution, insurrection, military or usurped power, damage from aircraft, sabotage, explosions, quarantine restrictions, beyond the control of either party.

It is understood and agreed between the parties here to that the rights and obligations of the parties shall be deemed to be in suspension during the continuance of the force majeure even as aforesaid and the said rights and obligation shall automatically revive upon cessation of the intervening force majeure event. The period within which the rights and obligations of the parties shall be in suspension due to force majeure event, shall not be considered as a delay with respect of the period of completion and/ or taking over work under the contract or otherwise to the detriment of either party.

Not withstanding the provision of the immediately foregoing clauses, it is further understood and agreed between the parties hereto that in the event of any force majeure persisting for an uninterrupted period exceeding 6 (Six) months, either party hereto reserves the right to terminate this contract upon giving prior written notice of 30 (thirty) days to the other party of the intension to terminate without any liability other than agreement for the completed work and/ or contractor's materials lying at site."

25. 12 (a) Condition 17, line 1
For: "The Contractor shall not"
Read: "The Contractor including Public Sector undertaking/ Government agency shall not"

26 13 (a) Condition 18, line 1
For: "The Contractor shall not"
Read: "The Contractor including Public Sector undertaking/ Government agency shall not"

27. 14 and 15 (a) Condition 25, the contents of subpara 2 after line 5 shall be substituted as under:-

(a)	For works costing between Rs. 50 lakh to 1000 lakh	A Degree holder in Engineering from a Govt recognized Institution or equivalent, with final or direct final passed of Sub Division II of the Institution of Surveyors (India) with at least 4 years practical experience of works.
(b)	For works costing between Rs. 7.5 lakh to 50 lakh	A Degree holder in Engineering from a Govt recognized Institution or equivalent, with final or direct final passed of Sub Division II of the Institution of Surveyors (India) with at least 2 years practical experience of works.
(c)	For works costing below Rs. 7.5 lakh	A Diploma holder in Engineering from a Govt recognized Institution with adequate practical experience of works.

Notes:-

(1) Engineers (Degree/ Diploma holders) employed should be of the relevant discipline to which nature of work pertains.

(2) The provision at Serial (b) & (c) above shall be applicable irrespective of the fact whether contractor himself is a Qualified Engineer or not.

(3) Contractor shall employ additional Engineers as directed by GE where there are scattered sites.

(4) For works costing more than Rs. 10 crore, the requirement of Engineering staff

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Amendment No	Page No	Particulars
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shall be as given in the tender documents.

(5) For specialist works/ services Accepting Officer may vary requirement of supervisory staff in tender documents.

(b) Condition 25, last subpara on page 15:
The contents of last subpara on page 15 shall be substituted as under:-

" The G.E. shall have full powers, to put the contractor on notice on account of default either for non-employment of Engineer(s) or absence of Engineer(s) from site and levy penalty @ Rs. 500/- per day per vacancy upto 30 days period. Thereafter GE shall have the option to either suspend the work or employ Engineer(s) at contractor's cost and recover the amount from contractor's dues."

28. 15 (a) Condition 26, last subpara, last line

For: "Rs. 50/-"
Read: "Rs. 5,000/- (Rupees five thousand only)"

29. 16 (a) Condition 31, third subpara, line 2

For: "at the *All India Flat Rate per 1,000 gallons which"
Read: "at the *All in cost Rate per 1,000 gallons subject to a minimum of Rs. 3.75 per every Rs. 1000/- worth of work done priced at contract rates which"

30. 17 (a) Condition 36, last subpara
Add the following at the end:-
" Fencing be provided wherever necessary as decided by GE to isolate the working area to make the area unrestricted from restricted"

31. 17 (a) Condition 44, subpara 3, line 1

For: "rupees one lakh"
Read: "rupees two lakh"

(b) Condition 44, subpara 3, line 2
For: "rupees five lakh"
Read: "rupees ten lakh"

32. 18 (a) Condition 46, subpara 2, line 11 to 14

For: "Provided always that the liability of the Contractor under this Condition shall not extend beyond the defects liability period except as regards workmanship which the G.E. shall have previously given notice to the contractor to rectify."
Read: "Alternatively, such work, if technically/ structurally acceptable, without detriment to the safety and utility of the item and the structure may be permitted to be accepted as devalued and recovery shall decided by competent authority (CWE in respect of contract concluded by himself and GE's and AGE(I) and CE in respect of contract concluded by him) or he may reject the work outright without any payment and/ or get it and other connected and incidental items rectified, or removed and re-executed at the risk and cost of the contractor. Whether any particular defect is due to unsound, imperfect or unskillful workmanship or due to normal wear & tear or user's negligence, decision of GE shall be final and binding. Provided always that the liability of the contractor under this Condition shall not extend beyond the defects liability period except as regards workmanship which the G.E. shall have previously given notice to the contractor to rectify. Govt. further reserves the right to get the work technically inspected during

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currency of the contract and also during defects liability period by the Additional Director General of Technical Examination and/ or his Officers or any other agency. The defects observed as a result of such technical examination shall be rectified by the contractor as notified by the GE. However, if the defects are not rectified, the devaluation of the work shall be carried out and recovery thereon shall be affected. ”

33. 19 (a) Condition 49, subpara 2, line 2,3 and 4

For: “within such period as may be notified by the Engineer-in-Charge, to the place of issue against written receipt from the Engineer-in-Charge.”

Read: “to the place of issue as stipulated in Condition 10(B) here-in-before against written receipts from the Engineer-in-Charge.”

(b) Condition 49, subpara 5, line 3,4 and 5

For: “before the completion of entire group, but for all purposes of the contract except for compensation for delay, the completion of the entire group shall be taken into account.”

Read: “before the completion of the entire group. In such event, the grouping (phasing) of items as catered for in the contract shall be deemed to have been amended accordingly.”

33. (Contd.) (c) Condition 49, subpara 7

Existing contents shall be substituted as under:-

“On receipt of notice from the Contractor that the work has been completed, the G.E. shall within seven days certify to the Contractor the Date(s) on which the items or group of items of works are completed and taken over and the state thereof or shall notify the details of incomplete items of work to the contractor. In case of dispute between G.E. and the Contractor over completion of work, the decision of Accepting Officer or CWE in case of G.E.’s contract shall be final and binding.”

34. 21 (a) Condition 55

Existing contents shall be substituted as under:-

“**55. Termination of Contract for Death**—Without prejudice to any of the rights or remedies under this contract, if the Contractor dies, the Accepting Officer shall have the option of terminating the Contract without compensation to the Contractor. If proprietor has nominated a person during his lifetime, the nominee will be allowed by the GE to complete the balance work. However if the nominee is not willing or in the opinion of Accepting Officer is not capable of completing the work as contracted for, he shall terminate the contract without any compensation to the nominee. The decision of the Accepting Officer whether the nominee is capable or not shall be final and binding.”

35. 21 and 22 (a) On page 2 against Sl 59, Blank, add following in description

“Determination of contract in the event of Force Majeure_”

(b) On page 22, against SL 59, Blank

Insert new condition 59 in lieu of **Blank** as under:-

“**59. Determination of Contract in the event of Force Majeure—** The contract may be determined at the option of either party by giving 30 days notice in writing to the other party should any event of ‘Force Majeure’ continue to prevail for an uninterrupted period of six months and no progress of work is achieved owing to such circumstances during theses six months. On receipt of notice from one party to other explaining circumstances of ‘Force majeure’ the Garrison Engineer and contractor will carry out joint inspection of works and an inventory of completed, incomplete works and the materials collected for

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incorporation in the work shall be prepared for processing of final bill. The contractor shall have no claim to payment of any compensation on account of any profit or advantage which he may have derived from the execution of work in full but which he could not derive due to determination of contract on account of 'Force Majeure."

36. 22 (a) Condition 61, Last but one para, line 6

For: "Rs. 500/-"
Read: "Rs. 5,000/-"

37. 24 (a) Condition 64, subpara1 and 2
Substitute existing contents with following:-

"64. Advances on Account-- The contractor may at intervals of not less than 30 days for contracts of value less than Rs. 50 lakhs and 15 days in case of contracts more than 50 lakhs submit claims on I.A.F.W. – 2263 provided the payment due is not less than Rs. 50,000/- and for Term Contracts, contractor may prefer not more than two claims for payment of advances on account of work done and of materials delivered in connection with Measurement and Lump Sum Contracts. However such claim for work done, which are required to be measured, shall be submitted only after recording joint measurements in the MES Measurement Book IAFW-2261."

(b) Condition 64, subpara 3, line 3

For: "Engineer-in-Charge:-"
Read: "Garrison Engineer:-"

(c) Condition 64, subpara 8 and 9
Insert a new subpara between subpara 8 and 9 as follows:-

" Provided further, the contractor may be paid advance on account to the full value of materials such as fittings and fixtures and other manufactured items as decided by the GE which do not lose their identity, brought on the site, on his furnishing Guarantee Bond(s) or Fixed Deposit Receipt(s) from Schedule Bank for the amount of retention money which should otherwise be recoverable from him under the contract. The Guarantee Bond and Fixed Deposit Receipt shall be executed and kept valid in a manner as described here-in-before."

38. 25 (a) Condition 66, line 2

For: "the period being"
Read: "the period to be"

(b) Condition 66, line 3

For: "by the G.E."
Read: "by the Engineer-in-Charge"

(c) Condition 66(a)

For: "Four months"
Read: "Six months"

(d) Condition 66(b)

For: "Six months"
Read: "Nine months"

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39.	27	Existing Condition 71 Jurisdiction of Courts shall be renumbered as Condition 72
40.	27	Add a new Condition 71 Conciliator as under:- "**71. Conciliator**—If a dispute (other than those for which the decision of the CWE or any person is by the contract expressed to be final and binding) of any kind whatsoever arises between the parties to the contract during the execution of the works, or after completion or after determination cancellation/ termination of the contract, including any disagreement by either party with any action, inaction, opinion, instruction, certificate or valuation by the Accepting Officer or his nominee, the matter in dispute shall, in the first place be referred to the Disputes Resolution Board (DRB) in case of contracts valuing Rs. 10 crore or more and to conciliation, by a sole conciliator, in case of contracts valuing less than Rs. 10 crore. In case of disagreement with the decision of such DRB or conciliator, any party may invoke arbitration clause. Procedure for the constitution/ appointment of DRB/ Conciliator shall be as laid down in the Contract Agreement."
41.	13	Condition 22, Security Deposit including all its sub para **"Security deposit** including all its sub para as existing be deleted in toto and read as BLANK."
42.	20	Condition 53, Three Paragraphs after Para 53(c) Substitute existing contents with following:- "Whenever the Accepting Officer exercises his authority to cancel the Contract under this condition, he may complete the Works by any means independently without risk and cost of the original contractor. On cancelation of contract, the performance security and retention money upto last paid RAR shall be forfeited. All T&P and material of contractor lying at site shall be confiscated by Government and shall be absolutely at the disposal of the president of India and No compensation whatsoever shall be allowed by the department. If the failed contractor is a Company, or a Firm then every member / partner / Director of Company, or Firm shall be barred from participating in the tender for the balance work either in his/her individual capacity or as a partner of any other Company/Firm."
43.	20 and 21	Condition 54, Paras after Para 54(d) Substitute existing contents with following:- "The Accepting Officer may without prejudice to any other right or remedy which shall have accrued or shall accrue thereafter to the Government, cancel Contract as a whole or only such Work Order(s) or items of Work in default from the Contract. Whenever the Accepting Officer exercises his authority to cancel the Contract under this condition, he may complete the Works by any means independently without risk and cost of the original contractor. On cancelation of contract, the performance security and retention money upto last paid RAR shall be forfeited. All T&P and material of contractor lying at site shall be confiscated by Government and shall be absolutely at the disposal of the president of India and No compensation whatsoever shall be allowed by the department. If the failed contractor is a Company, or a Firm then every member / partner / Director of Company, or Firm shall be barred from participating in the tender for the balance work either in his/her individual capacity or as a partner of any other

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		Company / Firm.”
44.	25 & 26	Condition 67, Recovery from contractor (i) Para 67(a), line – 10 For words : “Security Deposit or Security Bond amount” Read: “Performance Security amount (or from the Contractor’s Standing Security Deposit amount or Security Bond if Performance Security amount is not adequate)” (ii) Para 67(g), line – 5 For words : “Security Deposit or Security Bond amount” Read: “Performance Security amount (or from the Contractor’s Standing Security Deposit amount or Security Bond if Performance Security amount is not adequate)”
45.	26	Condition 68, Refound of Security Deposit including all sub paras Existing contents shall be substituted as under:- “**68. Refund of Performance Security:-** The Performance Security Deposit mentioned in Condition 19 above may be refunded to the Contractor after the expiration of the defects liability period (vide Condition 46) by the G.E. provided always that the Contractor shall first have been paid the Final Bill and have rendered a No-Demand Certificate (I.A.F.W.-451).”
46.	26	Condition 70, Para 3 Deleted the existing contets and shall be substitude as “BLANK”
47.	13	Condition 19 **For:** “BLANK” **Read: “19. Performance Security**:- 19.1. Within 28 days of receipt of the Letter of Acceptance, the successful contractor shall deliver to the Accepting Officer a Performance Security in any of the forms given below for an amount equivalent to 5% of the contract sum. (a) A Bank Guarantee in the prescribed form. (b) Government Securities, FDR or any other Government Instruments stipulated by the Accepting Officer. 19.2. If the performance security is provided by the successful Contractor in the form of a Bank Guarantee, it shall be issued by Nationalized/Scheduled Indian Bank but its confirmation shall be done only from the Head Office of the Bank. 19.3. Failure of the successful contractor to comply with the requirements of subclause 19.1 shall constitute sufficient grounds for cancellation of the award of work and forfeiture of the Earnest Money. In case of MES enlisted contractor,

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amount equal to the Earnest Money stipulated in the Notice Inviting Tender, shall be notified to the tenderer for depositing the amount through MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury.

19.4. All compensation or other sums of money payable by the contractor to the Government under the terms of this contract or under any other contract with Government may be deducted from, or paid by the sale of a sufficient part of the Performance Security or from the interest arising there from or from any sums which may be due or become due to the contractor by the Government on any account whatsoever and in the event of his Performance Security being reduced by reason of any such deduction, or sale as aforesaid, the contractor shall within ten days thereafter make good in cash or securities, endorsed as aforesaid, any sum or sums which may have been deducted from or realized by the sale of his Performance Security or any part thereof.

Government shall not be responsible for any loss of securities or any depreciation in the value of securities while in their charge nor for loss of interest thereon.

47. (Contd) 19.5. In the event of contract being cancelled, under Condition 52, 53 & 54 of General Conditions of Contract, the Performance security shall be forfeited in full and shall be credited into Consolidated Fund of India."

48. 17 and 18 <u>Condition 46, Inspection of the Works, Second Para, lines 1 to 5</u>

For: "Should the G.E. consider, at any time during the construction or reconstruction or prior to the expiration of period of a twelve calendar months after the works have been handed over to Government (hereinafter referred to as the "defects liability period") that any work has been executed with unsound, imperfect or unskilful workmanship or of a quality inferior to that contracted for or not otherwise in accordance with the Contract....."

Read: "Should the G.E. consider, at any time during the construction or reconstruction or prior to the expiration of period of:-

(a) Thirty six calendar months after the works have been handed over to Government (hereinafter referred to as the "defects liability period") for Runway works, Marine & Harbour works, High Altitude works & Specialist works of Hospitals & Medical Equipment/Medical Gas.

(b) Twenty four calendar months after the works have been handed over to Government (hereinafter referred to as the "defects liability period") for works other than mentioned in sub para (a) above.

(The period of defects liability period shall be specified in the tender documents.)

that any work has been executed with unsound, imperfect or unskilful workmanship or of a quality inferior to that contracted for or not otherwise in accordance with the Contract....."

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Amendment No	Page No	Particulars
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49.	17 and 18	Condition 46, Inspection of the Works, Second Para **For:** "Should the G.E. consider, at any time during the construction or reconstruction or prior to the expiration of period of:- (a) Thirty six calendar months after the works have been handed over to Government (hereinafter referred to as the "defects liability period") for Runway works, Marine & Harbour works, High Altitude works & Specialist works of Hospitals & Medical Equipment/Medical Gas. (b) Twenty four calendar months after the works have been handed over to Government (hereinafter referred to as the "defects liability period") for works other than mentioned in sub para (a) above." **Read:** "Should the G.E. consider, at any time during the construction or reconstruction or prior to the expiration of period of:- (a) Thirty six calendar months after the works have been handed over to Government (hereinafter referred to as the "defects liability period") for Runway works, Marine & Harbour works, High Altitude works & Specialist works of Hospitals & Medical Equipment/Medical Gas. (b) Twelve calendar months after the works have been handed over to Government (hereinafter referred to as the "defects liability period") for works of Periodical Services and Welcome Maintenance, Minor Works and Revenue Works and Works for 'Maintenance & Operation' including Repairs and Works for Maintenace & Repair/Replacement. (c) Thirty days or date of taking over of the Installation by MES, whichever is earlier, for works of 'Manning & Operation' & 'Maintenance & Operation' involving predominantly Outsourcing of Services through Deployment of Manpower/Work-force and not involving Repair/Replacement. (d) Twenty four calendar months after the works have been handed over to Government (hereinafter referred to as the "defects liability period") for works other than mentioned in sub para (a) to (c) above.

SIGNATURE OF CONTRACTOR
DATED :

Jt/Dy DIRECTOR/AAD (CONTRACTS)
FOR ACCEPTING OFFICER

The existing descriptions of Condition 70 of IAFW-2249 shall be substituted by under mentioned description:

70. Arbitration.

(a) Arbitration Where Applicability of Section 12 (5) of the Arbitration and Conciliation Act Has Been Waived Off:-

All disputes, between the parties to the Contract (other than those for which the decision of the CWE or any other person is by the Contract expressed to be final and binding) shall, after written notice by either party to the Contract to the other of them, be referred to the Arbitral Tribunal of a Sole Arbitrator (in case of contract sum less than or equal to Rs 100 Crore) or to Arbitral Tribunal of three Arbitrators (in case of contract sum exceeding Rs 100 Crore) from MoD Panel of Arbitrators/ Command Panel of Arbitrator. The officers so considered for appointment of Arbitrator, either as sole Arbitrator or for Arbitral Tribunal, shall be having degree in Engineering or equivalent or having passed Final/Direct Final Examination of Sub-Division II of Institution of Surveyor (India) or similar other Institutes recognized by the Government of India.

In case of arbitration by Sole Arbitrator, the Arbitrator shall be appointed by the Authority mentioned in the contract document within a period of thirty days of having received the notice from any of the parties to Contract, out of MoD Panel of Arbitrators/ Command Panel of Arbitrator. In case of Arbitral Tribunal consisting of panel of three Arbitrators, both the parties will be asked by the Appointing Authority to suggest at least two names out of MoD Panel of Arbitrators/ Command Panel of Arbitrator within thirty days. The Appointing Authority will appoint two Arbitrators, one Arbitrator each out of the suggested names. The two Arbitrators so appointed will select one Arbitrator from the MoD Panel of Arbitrators/ Command Panel of Arbitrator who will be the 'Presiding Arbitrator'.

The Serving Officer(s) so appointed as Arbitrator(s), either as Sole Arbitrator or as one of the three Arbitrators in the Arbitral Tribunal, can continue as Arbitrator even after retirement, provided both the parties to the Contract give written consent to this effect. In such case, however, the Arbitrator shall not be entitled for any fee even after retirement.

(b) Arbitration Where Applicability of Section 12 (5) of Arbitration & Conciliation Act Has Not Been Waived Off.

All disputes, between the parties to the Contract (other than those for which the decision of the CWE or any other person is by the Contract expressed to be final and binding) shall, after written notice by either party to the Contract to the other of them, be referred to the Arbitral Tribunal of a Sole Arbitrator (in case of contract sum less than or equal to Rs 100 Crore) or to Arbitral Tribunal of three Arbitrators (in case of contract sum exceeding Rs 100 Crore). The Officers so considered for appointment as Arbitrator, either as Sole Arbitrator or for Arbitral Tribunal, shall be having degree in Engineering or equivalent or having passed Final/Direct Final Examination of Sub Division II of Institution of Surveyor (India) or similar other Institutes recognised by the Government of India.

In case of arbitration by sole Arbitrator, the Arbitrator shall be appointed by the authority mentioned in the contract document from the MoD Panel of Arbitrators within a period of thirty days of having received the notice from any of the parties to Contract. In case of Arbitral Tribunal consisting of panel of three Arbitrators, both the parties shall be asked by the Appointing Authority to suggest at least two names out of MoD Panel of Arbitrators within thirty days. The Appointing Authority will appoint two Arbitrators, one Arbitrator each out of the suggested names. The two Arbitrators so appointed shall select one Arbitrator from the MoD Panel of Arbitrators who will be the 'Presiding Arbitrator'

(c) Common For All Arbitration.

Unless both parties agree in writing, such reference shall not take place until after the completion or alleged completion of the works or termination or determination of the contract under Condition Nos 55, 56 and 57 hereof.

Provided that in the event of abandonment of the works or cancellation of the Contract under Condition No 52,53 or 54 hereof, such reference shall not take place until alternative arrangements have been finalised by the Government to get the works completed by or through any other Contractor or Contractors or Agency or Agencies.

Provided always that commencement or continuance of any arbitration proceeding hereunder or otherwise shall not in any manner militate against the Government's right of recovery from the Contractor as provided in Condition 67 hereof.

If the sole Arbitrator or one or more Arbitrators of the Arbitral Tribunal so appointed resign(s) from his/her appointment or vacate(s) his/her office or in unable or unwilling to act due to any reason whatsoever, the Authority appointing him/her will appoint a substitute Arbitrator to act in his/her place in the manner specified hereinabove. In case the Arbitrator resigning in this manner is the Presiding Arbitrator, the other two Arbitrators of the Arbitral Tribunal shall appoint the substitute Presiding Arbitrator.

The Arbitral Tribunal may proceed with the arbitration, exparte, if either party, inspite of a notice from the arbitrator fails to take part in the proceedings. The Arbitral Tribunal may from time

to time with the consent of the parties, enlarge the time for making and publishing the award subject to the limit laid down in the Arbitration & Conciliation Act 1996 as amended upto the date on which arbitration proceedings commence.

The Arbitral Tribunal shall make the award within the period as provided in the Arbitration & Conciliation Act 1996 (as amended upto the date on which arbitration proceedings commence) from the date of entering on the reference or within the extended period as the case may be on all matters referred to it and shall indicate findings alongwith sums awarded separately on each individual item of dispute. The Arbitral Tribunal shall give reason for the award in each and every case irrespective of the value of claims or counter claims.

The venue of Arbitration shall be such place or place or places as may by fixed by the Arbitral Tribunal in its sole discretion.

The Award of the Arbitral Tribunal shall be final and binding on both parties to the Contract.

Jt/Dy DIRECTOR/AAD (CONTRACTS)
FOR ACCEPTING OFFICER

_____
SIGNATURE OF CONTRACTOR
DATED :

SCHEDULE OF MINIMUM WAGES

1. It is hereby agreed that the "**Schedule of Minimum wages**" as published by Govt of India/State Govt/Govt local authorities whichever is highest and which specifies minimum rates of wages for various categories of workman as applicable on the last due date of receipt of this tender shall form part of these tender document.

The minimum rates of wages shall consist of all inclusive rates and include also the wages for weekly day of rest.

My/our signature here under amount to my/our having signed the aforesaid documents forming part of this tender.

_____
SIGNATURE OF CONTRACTOR
DATED :

Jt/Dy DIRECTOR/AAD (CONTRACTS)
FOR ACCEPTING OFFICER

SPECIAL CONDITIONS (CONT....)

1. **GENERAL**

1.1 The following Special Conditions shall be read in conjunction with the General Conditions of Contracts, IAFW-2249 and IAFW-2159 (Revised-1947) including errata/amendments thereto. Any provision in these special conditions if at variance with that of the aforesaid documents shall take precedence there over. The term General specification used in any of the documents forming part of the contract refer to the specification contained in the MES Schedule.

2. **ADMISSION TO SITE BY CONTRACTOR AND RESPONSIBILITY TO ASCERTAIN HIS OWN INFORMATIONS**

2.1 Refer paras 11 and 15 of Notice of Tender (IAFW-2162) and Condition 4 of IAFW-2249. The tenderer shall be deemed to have satisfied himself before submitting his tender as to the nature of the ground and sub-soil (so far as it is practicable), the climatic conditions, the form and nature of the site, nature of work and the manner of access to the site and the accommodation he may require. In general, the tenderer shall be deemed to have himself obtained all necessary information's on inspection of site, as to risks, contingencies and other circumstances which may influence or effect his tender. No extra payment consequent on any mistake or misunderstanding or otherwise on this account shall be allowed.

2.2 The tenderer shall be deemed to have visited the site(s) before quoting rate and made themselves familiar with the working conditions, whether he actually inspect the site(s) or not.

3. **AVAILABILITY OF LAND FOR STORAGE OF MATERIALS, ACCOMMODATION FOR LABOUR ETC**

3.1 Delete the following from lines 5 to 9 of sub para 1 of Condition 24 of IAFW-2249 General Conditions of Contracts and insert as under in lieu thereof :-

"The contractor shall be permitted to store his materials including erecting temporary sheds thereof, temporary Workshops and the like at the area of land, marked for this purpose, on the site plan/shown by the GE free of rent, if it is on class A-1 land. For other types of land he shall be charged licence fee Re 1/- per year or part of a year for each and every separate area of land allotted to him. and shall be allotted to the contractor for construction of huts for accommodation of labour.

4. **CONDITIONS OF WORKING**

4.1 The work lies in **UNRESTRICTED AREA**. However, the contractor, his agents, employees, work people and vehicles may pass through the unit lines in which case the Engineer-in-Charge shall, at his discretion have the right to issue passes, control their admission to the site of work or any part thereof. The contractor shall on demand by the Engineer-in-Charge submit a list of personnel, etc. concerned and shall satisfy the Engineer-in-Charge as to the bonafides of such people. Passes shall be returned at any time on demand by the Engineer-in-Charge and in any case on completion of work.

4.2 The contractor and his agents, employees and work people shall observe all the rules promulgated by the authority controlling the area in which the work is to be carried out e.g. prohibition of smoking, lighting and fire precaution, search of persons at entry and exit, keeping to specified routes, restricted hours of working etc. Any person found violating the security rules laid down by the authority, shall be immediately expelled from the area without assigning any reasons what so ever and the contractor shall have no claim on this account. Nothing shall be admissible for any man-hours lost on this account.

4.3 **WORK ON HOLIDAYS:-**The contractor shall not carry out any work on gazetted holidays, weekly holidays and other non-working days except when he is specially authorized in writing to do so by the GE. The GE may at his sole discretion declare any day as holiday or non-working day without assigning any reason for such declaration. Nothing extra shall be admissible on this accounted for any man hours lost.

5. **CONTRACTOR'S REPRESENTATIVES AND WORKMEN**

5.1 Refer Condition 25 of IAFW -2249. The contractor shall employ only Indian Nationals as his representatives, servants and workmen and verify their antecedents and loyalty before employing them for the works. He shall ensure that no person of doubtful antecedents and nationality is in any way associated with works. If for reasons of technical collaboration, or other consideration employment of any foreign national is unavoidable, the contractor shall furnish full particulars to this effect to the Accepting Officer at the time of submission of his tender.

6. **SECURITY OF CLASSIFIED DOCUMENTS**.

6.1 Contractor's special attention is drawn to Conditions 2-A and3 of IAFW-2249 (General Conditions of Contracts). The contractor shall not communicate any classified information regarding the work either to subcontractors or others without the prior approval of the Engineer-in-Charge. The contractor shall also not make copies of the design/drawings and other documents furnished to him in respect of the work, and shall return all document son completion of the works or earlier on determination /Termination /cancellation of the contract. The contractor shall alongwith the final bill, attach a receipt of his having returned the classified documents as per condition 3 of IAFW-2249 (General Conditions of Contracts).

SPECIAL CONDITIONS (CONT....)

7. **MINIMUM WAGES PAYABLE**

7.1 Refer Condition 58 of IAFW-2249. The contractor shall not pay wages lower than minimum wages for labour as fixed by the Govt of India/State Govt/Union Territory under Minimum Wages Act or Contract Labour (Abolition and Regulation Act) whichever is higher.

7.2 The fair wage referred to in condition 58 of IAFW-2249 shall be deemed to be the same as the minimum wages payable as referred to above.

7.3. The contractor shall have no claim whatsoever, if on account of local factors and/or regulations, he is required to pay the wages in excess of minimum wages as described above during the execution of work.

8. **CO-OPERATION WITH OTHER AGENCIES**

8.1 The contractor shall permit free access and generally afford reasonable facilities to other agencies or departmental workmen engaged by the Govt to carry out their part of the work, if any, under separate arrangements. The contractor shall not be allowed any extra payment on this account.

9. **QUARRIES**

9.1 Quarries are not available on land which is in the charge of the MES authorities. Condition 14 of IAFW-2249 shall thus be treated as deleted.

10. **WATER**

10.1 Refer Condition 31 of General Conditions of Contracts (IAFW–2249).

10.2 Water will not be supplied by MES. The tenderer are advised to visit the site of works to ascertain the availability of water from civil sources, nearby natural sources outside Ministry of Defence land. The contractor shall be allowed, if he so desires to install hand pumps, tube wells at site of work at places as approved by Engineer-in-Charge without any charges from the contractor on this account. The contractor shall remove the hand pumps, tube wells as and when asked to do so by Engineer-in-Charge / GE and in any case on completion of the work and before issue of completion certificate, unless GE desires that these hand pumps, tube wells be left in position and the contractor agrees to do so without claiming cost thereof from Department. No compensation whatsoever shall be admissible to the contractor, if he is required to remove the pumps, tube wells before completion of work. Use of water from such sources shall only be permitted if found after testing, portable and fit for use in the work. The water from such sources shall be got tested by the contractor from laboratory approved by the GE, who shall after satisfying himself permit the contractor to use the water from such sources. Testing charges shall be borne by the contractor. The contractor shall however ensure that water drawn from such sources is clean and free from all impurities and is suitable for mixing of concrete, mortar, washing, aggregate and curing of concrete as specified in IS : 456.

11. **SUPPLY OF ELECTRICITY**.

11.1 Electricity will not be supplied by MES

11.2 Electricity will not be supplied by MES. The tenderer are advised to visit the site of works to ascertain the availability of Electricity from civil sources, nearby sources outside Ministry of Defence land. The contractor shall be allowed, if he so desires to install Electricity from the civil sources at site of work at places as approved by Engineer-in-Charge without any charges from the contractor on this account. The contractor shall remove the installed electric equipment as and when asked to do so by Engineer-in-Charge / GE and in any case on completion of the work and before issue of completion certificate, unless GE desires that these electrical equipment be left in position and the contractor agrees to do so without claiming cost thereof from Department. No compensation whatsoever shall be admissible to the contractor, if he is required to remove the electrical equipment after completion of work.

12. **SAMPLE OF MATERIALS**

12.1 Refer condition 10 of I.A.F.W. 2249 and Clause No.1.6 & 1.7 of MES Schedule Part I.

12.2 The tenderers are advised to inspect sample of the materials which are displayed in the office of the Garrison Engineer before submitting his tender. The tenderers shall be deemed to have inspected the samples and satisfied himself as to the nature and quality of materials, he is required to incorporate in the work, irrespective of whether he has actually inspected or not. The materials to be incorporated in the work by the contractor shall conform to or shall be superior in quality to the sample displayed and shall comply with the specifications given hereinafter.

12.3 The contractor shall not procure materials unless the samples are first approved by the Garrison Engineer.

12.4 The materials other than materials of local origins shall comply with the requirement of the latest IS.

12.5 The cost of testing of material shall be borne by the contractor while quoting their lumpsum. This aspect shall be kept in mind and nothing extra shall be admissible on this account.

SPECIAL CONDITIONS (CONT....)

13. **RECORD OF MATERIALS.**

13.1 The quantity of materials such as cement, steel, paints, water proofing compound, chemicals for antitermite treatment and the like, as directed by the Engineer-in-Charge (the quantity of which cannot be checked after incorporation in the works), shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.

13.2 Materials brought to site shall be stored as directed by the Engineer-in-Charge and those already recorded in Measurement Book shall be suitably marked for identification.

13.3 Contractor shall produce vouchers/invoices from the manufacturers and/or their authorized agents for the full quantity of all items for which payment is claimed in RAR as " material lying at site"the following materials as a pre-requisite before submitting claims for payment for advances on account-of the work done and/or materials collected in accordance with Condition 64 of General Conditions of Contracts-IAFW-2249 and for materials invariably of the followings :

(i) All items for which payment is claimed in RAR as " material lying at site"	(ii) Cement
(iii) Reinforcement/ Structural steel and hollow section	(iv) Lightning protection items
(v) GI barbed wire & Chain link fencing	(vi) All type of cable junction box
(vii) Roofing Sheet	(viii) Heater Water Storage Type Electric Geyser
(ix) Aluminum section for Door / Window / Partition / Structural Glazing	(x) Factory made Door / Window Shutter
(xi) PVC/ HDF/Flush Doors Shutter	(xii) APP membrane & primer
(xiv) Fibre Cement Board / Gypsum Plaster Board	(xv) Distemper / Cement base / Acrylic emulsion paint / Synthetic enameled Paint & Primer
(xvi) All type of Ceramic / Non-skid / Vitrified tiles / Chequered Cement Concrete Tiles / Precast Interlocking Paver Blocks	(xvii) All types of pipes & Valves/Taps
(xviii) Paving Bitumen	(xix) Glass for door & window and structural glazing / partition
(xx) ATT Chemicals	(xxi) All type Sanitary Appliance
(xxii) Pre-laminated Particle Board/Plywood	(xxiii) All items related to False Ceiling
(xxiv) PVC Water Tank	(xxv) All types of light fitting including LED
(xxvi) LT / HT XLPE Cable including jointing kit	(xxvii) Cable Trays
(xxviii) PVC cable/wire for 1100 volts	(xxix) PVC conduit/casing caping
(xxx) Steel Tubular Pole/Pressed concrete pole	(xxxi) All Switch Gears viz. ACB/MCCB/ MCB/DB/ Junction Box, etc.
(xvi) Electrical fittings / fixtures (Internal & External)/LT /HT Panels boards/ feeder pilar box etc	(xvii) Ceiling / Exhaust fan
(xviii) High mast light sturcture including lamp	(xix) Fire fighting equipments
(xx) Submersible pump and centrifugal pump, booster pump	(xxi) Fire alarm sounder
(xxii) Electric meter	(xxiii) Earthing items

13.4 The contractor shall ensure that the materials are brought to site, in original sealed containers/packing, bearing manufacturer's marking except in the case of the requirement of materials(s) being less than smallest packing.

13.5 The vouchers/invoices will clearly indicate the contract number and the IS No., specific alternative to which the material conforms in case of various alternative in IS.

14. **PHOTOGRAPHS**

14.1 The contractor shall provide minimum 1000 HD photographs (coloured) in duplicate with soft copy CD of the each stage of the work in progress as directed by GE/ Engg-in-charge. The contractor shall arrange the equipment or any other attachment for taking the photograph.

14.2 The photograph shall be postcard size and shall be sharp, neat and clean and of standard as approved by GE. Cost of photographs shall be borne by the contractor.

SPECIAL CONDITIONS (CONT....)

14.3 In case photography is prohibited in this area, the photographs shall not be taken and no price adjustment shall be made on this account. The contractor shall not allowed to carry the camera or the equipment connected with the photographs in the area/buildings without permission of the Engineer-in-Charge.

15. **VALIDITY OF TENDER**

15.1 The tender shall remain open for acceptance for a period of **90 (ninety days)** from bid submission end date as mention in web site. For the purpose of calculating this period, the day on which the tenders are due to be submitted shall be excluded.

16. **CPM NET WORK DIAGRAM**

16.1 The time and progress chart to be prepared as per condition 11 of General Conditions of Contracts (IAFW-2249) shall consist of detailed net work analysis and a time schedule. The critical path net work shall be drawn jointly by the GE and the contractor soon after acceptance of tender. The time scheduling of the activities shall be done by the contractor so as to finish the work within the stipulated time. On completion of the time schedule a firm calendar date schedule shall be prepared and submitted by the contractor to the GE who shall approve it after due scrutiny. The schedule shall be submitted in four copies within two weeks from the date of handing over the site. In case the contractor fails, to submit the CPM net work diagram, the network prepared by the GE shall be binding on him.

16.2 During the currency of the work, the contractor is expected to adhere to the time schedule and this adherence shall be a part of his/their performance under the contract. During the execution of the work, the contractor is expected to participate in the reviews and updating of the net work undertaken by the GE. These reviews may be undertaken at the discretion of the GE, either as a periodic appraisal measure or when the quantum of work ordered on the contractor is substantially changed through deviation orders or otherwise. Any revision of the time schedule as a result of the review, shall be submitted within a week by the contractor to the GE for his approval after due scrutiny.

16.3 The contractor shall adhere to the revised time schedule there after. In case of contractor disagreeing with revised schedule, the same shall be referred to the Accepting Officer whose decision shall be final, conclusive and binding. GE's approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to a grant of extension of time. Extension of time shall be considered and decided by the appropriate authority mentioned in condition 11 of IAFW-2249 and separately regulated.

16.4 Contractor is expected to mobilize and employ sufficient resources to achieve the detailed schedule within the broad frame work of the accepted methods of working and safety.

16.5 No additional payment will be made to contractor for any multiple shift work or other incentive methods contemplated by him in his work schedule, even though the time schedule is approved by the department.

17. **RELEASE OF PERFORMANCE SECURITY DEPOSIT**

17.1 Refer Conditions 19 and 68 of IAFW-2249 (As amended vide amendment No 47 and 45 to IAFW-2249).

18. **SITE FOR EXECUTION WORK**

18.1 Site for execution of work shall be available as soon as the work is commenced on the date as mentioned in the work order No 1. However, in case due to unavoidable circumstances it is not possible to make the entire sites available on the date of the commencement, the contractor shall have to arrange his programme accordingly. No claim whatsoever for not getting the entire site on the date of commencement of work & for giving the site gradually shall be tenable.

19. **RECOVERY CHARGES FOR TESTING OF MATERIALS**

19.1 Where testing of contractor's materials is carried out in zonal/National test house/SEMT wing/Engineering college by the department, the contractor shall be liable to pay the testing charges as stated in Appendix `F' to particular specifications and shall be recovered from the contractor's running payment/final bill as the case may be.

SPECIAL CONDITIONS (CONT....)

20. **SAFETY PRECAUTION**

20.1 The contractor shall take every/precaution to control trifurcation road keeping danger boards, necessary lighting arrangements, fencing and watchman to avoid any damage. In case due to excavation or others the road is to be blocked the contractor shall, without any extra cost to the Govt. provide separate bypass so that normal traffic is not disturb.

20.2 Any damages to the existing road/building etc, shall be made good by the contractor with the same specifications as per existing work, without any extra cost to the Govt. In the event of contractor not fully complying with the above provisions to the satisfaction of the GE, the GE may provide the same for which the expenses incurred shall be recovered from the contractor.

21. **RECORD OF CONSUMPTION OF CEMENT**

21.1 The contractor shall maintain a pucca bound register with serially numbered pages with all pages initialed by Engineer-in-Charge against numbering showing quantities of cement records and used in the work daily and balance at the end of each day. The form of record shall be as approved by Engineer-in-Charge. The register shall be signed daily by the representative of MES and the contractor in token of verification of its correctness and shall be checked by Engineer-in-Charge, at least once a week and on the days cement is issued to the contractor.

21.2 The register shall be kept at site safe custody of the contractor representative during the progress of the work and shall on demand, be produced for verification to the inspecting officers.

21.3 On completion of the work the contractor shall deposit the cement register with the Engineer-in-Charge for record.

22. **REIMBURSEMENT/REFUND ON VARIATION IN PRICES :-** Refer Condition 63 of the General Conditions of Contracts (IAFW-2249)/Re-imbursement/Refund on Variation of Prices. The Condition 63 of General Conditions of Contract shall be deemed to be modified to the extent mentioned hereinafter. Increase or decrease in prices of Cement, Steel and other materials shall be adjusted on the basis stipulated hereinafter irrespective of the actual variation in prices (to the contractor):

(a) Cement : The cement cost component for the contract as a whole shall be taken as K_C% of the value of works executed under the contract. Accordingly value/cost of cement consumed in the work as well as that lying at site for which reimbursement/refund is applicable shall be:

$$V_{MC} = \frac{(Kc \times Vg)}{100} + Vc$$

Variation in prices of cement shall be worked out by applying the following formula:-

$$EM_C = (V_{Mc2} - V_{MC1}) \times \frac{(C_1 - C_0)}{C_0}$$

EMc = Variation in prices of cement to be adjusted.

Kc = Constant representing the percentage cost of Cement as compared to the total value of work under the contract as a whole. The value of Kc for this work shall be **10.5 (TEN DECIMAL FIVE ZERO)** only.

Vg = Amount of work done priced at contract rates up to for the last date of the period of reckoning excluding amount payable to the contractor towards items on star rate and PC sum.

V_C = Cost of all cement lying at site for incorporation in the work excluding cement issued under Schedule 'B' and excluding cement brought and paid or payable to contractor under Price Cost Sum and/or Star Rate (s).

C_1 = Wholesale Price Index for Pozzolana Cement (Base Year 2011-12) published by Economic Adviser to the Government of India of the month preceding to the month in which the date of commencement of the period of reckoning. In case the original contract period is extended under the condition 11 of General Conditions of Contracts (IAFW-2249), the Price Index as applicable on the date of commencement of the last period of reckoning before the original completion date (s) (phase wise except where phasing has been done only for sample quarter/sample block) shall only be applicable during the extended period. If phasing has been done for only sample quarter/ sample block, the price index as applicable on the date of commencement of the last reckoning period before the original completion date of the project as a whole shall only be applicable during the extended period.

SPECIAL CONDITIONS (CONT....)

C_O = As per C_1 but the index as on the last due date of Bid Submission. In case of tenders for specialist works made in two parts viz "T" and "Q", Co shall be as on date of opening of "Q" Bid.

V_{MC2} = Amount of cement up to the last date of the period of reckoning for which variation is adjustable as worked out as per formula for V_{MC}.

V_{MC1} = As per V_{MC2} but as on date of immediate preceding period of reckoning.

(b) Steel: The Steel cost component for the contract as a whole shall be taken as Ks% of the value of works executed under the contract. Accordingly value/cost of steel consumed in the work as well as that lying at site for which reimbursement /refund is applicable shall be.

$$V_{MS} = \frac{(Ks \times Vg)}{100} + Vs$$

Variation in prices of steel shall be worked out by applying the following formula:-

$$EMs = (V_{MS2} - V_{MS1}) \times \frac{(S_1 - S_0)}{S_0}$$

EMs = Variation in prices of steel to be adjusted.

Ks = Constant representing the percentage cost of Steel as compared to the total value of work under the contract as a whole. The value of Ks for this work shall be **13.00 (THIRTEEN DECIMAL ZERO)** only.

Vg = Amount of work done priced at contract rates up to for the last date of the period of reckoning excluding amount payable to the contractor towards items on star rate and PC sum.

Vs = Cost of all steel lying at site for incorporation in the work excluding steel issued under Schedule 'B' and excluding steel brought and paid or payable to contractor under Prime Cost Sum and/or Star Rate (s).

S_1 = Wholesale Price Index for Mild Steel (Long Products) (Base Year 2011-12) published by Economic Adviser to the Government of India of the month preceding to the month in which the date of commencement of the period of reckoning lies. In case the original contract period is extended under condition 11 of General Conditions of Contracts (IAFW-2249), the price index as applicable on the date of commencement of the last period of reckoning before the original completion date(s) (phase wise except where phasing has been done only for sample quarter/block) shall only be applicable during the extended period. If phasing has been done for only sample quarter/sample block, the price index as applicable on the date of commencement of the last reckoning period before the original completion date of the period as a whole shall only be applicable during the extended period.

S_0 = As per S_1 but the Index as on the last due date of Bid Submission. In case of tenders for specialist works made in two parts viz "T" and "Q", S_0 shall be as on date of opening of "Q" Bid.

V_{MS2} = Amount of steel up to the last date of the period of reckoning for which variation is adjustable as worked out as per formula for V_{MS}.

V_{MS1}= As per V_{MS2} but as on date of immediate preceding period of reckoning.

(c) Other materials (except Cement & Steel): The material cost component except cement & steel for the contract as a whole shall be taken as K_{OM}% of the value of works executed under the contract. Accordingly value/cost of other materials consumed in the work as well as that lying at site for which reimbursement/refund is applicable shall be.

$$V_M = \frac{(K_{0M} \times Vg)}{100} + (V_{0M} - V_B)$$

Variation in price of materials (except cement & steel) shall be worked out by applying the following formula:-

$$E_M = (V_{M2} - V_{M1}) \times \frac{(W_1 - W_0)}{W_0}$$

E_M = variation in prices of material to be adjusted except cement & steel.

K_{0M} = Constant representing the percentage cost of other material except cement & steel as compared to the total value of work under the contract as whole. The value of K_{0M} for this work shall be **36.50(THIRTY SIX DESIMAL FIVE ZERO)** only.

SPECIAL CONDITIONS (CONT....)

Vg = Gross value of work done at contract rates upto the last date of period of reckoning excluding amount payable to the contractor towards items on Star Rate and PC sum.

V_{0M} = Value of all materials (except cement & steel) lying at site for incorporation in the work including materials (except cement & steel) issued under schedule 'B' and including materials (except cement & steel) brought and paid or payable to contractor under Prime cost Sum and/or Star Rate(s).

V_B = Value of all materials (out of Vg and V_{OM}) (except cement & steel) issued under schedule 'B' plus value of all materials (except cement & steel) brought and paid or payable to contractor under Prime Cost Sum and /or Star Rate (s).

W1= Wholesale Price Index for All Commodities (Base Year 2011-12) published by Economic Adviser to the Government of India of the month preceding to the month in which the date of commencement of the period of reckoning. In case the original contract period is extended under Condition 11 of General Conditions of Contracts (IAFW-2249), the price index as applicable on the date of commencement of the last period of reckoning before the original completion date (s) (phase wise except where phasing has been done for only sample quarter/block) shall only be applicable during the extended period. If phasing has been done for only sample quarter/sample block, the price index as applicable on the date of commencement of the last reckoning period before the original completion date of the period as a whole shall only be applicable during the extended period.

W_O= As for W1 but the index as on the last due date of bid submission.

V_{M2}= Value of material (except cement & steel) upto the last date of the period of reckoning for which price variation is adjustable as worked out as per formula for V_M.

V_{M1}= -do- but as on date of immediate preceding period of reckoning.

Note:-

1. No adjustment, whatsoever, due to variation in prices of materials on accounts of coming into force of any fresh law or statutory rule or order as provided in Condition 63 of IAFW-2249 or otherwise than provided in this condition shall be made.

2. No adjustment in prices shall be made for any work done with materials brought at site after the original date of completion of the work as mentioned in work order No 01 under contract except as contemplated under definition of C_1, S_1 and W_1 hereinbefore.

3. Periodicity of working out the escalation on account of variation in prices will be three months. The last calculation shall however be done for the value of work at contract rates and materials lying at site for incorporation in the work as on date of completion or extension there of as mentioned in Note 2 above. Valuation of RARs is to be timed in such a manner that relevant date required for quarterly calculation under condition is available from RARs. In case on these dates no RAR is preferred by the contractor, dummy RAR would be prepared & shall be kept on record duly technically checked and audited. Amount payable relevant to work done and materials collected in quarter will be worked out after firm whole sale price indices for the relevant quarter are available. Once the amount adjustable for any quarter is worked out, the same shall be adjusted as and alongwith advance on account payment in the subsequent RAR(s).

4. Any dispute arising out of interpretation or application of this Special Condition shall be referred to the Accepting Officer whose decision shall be final and binding.

5. for the purpose of calculation of retention money liquidated damages, GST on works contracts , deduction of income tax at source and recovery of water charges (in case of un- metered supply) the value of contracts as revised by the above price variation will be taken into account.

6. In cases, where value of $V_{M2} - V_{M1}$ works out to minus on account of higher utilisation of schedule 'B' stores (i.e value of Sch B store under contract as a whole is higher than K_{OM} value) and the reimbursement on account of variation in prices of materials (except Cement & steel) works out to be negative in spite of the Wholesale price Index for All Commodities published by economic Adviser to Government of India going up for W_O, reimbursement on account of variation in prices of materials shall be treated as "Nil".

(d) <u>REIMBURSEMENT/REFUND ON VARIATION IN PRICES-WAGES OF LABOUR</u> :- Refer Condition 63 of IAFW-2249). The condition 63 of the General Condition of contract shall be deemed to be modified to the extent mentioned here-in-after. Increase or decrease in prices consequent on variation in wages of labour, shall be adjusted on the basis stipulated hereinafter, irrespective of the actual variation in price wages of labour to the contractor.

SPECIAL CONDITIONS (CONT....)

LABOUR.

The labour component for the work under the Contract as a whole shall be taken as KL % of the value of the work executed under the contract. Variation in the labour wages shall be worked out by applying the following formula **:-**

$$E_L = \frac{K_L \times Vg_1}{100} \times \frac{L_1 - L_0}{L_0}$$

Where

E_L = Variation in wages of labour reimbursement to be made to the contractor or refund to be made by the contractor.

K_L = Constant representing the percentage cost of labour element as compared to the total value of the work under the contract as a whole. The value of K_L for the work shall be **20.0(TWENTY DECIMAL ONLY)**.

V_{g1} = Gross value of work done at contract rates during the period of reckoning less value of work paid or payable to the contractor based on actual cost (eg. star rate (s), work executed under prime cost sum etc) during the period of reckoning.

L_1= Minimum wage in rupees of an unskilled adult male Mazdoor as fixed under any law, statutory rule or order as on the date of commencement of the period of reckoning.

L_0= Minimum wage in rupees of an unskilled adult male Mazdoor as fixed under any law, statutory rule or order as on the last due date for receipt of tender i.e bid submission end date as mention in web site. If labour wage on the date of receipt of tender are increased afterwards with retrospective effect, the value of L_0 shall be fixed keeping in view the following aspects :-

(a) If the increase/decrease in wages of labour are made known to the public by any means of media before receipt date of tender but the same is officially notified thereafter giving retrospective effect, the value of "L_0" shall be as per notification though made subsequently.

(b) If a net wage comprises a fixed basic wage and the living allowance revised from time to time based on consumer price index (CPI) and increase in CPI is made known to the public by any means before the date of receipt of tender, the "L_0" will be revised wages corresponding to revised CPI, though the formal notification for the net wage (considering the revised living allowance corresponding to revised CPI) is made subsequent to date of receipt of tender.

(c) In case the labour enforcement officer makes the announcement before date of receipt of tender but Gazette notification is made subsequently making wages applicable with retrospective effect, the value of "L_0" shall be as per Gazette Notification subsequently made.

(d) If the increase/decrease in wages of labour is notified/announced subsequent to receipt of tender with retrospective effect without making the same publicly known by means of publicity/media prior to the date of receipt of tender, then, the value of "L_0" shall be as per wage known at the time of receipt of tender.

NOTES

1. The contractor shall within reasonable time of his becoming aware of any alteration to the payment of wages of labour consequent on fixation of minimum wages under any law, statutory rule or order, give written notice thereof to the GE stating that the same is given pursuant to this special conditions together with all information relating thereto which he may be in a position to supply.
2. Irrespective of the variation in minimum wages for any category of labour, for the purpose of adjustment under this special condition, the variation in minimum wages fixed under any law, statutory rule or order for an unskilled adult male Mazdoor, if any, shall only form the basis.
3. Periodicity of working out the variation in wages of labour will be three months commencing from the last due date for receipt of tender. The last adjustment for variation in wages of labour shall however, be done for the period up to the date of completion or extended date of completion. Valuation of price adjustment due to increase/decrease in minimum wages under any law, statutory rule or order for the purpose of making reimbursement/refund in RARs, will be timed in such manner that relevant data required for quarterly calculation under this special condition is available from the RARs. The first price adjustment in respect of variation in

wages of labour will be worked out for the relevant quarter during which alteration to the wages of labour took place. For implementing this provision, the period of reckoning in such quarter will have to be divided into two periods i.e., the first period upto the RAR payable immediately after the date of variation and the other upto the end of the quarter. Value of L_1 at the beginning of the other period shall be the altered wage. If there are more than one change in wages in a quarter, there will be more than two periods of the reckoning on similar basis. Amount payable relevant to work done for any quarter will be worked out after the minimum wage of an unskilled adult male Mazdoor as fixed under any law, statutory rule or order for the relevant quarter is available. Once the amount adjustable for any quarter is worked out, the same shall be adjusted in subsequent RAR as 'advance on account' adjustments along with adjustment for "Material".

4. No adjustment in prices shall be made for any work done after the due date of completion or extension of time granted under Condition 11 of IAFW-2249 (whichever is later) for the work under the contract.
5. No adjustment, whatsoever, due to variation in wages of labour on account of coming into force of any fresh law or statutory rule or order as provided in Condition 63 of IAFW-2249 or otherwise, than provided in this special condition shall be made.
6. Any dispute arising out of interpretation or application of this special condition shall be referred to the Accepting Officer whose decision shall be final and binding.

For the purposes of calculation of retention money, liquidated damages, sales tax/service tax on works contract, deduction of income tax at source and recovery of water charges (in case of unmetered supply), the value of contract as revised by the above price variation will be taken into account

23. **RETENTION MONEY/COMPENSATION FOR DELAY**

23.1 Refer condition 64 of IAFW-2249 (Advances on account) and Condition 50 of IAFW-2249 (Compensation for Delay).

23.2 For the purpose of calculating retention money and liquidated damages under condition 64 of IAFW-2249 and compensation for delay in completion of work under condition 50 of IAFW-2249, the value of contract as revised by above mentioned price variation under modified condition 63 of IAFW -2249 shall be taken into account.

24. **OFFICIAL SECRET ACT**

24.1 Refer Condition 24 of General Condition of Contract IAFW-2249. The contractors attention is invited to Indian Official Secret Act 1923 (xxx of 1923) particularly Sec 5 thereof. The contractors shall be bound by the provision of this Act.

25. **DAMAGE TO EXISTING STRUCTURE/BUILDING**

25.1 Any damage caused to the existing structure during the execution of work shall be made good by the contractor at his own cost and the site of work left clean and tidy on completion. Rectifications/ making good etc. shall conform to the standard of materials originally used in the work and finished work shall match with existing work in all respects to the entire satisfaction of the GE. In case of any dispute on this account the matter shall be referred to the CE whose decision in writing shall be final, conclusive and binding.

26. **ROCK MET IN EXCAVATIONS**

26.1 If during excavation, rock is met with GE shall be informed of this in writing by the contractor. The nature/type of rock met with shall be decided by the GE, whose decision is final and binding. If the nature of rock met with is `Hard Rock' the same shall be priced and paid as excavation in hard rock at the rates as applicable to this contract. Hard rock obtained from excavation shall be properly stacked, measured and recorded in measurement book and will be the property of the Contractor. Credit at the rate of **Rs. 500.00** per cubic metre of hard rock obtained sall be recovered from the contractor. The measurement shall be signed by Engineer-in-charge and contractor. Hard rock obtained thus can be reused in the execution under this contract subject to fulfillment of requisite specification specified here-in-after and after written approval of Garrison Engineer.

SPECIAL CONDITIONS (CONT....)

27. **ADVANCE ON ACCOUNT OF MATERIALS WHICH DOES NOT LOSS IDENTITY (APPLICABLE FOR CONTRACT OF VALUE Rs.50 LAKHS AND ABOVE)**

27.1 Refer condition 64 of the General Condition of Contract (IAFW- 2249) Advance on Account.

27.2 The Bank Guarantee Bonds shall be executed for a period of 24 months and on a form as directed by the Accepting Officer. The contractor shall further arrange to extend the period of Guarantee Bond if and when necessary, as directed by the Accepting Officer or shall furnish fresh guarantee bond(s) of similar value in lieu. It shall be noted that advance on account to the full value of materials brought on the site is permissible only in respect of fittings and fixture and other manufactured items which do not lose their identity materials.

28. **FEMALE LABOUR**

28.1 If the contractor desires to employ female labour on works to be carried out inside the area of a Factory, Depot, Park etc and a female searcher is not borne on the authorised strength of the Factory, Depot, Park etc, at the time of submission of the tender, he shall be deemed to have allowed in his tender for pay and allowances etc. for a Female Searcher (Class (IV) servant/GP `D' servant) calculated for the period, female labour is employed by him inside that area. If more than one contractor has/have to employ female searcher in addition to the autorised strength of the Factory, depot, park etc the salary and allowances paid to additional female searcher(s) shall be distributed on an equitable basis between the contractors employing female labour taking into consideration the value and period of completion of their contracts. The GE`s decision in regard to the amount recoverable on this account from any contractor shall be final and binding.

29. **GUARANTEE FOR ANTITERMITE TREATMENT**

29.1 Should the GE, at any time during construction or reconstruction or prior to the expiration of period of ten years after the anti-termite treatment has been completed as per contract finds that buildings have been infested with termites, the contractor shall on demand in writing from the GE, specifying the building(s) complained or not with standing the same may have been in advertantly passed, certified and paid for forthwith under take to carry out such treatment as may be necessary to render (the said building) free from termite infestation at his own expense for period of ten years from the certified date of completion and in the event of failing to do so with in a period to be specified by the GE in his demand aforesaid, the GE shall under take such treatment at the risk and expenses in all respect of the contractor. The liability of the contractor under this condition shall not extend beyond the period of ten years from the certified date of completion unless the GE has previously given notice to the contractor. Conditions of contract.

29.2 The Security Deposit referred to be held from the contractor against the guarantee period for anti-termite treatment shall be calculated as per the scale laid down for calculating Earnest Money Deposit enhanced by 25% on the amount of an equal to 10 % of the cost of anti termite treatment at contract rates for the buildings for which treatment is to be done and may be refunded to the contractor after the expiration of the period of ten years from the certified date of completion by the GE provided always that the contractor shall first have been paid the final bill and have rendered a No Demand Certificate (IAFA-451). The facility of furnishing fixed deposit receipt in lieu of the sum to be retained as security may be accepted.

29.3 Condition 10, 46 & 68 of the General Conditions of Contracts (IAFW-2249) shall be deemed to be amended to the extent mentioned above.

29.4 In addition to the above special conditions, following points shall also be ensure at the time of execution:

(a) It shall be ensured that the agency executing the work procures the chemicals only from the manufacturers or their authorised agents/dealers.

(b) Chemicals brought to site only in sealed container bearing ISI certification marks shall be permitted to be used.

(c) Chemicals shall be stored carefully at site. Seals of containers shall be broken only in the presence of the Engineer-in-Charge. Empty containers shall be removed off the site promptly. If on any particular day the contents of one full container could not be used in the work, the container shall be sealed at the end of the day in the presence of the Engineer-in-Charge and opened when required in the presence of the Engineer -in -Charge.

(d) The Engineer-in-Charge shall ensure that paid vouchers are produced by the agency executing the work for the full quantity of the chemical required and brought to the site and a record of such vouchers, dully defaced, shall be kept by the Engineer-in- Charge.

(e) Tests of the chemical brought by the agency executing the work may be carried out in a recognised laboratory or test house at the discretion of the GE to satisfy that spurious materials are not being used.

(f) The EIC shall work out, on his own, the total requirements of chemicals required for the work as per specifications and ensure that full quantity is brought to site and used in the work. For this purpose, entries shall be made in the Measurement Book (not for payment-not to be abstracted) indicating the brand name, quantity brought etc and signed by EIC as well as representative of the agency executing the work.

(g) Work of anti termite treatment shall be done through an approved specialized agency holding valid licence as per clause 13 of Insecticides Act 1968 as amended from time to time. Persons employed to carry out anti termite treatment shall be qualified as per Rule 10 promulgated under Insecticides Rules 1971 as amended from time to time.

(h) GEs/CsWE during their inspection of the work shall specifically examine the Anti-termite treatment.

(i) A copy of IS 6313 (Part-II) shall be available at site with every Engineer-in-Charge who gets anti-termite treatment works executed.

30. **GUARANTEE FOR WATER PROOFING TREATMENT**

30.1 The period of Guarantee for the water proofing treatment shall be 10 (ten) years from the certified date of completion. The contractor shall be responsible for effectiveness of the water proofing treatment during this period and any leakage noticed therein during this period shall be made good by him at his own cost.

30.2 The contractor shall furnish guarantee in favour of the Garrison Engineer for the efficacy of the water proofing treatment during the guarantee period.

30.3 An amount of security deposit equal to the Earnest Money Deposit enhanced by 25% on the amount of water proofing treatment at contract rates for the building for which treatment to be carried out shall be retained from the contractor's final bill as security deposit for the water proofing treatment which shall be released to the contractor only after expiry of guarantee period. The contractor may alternatively furnish a fixed deposit receipt in favour of GE in lieu of sum to be retained as security deposit for water proofing treatment from any approved Schedule Bank.

30.4 Condition 46 of the General Conditions of Contracts (IAFW 2249) shall be deemed to be amended to the extent mentioned above.

31 **RE-IMBURSEMENT/REFUND ON VARIATION IN "TAXES DIRECTLY RELATED TO CONTRACT VALUE**

(a) The rates quoted by the contractor shall be deemed to be inclusive of all taxes(including GST on material, GST on Works Contract, Labour welfare cess/tax etc), duties, Royalities Octroi & other levies payable under the respective statutes. No reimbursement/refund for variation in rates of taxes, duties, Royalities, Octroi & other levies, and/or imposition/abolition of any new/existing taxes, Duties, Royalities, Octroi & other levies shall be made except as provided in sub para (b)(i) here-in-below.

(b) (i)The taxes which are levied by Government at certain percentage rates of Contract Sum/Amount shall be termed as "taxes directly related to Contract value" such as GST on material, GST on Works Contract, Labour welfare cess/tax and the like but excluding Income Tax. The tendered rates shall be deemed to be inclusive of all "taxes directly related to Contract value" with existing percentage rates prevailing on the bid submission end date. Any increase in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on the bid submission end date shall be reimbursed to the contractor and any decrease in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on the bid submission end date shall be refunded by the Contractor to the Govt/deducted by the Government from any payments due to the Contractor. Similarly imposition of any new "taxes directly related to Contract value" after the bid submission end date shall be reimbursed to the Contractor and abolition any "taxes directly related to Contract value" prevailing on the bid submission end date shall be refunded by the Contractor to the Govt /deducted by the Government from the payments due to the Contractor.

(ii) The contractor shall, within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further "taxes directly related to Contract value" give written notice thereof to the GE stating that the same is given pursuant to this Special Condition, together with all information relating thereto which he may be in a position to supply. The Contractor shall also submit documentary proof/ information as the GE may require.

(iii)The Contractor shall, for the purpose of this condition keep such books of account and other documents as are necessary and shall allow inspections of the same by a duly authorized representative of Government, and shall further, at the request of the GE furnish, verified in such a manner as the GE may require, any documents so kept and such other information as the GE may require.

(iv)Reimbursement for increase in percentage rates/imposition of "taxes directly related to Contract value" shall be made only if the Contractor necessarily and properly pays additional "taxes directly related to Contract value" to the Government, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Government Authority and submits documentary proof for the same as the GE may require.

32. **OUTPUT OF ROAD ROLLER**

32.1 Refer CONDITION 15 OF IAFW - 2249.

32.2 Where road roller(s) is/are hired by the Department to contractors, a log book for each road roller shall be maintained by the Department for recording hours of working of the road roller. In case, however, when the contractor shall arrange road rollers from sources other than the Department. A log book for each road roller shall be maintained by him for recording hours of working of the road roller. Entries in the log book shall be signed by the Contractor or his authorised representative and by the Engineer-in-Charge.

32.3 To ensure proper consolidation, roller must work for at least the number of days assessed, on the basis of output given hereinafter. If the roller has not worked for the number of days so assessed, recovery shall be effected from the contractor for the number of days falling short of the days assessed on the basis of output stipulated. The recovery shall be effected as under :-

(a) Where road roller is hired out only by the department to the contractor; at rates given in Schedule 'C'.

(b) Where road roller is hired by the contractor only from sources other than the department; @Rs.1400/- per working day of 8 hours for 8 to 10 tonne roller.

(c) Where road roller is hired by the contractor from the department and also from sources other than the department; at higher of the two rates given in Schedule 'C' of the contract and para- (b) above.

32.4 The above provision shall not, however, absolve the contractor of his responsibility of properly consolidating surfaces as required under the provisions of the contract.

32.5 Output of Road Roller per day of Eight hours work of power roller (8 to 12 tonne).

(i)	Consolidation of formation surfaces / subgrade	1850 Sq.Metres.
(ii)	Consolidation of stone soling 22 to 23 cm thick with 8 to 10 tonne roller.	518 Sq.Metres.
(iii)	-do- but 15 cm thick.	800 Sq.Metres.
(iv)	Consolidation of water bound macadam (stone metal) /WMM 7.5 cm consolidated spread thickness including spreading and consolidation with binding material.	372 Sq.Metres
(v)	-do- but 10 cm consolidated thickness	279Sq.Metres.
(vi)	-do- but 11 cm consolidated thickness	248Sq.Metres.
(vii)	Consolidation of single coat surface dressing	774 Sq.Metres.
(viii)	Consolidation of two coats surface dressing.	558 Sq Metres.
(ix)	Consolidation of 4 cm thick premixed carpet including seal coat.	118 Sq Metres.
(x)	Consolidation of 3 cm thick premixed carpet including seal coat.	446 Sq Metres
(xi)	Consolidation of 2.5 cm thick premixed carpet including seal coat.	595 Sq Metres.
(xii)	Consolidation of 2 cm thick premixed carpet including seal coat.	744 Sq Metres
(xiii)	Consolidation of bituminous mixture, 2 parts of broken stone metal and one part of sand bitumen consolidated thickness 4 cms.	372 Sq Metres

(xiv)	Consolidation of 15 cm thick earth berms.	1800 Sq Metres
(xv)	Consolidation of 150 mm thick broken stone base.	500 Sq Metres.
(xvi)	Premix macadam 5 cm thick	500 Sq Metres.

33. **CONTRACTOR SITE LABORATORY AND TESTING CHARGES** :-

33.1 A site laboratory shall be set up by the contractor for carrying out the tests as required under the contract. The laboratory shall fully equipped and shall be approved in writing by the Garrison Engineer. The calibration of equipment shall be periodically checked by the Garrison Engineer.

33.2 The types of test and frequency of tests to be carried out in such site laboratory are attached as Appendix 'E". However keeping in view the nature of work, Garrison Engineer may add additional test checks to this list or modify the list as required.

33.3 The contractor shall be responsible for such tests to be carried out and for that they shall employ a competent technical representative as approved by the Garrison Engineer and all such tests shall be carried out in the presence of Engineer-in-Charge.

33.4 The charges for the tests carried out in site laboratory of the contractor shall not be recovered, as the test shall be carried out by the contractor at his own cost in his site laboratory. However, the charges for test shall be recovered from the contractor only if he does not carry out the tests catered for in the contract and these are got done at MES laboratory.

33.5 Where testing of contractor's materials is carried out in Zonal Laboratory/ Command Testing Laboratory testing charges shall be recovered from the contractor's running payment/final bills as the case may be.

33.6 A percentage/selected checks as decided by the Accepting Officer, shall got done independently in the Zonal Laboratory/ Command Testing Laboratory, National Test House, SEMT Wing CME, Regional Research labs, IIT, NABL approved Labs, Govt Engineering Collage or Any Govt Lab and expenditure for such tests shall be borne by the department.

33.6A The lump sum amount quoted by the contractor is deemed to be inclusive of the expenditure towards tests to be carried out as per Appendix ‘E’ enclosed here-in-after

33.7 Level of testing shown in legend as "A", "B" & "C" are defined as under :

LEVEL ‘A'

(a) "Site lab" means own site lab established by Contractor at the work site for such tests. This lab shall house all the facilities including T& P, machinery, equipment, manpower, etc required for conducting tests. This lab shall be operative for the entire duration of the contract till its completion. Tests shall be carried out in the presence of Engineer-in-Charge to be nominated by GE or any other departmental official to be nominated by the GE. Random check of compliance of frequency of testing shall be done by GE.

(b) Records shall be maintained at work site. These test results shall be signed by Contractor or this authorised representative and aforesaid departmental official.

(c) Within 15 days of placement of work order No. 1, site lab shall be established and fact reported by the contractor to GE in writing who will verify the fact and satisfy himself of the facilities provided. There after GE shall issue a certificate to this effect in writing listing out equipment particulars etc of each material test. Only after issue of this certificate by GE the tests shall be carried out and materials so approved shall be incorporated in the work.

(d) Manpower, material and infrastructure like electricity, water etc required for conducting these tests shall be provided by the Contractor.

(e) Tenderer is deemed to cater for above provisions in his quoted lump sum.

(f) Remedial measures, if any, required to achieve/obtain desired results for each test shall be taken promptly by Contractor.

(g) In case, during any point of time, Contractor fails to adhere to the laid down frequency of the tests for any reasons whatsoever, the Contractor shall get the tests done from any other lab whose name shall be approved by GE in writing. The testing charges payable by Contractor to such labs shall be paid by the Contractor. Contractor's quoted lum sum is deemed to include for this eventuality and nothing extra shall be payable to the Contractor. No extension of time shall be admissible on this account.

LEVEL 'B'

(a) " Zonal Laboratory/ Command Test Laboratory " means lab of **CE Eastern Command Test Laboratory, located at SE Falls, Shillong** or at any other location within the Eastern Command as notified by the Accepting Officer.

(b) The tests shall be conducted as per frequencies laid down for these tests in these labs for which Contractor shall provide all requisite facilities like samples, cubes, material etc, transportation to these labs for testing purpose. It will be Contractor's responsibility to adhere to the laid down frequency of testing. Test results shall be sent by lab to the GE whose copies can be made by Contractor at his own expense. Testing charges for the tests so conducted shall be recovered at the rates indicated in Appx `E' from the running payments. The Contractor's quoted lump sum is deemed to quote for above provision.

LEVEL 'C'

(a) Level 'C' lab stands for National Test House, SEMT Wing CME, Regional Research labs, IITs, National Institutes of Technology,NABL Labs, Command Test Laboratory ,Govt Engineering Collage or Any Govt laboratories where such facilities exists.

(b) Rest provision contained in para (b) of Level `B' above shall be applicable here except that Contractor shall obtain test results from these labs and handover to GE. The testing charges payable to these labs for conducting these tests shall be borne by the Contractor and his quoted lump sum is deemed to include this provision.

34. **TRAINING OF MES STAFF**

34.1 The contractor shall give training to MES staff as detailed by the Engineer-in-Charge for 15 days period regarding operation, running and maintenance of various equipments, installed by him, as desired by the Engineer-in-Charge. Unit rates lump sum quoted by the contractor shall be deemed to have include the rates for the same and no extra payment shall be made to the contractor on this account.

35 **QUALIFIED TRADESMEN**:

35.1 The compliance with the condition 26 of IAFW-2249 (General Conditions of Contract), the contractor shall employ skilled/semi skilled tradesmen who are qualified and possessing certificate in particular trade from Industrial Training Institute (ITI)/National Institute of Construction Management and Research (NICMAR)/ National Academy of construction (NAC) Hyderabad Similar reputed and recognised Institute by State/Central Government to execute the works of their respective trade. The number of such qualified tradesmen shall not be less than 25% of total skilled/semi skilled tradesmen required in each trade. The contractor shall submit the list of such tradesmen alongwith requisite certificates to Garrison Engineer for verification and approval. Notwithstanding the approval such tradesmen by GE, if the tradesmen are found to have inadequate skill to execute the work of their trades, leading to unsatisfactory workmanship, the contractor shall revoke such tradesmen within a week after written notice to this effect by the GE and shall engaged other qualified tradesmen after prior approval of GE. GE's decision whether a particular tradesmen possesses requisite qualification, skill and expertise commensurate with nature of work, shall be final and binding. No compensation whatsoever on this account shall be admissible.

36. **CLEANING DOWN**

36.1 Refer condition 49 of IAFW-2249 (General conditions of contracts).

SPECIAL CONDITIONS (CONT....)

36.2 The contractor shall clean all floor, walls, remove cement/lime/paint marks/drops, etc clean the joinery glass panes etc, touch up all painters work and carryout all other necessary items of work in connection herewith and leave the whole premises clean and tidy before handing over the building(s).

37. **GRANITE STONE PLATE FOR GUARANTEE**

37.1 The contractor, within four weeks from the certified date of completion, shall provide a black granite stone plate of size 90 cm x 60 cm x 1.5 cm at a prominent place/location of each building as directed by the GE having engraved the following information/details on the plate :-

(a) Job No
(b) CA No & year
(c) Name of work
(d) Name of contractor
(e) Name of GE
(f) Name of Engineer-in-Charge
(g) Date of commencement
(h) Date of completion (phase-wise) (As applicable)
(j) Date of expiry of defect liability period
(k) Date of expiry of guarantee period Given against ATT, water proofing treatment over roof/terrace
(l) Name of building

37.2 Granite stone plate shall be fixed over 15 mm thick cement screed in cm (1:3)

37.3 All figures /words engraved shall be painted with golden paint of approved make.

37.4 The cost of above is deemed to included in the lump sum amount of quoted by the contractor against Schedule 'A' Part-I.

38. **CONCILIATION**

38.1 **CONCILIATION BY DRB OR CONCILIATORS:**-In this connection please refer condition 71 of IAFW 2249 amended as per amendment No. 40 vide Govt of India, MOD, New Delhi letter No. PC-1 to 33487/ IAFW-2249/R/01/2011/D (works-II) dt 24 Mar 2015.

38.2 **APPOINTMENT OF DRB / CONCILIATOR:**- During execution of the works or after completion or after determination/cancellation/termination of the contract all dispute between the parties to contract arising out of the contract (except those for which decision of Accepting Officer or any other officer (CWE and/or GE) is expressed to be final and binding), including any disagreement by either party with any action, inaction, opinion, instruction, certificate or valuation by the Accepting Officer or his nominee, the matter is dispute shall, in the first place be referred to the Dispute Resolution Board (DRB) (for Contracts valuing more than Rs 10 crore) and to sole conciliator (for the Contracts valuing less than Rs 10 crore) viz Serving Officer having degree in **Engineering** or equivalent or having passed final/direct final examination of sub division-II of Institution of Surveyors (India) to be appointed by the Engineer-in-Chief, Army HQ, New Delhi or in his absence the officer officiating as Engineer-in-Chief or Director General of Works specifically delegated by the Engineer-in-Chief in writing. In case of disagreement with the decision of such DRB or conciliator, any party may invoke arbitration clause.

38.2.1 The constitution of the DRB shall be a three member body as under :-

(i) **Chairman** : Joint DG (Contracts) of the concerned Command Chief Engineer. Where JT DG (C) is not posted in the Comd, any other Chief Engineer/Brig level Officer posted in CE Comd shall be nominated by Comd CE at his sole discretion.

(ii) **Member 1** to (iii) **Member 2** } Col/Director rank officers of Comd CE or any other Zonal CE to be nominated by Comd CE.

38.2.2 The name of Chairman and Members shall be notified by the Accepting officer within one month of the date of acceptance of contract.

SPECIAL CONDITIONS (CONT....)

38.2.3 Once the DRB is constituted the members and Chairman shall disclose in writing their neutrality and impartiality about any personal interest in the work.

38.2.4 The dispute shall be referred to the Chairman of the DRB by the concerned party after giving notice to the other party for invoking of this clause.

38.2.5 The DRB shall decide the dispute in accordance with the terms of the contract, principle of natural justice, equity and fair play.

38.2.6 The DRB may fix oral hearing at a place, date and time as decided by the Chairman.

38.2.7 The requisite administrative support to the DRB shall be provided by the Accepting Officer.

38.2.8 All the contract documents pertaining to the case shall be provided by the Accepting Officer for reference by the DRB.

38.2.9 DRB shall give its decision on the disputes within three months of notice from any party invoking the DRB clause. This period can be extended by one month with the consent of the parties.

38.2.10 All the decisions given by the DRB shall be by majority and such decisions shall be communicated in writing by Chairman to the parties.

38.2.11 If the decision of the DRB is not to the satisfaction of either party or if the DRB fails to give decision within the laid down time either party shall indicate his reservations on the decision to Accepting Officer within 30 days of such decision and to refer that dispute for arbitration within the provision of condition 70 of IAFW 2249 General Conditions of Contract.

38.2.12 It shall be mandatory for the party invoking arbitration on any particular dispute to have first exhausted the remedy provided under the DRB clause for that particular dispute.

38.2.13 The mandate of the DRB shall terminate on completion of one year from the date of completion/determination/cancellation/termination of the contract.

38.2.14 If any member or Chairman of the DRB is unable to function due to any reason whatsoever, or he resigns his appointment, Chief Engineer Command as the case may be, shall fill the vacancy so caused within 15 days of happing of such vacancy.

38.2.15 Any dispute referred to the DRB and having been decided by the DRB and not objected to by either party within 30 days shall attain finality and shall not be referable to arbitration.

38.2.16 Accepting officer shall ensure implementation of the decision of the DRB which attain finality, i.e, except those which are objected by him or by contractor within 30 days as per Para 38.2.11 above.

38.2.17 Findings and decision of DRB shall be admissible as evidence, to the extern permissible as per law, in the subsequent Arbitration and /or litigation.

38.2.18 DRB Chairman/Members shall not, in any case, be liable to be called as witness or to produce any evidence in any Arbitration or departmental proceedings of any kind.

38.2.19 During execution of work the disputes may be referred to the DRB as per the requirement of each party after having exhausted the decision making process provided in the contracts. In case of completion of work or after determination/cancellation/termination of the contract all the disputes including payment/non-payment/delay in final bill shall be simultaneously referred to the DRB within six months of completion/determination/cancellation/termination of the contract.

38.2.20 The department case before the DRB shall be presented by Accepting Officer himself and/or Dir (Contract) of CE Zone assisted by CWE and his DCWE (Contract), GE and his AGE (Contracts) and any other officer and legal counsel nominated by Accepting Officer. The Contractor may present his case by himself and/or by his nominated reps & authorized legal/technical counsel.

38.3 **STATUS OF EFFECT OF SETTLEMENT AGREEMENT** :- The settlement agreement signed by the parties as a result of conciliation proceedings shall have the same status and effect as it is an arbitral award on agreed terms.

SPECIAL CONDITIONS (CONT....)

39. **PERFORMANCE EVALUATION AND MONITORING OF WORKS:-**

(a) The works shall be inspected by Accepting officer and performance evaluation and monitoring of works shall be done as under:-

(i) First Evaluation	:	Within two months from the date of Commencement of work as per Work Order No. 1
(ii)Second/Intermediate Evaluation	:	Every six months from the date of Commencement of work as per Work Order No. 1
(iii) Third Evaluation	:	Evaluation one month prior to original / extended date of completion.
(iv) Final evaluation	:	One month after actual date of completion

(b) The details of performance evaluation and monitoring of works to be exercised have been issued vide IHQ of MOD E-in-C's Branch letter No. A/37696/OSDPL/pol/E2W (PPC) dt 13 Aug 2014, A/37696/OSDPL/pol/E2W (PPC) dt 25 Aug 2014 and A/37696/OSDPL/pol/ E2W (PPC) dt 29 Dec 2014 & 66546/Manual/18/E8 dt 10 Dec 2018. These letters can be seen in any of the MES formations and are also available on MES web site www.mes,gov.in.

(c) Contractor is advised to be fully aware with these instructions and the requirements.

40. **FEES CHARGABLE BY POLICE AUTHORITIES FOR VERIFICATION OF ANTECEDENTS**

40.1 Verification of antecedents of Proprietor/Partners/Directors of the firm in connection with issue of tender and of Enlistment shall be Govt responsibility and accordingly payment to police authorities towards verification of antecedents shall be borne by Govt.

40.2 Verification of antecedents of Contractor's representatives/labour deployed at site in connection with execution of work under the contract, as per security requirement of User Unit/Installation shall be the responsibility of the contractor and all expenses in connection with verification of antecedents by Police Authority/Security Agency shall be borne by the contractor.

41. **APPROACHES** :- The contractor shall make arrangements for and provide at his own cost all temporary approaches, if required to the site(s), after obtaining approval in writing of the GE to the layout of such approaches.

42. **DAMAGE TO EXISTING STRUCTURES** Any damage to the existing structures, any existing road etc, during the execution of work shall be made good by the contractor at his own expense. Rectification, replacement, making good any touching up etc shall be carried out, conforming to the materials and workmanship originally provided and to the satisfaction of the Engineer-in-Charge. In case of any dispute on this account, the decision of the GE shall be final, binding and conclusive.

43. **Location of Buildings and Works**. There may be some changes in location/siting of building shown in site (layout) plan (s) to suit local conditions and/or departmental requirements. The contractor shall have no claim what-so-ever consequent to such changes in the location/sitting of works.

44. **Handing Over of Site**. Site for execution of work will be available as soon as the work is awarded. In case it is not possible to make the entire site available on the award of work, the contractor will have to arrange his working programme accordingly. No claim whatsoever, for not giving entire site on award of work and for giving site gradually, will be tenable.

45. **CONDITION 25 OF IAFW-2249 (GENERAL CONDITION OF CONTRACTS) AND AMENDMENT NO 27.:-** Refer Amendment No 27 Of IAFW 2249

(a)	For contract amount more than Rs. 1000 lakh and upto 2500 lakh	(a) 1 no Graduate Engineer from a Government recognized institution with minimum experience of 5 years. (b) 2 nos Diploma Engineers from Government recognized institution with minimum experience of 8 years.
(b)	For contract amount more than Rs. 2500 lakh and upto 5000 lakh	(a) 2 nos Graduate Engineers from a Government recognized institution with minimum experience of 5 years. (b) 3 nos Diploma Engineers from Government recognized institution with minimum experience of 8 years. (c) Any one of the Engineers should have capability to use Project Management Software like MS Projects/Primavera in

		all projects
(c)	For contract amount more than 5000 lakh	(a) 3 nos Graduate Engineers from a Government recognized institution with minimum experience of 5 years. (b) 3 nos Diploma Engineers from Government recognized institution with minimum experience of 8 years. (c) Any one of the Engineers should have capability to use Project Management Software like MS Projects/Primavera in all projects

46. **REQUIREMENT OF T&P**

46.1 Contractors to note that they will be required to strictly ensure deployment of 'T&P, Machinery & Transport' as given below. Inadequate deployment of T&P shall be considered as serious lapse attracting ban/removal/downgrading/debarment of the firm/company or any other action as deemed fit will be taken by the Accepting officer/Department.

Sl No	Special T&P/ Machinery/ Transport	Minimum Nos. Required	Remarks
a (i) Building & road category			
1.	One bag capacity concrete mixer (diesel)	-	-
2.	Ready mix plant and fully automatic batching plant	1	
3.	Digital Weight bridge	1	
4.	Stone crusher with different sizes of sieves	1	
5.	Vibrators (Needle and Plate type)	20	-
6.	Surface vibrators	05	
7.	Nozzle vibrator	10	
8.	Tower/builder's hoist	4	-
9.	Steel shuttering with spans, props etc.(sqm)	6000	-
10.	Trucks/Tippers	20	-
11.	DG Sets 5/10 KVA	3	-
12.	Total Stations	3	-
13.	Digital theodolite	2	
14.	Prismatic compass	6	
15.	Leveling staff	6	
16.	Sprit level	15	
17.	Concrete cube testing machine (Hydraulically operated)	3	-
18.	Fully automatic concrete batching plant	1	-
19.	Transit Mixers	3	-
20.	Concrete Pumps	3	-
21.	Concrete mobile weigh batchers	3	-
22.	Digital steel bar bending machine	2	
23.	Digital rebound hammer test machine	2	
24.	Cranes	2	-
25.	Excavators (Power shovels/draglines)	6	-
26.	Bull dozers	2	-
27.	Vibrator Road Rollers	2	-
28.	Tandem roller	2	
29.	Bitumen paver with min 3.65 width	1	

SPECIAL CONDITIONS (CONT....)

30.	Drilling machines	3	-
31.	Boring rig	5	Optional in case of tube well boring work
32.	Pile driving equipment (sets)	5	Optional in case of pilling work
33.	Pile testing equipment (sets)	3	Optional in case of pilling work
34.	Steel/Aluminium Ladders 1.5 m to 8m	10	
35.	Chase cutting machines	3	
36.	Torque wrench for nuts/bolts/ screws	3	
37.	Primary and secondary injection equipment	3	
38.	Dielectric strength testing eqpt	3	
39.	Magnetic dial indicator for alignment	3	
40.	Conduit die set	3	
41.	Pipe vice	3	
42.	Bench vice	3	
43.	LT Meggar 500 volts	3	
44.	HT Meggar 5000 volts	3	
45.	Tong tester	3	
46.	Hydraulically operated & hand operated crimping machines	3	
47.	Earth tester	3	
48.	Portable drilling machines	3	
49.	Over head conductor puller	3	
50.	Test bench for light fittings	3	
51.	Cable fault location meggar	3	
52.	Oil testing set	3	
53.	Multi-meter	3	

Note :-

1. Any other T&P /Machinery / Transport required for executing the work even though not categorically mentioned above but technically required shall be deemed to be proved and deployed at site.
2. Stage wise deployment of T&P /Machinery / transport shall be ensured.
3. The above list is for guidance and GE will be the sole authority to decide the optimum requirement of minimum Nos & type of T&P /Machinery / Transport at site at a particular stage of work ensuring that no delay is encountered due to any shortage of T&P/ Machinery/ Transport.

SIGNATURE OF CONTRACTOR **DY DIR/AAD (CONTRACTS)**

DATED : **FORACCEPTING OFFICER**

PARTICULAR SPECIFICATIONS

1. GENERAL

1.1. GENERAL REQUIREMENTS

1.1.1. The Work under this CONTRACT shall be carried out in accordance with SCHEDULE 'A' (BOQ), PARTICULAR SPECIFICATIONS, DRAWINGS and GENERAL SPECIFICATIONS and other provisions in MES STANDARD SCHEDULE OF RATES 2009 (PART I - SPECIFICATIONS) and MES STANDARD SCHEDULE OF RATES 2020 (PART II - RATES) (here in after called as MES SCHEDULE) read in conjunction with each other.

1.1.2. General Specifications referred to/mentioned in GENERAL CONDITIONS OF CONTRACTS (IAFW 2249) (1989 - PRINT) and in other documents mean Specifications including other provisions given in MES Schedule including Amendments and Errata as applicable.

1.1.3. Materials & Workmanship required for these Works/Services shall be as described against 'Specifications & Workmanships' in the various Trade/Sections of the MES Schedule duly modified by these Particular Specifications here-in-after.

1.1.4. Reference to some Paragraphs of MES Schedule has been made in these Particular Specifications, but other Paragraphs and Provisions as applicable are also to be followed e.g. reference to Clause No. pertaining to General Workmanship for Brick Masonry, Joinery, Iron and Steel Work, etc. have not been made but provisions therein as required for the work are applicable.

1.1.5. Provisions contained in the MES Schedule Part II in the Preambles to the relevant Items of SCHEDULE 'A' (BOQ) shall also be read in conjunction with the Provisions contained in these documents.

1.1.6. Where Specifications/Provisions for any Item of Work given in these Particular Specifications are at variance with the Provisions/Specifications given in MES Schedule, Specifications/Provisions given in these Particular Specifications shall be followed. However for pricing of Deviation Orders, Rates given in MES Schedule Part II shall be adopted.

1.1.7. Where Specifications for any Item of Work are not given in MES Schedule or in these Particular Specifications, Specifications given in relevant Indian Standard Specifications or Code of Practice shall be followed.

1.1.8. Particular Specifications given here-in-after shall be read in conjunction with the Provisions in the MES Schedule and relevant **Bureau of Indian Standards** referred to thereto and in these Particular Specifications. In case of any discrepancy, the Provisions in these Particular Specifications shall take precedence.

1.1.9. Materials/Accessories/Equipments for which MAKE have not been specified in these Particular Specifications, shall be of BIS Makes and shall strictly comply with Current Appropriate BIS Specifications, for which BIS Specification has not been issued/available, they shall comply with the Currents BS Specifications or as approved by GE.

1.1.10. These Particular Specifications shall deem to be included the APPENDICES attached here-in-after.

1.2. SCOPE OF WORK

1.2.1. This Contract includes for the full, final and entire completion of Works described in SCHEDULE 'A' (BOQ) including Schedule 'A' Notes thereon and specified in these Particular Specifications forming part of the Tender documents.

1.2.2. Tenderers are advised to visit the SITE(S) and ascertain for themselves the exact scope of Work and its Working Conditions, viz Working Hours, Availability of Site(s), etc. and quote their Tenders accordingly. Any Claim whatsoever, on such/these accounts will not be entertained by the Government at a later Date/Stage.

1.2.3. The contractor shall submit **Quality Control Plan Part I as per Appendix 'A' to the GE within 30 days of the commencement of work**. The Part II of Quality Control Plan shall be completed by GE and shall be forwarded to CWE for approval. **The quality Control Plan shall be approved by CWE and a copy thereof shall be forwarded to Accepting Officer**.

1.3. SITE OF WORK

1.3.1. The Works shall be carried out at different Locations/Site(s). The Contractor shall finalize the programmed with the Engineer-in-Charge well in advance in such a way that neither the USERS feel inconvenience nor Work is delayed.

1.3.2. The proposed Location as shown on site plan of Various Works shall be as directed by the Engineer-in-Charge at Site(s) of Works.

1.4. VISIT TO SITE

1.4.1. The Tenderers are advised to Contact GE on any Working Day during Working Hours, who will arrange for their Visit to Site of Work.

1.4.2. The Tenderer shall be deemed to have visited the Site of Work before Quoting their UNIT RATES and no Claim for any EXTRA PAYMENT on account of any misunderstanding, etc. will be entertained by the Department, irrespective of the fact whether they have actually visited the Site or not.

2. MATERIALS

2.1. GENERALLY

2.1.1. All Materials to be supplied by the Contractor for incorporation in the Work shall conform to relevant IS Specifications. In case Specification of any Materials needed for incorporation in work is not contained in any of the Contract Documents, the Specification of such Materials/Articles to be incorporated in work shall be got approved in writing from the GE before their incorporation in work by the Contractor without any Price Adjustment.

2.1.2. The relevant **Bureau of Indian Standards** shall be of Latest Revision including Amendments if any. The Contractor shall incorporate in the work Materials complying with the requirements of relevant Bureau of Indian Standards of the Latest Publication (Edition) **i**ncluding all amendments/revision issued by **Bureau of Indian Standards on and before** bid submission end date, thereof without any Price Adjustment in the Quoted Rate.

2.1.3. As far as practicable all manufactured Materials/Articles other than those manufactured in Contractor's Workshop at Site shall bear IS Certification Marks. In case, any Article not bearing IS Certification Marks but conforming to relevant Specification is proposed to be incorporated in the Work, Samples of the same shall be got approved in writing from GE before their incorporation in work. The Contractor shall submit sufficient evidence to the GE to show that such Articles conform to the relevant Specifications and no Price Adjustment shall be made on this account.

2.1.4. Manufactured Materials/Articles shall be brought at Site in Original Sealed Containers/Packing bearing Manufacturer's marking unless the quantity required is a fraction of smallest packing.

2.1.5. Materials of Proprietary nature such as Paints, Water Proofing Compound, Chemicals for Anti-Termite Treatment and the like, Quantity of which cannot be Checked after incorporation in the Work shall be Measured and Recorded in the MEASUREMENT BOOK as soon as those are brought at Site. These Measurements shall be signed both by the Engineer-in-Charge and the Authorised Representative of the Contractor. ORIGINAL STAMP RECEIPTED BILLS alongwith the relevant INVOICES from the Manufacturers or their AUTHORISED DEALERS (if any) in support of having brought the Full Quantity required for incorporation in the Work, shall be produced to Engineer-in-Charge.

2.1.6. The Contractor shall carry out such Instructions that are given to him in Writing by the Engineer-in-Charge to ensure that Full Quantity of such Materials goes in to Work.

2.2. APPROVED MAKES/BRAND OF MATERIALS

2.3. Materials/Accessories/Equipments for which MAKE have not been specified in these Particular Specifications, shall be of Standard Makes and shall strictly comply with Current Appropriate **BIS** Specifications, for which IS Specification has not been issued/available, they shall comply with the Currents **BIS** Specifications or as approved by GE.

2.4. A list of certain items for which specific manufacturers/ make only are to be provided in this contract (other than steel and cement) are given in Appendix `B'.

2.5. APPROVED SOURCES OF LOCAL ORIGIN MATERIALS

2.5.1. Materials such as Bricks, Aggregates, etc., shall generally conform to the Samples kept in GE's Office in addition to their conformity with relevant Specifications given in the Tender documents. The Sources and Sample for all such Materials shall be got approved from GE in writing before these Materials are brought at Site in bulk.

2.5.2. The sources of some of the Materials as applicable to the place of execution of the work are indicated in **APPENDIX 'C'** given here in after for the guidance of the Tenderer.

2.5.3. Contractor at his own may procure materials from any other sources without any extra cost to the Department provided that the same conform to respective IS as stated here-in-before.

2.6. APPROVAL OF SAMPLES/MATERIALS

2.6.1. Approval of Samples/Materials by GE shall be recorded in a Register labeled as 'SAMPLE APPROVAL REGISTER' to be maintained by the Engineer-in-Charge which interalia shall mention Sources of Supply, Name of Manufacturer, Trade Name/Brand (if applicable) and reference to Clause of Tender Documents containing Specifications of Particular Materials.

2.7. Letters conveying Approval of Samples/Materials by Garrison Engineer shall interalia mention sources of Supply, Name of Manufacturer, Trade Name/Brand (if applicable) and reference to Clause of Tender documents containing Specifications of Particular Materials.

2.8. STANDARD OF QUALITY & WORKSMANSHIP

2.8.1. GENERAL

2.8.2. The Work shall strictly comply with the provisions contained in the Latest Edition of INDIA STANDARD CODE OF PRACTICE and/or IS SPECIFICATION as applicable, Works except where such Regulations and Rules are modified by these Particular Specifications.

2.9. All Works shall be carried out by properly Skilled Trademan. The Contractor on demand shall produce such evidence of Qualifications of his Workmen/Skilled Trademan/Supervisors / Engineers, either at the commencement of Work or at any time thereafter/during currency of the Contract. The entire Work shall be High Class with the best Workmanship and to the entire satisfaction of Engineer-in-Charge/ GE.

3. EXCAVATION & EARTH WORK

3.1. GENERALLY

3.1.1. The Lump Sum Amount against Schedule 'A' Part-I(A) shall be deemed to be included for all Excavation, Earth-work except raising of existing ground level to made-up-ground level/ Road level etc. in Foundation in **soft/loose Soil**. If during execution, any other type of strata is met with, the same shall be ordered through a Deviation order.

3.2. HARD ROCK:-

3.2.1. If Hard Rock is met with in Excavation, the same shall be intimated to the GE in writing by the Contractor and shall be entered in the Measurement Book duly signed by Engineer in Charge and Contractor. Hard Rock / Boulders obtained from Excavation shall be properly stacked, measured and recorded in Measurement Book and will be the property of the Contractor. Credit at the Rate of Rs **500.00 PER CUBIC METRE** of hard rock/Boulder obtained shall be recovered from the Contractor. The measurement shall be signed by the Contractor and the Engineer-in-Charge. Hard rock thus obtained can be reused in the execution under this Contract subject to fulfillment of requisite specification specified here-in-after and after written approval of GE.

3.2.2. Mode of measurement of stones/boulders for the purpose of recovery shall be as follows:

(a) Refer **IS 1200 part-I**

(b) Boulders **(**size as per **IS 1200 part-I)** obtained during excavation from hard/dence soil and boulder (size as per IS 1200 part-I) shall be neatly stacked at site by the contractor without any extra cost to the Department. A deduction of 40% (Maximum) shall be made for voids in stack measurements and accordingly quantity of stones/boulders shall be worked out.

(c) In case hard rock is met at site, complete quantity of hard rock obtained during excavation shall be issued to the contractor. Quantity of hard rock for issue purpose shall be considered same as that of excavation done in Hard Rock.

(d) The hard rock/stone boulders shall be issued to the extent available from the excavation.

3.2.3. In case due to encountering of hard rock /boulders during excavation, extra earth work is required for returning filling in foundations as per drawing, the contractor shall re-use any surplus earth retrieved from sch 'A' Part-II. However, earth required for returning, filling beyond the surplus earth retrieved from sch 'A' Part-II, may be obtained from outside Defence land which the same will be regularized through deviation order (DO).

3.2.4. If Soft/Disintegrated Rock or any other type of Soil is met with the same shall be disposed off/reused in filling as and where permitted by the GE. The decision of GE in this regard shall be Final, Conclusive and Binding.

3.2.5. If hard strata are met with, which can neither be treated as the Soft rock nor Hard rock within the definition of MES Schedule, the same shall be treated as Hard soil.

3.2.6. **Blasting of rock is not permitted during execution of Work at Site.**

3.3. SITE CLEARANCE AND SURFACE DRESSING

3.3.1. Site Clearance and Surface Dressing for the areas covered by Buildings extending upto 3.0 Metre around the Outer Perimeter of the Plinth Protection or External Wall of the Building (wherever Plinth Protection is not shown or not to be provided) and/or upto 3.0 Metre from the outer most execution details as per Drawings for works other than Buildings shall be carried out as per Clause No. 3.6 and 3.10 of MES Schedule Part-I to the entire satisfaction of the Engineer-in-Charge. All Rubbish obtained from Site Clearance and all Spoil obtained from Surface Dressing shall be removed, spread and leveled as directed by Engineer-in-Charge. The Cost of Site Clearance and Surface Dressing shall be deemed to be included in the quoted Lump Sum Amount against Schedule 'A' Part I.

3.4. EXCAVATION IN TRENCHES AND OVER AREAS

3.4.1. Excavation in Trenches and Over Areas shall be carried out as described in Clause No. 3.13.1 to 3.13.5 of MES Schedule Part-I.

3.4.2. Excavation shall be restricted to Dimension as shown on Drawings and as specified in MES Schedule. Excavation made if any, in excess of required Depth/ Authorised Width shall be made good by the Contractor with Cement Concrete 1:5:10 (40 mm graded stone aggregate) without any Extra Cost to the Govt.

3.4.3. In case of Excavation in Hard Rock, Extra Depth of Excavation in excess to the required Depth shall be made with Cement Concrete 1:4:8 (40 mm graded stone aggregate) without any Extra Cost to the Govt.

3.4.4. Beds of the Trenches/Column shall be watered and well rammed and any depression thus formed shall be filled with approved earth as required and level and slope as directed by the Engineer-in-Charge.

3.4.5. After construction and before handing over any Building the Area around shall be dressed without any Extra Cost to the Govt.

3.4.6. For the purpose of reckoning the depth of Foundation, the Average Level of the Ground after Surface Dressing shall be considered.

3.4.7. If Rock (Soft/Disintegrated and Hard) is met at Site, Contractor shall immediately notify the facts to GE in writing.

3.5. BAILING AND PUMPING

3.5.1. A Tenderer shall ascertain himself the Level of Sub-Soil Water at Site of Work to allow in his Tender for any Bailing and Pumping out of Water, etc. as per Site Condition. Bailing and Pumping out of Water, if required, will be done as described in Clause No. 3.17 of MES Schedule Part-I. No Extra Payment shall be admissible for the above mentioned operations. In the event of Deviation, no Adjustment shall be made for Cost of Bailing, Pumping and De-watering as specified here-in-before.

3.6. FILLING IN TRENCHES, PLINTH AND UNDER FLOORS :-

3.6.1. Refer Para 3.19 and sub paras of 3.19 of MES Schedule Part-I(2009).

3.6.2. Soil obtained from excavation (except earth / spoil obtained from surface dressing/ surface excavation) may be used for filling under floor & filling around foundation, if only the earth so obtained is of **non-expansive type** and same shall be approved by GE & CWE in writing.

3.6.3. Approved soil as obtained from excavation (other than **expansive soil**) shall be utilized for returning, filling in as well as for filling under floors and in any other situations after removing big stones, grass, roots, vegetables and other organic matters. Filling under floors/sides of trenches shall be in layers not exceeding 250 mm and each layer shall be compacted by rammers of 7 to 10 Kg weight. Earth filling shall be adequately watered for achieving maximum compaction. Thickness shown on drawings shall be consolidated thickness. In case due to actual site conditions excess earth is required to make up height of finished floor level including raising ground level as shown in drawings and as specified in these tender documents the contractor shall arrange earth at his own cost from outside Defence land and his rate against schedule 'A' Part I shall be deemed to include for the same. In case of pricing of deviation the approved earth/sand obtained from outside Defence land shall be priced at applicable rates of **MES Schedule Part II(2020)** adjusted by contractor's percentage for rough excavation in soft/loose soil and removal. The distance shall be considered as exceeding 1.5 KM but not exceeding 5 KM. No royalties for earth obtained from outside MOD land

shall be paid extra. However, in case the hard rock is met with at site, the provisions as per para 3.2.3 here-in-before shall be applicable

3.7. In case due to encountering of **expansive soil** during excavation , extra approved earth is required for returning filling in foundations as per drawing, same shall be obtained from outside Defence land which shall be regularized though deviation order after approval of Accepting Officer.

3.8. EARTH FILLING/CUTTING FOR SITE/AREA DEVELOPMENTS/ TRAVERSES:-

3.8.1. Refer Para 3.22 and subsequent Clauses of MES Schedule Part-I(2009).

3.8.2. The excavation shall be carried out as under :-

3.8.3 A Board of Officers shall be **ordered by CWE** to take the initial level of the **original existing ground of proposed entire site(s) as per site plan**. Further the final formation levels of earth filling site(s) shall be taken. The formation quantities shall be worked out based on the levels. Initial and final levels of proposed earth fill/ cutting site(s) shall be worked out based on the level sheet prepared by the Board of Officers and quantity shall be computed using software for Total Station,random check of quantity computed by the software shall be made by the GE as per method given here-in-after to satisfy himself for the correctness of the quantity computed by the software.

3.8.4 Before commencing the work, the existing ground levels of the whole area under proposed work shall be taken by Board of Officers as under :-

3.8.5 Existing ground levels shall be taken at 3m grid and upto 3m beyond the outer edges of the proposed work.

3.8.6 All these levels will be recorded in level books and contour shall be plotted on graph papers. The level book and graph papers shall be signed by the Contractor and BOO.

3.8.7 Contractor shall provide Total Station and other instrument as required with operators of total station for taking the initial level and final levels for Board of Officers without any extra cost to the Govt. and quoted lump sum in by the bidder/tenderer shall be deemed to inclusive of this provision.

3.8.8 Detailed photographs of site shall be taken including permanent / intermediate bench mark locations and prints made for record showing the existing site duly marking the layout of bldgs with flags on all boundaries /four corners.

3.9 FINALISATION AND APPROVAL OF INITIAL LEVELS :- Recording of initial levels at site as mentioned in here-in-above shall be completed within a period of 30 days from the date of commencement of work as per BOO ordered by CWE. The level sheet alongwith the board proceedings and propose modified ground level and formation level of road shall be forwarded by GE duly recommended, immediately to CWE for approval /concurrence **under intimation to AO**. Only after approval of Initial levels and propose modified ground level and formation level of road by CWE, any earth work shall be commenced and the original existing ground levels shall not be disturbed till approval of level sheet by CWE.

3.9 The work in cutting and filling shall be carried out as instructed by the GE.

3.11 METHOD OF MEASUREMENTS AND CALCULATION FOR EXCAVATION AND EARTH WORK

3.11.1 The area of cutting / filling in each cross sections shall be separately computed by Simpson's Rule as described here-in-after. The volume (Cubic contents) of cutting and filling shall be separately computed by Prismoidal Formula.

3.11.2 For computing area of cross sections the interval between two ordinates shall be as decided by the Engineer-in-Charge to suit the site conditions but in no case it should exceed 3 m.

3.12 REMOVAL OF SURPLUS SOIL

3.12.1 Surplus Soil and/or Rubbish obtained from Excavation shall be disposed unless otherwise specified in Particular Specifications here-in-after, upto a distance of exc. 50 Metre and not exc. 100 Metre at the Location(s) as directed by Engineer-in-Charge, deposited, spread and levelled as specified in MES Schedule Part-I(2009).

3.12.2 Soil/earth obtained from excavation shall be used for returning filling in places where any other type of filling like moorum/sand is not specified or shown in drawings after separating it from grass, vegetations and other foreign materials. Surplus Soil and / or Rubbish obtained from Excavation shall be disposed unless otherwise specified in Particular Specifications here-in-after, upto a distance of exceeding 50 Meter and not Exceeding 100 Meter at the Location(s) as directed by Engineer-in-Charge, deposited, spread and leveled as specified in MES Schedule Part-I(2009)

3.12.3 Filling under floors, around foundation in trenches, around plinth beam & in other situations shall be as shown on drawings.

3.12.4 **APPROVED EARTH FROM OUTSIDE MOD LAND**

(a) Filling of earth shall as specified in clause 3.22.6 of MES Schedule Part-I.

(b) The properties of approved earth shall be as under

(i) Unit weight of soil:- 1.8 gm/cm3.

(ii) Cohesion (Cu)- 0.25-0.50 Kgf/cm2

(iii) Angle of internal friction(Ø-15° to 28°

3.13 HARD CORE

3.13.1 Refer Clause No. 3.27 and subsequent Clauses thereof of MES Schedule Part-I(2009) as applicable.

3.13.2 The material for Hardcore wherever mentioned in Particular Specifications here-in-after or as indicated in Drawings shall be of Crushed/Broken Stone Aggregate (at the discretion of Contractor without any Price Adjustment) of gauge not exc. 63 mm conforming to IS-383. In case Coarse Aggregate does not conform to CA provision, the same shall be replaced by the Contractor, with out any Extra Cost to the Govt., so as to conform the Aggregate to the relevant IS and CA provision.

3.13.3 The Hardcore shall be spread, watered and rammed. The thickness of Hardcore shown in Drawings is the consolidated thickness. Where thickness is not indicated in Drawings, it shall be 150 mm thick consolidated.

3.14 PRECONSTRUCTION ANTITERMITE CHEMICAL TREATMENT

3.14.1 Pre-constructional anti-termite chemical treatment shall be done as per Para 3.26 of MES Schedule Part-I(2009) and IS: 6513 Part-1 and Part-2:2013/2016(latest).

3.14.2 Anti-termite treatment shall be got done through approved specialist agency which is a registered by the Pest Control Association of India (PCAI) / Indian Pest Control Association (IPCA). Persons employed to do the anti-termite treatment in this work shall be qualified as per rule 10 of the rules framed under the inspection rule 1971. The specialist agency may be any one from list approved by E-in-C Branch/Headquarter Chief Engineer Eastern Command/ Headquarter Chief Engineer Shillong Zone(SE Falls), which are located and active in the area with prior approval of GE.

3.14.3 Chemical for ATT shall be **Chloropyriphos** EC 20% (as per IS: 8944-2005) or **imidacloprid** 30.50 SC (as per IS: 16131) all as specified in **Table 1 on page 2 of IS 6313 (Part 2)** : 2016. The chemical shall be brought in original sealed containers & should be purchased from valid BIS license holder(CML) or their authorized agents and any other firm manufacturing ISI marked chemical after taking written approval from GE .The Contractor shall submit test certificates of the chemical from manufacturers. The contractor shall be required to get the chemical tested from any laboratory approved by GE for its properties and the cost of chemical for testing, testing charges and any other charges such as transportation charges etc. shall be borne by the contractor irrespective of its test results.

3.14.4 Concentration of the chemical as emulsifiable concentration is indicated on the sealed containers. For obtaining specified concentration, chemical shall be diluted with water in the required quantity before it is used. Graduated containers shall be used for the dilution of the chemical.

4. CEMENT CONCRETE WORK

4.1. MATERIALS

4.2. CEMENT :- Refer paras 4.3 & 4.3.1 of MES Schedule Part I.

4.2.1.1. TYPE OF CEMENT : Cement shall be Ordinary Portland cement Grade 43 conforming to IS: 8112 of 2013 or Portland Pozzollana cement conforming to IS 1489.

4.2.1.2. The following checks and procedures shall be followed before the cement supplied by the contractor is accepted and is approved for incorporation in the works.

4.2.2. PROCUREMENT.

4.2.2.1. Cement supplied by the contractor shall be procured from main producers of cement

4.2.2.2. The following are the approved main cement manufacturers company of OPC and PPC.

PARTICULAR SPECIFICATIONS (CONTD…..)

Ser No	Company Name	Address	Approved grade of cement
1.	The Associated Cement Companies Ltd **Brand : "ACC"**	414-421 , Splendor Forum (4th Floor}, 3, District Centre, Jasola, New Delhi – 110044, Ph- 011 46583600	All
2.	The India Cement	Dhun Building,827,AnnaSalai,Chennai, Pin- 600002	All
3.	Ultra Tech Cement Ltd **Brand : "ULTRATECH"**	'B' Wing, 2nd Floor, Mahakali Caves Road, Andheri (East), Mumbai- 400093 Ph – 022 – 66917800	All
4.	Dalmia Cement (Bharat) Ltd **Brand : "DALMIA INFRA PRO"**	Dalmiapuram, Distt- Tiruchirappalli, Tamil Nadu - 621651	All
5.	Century Cements **Brand : "CENTURY"**	Industry House, 159 Church gate Reclamation, Mumbai -400020 , Ph-22-22023936	All
6.	SaurashtraCement **Brand :"SAURASHTRA"**	Gala No A-1, Ground Floor. Udhyog, Sadan No 3 MIDC, Central Road, Andheri (East), Mumbai- 400093 Ph – 022 – 32955557 / 67, Mo- 9320290081	All
7.	The Ramco Cements Ltd (Formerly Madras Cement) **Brand : "RAMCO"**	Auras Corporate Centre, 98-A, Dr., RadhakrishnanSalai, Mylapore, Chennai – 600004, Ph-044 28478666	All
8.	Mangalam Cement Ltd **Brand :"MANGALAM"**	PO Adityanagar, Morak, Dist- Kota,Rajasthan-326520 Ph-9351468076	All
9.	Birla Corporation Ltd **Brand :"BIRLA"**	Birla Building (3rd &4th Floor) ·9/1 , R.N Mukherjee Road, Kolkata -700001, Ph-033-30573700	All
10.	Orient Cement **Brand : "BIRLA-A1"**	5-9-22/57/0, 2nd and 3rd Floor, GP Birla Centre, Adarsh Nagar, Hyderabad-500063, Pin-044 23688600	All
11.	Nuvoco Vistas Corporation Ltd. (Formerly Lafarge Cement) **Brand : "NUVOCO"**	Equinox Business Park Tower-3, East Wing, 4th Floor LBS Marg, Kurla (West),Kurla, Mumbai, Maharashtra-400070	All
12.	Shree Cement **Brand : "SHREE"**	Bangur Nagar, Beawar, Dist:-Ajmer, Rajasthan-305901 Ph -01462 228101-06	All
13.	J K Cement **Brand : "J K"**	Kamla Tower Kanpur - 208001	All
14.	J K Lakshmi Cement Ltd **Brand : "J K LAKSHMI"**	Jaykaypuram, Dist-Sirohi, Rajasthan Ph-02971 244409/10	All
15.	Jaypee Rewa Cement **Brand : "JAYPEE"**	Jaypee Nagar, PO – Jaypee Nagar Rewa – 486450 (MP)	All
16.	Ambuja Cement Ltd **Brand : "AMBUJA"**	Kodinar, PO-Ambujanagar, Taluka-Kodinar, Dist- Junagadh, Gujarat-362715 Ph-02795 237000	All
17.	M/S Kesoram Industries Ltd.	Cement Division, Office No 613 to 616 White house, block-III, VI Floor,6-3-1192/1/1,	(a) OPC 43 (b) PPC

PARTICULAR SPECIFICATIONS (CONTD…..)

	Brand : "Birla Shakti	Kundanbagh, Begumpet, Hyderabad-500016, Tele-040-4334-4555	(c) OPC 53
18.	M/s Sagar cement Ltd **Brand "SAGAR"**	M/s Sagar Cements Ltd, Plot No 111, Road No 10, Jubilee Hills Hyderabad-500033 Tele-040-23351571	(a) OPC 43 (b) PPC (c) OPC 53 (d) PSC
19.	M/S Star Cement Meghalaya	Lumshnong, PO-Khliehriat, Dist-Jantia Hills, Meghalaya, Ph-03655 278215/16/18	(a) OPC (b) PPC
20.	M/S Meghalaya Cements Ltd **Brand : "TOP CEM"**	Lohia House, M. G Road Factory Bazar, Guwahati-781001. Tel 0361 -260367/671	(a) OPC (b) PPC
21.	M/s Hill Cement Company Ltd. **Brand: "TAJ CEMENT"**	Shivam Complex, Bharalmukh, A T Road, Guwahati, Assam-781009, Mob- 7086085118 Ph No. (0361) 2735527, 2735552, Email: kchakraborty@tajcement.com	(a) OPC 43 (b) PPC (c) OPC 53
22.	M/s Green Valliey industries Ltd. **Brand: "MAX cement"**	M/s Green Valliey industries Ltd. 4th Floor, LB Plaza GS Road, Bhangagarth Guwahati-781005 Tele +91-361-2465482/84 Fax +91-361-2465481 Email: info@greenvalliey.com Website- www.maxcement.co.in	(a) OPC 43 (b) PPC (c) OPC 53

Note : Proper validity of approval by E-in-C's branch has to be ensured before approval of make by GE..

4.2.2.3. The following conditions shall have to be satisfied before PPC shall be allowed to be used in the work:-

(a) **PPC should meet the strength criteria of 43 Grade OPC as laid down in IS-8112-2013.**

(b) The minimum period before striking off form work while using PPC shall be as given in relevant PS clause here-in-after.

(c) In cold climate regions where temperature is lower than 15° C only OPC shall be used.

(d) Mixing of OPC & PPC shall not be allowed in a work. However with the strict control of ground executives different buildings can have different types of cement.

(e) While procuring PPC, the following requirements are to be ensured and certificate to the effect is to be obtained by the contractor for each batch from the manufacturer and the same shall be submitted to the GE for approval to procure PPC :-

(i) The quality of fly ash is strictly as per IS-1489(Part-I)- 2002.

(ii) Fly ash is inter-ground with clinker not mixed with clinker.

(iii) Dry fly ash is transported in closed containers and stored in silos. Only pneumatic pumping have been used.

(iv) The fly ash received from thermal power plants using high temperature combustion above 1000 ^{o} C have been used.

4.2.2.4. The particulars of the manufacturer /main producer of cement along with the date of manufacture shall be submitted by the contractor for every lot of cement separately. The documents in support of purchases of cement shall be verified by the site staff and GE.

4.2.3. **STORAGE**

4.2.3.1. There will be two separate godowns provided by contractor at site of works exclusively for storage of cement **:-**

4.2.3.1.1. "First Godown" will have the cement lot(s) for which independent testing is to be got done and/or under process.

4.2.3.1.2. "Second Godown" will have the cement lot(s) whose independent testing has been got done and which after successful testing stands approved by GE for incorporation in the work.

4.2.3.2. Cement shall be stored over dry platform with gap of 20 cm from walls around and at least 20 Cm high from floor level. Compliance of provisions in clause 4.3.1 of MES Schedule Part I shall be

ensured. Godowns shall have easy access for proper inspection and cement shall be protected from dampness/moisture to minimise storage deterioration or intrusion of foreign matter**.**

4.2.3.3. Both the godowns shall be provided with two locks on each door. The key of one lock at each door shall remain with the Engineer-in-Charge or his representative and that of the other lock with the contractors' authorised agent at the site of works so that cement is removed form the godown only according to the requirements for independent testing purposes or for daily consumption purposes with the knowledge of both the parties.

4.2.4. TESTING OF CEMENT

4.2.4.1. Contractor shall submit particulars of manufacturer of cement along with the date of manufacture for every lot of cement separately. The manufacturer is to carry out inspections and testing of cement in accordance with the relevant BIS provisions. Contractor shall also submit manufacturer's test certificates in original alongwith the test sheet or an authenticated copy thereof giving results of each physical test as applicable and the chemical composition of cement. The documents giving above particulars shall be verified by site staff and GE in support of the purchases of cement and these details shall be recorded in the Cement Acceptance Register as per proforma given in **Appendix `D'** to the tender documents here-in-after.

4.2.4.2. The GE shall organise independent testing of random samples of cement drawn from the various lots which shall be got tested by the contractor under his own arrangements and the cost of both materials and labour required for sampling, testing, packing, transportation of samples to the testing places and testing charges thereof, and for subsequent identification of cement sampled, shall be borne by the contractor. The independent testing shall be got done from any of the following where facilities exist for all the tests required, viz, National Test House, SEMT Wing CME, Regional Research labs, IITs, National Institutes of Technology, Zonal Laboratory/ Command Testing Laboratory testing, NABL laboratories, Govt Engineering Collage or Any Govt laboratories.

4.2.4.3. The independent testing shall be got done as per :-

(a) IS--3535 : Method of sampling hydraulic cement.

(b) IS-4031 : Methods of Physical test for hydraulic cement.

(c) IS-4032 : Method of chemical analysis of hydraulic cement.

The other relevant IS shall also be followed. However the tests shall be restricted to those indicated in Appendix `E' to the tender documents here-in-after.

4.2.4.4. The above tests shall be independent tests. The samples shall be taken immediately within one week of delivery at the site of works and all the tests got carried out by contractor within one week of sampling. In case it is not possible to test the samples within one week, the samples shall be packed and stored in air tight containers as decided by GE till such time that they are tested.

4.2.4.5. The entire lot(s) of Cement shall stand rejected; if it does not comply with any of the requirement of IS specifications and the test results on samples are not within the acceptable limits. The rejected lot(s) shall be removed from the site of works by the contractor under their own arrangement and cost. The cost of tests shall be borne by the contractor irrespective of the results.

4.2.4.6. The contractor shall use only one type brand of cement for entire structural/reinforced concrete work. In the event of contractor desiring use of any different brand of cement, fresh design mix shall be carried out and submitted for approval without any extra cost to the Govt.

4.2.5. DOCUMENTATION

4.2.5.1. The contractor shall submit original paid vouchers from the main manufacturers for the total quantity of cement supplied under each consignment brought to the site of work.

4.2.5.2. All consignments so received at the work site shall be inspected by GE alongwith the relevant documents for permitting storage in the "first godown" defined here-in-before.

4.2.5.3. The original vouchers and the test certificates shall be kept in record in the office of GE duly authenticated and with cross reference to the Control Number recorded in the Cement Register as per format given in **Appendix `D'** in the tender, maintained for "first godown" duly signed by JE (Civ), Engineer-in-charge, GE and the contractor.

4.2.5.4. After the independent testing has been got done by GE and satisfactory results obtained and kept on record, the consignment so got tested shall be removed from the "first godown" and transferred to "second godown" as defined here-in-before. Original vouchers and the test certificates submitted by the contractor alongwith the consignment and test certificates of independent testing shall be defaced by Engineer-in-charge and kept on record of GE duly authenticated and with cross reference to the control Number rendered in the cement register maintained for the "Second

Godown". This second register shall be termed as "Cement Acceptance Register" which shall be signed by JE (Civ), Engineer-in-Charge, GE and the contractor. It shall be as per format given at **Appendix `D'** to this tender as an extra provision similar to "Details of test certificates" and termed as "Details of Independent test certificate". Besides this the quantity of cement shall be also suitably recorded in the Measurement Book for record purposes before incorporation in the work and shall be signed by Engineer-in-charge and the contractor.

4.2.5.5. The Accepting Officer may order a Board of officers in which contractor shall also be a member for random check of cement and verification of connected documents.

4.2.5.6. Daily inspection shall be carried out by Engineer-in- charge or his representative and contractor's authorised agent and separate register shall be maintained in both the godowns duly indicating on daily basis the inspection carried out and indicating holding in godown at beginning of the day, inward supplies in the day, removal in the day and balance at the close of the day duly signed by above persons. In the removal column particulars of lot removed and purpose for which removed shall be indicated.

4.2.5.7. The purchase voucher and test certificate of manufacturer shall be linked.

4.2.6. WEIGHING AND PAYMENT FOR CEMENT.

4.2.6.1. Random sample of 5 bags per 100 bags shall be got weighed by Engineer-in-charge in presence of representative of contractor from each consignment to arrive at average weight per bag for calculating total quantity of cement brought at site for incorporation in the work. All arrangement for weighing of cement bags including labour etc shall be provided by the contractor at their own cost.

4.2.6.2. The quantity of cement arrived at as aforesaid shall be considered for payment as applicable in accordance with IAFW-2249, General Conditions of Contracts.

4.2.6.3. Only when the test results of independent tests carried out on the cement are found satisfactory and it is permitted to be stored in the "Second Godown" and quantity entered in MB, then only this lot shall be considered for payment in RARs.

4.2.6.4. Payments shall be restricted as per provisions in clause "Advance on Account" of General Conditions of Contracts (IAFW-2249).

4.2.6.5. **CURING AND MINIMUM PERIOD FOR STRIKING OFF FORM WORK WHEN** PPC is used for RCC works in various locations , the minimum period for curing and striking off form work shall be as under in lieu of that specified in clause 4.11.6.3 of MES Schedule Part-I.

(a) Curing : -

(i) Structural RCC work, : 21 days

(ii) PCC work, plastering etc : 14 days

(b) Striking off form work :-

(i) Walls ; columns and vertical sides of beams : 3 to 4 days.

(ii) Slabs (Props left under) : 8 days.

(iii) Beams soffits (Props left under) : 9 days.

(iv) Removal of props to slab/beam :-

(a) Spanning upto 6 M Span : 16 days

(b) Spanning beyond 6 M Span : 22 days

(v) For cantilever portion of slab form work with support shall be retained until the completion the casting of the entire frame work of the building.

4.2.6.6. In case of bad weather, periods mentioned above may be revised at the discretion of the Engineer-in-Charge. The contractor shall be deemed to have considered the above provision before tendering and quote lump sum accordingly. No claim will be entertained if longer periods required for striking off form work and curing and all such effected matters are looked into upon the use of pozzolana cement as a result being issued to the contractor in full or part of for bad weather.

4.2.6.7. Curing shall be carried out using pump of suitable capacity at no extra cost to the Govt all as directed by Engineer-in-Charge.

4.3. COARSE AND FINE AGGREGATE FOR PLAIN & REINFORCED CEMENT CONCRETE

4.3.1. Coarse aggregate for all concrete shall be graded crushed hard granite, trap or basalt stone as approved by the GE and shall conform to the requirements laid down in clause 4.4 of MES

Schedule Part I. Hand broken stone aggregate shall not be permitted for use in the work. Size and grading of aggregate shall be as per clause 4.4.7.1 of MES Schedule Part I.

4.3.2. GRADING OF AGGREGATES

4.3.2.1. GRADING OF COARSE AGGREGATE - Grading of Coarse Aggregate for Plain and Reinforced Concrete i.e. Stone Aggregate shall be as per Clause No. 4.4.7 (1) of MES Schedule Part I.

4.3.2.1.1. Grading of Coarse Aggregate unless otherwise specified shall be as follows :-

(a)	For Reinforced/Structural Element of depth/thickness 50mm or under	:	12.5mm graded
(b)	For Reinforced/Structural Element of depth/thickness over 50mm	:	20mm graded
(c)	For Plain Cement Concrete thickness exc. 25mm but not exc. 75mm	:	20mm graded
(d)	For Plain Cement Concrete thickness 75mm and over	:	40mm graded
(e)	Lime Concrete	:	40mm graded

4.4. Fine Aggregate: Fine aggregate for all concrete work shall be naturally occurring river sand conforming to the specification in Para 4.4.7.2 of MES schedule Part I. Use of sand conforming to grading Zone IV of IS 383 (1970 second revision or as amended) shall not be allowed for RCC works.

4.5. WATER PROOFING COMPOUND

4.5.1. Refer Clause No. 4.8 of MES Schedule Part I.

4.5.2. Integral water proofing compound shall conform to IS: 2645. The brand thereto shall be got approved in writing from the GE before they are brought to site. Proportion of mix and method of mixing shall be as per manufacturer's printed instructions.

4.6. WATER

4.6.1. Water shall conform to the requirement stipulated in IS-456 and as per Clause No. 4.9 and 4.9.2 of MES Schedule Part-I.

4.7. PLASTICISER

4.7.1. Plasticiser shall be used in Design Mix Concrete as per the Manufacturer's instruction without any Extra Cost to achieve the desired Results Strength and Workability. Plasticiser should confirm to IS-9103.

4.8. WORKMANSHIP

4.8.1. GENERAL

4.8.1.1. Refer Clause No. 4.11 of MES Schedule Part-I as applicable and as directed by the Engineer-in-Charge.

4.9. MIX OF CONCRETE

4.9.1. VOLUMETRIC MIX CONCRETE

4.9.1.1. Wherever the Concrete is specified / ordered in Volumetric Proportion the same shall be provided by Volumetric Mix. Volumetric Mix shall be done all as specified in Clause No. 4.11.2.4 of MES Schedule Part-I.

4.9.2. DESIGN MIX CONCRETE

4.9.2.1. Refer Clause 4.11.2.1 of MES Schedule Part-I. M-25 and M-30 Mix Design shall be carried out as per provisions of IS-456 :2000/IS 3370/IS 2911where as applicable and guidelines given in IS-10262 and SP 23 with the following conditions unless otherwise specified :-

(i)	Exposure Condition for Mix Design M-25	:	Moderate
(ii)	Degree of Quality Control for Mix Design M-25	:	Good.
(iii)	Exposure Condition for Mix Design (M-30)	:	Severe
(iv)	Degree of Quality Control for Mix Design (M-30)	:	Good.

4.9.2.2. For Design Mix Concrete of M-25 and M-30 Grade, the Minimum Cement Content and water cement ratio shall be as per IS 456/IS 3370/IS 2911. The Contractor is required to Design each Grades of Concrete as per IS 456/IS 3370/IS 2911and IS-10262 to work out the Cement Content of the above Grade of Design Mix.

4.9.2.3. Design Mix shall be got designed from National Test House, SEMT Wing CME, Regional Research labs, IITs, National Institutes of Technology, **Command Test Laboratory,** Govt Engineering College.

4.9.2.4. In case Design Mix got other than Command Test Laboratory, the Design Mix Concrete shall be approved by the Accepting Officer prior to incorporation in the work.

4.9.2.5. The Design Mix furnished by the laboratory shall include the following details :-

(i) Grading of individual coarse and fine aggregate used in Design Mix.
(ii) Proportion by weight of each component of aggregate of Design Mix.
(iii) Water cement ratio for recommended proportions of aggregates.
(iv) Slump and compaction factor values for the recommended Design Mix.
(v) Use of plasticiser/ super plasticiser is permitted to the extent allowed in clause no 10.3.3 of IS 456-200
(vi) Initial standard deviation to be adopted for acceptance criterion as per IS-456 of 2000.
(vii) 28 days Compressive strength of cement supplied for Design Mix.
(viii) Actual cement content in the Design Mix.
(ix) The maximum free water cement ratio shall be as per table 5 of IS 456-2000.Sampling and acceptance of concrete shall be as per clause 15 & 16 of IS-456 of 2000. The maximum size of aggregate for reinforced cement concrete shall be 20mm.
(x) Recommendation of mixed proportions at site.

4.9.2.6. Cost of designing the mix shall be borne by the Contractor. All material required for carrying out the design shall be provided by the contractor at the laboratory. In case of design mix is carried out from Command Test Laboratory and cost testing charge shall be recovered from RARs bill.

4.9.2.7. In case contractor fails to submit the samples of design mix soon after commencement of work, the delay shall solely be attributable to the contractor and no claim of whatsoever nature shall be admissible on this account.

4.9.2.8. The Tenderer shall quote their Rate for the applicable Items considering the Actual Quantity of Cement required to be incorporated in the Work which is to be assessed by them.

4.9.2.9. No Plus DO (Deviation Order) shall be admissible even if the required/approved quantity of Cement is more than specified in IS 456/IS 3370/IS 2911. Contractor shall be responsible / guarantor to that provides M-25 and M-30 Design Mix Concrete. This shall not be considered as misrepresentation of fact and no claim what so ever on this provision shall be admissible.

4.9.2.10. The Contractor is advised to assess the realistic quantity of Cement for the Design Mix specified above before quoting. If considered necessary, they may get the Design Mix done on their own before quoting the Tender. However, the Design Mix done prior to quoting of Tender shall not be considered for incorporation in the Work after conclusion of Contract and shall have to be done as specified above.

4.9.2.11. A re-verification of Mix proportions for Design mix concrete shall be essential for any different brand of cement. With any change in aggregate source, mix design or grade of cement, fresh design mix shall be done by the contractor. Nothing extra shall be paid on this account. The decision of GE regarding the necessity for re verification or redesign of mix will be final and binding.

4.9.3. READY MIX CONCRETE (RMC).

4.9.3.1. All Design Mix and Reinforced Cement Concrete work covered under the Sch'A'/BOQ for pile foundations, walls, columns, beams, slabs and staircase etc. shall be with Ready Mix Concrete (RMC) from approved manufacturers having fully automatic batching plants installed of capacity not less than 20 cum/hr and from fully automatic batching plant installed at site of capacity not less than 15 Cum per hour with tested and calibrated water meter, control panel and provision for computer generated batch report all as per IS-4926 (amended up to date). However, for smaller quantity i.e. for concreting to lintels, chajjas, shelves, etc mechanical mixer may be used with prior approval of the GE and mixing of concrete shall be as per clause 10.3 of IS:456-2000.

4.9.3.2. The relevant details of Design Mix Concrete as specified earlier shall be applicable along with the following specific details for Ready Mix Concrete. The contractor shall use ready mix concrete (RMC) of grades as specified. RMC shall conform to the requirements as per IS-4926 (amended up to date) and IS 456-2000.

4.9.3.3. In proportioning design mix concrete, the quantity of both cement and aggregate shall be determined by mass and all in accordance with clause 4.3 and sub clauses thereon of IS-456-2000 as applicable. The mix design shall be revised with change in properties of cement and aggregate.

All the materials shall be tested at a frequency as laid out in relevant IS. Use of weigh batcher is compulsory for all grades of concrete. The Contractor shall ensure that concrete is manufactured under computer-controlled operations and pumped and placed at site using sophisticated equipments and methods. For that the various operations of production of concrete like weigh batching of all ingredients, adjustment for moisture, water cement ratio control, admixture dose if required, auto timer for mixing, delivery slips etc. shall be fully computerized. This record of quality assurance shall be submitted to the GE and shall be kept on record. Fully automatic batching plant at site shall conform to relevant IS and shall be of standard make as approved by Accepting Officer. The concrete produced from the batching plant shall be pumped using pumps of suitable capacity. Admixtures / plasticizers of make as given below can be added (as per mix design) to make concrete plastic for facilitating pumping.

4.9.3.4. Specification for all materials / ingredients, etc as specified here-in-before shall hold good for RMC.

4.9.3.5. **ADMIXTURE / PLASTICIZERS** : Admixtures / plasticizers @ not more than 1% of weight of cement at no extra cost to the Department may be used conforming to IS 9103-1999 (Revised).

4.9.3.6. Initial mix design from RMC manufacture shall be submitted by the contractor in quadruplicate with complete data adopted for mix design along with test results of all materials and concrete. The initial mix design should take into account the aspects such as loss of workability and strength during transportation, till placement of concrete. The design mix concrete shall be approved by Accepting Officer prior to incorporation in the work.

4.9.3.7. RMC manufactured and supplied shall be in conformity with the initial mix design approved by the Accepting Officer. Any change shall be made only with approval of Accepting Officer after necessary revision in design mix.

4.9.3.8. The contractor shall ensure that all facilities are made available for the departmental officials to inspect the materials incorporated, test carried out for all materials, concrete etc. Copies of all tests carried out for materials used / concrete shall also be made available to the department.

4.9.3.9. The concrete shall be transported in concrete transit agitators conforming to IS-5892.

4.9.3.10. The concrete shall be delivered and placed by pumping using BOOMER. The decision as to whether the RMC can be placed by pumping or not shall be decided by GE whose decision shall be final and binding. The tenderer will have no claim any extra on this account.

4.9.3.11. The pump and pumping arrangement shall be inspected and approved by GE.

4.9.3.12. No water / admixture shall be allowed after initial mixing of concrete at the plant. The use of PPC / Fly ash / mineral based admixture in RMC shall not be permitted.

4.9.3.13. Slump test shall be carried out at site by the department in the presence of contractor or their representative. The concrete shall be placed in position within the designed initial setting time and at the end of initial setting time, the remaining concrete shall be rejected.

4.9.3.14. Sampling and testing of concrete shall be carried out at the site before delivery as per IS-456-2000 by the department along with the representative of the contractor. However, the cost of testing shall be borne by the contractor. The concrete, which does not meet the requirement of acceptance criteria as per IS-456-2000 shall be rejected and the contractor shall make good the same at his risk and cost.

4.9.3.15. The records indicating following details duly signed by the competent representative of contractor and GE shall be maintained.

(a) Date of delivery

(b) Time of mixing

(c) Time of delivery at site.

(d) Quantity and Grade of RMC.

4.9.3.16. **SAMPLING AND ACCEPTANCE CRITERIA FOR CONCRETE:**

The criteria for sampling and acceptance of the concrete shall be as per Para 15.1 to 16.1 of IS-456:2000, (Code of practice for plain and reinforced concrete). The minimum frequency for the work tests shall be as specified therein and as specified here in after.

4.9.3.17. **PACKING AND TRANSPORTATION OF SAMPLES**

The Contractor must, at his own expenses, properly pack the samples. The Contractor shall bear the cost of transportation of the samples required to be tested from site of work to the laboratory as approved by Accepting Officer.

4.9.3.18. **TESTS:**

The following tests shall be carried out during the execution of work. Tests shall be carried out in accordance with IS-516 and IS-1199. The contractor shall provide all facilities and equipment for casting and curing of test cubes.

4.9.3.19. **SLUMP TEST**. The frequency of the test shall be decided by the GE. The slump for the vibrated concrete shall be as specified in IS-456. The GE, however, reserves the right to vary the limit which will be ascertained at the time of deciding the mix design for each grade. Any batch from which a slump test is being made shall not be transferred to the places of laying until the slump in excess of the required for the particular batch is not removed from the site.

4.9.3.20. **COMPRESSIVE STRENGTH TEST**. Three cubes shall be tested at 7 days and three at 28 days for crushing and the remaining three cubes shall be kept as reserved exclusively for Govt use for subsequent testing (if so desired by Govt) and preserved for one year from certified date of completion of work under the contract. The contractor cannot object testing of other cubes for the purpose as a matter of right.

4.9.3.21. For the purpose of subsequent identification of the work test cubes, the concrete to which these pertains shall be cross referred and record of this maintained and signed by the GE and contractor or his authorized representative.

4.9.3.22. **TRANSPORTING, PLACING, COMPACTION AND CURING:**

(a) It should be as per clause 13 of IS 456:2000. Concrete shall be transported without delay and incorporated in works at the position of laying within 20 minutes from the time of discharge from the mixer.

(b) Mixed concrete shall be deposited in final position and solidly packed around reinforcement, carefully poured and consolidated by means of portable vibrators or mechanically operated and of the kind as suitable for a particular situation as directed by the GE. Care shall be exercised that no voids or honey comb pockets are formed. The concrete shall not be laid in position for more than 1 metre in height in one concrete operation. The compaction of concrete in RCC roof and floor slabs shall be done with plate vibrators.

(c) Exposed surfaces of concrete shall be kept continuously in a damp or wet Condition by pounding or by covering with a layer of sack, canvas, Lenin and kept constantly wet for at least 14 days. The concrete shall be protected from premature drying for at least 8 days after pouring and shall be cured as directed by the GE for a period of not less than 14 days.

4.9.4. **SAMPLING AND TESTING OF CONCRETE OF GRADE M-30 AND M-25 (DESIGN MIX)**

4.9.4.1. Refer Clause No. 4.11.17 of MES Schedule Part-I and as per IS-456. Testing of Concrete shall be carried out by the Contractor as directed by the GE without any Extra Cost to the Department and usages of following Test Equipment at Site shall deemed to be included in Unit Rate of Schedule 'A' (BOQ) **:-**

(a) A set of IS Sieves for Fine Aggregate Sieve Analysis (Reference IS-2386).
(b) A set of IS Sieve for Coarse Aggregate Sieve Analysis (Reference IS-2386).
(c) Two graduated Jars of 500 ml capacity for Silt Test (Reference IS-2386).
(d) A set of 6 No. Cube moulds of 7.06 cm size for Testing Strength of Cement.
(e) Vicat Needle Apparatus for testing Initial and Final Setting Time.
(f) A Physical Balance.
(g) Three set of 6 Standard 15 Cm Cube Moulds for testing Compressive Strength.
(h) Slump Test Apparatus.
(j) Compressive Strength Testing Machine (Automatic operated).
(k) Electric Oven.
(l) Flakiness Index Testing Apparatus

4.10. **FREQUENCY OF TESTING**

4.10.1. Frequency of Testing of Concrete is given in APPENDIX 'E' here-in-after. Following further requirements are to be met with as regards Testing of Concrete ;-

(i) Each Sample shall consist of 3 Cubes.

(ii) At least One Sample shall be taken for each shift.

(iii) In addition to number of Samples specified in APPENDIX 'E' for Testing 28 Days Compressive Strength One Additional Sample for 7 Days Strength shall be casted and tested for each Shift.

4.11. **WORKABILITY OF CONCRETE** – Refer Clause No. 4.11.4 of MES Schedule Part-I and Clause No.7 of IS-456-2000 and as directed by the Engineer-in-Charge.

4.12. **MIX & GRADE OF CEMENT CONCRETE**

4.12.1. Refer Clause No. 4.11 of MES Schedule Part-I.

4.13. **FORM WORK & SCAFFOLDING**

4.13.1. **GENERAL**

4.13.1.1. Form work shall comply with the requirements of clause No. 4.11.6 & 7.15.1 to 7.15.12 of MES Schedule Part I. The shuttering shall be of steel only except at places where it is not feasible to use steel shuttering on technical grounds plywood shuttering can be used with written approval of GE.

4.13.2. **CLEANING & TREATMENT OF FORMS**

4.13.2.1. Refer Clause No. 4.11.6.2 of MES Schedule Part-I (as applicable) and as directed by the Engineer-in-Charge.

4.13.3. **STRIPPING TIME WHERE PORTLAND POZZOLLANA CEMENT IS USED**

4.13.3.1. Refer Clause No. 4.11.6.3 of MES Schedule Part-I. Where Pozzolana Cement is used for RCC Works in various Locations, the Minimum Period for Striking Off Formwork shall be as under in lieu of that specified in MES Schedule Part-I :-

Walls ; Columns and Vertical Sides of Beams	:	3 to 4 Days
Slabs (Props left under)	:	8 Days
Beams Soffits (Props left under)	:	9 Days
Removal of Props to Slab/Beam **:-**		
(aa) Spanning upto 6.0 METRE Span	:	16 Days
(ab) Spanning beyond 6.0 METRE Span	:	22 Days

For Cantilever Portion of Slab Formwork with Support shall be retained until the completion the Casting of the Entire Frame Work of the Building.

4.13.4. **REMOVAL OF FORMWORK**

4.13.5. Refer Clause No. 4.11.6.4 and 4.11.6.5 of MES Schedule Part-I and as directed by the Engineer-in-Charge.

4.13.6. In case of Bad Weather, Periods mentioned here-in-above may be revised at the discretion of the Engineer-in-Charge. The Contractor shall be deemed to have considered the above provision before tendering and Quote Rate accordingly. No Claim will be entertained if Longer Periods required for Striking Off Formwork and Curing and all such effected matters is looked into upon the use of Pozzolana Cement as a Result being issued to the Contractor in Full or Part of for Bad Weather.

4.13.7. **FINISHES**

4.13.8. Refer Clause No. 4.11.16 of MES Schedule Part-I (as applicable) and as directed by the Engineer-in-Charge.

4.14. **ASSEMBLY OF REINFORCEMENT** – Refer Clause No. 4.11.7 of MES Schedule Part-I (as applicable) and as directed by the Engineer-in-Charge.

4.15. **COVER TO REINFORCEMENT** – Refer Clause No. 4.11.8 of MES Schedule Part-I (as applicable) and as directed by the Engineer-in-Charge.

4.16. **TRANSPORTING –** Refer Clause No. 4.11.9 of MES Schedule Part-I (as applicable) and as directed by the Engineer-in-Charge.

4.17. **PLACING –** Refer Clause No. 4.11.10 and 4.11.10.3 of MES Schedule Part-I (as applicable) and as directed by the Engineer-in-Charge.

4.18. **COMPACTION –** Refer Clause No. 4.11.11 of MES Schedule Part-I. Compaction of Concrete in RCC Slab, Beam Walls and Columns shall be done with Approved Mechanical Vibrator. In other Locations Hand Compaction may be permitted by Tamping or Roding as approved by the GE.

4.19. **CURING :-**Refer Clause No. 4.11.13 of MES Schedule Part-I (as applicable) and as directed by the Engineer-in-Charge.

4.20. **PROTECTION –** Refer Clause No. 4.11.14 of MES Schedule Part-I and as directed by Engineer-in-Charge.

4.21. **INSPECTION –** Refer Clause No. 4.11.15 of MES Schedule Part-I and as directed by Engineer-in-Charge.

5. BRICK WORK

5.1. MATERIALS

5.1.1. **CEMENT** – Cement shall be as specified for Cement Concrete Work here-in-before.

5.1.2. **SAND** – Sand for Mortar shall be as specified in Clause No. 5.4, 5.4.1 of MES Schedule Part I.

5.1.3. BRICKS

5.1.4. **BRICKS:-**Brick for brick work shall the burnt bricks of normal size of bricks shall be specified in Appedix ‘C’ and should have minimum crushing strength of 75 Kg/Cm2 when tested in accordance with clause 26 of IS-1077. The dimension of bricks when tested in accordance with clause no 6.2.1 of IS-1077 shall be within the following tolerance limits per 20 bricks:

		For 250x 125x 75 mm	For 230x 115x 75 mm
(a)	Length	4920 mm to 5080mm (5000±80 mm)	4520 mm to 4680mm (4600±80 mm)
(b)	Width	2460 mm to 2540mm (2500±40 mm)	2260 mm to 2340mm (2300±40 mm)
(c)	Height	1460 mm to 1540mm (1500±40 mm)	1460 mm to 1540mm (1500±40 mm)

5.1.5. Compressive strength: The bricks when tested in accordance with the procedure laid down in IS-3495 shall have a minimum average compressive strength as 75 Kg/cm2.

5.2. All brick work shall be built in English bond except in half brick walls which shall be in Stretcher bond.

5.3. **WATER** – Refer Clause No. 5.18 of MES Schedule Part I. 5.12

5.4. DAMP PROOF COURSE

5.4.1. Damp Proof Course for the Buildings shall be 40 mm thick M-15 using 12.5 mm graded stone aggregates mixed with water proofing compound (IS. 2645) as per manufacturer's instructions and apply evenly two coats of hot blown bitumen grade VG-10 @ 1.20 kg per sqm blended with clear sand @ 0.05 cum/sqm. each coat and fixing polythene film 150 micron over coat of bitumen over full width of wall including thickness of internal and external plaster.

5.4.2. The width of the DPC shall be equal to the width of Masonry Wall Plus Thickness of Internal and External Plaster. The Plaster of Masonry Wall below DPC will be done first and the DPC will be laid thereafter. The Plaster shall not cover DPC layer from sides.

5.4.3. Where Plinth Beam/Band are shown on Drawings at Plinth Level, Damp Proof Course as specified here-in-before shall not be provided.

5.4.4. In the event of deviation, pricing shall be done taking Water Proofing Compound at the rate of 3% by Weight of Cement.

6. WOOD WORK (CARPENTER’S & JOINERY WORK)

6.1. MATERIALS

6.1.1. **TIMBER** – Refer Clause No. 7.3 of MES Schedule Part-I.

6.1.1.1. **DEFECTS IN TIMBER** – Refer Clause No. 7.4 of MES Schedule Part-I for ‘STRUCTURAL & CARPENTER’S WORK’ and Clause No. 7.5 of MES Schedule Part-I for JOINER’S WORK.

PARTICULAR SPECIFICATIONS (CONTD…..)

6.1.1.2. **SEASONING –** Refer Clause No. 7.6 of MES Schedule Part-I.

6.1.1.3. **MOISTURE CONTENT** – Refer Clause No. 7.7 of MES Schedule Part-I. Moisture Contents shall be as specified for Zone.

6.1.1.4. **SPECIES** – Refer Clause No. 7.8 of MES Schedule Part-I. Timber for 'STRUCTURAL & CARPENTER'S WORK & JOINER'S WORK' (except Particle Board and Block Board) shall be of Second Class Hardwood of the following Species as approved by GE, if otherwise not specifically indicated here-in-after/Drawings :-(i) BONSUM (ii) HOLLOCK (iii) TITA CHAMPA (iv) KOKKO (v) KALA SIRIS (vi) CHAPLASH

6.2. **WOODEN CHOWKATS OR FRAME (WHEREVER SPECIFIED / SHOWN IN DRAWINGS)**

6.2.1.1. Refer Clause No. 8.18 of MES Schedule Part–I for Timber Frames and Chowkats and shall be of Second Class Hard Wood of the species mentioned here-in-before and as approved by GE.

NOTE: The Surfaces of Wooden Chowkats or Frame. coming in contract with Concrete or Brick Works, etc. shall be treated with Two Coats of Bituminous Paint after thoroughly cleaned from Dirt, Oil, etc. The cost of the same shall deem to be included in the quoted rate by the Tenderer.

6.3. **FLUSH DOOR SHUTTER**

6.3.1. Flush door shutter shall be provided complete all as specified in Schedule 'A' (BOQ) and shown/ mentioned on TD drawings No. CESZ/2015/TD/003 and other as specified in para 8.25 of MES Schedule Part-I and as per IS 2202 (Part **:** I -1999).

6.3.2. Flush door shutter shall be solid core types with block core, and shall conform to IS 2202 (Part I) - 1999, Specification for wooden flush door shutters (solid core type) Part I.

6.3.3. All timber used shall be well seasoned and chemically treated. Adhesive shall be phenol formaldehyde synthetic resin BWP type specified in IS:848-2006. All dimensions shall be finished dimension and Manufacture's Test Certificate for test specified in IS: 2202 (Part I) shall be rendered**.**

6.4. **FACTORY MADE SHUTTERS**

6.4.1. Refer Clause No. 8.20.8 of MES Schedule Part-I.

6.4.2. All Panelled and Wire Gauge Shutter for doors shall be factory made conforming to IS-1003 manufactured from well selected kiln seasoned, Second class Hardwood chemically treated as per IS-401 for antitermite. Samples of shutter shall be got approved by the GE before placing the bulk order.

6.4.3. Factory Made Shutters shall be from any of the Approved Manufacturers listed in relevant APPENDIX to this Tender documents here-in-after.

6.4.4. The door shutter shall be tested in the reputed Testing House/Lab as approved by GE as per IS-1003. The number of door samples shall be as required by relevant IS-1003. In case, the results are not satisfactory, the entire lot shall be rejected. The cost of the testing and the cost of the door shutter sample is deemed to be included in the rates quoted by the tenderer.

6.5. **PLYWOOD**

6.5.1. Plywood shall conform to IS-303 and shall be ISI marked. Plywood wherever shown on drawing shall be of BWR grade, type 'BB' of thickness as mentioned in drawings and number of ply shall be as specified in MES Schedule Part-I.

6.5.2. Refer Clause No. 12.10 and subsequent Clause No. thereof of MES Schedule Part I for Plywood as applicable.

6.6. **VENEERED PARTICLE BOARD**

6.6.1. Refer Clause No. 8.8 of MES Schedule Part-I.

6.7. **NAILS, WOOD SCREWS & GLUE**

6.7.1. Refer Clause No. 7.10 to 7.12 of MES Schedule Part–I.

6.8. **WORKMANSHIP**

6.8.1. **GENERALLY** – Refer Clause No. 7.13 of MES Schedule Part-I for 'STRUCTURAL & CARPENTER'S WORK' and Clause No. 8.15 of MES Schedule Part-I for JOINER'S WORK.

6.8.2. **LENGTHENING OF MEMBERS** – Refer Clause No. 8.16 of MES Schedule Part–I.

6.8.3. CHEMICAL/PRESERVATIVE TREATMENT

6.8.3.1. Before painting or French polishing all joinery and timber incorporated in the work to be manufactured at site shall be treated as mentioned in IS-401 by modified hot and cold treatment, the minimum quantity of chemical impregnated in each cubic meter of timber shall be at the rate as specified in table 2 group 5 of said IS and all joinery manufactured at site, which has been exposed by chiseling/planning/cutting after treatment by hot and cold process, except those in factory made paneled shutters, which shall be chemically treated in factory, shall be treated with two coats of any of the approved chemicals such as ASCU etc. as directed by GE for protection against termite.

6.8.3.2. Cost of Antitermite Treatment to Wood Work shall be deemed to have been included in the quoted rate. The quantity and application of chemicals shall be as per manufacturer's instructions.

6.8.3.3. Wood work shall be painted or polished as specified in Schedule 'A' (BOQ) after second coat of antitermite treatment has completely dried.

6.8.4. FIXING OF TIMBER FRAMES & CHOWKATS

6.8.4.1. Fixing of Chowkhat Frame is permitted as per 'Prepared Opening Method' without any Price Adjustment as specified in Clause 8.19.2 of MES Schedule Part I.

6.8.5. FIXING OF SHUTTERS

6.8.5.1. **GENERALLY** – Refer Clause No. 8.20 and 8.21 of MES Schedule Part-I as applicable and as directed by Engineer-in-Charge.

6.8.5.2. **STANDARD DIMENSIONS OF SHUTTERS FOR DOORS, WINDOWS, ETC.** – Refer Clause No. 8.24 of MES Schedule Part-I as applicable for Panelled, Glazed, Gauzed or Partly Panelled and Glazed Shutters.

6.9. FABRICATION

6.9.1. Refer to relevant Clause No. of MES Schedule Part-I and as directed by Engineer-in-Charge.

6.9.2. Styles and Rails including Lock Rail of Shutters shall be One Piece only and shall be Jointed to each other by Mortise and Tenon Joints at Right Angles. Rails more than 150 mm width shall have Two Tenons. The Bottom, Lock and Up Rails shall be inserted 25mm short of the width of the Styles.

6.9.3. The Edges of the Panel shall fit into groove to the full depth leaving an Air Gap of 1.5mm. Edges of the Panel insert shall be sealed by application of Glue.

6.9.4. All Members of the Shutter shall be straight without any Warp or Bow. They shall have smooth well planned surfaces at Right Angle to each other. The Right Angles of the Shutter shall be checked by measuring the True Diagonals from One Extreme Corner to the Opposite One and the surfaces of Mortise and Tenon Joints shall be Glued before putting together. All Joints shall be Pinned with Bamboo Pins Two per Tenon in staggered position.

6.9.5. PRESSED STEEL DOOR FRAME

6.9.5.1. Pressed steel frames shall be provided complete all as specified in Schedule 'A' (BOQ) and shall confirms to IS-4351 and as specified in clause 10.27 of SSR Part I. The tolerance over the profile size shall be $\pm$2 mm. The thickness of sheet shall be 1.6mm irrespective of whatever is shown on drawings and it shall be bent to shape as per drawing. The inner surface of the frames shall be treated with a coat of tar. Shop coat and red oxide shall be applied outside before erecting in position. Mild Steel Tie Bar of 16 mm dia Round Bars shall be provided at the bottom of frames by welding to the frame and this tie bar shall be embedded and left inside floor as per drawing/IS. Cold rolled steel hinges as per IS shall be bolted to pressed steel frame.

6.9.5.2. Mild Steel Conduit 20 mm dia of 16 gauge closed at one end shall be welded to inner face of frame for receiving shoot of bolt. M.S. sockets for housing shoot of tower bolt shall also be provided as directed by Engineer-in-Charge.

6.9.5.3. The fixing lugs and inner surfaces of door frames coming in contact with concrete shall be treated with two coats of bituminous paint. The door frame shall be thoroughly cleaned, made free from rust, dirt, oil, etc.

6.9.5.4. The fixing of Pressed Steel Door Frames shall be done as per Clause No. 5.34.5(A) of MES Schedule Part -I.

6.9.6. FACTORY MADE SOLID PANEL PVC DOOR SHUTTERS

6.9.6.1. Factory made solid PVC moulded door shutters with frames shall be provided complete all as specified in Schedule 'A' (BOQ) and as per drawing and all as specified in SSR Part-I & Part II.

6.9.6.2. The PVC door shutter shall be tested in the reputed testing house/lab as approved by GE as per IS-4020. The numbers of door samples shall be as per requirements laid down in IS-4020. In case, the results are not satisfactory, the entire lot shall be rejected. The cost of the testing and the cost of the door shutter samples is deemed to be included in the rates quoted by the tenderer.

6.9.7. SOLID PVC DOORS FRAMES :- Solid PVC doors frames shall be provided complete all as specified in Schedule 'A' (BOQ) and all as specified in SSR Part-I & Part II.

7. BUILDER'S HARDWARES

7.1. GENERAL

7.1.1. Unless otherwise specified all articles of builder's hardware (other than for steel windows) except butt hinges shall be of aluminum anodized bearing ISI mark as specified here in below.

7.1.2. BARREL TOWER BOLTS :- Refer Clause No. 9.3 of MES Schedule Part-I.

7.1.3. SLIDING DOOR BOLTS :- Sliding Door Bolt shall be provided complete all as specified in Schedule 'A' (BOQ) and all as specified in Clause No.9.5 of MES Schedule Part I, ISI marked.

7.2. BUTT HINGES :-

7.2.1. Refer Clause No. 9.7.2 of MES Schedule Part-I.

7.2.2. Butt hinges shall be of cold rolled mild steel medium weight bearing IS marked.

7.3. CONTINIOUS PIANO HINGES

7.3.1. Refer Clause No. 9.7.2 of MES Schedule Part-I.

7.4. DOOR HANDLES :- Door handle shall be provided with complete all as specified in Schedule 'A' (BOQ) and all as specified in clause No. 9.11 of MES Schedule Part I. However, Aluminum alloy handle shall be of `D' type with base plate cast type, ISI Marked.

7.5. DOUBLE ACTION SPRINGS :- Refer Clause No. 9.7.7 of MES Schedule Part I. These shall be of mild steel double action spring hinges with steel wire spring.

7.6. HYDRAULIC DOOR CLOSERS :- The hydraulic door closer provided with complete all as specified in Schedule 'A' (BOQ). These shall be ISI marked and as per Para 9.16 of MES Schedule Part I. Make Godrej or equivalent.

7.7. BALL CATCH :- This shall be of with 8 mm dia steel ball and brass body and all as specified in Clause No. 9.14 of MES Schedule Part I.

7.8. HASP AND STAPLE :- This shall be of safety type made of aluminium anodized as specified in Clause No. 9.10 of MES Schedule Part-I.

7.9. RAT TAIL DOOR SPRINGS (WHERE INDICATED) :- Refer Clause No. 9.7.8 of MES Schedule Part I. These shall be of mild steel.

7.10. DRAPERY ROD :- Curtain rod(CR) / drapery rod (DR) shall provided with complete all as specified in Schedule 'A' (BOQ). The finish of the rod shall be as approved by GE and all as specified in TD no CESZ/2015/TD/013.

7.11. STAINLESS STEEL WIRE CLOTHING:-

7.11.1. Refer Clause No. 9.25 and 9.32 of MES Schedule Part I.

7.11.2. Wire Cloth shall be Stainless Steel Wire of 0.36 mm dia and avg aperture 1.40mm & grade 304 fixed with Stainless Steel Wire Clips.

8. STEEL & IRON WORK

8.1. GENERAL

8.1.1. Refer relevant Clause No. of MES Schedule Part-I and as approved by GE. All Steels require to incorporate in the Work shall be Contractor's Supply.

8.2. TYPES OF STEEL

PARTICULAR SPECIFICATIONS (CONTD…..)

8.2.1. **REINFORCEMENT STEEL FOR ALL LOCATIONS :-** Steel bars for concrete reinforcement wherever shown on drawings or as specified in Schedule 'A' (BOQ) shall be TMT bars produced by Thermo Mechanical Treatment Process and grade of steel shall be mention on drawing or specified in Schedule 'A' (BOQ), unless other wise the grade of steel is not specified on drawing or specified, the grade of steel shall be (Grade Fe 500D) and meeting all requirement of IS-1786-2008. Where ever mild steel reinforcement bars are shown on drawings these shall be amended to read as TMT bars of same dia or nearest higher dia and accordingly to be provided.

8.2.2. **STRUCTURAL STEEL**

8.2.2.1. **STANDARD QUALITY :-** Structural steel except hollow steel sections shall be conform to IS-2062-2006, E-250(FE 410W). This steel shall be provided in the location mentioned in the drawings or as specified in Schedule 'A' (BOQ) and as specified in Clause 10.4.1 of MES Schedule Part-I. . In case the quality of structural steel i.e. A, B or C is not mentioned in drawings then quality A shall be used.

8.2.2.2. **ORDINARY QUALITY :-** Unless otherwise specified, Mild Steel Members in Grills/Guard Bars, Holdfasts, Door and Window, Frames, Railing, Steel Windows, Steel Doors and the like shall be ordinary quality of Steel conforming to IS-2062-2006, E-165 (Fe 290).

8.2.2.3. **HOLLOW STEEL SECTIONS FOR STRUCTURAL USE :-** Hollow steel sections for structural steel section shall be conform to IS-4923-1997 and grade of steel shall be YSt 310. This steel shall be provided in the location mentioned in the drawings or as specified in Schedule 'A' (BOQ).

8.2.3. **SOURCE OF PROCUREMENT**

REINFORCEMENT STEEL (TMT STEEL Fe- 500D) :- TMT steel bars of all sizes shall be procured directly from following primary producer.

Sl No	Company Name	Adress	Steel Grades / Sizes
1.	Rashtriyalspat Nigam Limited (RINL)Brand: "RINL"	Visakhapatnam Steel Plant Visakhapatnam- 530 031, India, Tel: (91 891) 518226. 518376 Fax: (91 891) 518316,Email: cmdvsp@itpvis.ap.nic.in	All
2.	Tata Iron & Steel Company (TISCO, or Tata Steel) Brand: "TATA"	Bombay House, 2, 4 Homi Modi Street, Mumbai - 400 001. India,Tel: (91 22) 204 9131, Fax: (91 22) 204 9522. 287 0840 Email: corpcomm@ jsr.tatasteel.com	All
3.	Steel Authority of India Limited(SAIL) Brand: "SAIL"	Central Marketing Organization. Northern Region, 17th Floor, scope Minar, Laxmi Nagar Distt. Centre. Delhi- 110092	All
4.	M/S Jai Balaji Industries Ltd Brand: "BALAJI SHAKTI"	5 Bentek street, Kolkata- 700001 Delhi Office, 510, Block-b, Navraung House 21 Kasturba Gandhi Marg, New Delhi-110001 011-43620219,43620220, Mob: 7838272772/9958936103 E-mail- 1nfo@jaibalaj1group.com	TMT Bars of Gde Fe 500 & Fe 500D
5.	M/S Shyam Steel Industries ltd Brand: "SHYAM "	Shyam Towers EN-32, Sector-V, Salt Lake. Kolkata-700091,Tel 033-40074007, Fax-033-40074010, E-mail-marketing@shyamsteel.com	TMT Bars of Gde Fe 500, Fe 500D & CRS
6.	M/S Steel Exchange India ltd Brand: "SIMHADRI TMT"	My Home Laxminivas Apartments. Ameerpet, Hyderabad-500016, A.P. Tel-040-23403725, Fax-040-23413267,E-mail-info@seil.co.in	TMT Bars of Gde Fe 500, Fe 500D & HSCRM
7.	M/S Shri Bajrang Power &lspat ltd. Brand: "GOEL TMT"	Viii - Borjhara, Urla Industrial Area. Raipur- 493 221, Chhattisgarh. Tel : 0771 4288019/29 / 39	TMT Bars of Gde Fe 500 & Fe 500D (Size 8-32 mm)
8.	M/S Real Ispat & Power Ltd Brand: "G K TMT"	Vrindavan, Near IDBI Bank, Civil Lines, Raipur-492001, C.G, Tel: +91-771-4224000, Fax-+91-771-4224010,Email- real@realispat.com, Website : www.realispat.com	TMT Bars of Gde Fe 500, Fe 500D(Size 8-36 mm)
9.	M/S Super Smelters Ltd, Kolkata Brand: "SUPER SHAKTI"	Premlata, 39, Shakespeare Sarini, 3rd Floor, Kolkata-700017, Phone-033-2289-2734/36 Email-: info@supershakti.in Website :www.supershakti.in	TMT Bars of Gde Fe500, Fe500D & Fe 550, (Size 8-32 mm)

PARTICULAR SPECIFICATIONS (CONTD…..)

10.	M/s Shyam Metalics & Energy Ltd Brand: "SEL"	Viswakarma, 1st Floor, 86 C, Topsi Road, Kolkata-700046 Ph: +91 33 2285 2212, Website: www.shyamgroup.com	TMT Bars of Gde Fe500 & Fe 500 D (Size 8-32 mm)
11.	M/s Electrotherm (India) Ltd Brand " ET TMT"	Survey No 72, Palodia, Vil-Thaltej, Ahmedabad, Gujrat-382115, Tele-+91-2717-660649/660550, Fax-+91-2717-234866, Website-www.electrotherm.com	TMT Bars of Gde Fe 500, Fe500 D & CRS (Size 8-36 mm)
12.	M/s Gallantt Metal ltd Brand "GALLANTT TMX'	Ward 10 BC, Plot no 123, Ground Floor, Gandhi Dham, Kutch, Gujrat-370201,Tel:+91-2836-228164, Fax: +91-2836-235787, E-mail: gml@gallantt.com Website : www.gallantt.com	TMT Bars of Gde Fe 500, Fe500 D & CRS (Size 8-32 mm)
13.	M/s Rashmi Metaliks Ltd Brand: "RASHMI TMT"	Premlata Building, 39, Shakespeare Sarani, 6th Floor, Kolkata-700017,Tel-033-2289-4255/56, Fax: 033-22894254, Email-: mkt.domesticdip@rashmigroup.com Website :www.rashmigroup.com	TMT Bars of Gde Fe 500 (Size 8-32 mm), 500D,Fe 550D (Size 8-25 mm)
14.	M/s MSP Steel & Power Ltd.	16/S, Block-A, New Alipore Kolkata- 700053 Tele- 033 4005 7777/2398 2239 Mob- 7381028976, Email- contactus@mspsteel.com Website- www.mspsteel.com	TMT Bars of Gde Fe500D (Size 8-32 mm)

Note : Proper validity of approval by E-in-C's branch has to be ensured before approval of make by GE..

Note :- Incase the total quantity of reinforcement steel (TMT steel Fe 500D) required in the work is less than 1000 Kg then the contractor may purchase the same through authorized dealer of primary producers. However necessary test certificate to be submitted by contractor and independent testing has to be carried out by GE to confirm quality.

8.2.3.1. **STRUCTURAL STEEL(EXCEPT HOLLOW STEEL SECTIONS) :- S**tructural steel any section/plate (except hollow steel sections) shall be procured from following main producers.

1. M/s Rashtriya Ispat Nigam Limited (RINL)
Visakhapatnam Steel Plant
Visakhapatnam-530031, India
Tel: (91 891) 518226, 518376
Fax: (91 891) 518316
Email: cmdvsp@itpvis.ap.nic.in
Brand: "RINL"
Structural Steel (Angle, Beam, Column, Channel, Plate)

2. Tata Iron & Steel Company (TISCO or Tata Steel)
Bombay House, 2, 4 Homi Modi Street, Mumbai-400 001, India
Tel: (91 22) 204 9131
Fax: (91 22) 204 9522, 287 0840
Email: corpcomm@jsr.tatasteel.com
(Br office for North : Jeevan Tara Bldg, Patel Chowk, New Delhi)
Brand: "TATA"
Structural Steel (Angle, Beam, Column, Channel, Plate)

3. Steel Authority of India Ltd (SAIL)
Central Marketing Organization, Northern Region, 17th Floor, Scope Minar, Laxmi Nagar Distt Centre, Delhi-110 092
Brand: "SAIL"
Structural Steel (Angle, Beam, Column, Channel, Plate)

4. M/s Jindal Steels and Power Ltd
Jindal Centre,
Plot No. 2, Sector-32,
Gurgaon-122 001, Haryana
Tel: 0124 661 5000
Fax: 0124 661 2125
Website : www.jindalsteelpower.com
Brand: "JINDAL"
Structural Steel (Angle, Beam, Column, Channel, Plate)

8.2.3.2. However in case of its non availability with above mentioned primary producers, the same can be procured from approved secondary producers with a reduction of 5% (Five percent) of the accepted rates of structural steel. In case the desired section of structural steel is not rolled / manufactured by above mentioned primary producers, there shall be no price adjustment in use of structural steel procured from approved secondary producers for such sections of structural steel. In case the quality of structural steel i.e. A, B or C is not mentioned in drawings then quality A shall be used.

8.2.3.3. Steel section for railings, gate, fencing, guard bars, grills, steel chowkhat, holdfasts etc which do not constitute structural members, can be procured from main producers/ secondary producer/ BIS marked manufacturers or their authorized dealers at the option of contractor without any price adjustment. Tests will not be insisted upon for such steel sections.

8.2.3.4. **STRUCTURAL STEEL (HOLLOW STEEL SECTIONS) :-** Structural hollow steel sections shall be directly procured from TATA (STRUCTURA) /JINDAL STAR/ ISI Marked .

8.2.4. **TESTING OF STEEL**

8.2.4.1. The manufacturer is to carry out inspections and testing of steel in accordance with the relevant BIS provisions. The contractor shall submit the manufacturer's test Certificate in original along with the test sheet giving the results of each mechanical test as applicable and the chemical composition of the steel or authenticated copy thereof, fully signed by the manufacturer with each consignment. The Engineer-in-Charge shall record these details in Steel Acceptance Register, as given at **Appendix 'F'** after due verification and send a certified true copy of test sheet to GE for his records. The GE/CWE shall also organize independent testing of random samples of steel drawn from various lots from National Test House, SEMT Wing CME, Regional Research labs, IITs, National Institutes of Technology, NABL Approved laboratories, Command Testing Laboratory, Govt Engineering Collage or Any Govt laboratories as per the recommended minimum frequency shown in para here-in-after. Samples from each lot should be tested for quality and elongation. The elongation shall not be less than 18%. Cost of samples, transportation and testing shall be borne by the Contractor. The records of such checks would be maintained in the steel test register.

8.2.4.2. Nominal mass of any size of finishes bar/section of steel shall be checked as specified in relevant IS code. The nominal mass so determined shall be recorded in steel testing register giving cross reference to consignment number. Nominal mass of any size of finished bar/section of steel if found to be beyond the tolerance limits on minus side as specified in relevant IS code, the same shall be rejected and the contractor shall remove the same at his own cost without any extra cost to the Government. However, if the weight of steel section is beyond the tolerance limit on higher side the same can be provided with approval of the GE but without any extra cost to the Government.

8.2.4.3. If any Test Result of any particular size of Bar/Section of steel of any consignment is not found satisfactory as specified in relevant IS, the contractor shall remove the same at his own cost and no claim of contractor shall be entertained on this account.

8.2.4.4. **TYPE OF TESTING**

8.2.4.4.1. **REINFORCEMENT STEEL**

(a)	High Strength Deformed Steel Bars	– Nominal Mass Test, Tensile Test, Bend Test and Re-bend Test shall be carried out as per Clause No. 8 of IS-1786-1985 and Test Specimens shall be as per Clause No. 10 of IS-1786-1985.
(b)	Mild Steel Bars	– Nominal Mass Test, Tensile Test and Bend Test shall be carried out as per Clause No. 9 of IS-432 (Part-I)-1982. Re-bend Test is not required to be carried out for Mild Steel Bars.

8.2.4.4.2. **STRUCTURAL STEEL**

(a)	Structural Steel (Standard Quality)	– Tensile Test and Bend Test shall be carried out as per Clause No. 6, 7 and 8 of IS-226-1975 for Standard Quality of Steel Sections.
(b)	Structural Steel (Ordinary Quality)	– Tensile Test and Bend Test shall be carried out as per Clause No. 6, 7 and 8 of IS-1977-1975 for Ordinary Quality of Steel Sections.
(c)	Structural Steel (hollow steel sections)	– Tensile Test shall be carried out as per IS-4923-1997 for Standard Quality (Hollow Steel Section) of Steel Sections.

8.2.4.4.3. **FREQUENCY OF TESTING**

8.2.4.4.4. **REINFORCEMENT STEEL:-**Normal Mass Test, Tensile Test, Bend and Re-bend Test shall be carried out as per Minimum Frequency given below :–

NOMINAL SIZE OF BARS		FREQUENCY
(a) Bars size less than 10mm	–	1 SAMPLE (3 SPECIMENS) for Each Test for Every 25 TONNES or Part thereof.
(b) Bar size 10mm to 16mm	–	1 SAMPLE (3 SPECIMENS) for Each Test for Every 35 TONNES of Part thereof.
(c) Bar size over 16mm	–	1 SAMPLE (3 SPECIMENS) for Each Test for Every 45 TONNES of Part thereof.

8.2.4.4.5. **STRUCTURAL STEEL(FOR ALL GRADE):-** Tensile Test and Bend Test shall be carried out as per Minimum Frequency given below **:–**

NOMINAL SIZE OF BARS		FREQUENCY
(a) TENSILE TEST	–	1 SAMPLE (3 SPECIMENS) for Each Test for Each Individual Section for Every 25 TONNES of Steel or Part thereof.
(b) BEND TEST	–	1 SAMPLE (3 SPECIMENS) for Each Test for Each Individual Section for Every 25 TONNES of Steel or Part thereof.

*NOTE : For various Tests, Acceptance Criteria, Tolerance, etc. refer to STEEL SUPPLY & ACCEPTANCE REGISTER given in **APPENDIX 'F'** and relevant BIS Codes.*

8.2.4.5. **TESTING CHARGES :-** The Unit Rate/Amount quoted by the Contractor in the Tender shall be inclusive of Cost of all sorts of Testing to the extent as specified here in before and in relevant Indian Standards.

8.2.5. **DOCUMENTATION :-** The contractor shall submit Original Purchase Vouchers and Test Certificate from the Main Producer/ Secondary Producer/BIS marked manufacturer/authorized dealer as applicable for the total quantity of steel supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by the GE alongwith the relevant documents before acceptance. The Original Purchase Vouchers and Test Certificates shall be defaced by the Engineer in Charge and kept on record in the office of the GE duly authenticated and with cross reference to the consignment number recorded in the Steel Acceptance Register. The steel Acceptance Register will be signed by the JE (Civil), Engineer-in-Charge, GE and the, Contractor. The Accepting Officer may order a Board of Officers for random check of steel and verification of connected documents. The entire quantity of steel items shall also be suitably recorded in the measurement book for record purposes as "Not to be abstracted" before incorporation in the work and shall be signed by the Engineer in Charge and the contractor.

8.2.6. **STORAGE, ACCOUNTING, PRESERVATION & MAINTENANCE OF STEEL:** The storage, accounting, preservation and maintenance of steel supplied by the contractor shall be done as per standard engineering practice till the same is consumed in the work and the cost of the same shall be deemed to be included in the unit amount quoted by the tenderer. The GE shall inspect at regular interval to verify that steel lying at site are stored, accounted, preserved and maintained as per the norms. The steel shall be stored so as to differentiate each consignment separately. If the GE is not satisfied with the storage/preservation of any size of bar/section of steel, he may order for any test (s) of steel as applicable for that size of bar/section of steel and as specified in tender documents and relevant Indian Standard to recheck the acceptability criteria for the same. The contractor shall bear the cost of necessary testing(s) in this regard and no claim whatsoever shall be entertained by the Govt.

8.2.7. **MEASUREMENTS AND PAYMENT OF STEEL:-**

8.2.7.1. The entire quantity of all steel items shall also be suitably recorded in the Measurement Book for record purposes as **"Not to be Abstracted"** before incorporation in the work and shall be signed by the Engineer in Charge and the contractor.

8.2.7.2. The nominal mass conversion factor for various steel sections/size of finished bars as given in relevant IS codes shall be considered standard for measurement.

8.2.7.3. The payment of steel shall only be allowed after production of Original Purchase Vouchers, Test Certificates by the contractor for each consignment of steel and results of testing carried out by the Department are found satisfactory after testing.

8.3. STEEL WINDOWS/VENTILATOR & STEEL DOOR

8.3.1. STEEL WINDOWS/VENTILATOR

8.3.1.1. Refer Clause No. 10.25 of MES Schedule Part-I.

8.3.1.2. **PRESSED STEEL WINDOW/ VENTILATOR**: Pressed steel windows/ventilators shall be provided complete all as specified in Schedule 'A' (BOQ) and as per shown/mentioned TD drawings no CESZ/2015/TD/006 and as specified following **:-**

(a) Steel required for steel window shall be of cold reduced low carbon steel sheet of designation CR2 and grade D as per IS 513 :2008. MAKE of sheet : **SAIL, Rashtriya Ispat Nigam Ltd, IISCO, TISCO, M/S Jindal Steels and Power Ltd.**

(b) While supplying, steel windows shall have one shop coat of anticorrosive paint (Red oxide zinc chrome primer).

(c) Window shall be given another coat of anticorrosive priming and two coats spray painting before fixing of glass and wire mesh.

(d) Steel windows shall be fixed as per IS-1081 embedding in cement concrete block of appropriate size.

(e) All steel windows, ventilators shall be factory made and of approved make and shall be procured from any of the approved firms listed in Appendix `B' to Particular Specification.

(f) Pressed steel frame shall conform to IS 4351.

(g) Glass and frame shall be joint with 6x6 mm silicone polymer base sealant.

(g) The fitting shall be provided as following **:-**

(i)	70 mm long MS Box type hinge with two nos stainless steel washer(each end) all as specified in clause no 6.2.1.1 of IS 1038	:-	02 nos each shutter (both glazed shutter and gauged (fly proofing) shutter)
(ii)	250mm long 1.5 mm thick MS telescopic stay with lock screw as shown on drawing	:-	01 no each gauged (fly proofing) shutter.
(iii)	MS spring Latch	:-	01 no each glazed shutter
(iv)	75 mm long 'C' section MS handle	:-	01 no each shutter gauged (fly proofing) shutter.
(v)	75 mm long MS tower bolt	:-	02 no each gauged (fly proofing) shutter
(vi)	Any other fitting as per manufactures requirement		

8.3.1.2.1. **FABRICATION :-**Refer Clause No. 10.25.2 of MES Schedule Part-I and relevant Clause of IS-1038.

8.3.2. GUARD BARS TO STEEL WINDOWS & VENTILATORS:-

8.3.2.1. All steel windows shall be provided with guard bars as shown on drawing. However, in case of double windows in the same opening guard bars shall be provided to glazed windows only.

8.3.2.2. Guard bars to steel windows unless otherwise shown on drawings shall be mild steel 12mm square. The guard bars shall be integrated during the manufacture of the windows.

8.3.2.3. The process of welding adopted shall be of flush butt welding. Prior to procurement of entire quantity of steel windows and ventilators, the contractor shall produce a sample to the GE who shall after its inspection /examination and after satisfying himself with regards to above mentioned requirements, approve the samples and convey in writing to the contractor.

8.3.3. **TESTING :-** The number of steel windows/vents to be subjected to independent Destructive testing in reputed Testing House/ Laboratory by GE shall be as per Appendix to IS-1038. In case results are not satisfactory, the entire lot shall be rejected. Cost of Destructive testing including the requisite number of steel windows/ vents required as per lot size is deemed to be included in the quoted rate by the contractor.

8.4. STEEL DOORS

8.4.1. Steel doors shall conform to IS-1038 and the section particulars shall be provided as per TD drawing/main drawing and clause 10.25 of SSR Part-I.

8.5. WORKMANSHIP

8.5.1. GENERAL

8.5.1.1. Aluminium Doors and Windows shall be Fabricated, Erected and fixed in Position and as per the Details /Size all as directed by Engineer-in-Charge.

8.5.1.2. The thickness and Type of Glazing for Aluminum Doors and Windows shall be all as shown in Drawings. Neoprene Crest Glazing Gasket shall be provided with Snap or Beveled Glazing Beads. Glass shall be fixed with PVC/Rubber Gasket so as to avoid Metal to Glass contact and no projection at any bending shall be visible thereby making it Flush Glazed.

8.5.1.3. The Cleats for Mechanical, Vertical/ Horizontal Joints shall be of Extruded Aluminium Section so as to avoid any play between Jointed Members.

8.6. STEEL ROLLING SHUTTER

8.6.1. GENERALLY :Refer Clause No. 10.23 of MES Schedule Part I, 2009.

8.6.2. Steel rolling shutter shall be of approved make . The size of rolling shutters shall be all as per drawings. The position of fixing of the rolling shutters shall be as decided by the Engineer-in-charge.

8.6.3. Rolling shutters shall be self coiling type without ball bearing, push and pull manually operated upto a maximum of about 8 Sqm clear area and upto clear area of about 12 Sqm with ball bearing.

8.6.4. CURTAINS:-The curtain shall be built up of interlocking lath sections made from cold rolled steel strips 1.25mm thick. The lath sections shall be rolled so as to have interlocking curls at both edges and a deep corrugation at the center with a bridge depth of not less than 12 mm. Each lath section shall be continuous single piece without any welded joints. When inter lock the lath section shall have a distance of 75 mm between rolling centers. Each alternate lath section shall be fitted with malleable cast iron or mild steel clips securely riveted at either ends.

8.6.5. LOCK PLATE :Refer Clause No. 10.23.3 of MES Schedule Part I.

8.6.6. GUIDE CHANNELS :-Refer Clause No. 10.23.4 of MES Schedule Part-I, but the depth of guide channel shall be 65 mm upto 3.5 Metre clear width of shutter.

8.6.7. BRACKET PLATE :- Refer Clause No. 10.23.5 of MES Schedule Part-I.

8.6.8. ROLLER:-The size of pipes for suspension shafts for rolling shutter shall be 50 mm (Nominal) bore medium grade for shutter width upto 6 meter. The suspension shaft shall be made of steel pipe conforming to IS -1161 and shall be provided with steel or preferably cast iron pulleys and helical wire spring or flat spiral springs for counter balancing the weight of the shutter adequately.

8.6.9. Hood Covers, Gears, Safety Device, Safety Lever Locks, etc, the specification shall be all as per Clause No. 10.23.7, 10.23.8, 10.23.9 and 10.23.11 of MES Schedule Part I.

8.7. STEEL TRUSS/HOLOW TRUSS/TUBULAR TRUSS & PURLINS

8.7.1. GENERALLY

8.7.1.1. The steel member used for trusses and other places shall be manufactured as per specification given in relevant IS. Tolerance in thickness and weight shall be as per IS. The steel shall be procured from any of the manufacturers mentioned here-in-before. Steel member exposed to view shall be painted with two coats of synthetic enamel paints over a coat of primer. Steel members not exposed to eye shall be painted with two coats of red oxide primer.

8.7.1.2. COMPONENT MEMBERS: All component member shall be neatly fabricated true to shape according to dimensions/size shown on respective drawings. A marking diagram allotting distinct identification mark to each separate piece of steel work shall be prepared to ensure correct assemble and erection at site. The component parts shall be assembled in such a manner that they are neither twisted nor otherwise damaged.

8.7.2. Hollow steel sections shall be designated by their nominal bore. These shall be light, medium or heavy as specified depending upon the wall thickness. The standard size and weights of tubes are listed in Appendix C. Hollow sections shall be as per IS 4923.

8.7.3. Tubes shall be clean finished and reasonably free from scale. They shall be free from cracks, surface flaws, laminations and other defects. The ends shall be cut clean and square with axis of tube

8.7.4. **DRILLING HOLES:**-Work shall be carried out all as specified in Clause No. 10.11 of MES Schedule Part-I and as per IS. Holes shall not be formed by gas cutting process.

8.7.5. **CUTTING**

8.7.5.1. Cutting may be effected by shearing, cropping or sawing gas cutting by mechanically controlled torch will be permitted but after gas cutting edges shall be cleared of slag, burnt metal and made free from local protection and depressions exceeding 1mm by grinding. Hand frame cutting may be permitted subject to the approval of Garrison Engineer.

8.7.5.2. Shearing, cropping and gas cutting shall be clean, free any distortion and if it is not so, the edge shall be ground without any additional payment. Holes where required in steel members shall be done using suitable drilling machine and punches. Making holes with gas cutting is not permitted.

8.7.6. **CONNECTIONS :-** All connections of steel work shall be welded except bolting where specifically shown on drawings.

8.7.7. **JOINTS:-** The joints surface shall be prepared as specified in IS. Every member of the various types of steel trusses, roof girders, stanchions etc shall be in one piece without joint. However, additional splice joints, if required due to site conditions shall be carried out on written approval of Garrison Engineer without extra cost to Government. Splicing of members shall be done all as shown on drawings.

8.7.8. **FABRICATION**

8.7.8.1. Fabrication of all components including notching, cutting, punching/drilling and welding shall be carried out exactly in accordance with the detailed drawings and with the best workmanship. Sufficient steel work shall be trial fitted by the contractor in his workshop at site to ensure accuracy of various components, as work failing to fit accurately at site will be liable for rejection by the Garrison Engineer.

8.7.8.2. All fabrication shall be done at site and in position where necessary.

8.7.8.3. Erection shall be carried out as specified in Clause No. 10.16 of MES Schedule Part-I.

8.7.8.4. Contractor's particular attention is drawn to IS-7205 (Safety Code for erection on structural steel work). Anchor bolts for fastening of steel structure shall be set in designed position and grouted alongwith foundation. Also refer Clause No. 10.16.2 and 10.16.3 of MES Schedule Part-I.

8.7.8.5. Damaged structural members shall be examined and rectified or replaced as directed by Engineer-in- Charge.

8.7.8.6. The erected parts of structure shall be stable during all the stage of erection and the structural elements to be erected, shall be stable and strong to bear erection load.

8.7.8.7. Working on the already erected structure is permitted only after they are finally fixed. Erection of structures of each tier of high structure shall be executed only after the relevant fastening of lower ties by the permanent or temporary fastening devices as clarified for safety.

NOTE : *The contractor shall at his option may get the truss components fabricated at the manufacturer's shop and transport the same to the work site without any extra cost. However the fabricated truss will be erected only after the same is approved by Garrison engineer.*

8.8. **ERECTION OF TRUSSES:-** The trusses shall be lifted only at nodes. The trusses shall not be aligned at the Apex. It shall be lifted by slinging at two mid points of rafter, which shall be temporarily braced by suitable section. After trusses are placed in position, purlins and wind bracing shall be fixed as soon as possible.

8.9. **PRECAUTION DURING ERECTION :-**

8.9.1. Contractor shall maintain accuracy in dimensions of fabricated members all as shown on drawings.

8.9.2. Suitable equipment for erection/hoisting of members shall be used. Under no circumstances, members already erected shall be used/taken support to erect new members.

8.9.3. Contractor shall take all precaution to avoid damage to members or injury to workers.

8.9.4. **WELDING**

8.9.4.1. **GENERALLY**

(i) Refer Clause No. 10.15 of MES Schedule Part-I.

(ii) Welding shall be done by metal arc process in accordance with the standard specified in IS-816 and IS-9595. All welding shall be turned point upto a length not less than twice the thickness of the weld.

(iii) The work shall be positioned for down hand welding wherever possible/practicable.

(iv) Joint surfaces shall be smooth, uniform and free from fins, tears, laminations and such other defects which adversely effect quality of weld and workmanship.

(v) All welds shall be cleaned of slab and other deposit after completion. The specified size of welds shall be provided for the full length of each weld as shown on drawings.

(vi) Skilled welders shall only be employed for the work and the workmanship shall be of high standard.

8.9.4.2. **ELECTRODES FOR WELDING** :-Electrodes used for welding shall conform to IS-814 and shall be ISI marked. The contractor shall supply at his own cost necessary electrodes, for getting them tested if Garrison Engineer so desired.

8.9.4.3. **INSPECTION/TESTING WELDS:-** The Garrison Engineer or his representative shall inspect and examine the electrodes for welding. Garrison Engineer shall have the welds tested for soundness using any of the non destructive tests and the testing charges shall be borne by the contractor.

8.9.4.4. **PRECAUTION/SAFETY AND HEALTH:-** The contractor shall ensure that the safety requirements and health requirements in electric and gas welding and cutting operation are complied with during operations. The contractor shall also provide equipments for eye and face protection during welding as laid in IS. Fire precautions shall be taken in accordance with IS-3016, Code of practice for Fire Precaution in welding and cutting operations.

8.9.4.5. **EXPANDED METAL:-** Where expanded metal is shown and its size is not given on the drawings, it shall be 20mm short way by 50mm long way mesh, wire of nominal size 2.5mm x 1.25mm, weight not less than 2.183 Kilogram per Square Metre.

8.9.4.6. **WELD MESH:-** Weld mesh for reinforcement where indicated in drawings shall be of mild steel wire 4mm dia and 100mm c/c both ways, weighing 1.98 Kilogram per Square Metre.

8.9.4.7. **ANGLE IRON NOSING TO STEPS :-**Nosing of steps shall be of 40x40x3mm angle iron. Fixing details shall be as shown on drawings. Angle iron nosing shall be provided to all PCC finish steps and other location as shown in Schedule of Finishes drawings.

9. ROOF COVERING

9.1. PRE-PAINTED GALVALUME STEEL SHEET

9.1.1. Refer Clause No. 10.39 &11.3 of MES Schedule Part-I and IS 15961-2012 & 15965-2012.

9.1.2. **PRE-PAINTED GALVALUME STEEL SHEETS :-** The galvalume steel roofing sheet shall be **0.50 mm thick (Total Coated Thickness)** and shall have 550 Mpa minimum yield strength. The base metal steel plate shall be coated with minimum 150 gram/sqm (total on both sides) coating of hot dip alloy consisting of 55% Aluminium, 43.5% Zinc and 1.5% Silicon confirming to IS 15965-2012. The profile of the sheet shall be trapezoidal as defined in SSR Part 1, Clause No. 10.39 or similar profile as commercially available and approved by GE. The end laps shall be provided as per manufacturer's guidelines to eliminate any possibility of leakage of roof. Penetrations and laps in sheet shall be sealed as per manufacturer's instructions and literature. HDPE fillers shall be provided wherever required to close voids between sheets, sheet & fasteners etc. The color of the sheets shall be as approved by GE.

9.1.3. Side laps shall be one corrugation.

9.1.4. Manufacturer's name, length, width, thickness of BMT /TCT and number of corrugation and material identification (yield strength, coating) shall legibly be marked on top of each sheet.

9.1.5. The contractor shall submit the manufacturer's test Certificate in original along with the test sheet giving the results of each mechanical test as applicable and the chemical composition of the steel or authenticated copy thereof, fully signed by the manufacturer with each consignment.

9.1.6. Single length sheet shall be provided in each side of the roof and all as directed by GE.

9.2. CORRUGATED/PLAIN GALVANISED SHEET ROOFING

9.2.1. Refer Clause No. 10.29 and subsequent Clause No. thereof MES Schedule Part I.

9.2.2. These shall be procured directly from main producers i.e. SAIL, Rashtriya Ispat Nigam Ltd, TISCO, JINDAL.

9.2.3. Refer Clause No. 11.2, 11.22 and subsequent Clause No. thereof MES Schedule Part I.

9.3. ALUMINIUM WORKS (DOORS, WINDOWS, VENTILATORS, PARTITIONS, ETC.)

9.3.1. MATERIALS

9.3.1.1. ALUMINIUM SECTIONS & BEADS, ETC.

9.3.1.2. Refer Clause No. 10.37 of MES Schedule Part-I as applicable and as approved by GE.

9.3.1.3. Aluminum Door/window shall be provided as per main drawings and TD drawings. Specification for aluminium door and window shall be 15 micron anodized (colored) aluminum section having embossment of the name of the manufacturer on each section at every metre and thickness of colour anodizing shall be not less than 15 micron. The section particulars shall be provided as per drawing and clause 10.37 of SSR Part-I, 2009. In case of non availability of any particular section mentioned the next available section (having higher per Kg/Rm weight) shall be provided without any price adjustment. The test certificate from Govt authorized testing laboratory shall be produced by the contractor to conform thickness of anodizing.

9.3.1.4. **ALUMINIUM FITTINGS**, ETC.: Aluminium Doors and Windows shall be provided with the Standard Aluminium Fittings as specified in relevant Clause No. of MES Schedule Part-I, bearing ISI Mark and as approved by GE

9.3.1.5. **SHEET GLASS**:-Refer to relevant Clause No. of MES Schedule Part-I and as approved by GE.

9.3.1.6. ALUMINIUM DOORS/WINDOWS/VENTS

9.3.1.7. Aluminium Doors/Windows/Vents shall be fabricated, erected and fixed in position and as per the details/ size shown in TD drawings. These shall be in conformity with the standard aluminium fittings conforming to the specifications of relevant IS.

9.3.1.8. The cleats for mechanical, vertical/ horizontal joints shall be of extruded aluminium section so as to avoid any play between jointed members.

9.3.1.9. Glazing for aluminium doors shall with 8 mm thick toughened solar reflective glass and for windows/vents shall be with 4mm thick float glass excepted aluminum sliding window . Neoprene crest glazing gasket shall be provided with snap or beveled glazing beads. Glass shall be fixed with PVC/rubber gasket so as to avoid metal to glass contact and no projection at any bending shall be visible thereby making it flush glazed.

9.3.1.10. Door shutter shall be provided with Floor Spring Door Closure of standard door pattern with all necessary fixing arrangement and all doors shall be provided with mortice lock of Godrej make.

9.3.2. ALUMINIUM WINDOWS / VENTS

9.3.2.1. All aluminium work shall be deemed to include in various items to complete work including Fittings, Fixtures, Stays, Locks, Handle, Special Hinges, Floor Springs, Neoprene/Rubber Linings, Gaskets, Bushes, Rollers, Sealants, etc. as directed and approved by the GE.

9.3.2.2. MS grill with powder coated shall be provided as per shown on TD drawings

9.3.2.3. The items of Aluminium Glazing, Windows, Vents and Louvers Window Work shall include the provision of mullions, coupling bars, as required to join various units of Glazing, Windows and Louvers etc to form larger glazing, windows vents or louvers cum window/glazing area as per drawings. Also the necessary jointing through Cleats, Glazing Clips, Rubber Packing, Anodized Aluminium Snap Beading, Screws, etc. complete all as specified. No extra payment shall be made for Coupling Members, Mullions, etc.

9.3.2.4. Aluminium Sections used for Windows, Vents, Louvers and fixed glazing shall be of weight as specified on drawing. This shall be suitable for use to meet Architectural designs of relevant works and shall be subject to approval of GE who would require to be satisfied about their being appropriate on Technical, Structural Functional and Visual Considerations.

9.3.2.5. All joints shall be accurately fabricated and be hair line in appearance. The finished surface shall be free from visible defects.

9.3.2.6. Each aluminum frame shall be Tailor Made as per openings at site. No cutting and making good of concrete surfaces shall be permitted.

9.3.2.7. Frames shall be fixed to concrete or brick work with approved dash fastners. Method of fixing shall be got approved by the GE before installation.

9.3.2.8. Samples of typical glazing shall be made and got approved from GE before mass fabrication.

9.3.2.8.1. All hardware items shall conform to the relevant IS Specifications. Design, quality type, number and fixing of hardware shall be got approved by the GE.

9.3.2.9. No visual variation in shade shall be permitted. The fabricator shall clearly indicate the shade variation tolerance as measured by standard equipment.

9.3.2.10. The requirements, provisions, for all aluminium work shall conform to requirements and specifications and Clause No.meters given in this tender drawings / instructions and shall at least provide for or conform to fabrication, finishing, erection installation etc.

9.4. ALUMINUM GRILLS:

9.4.1. The aluminum grill shall be provided as per shown on Main / TD drawings.

9.4.2. The type of aluminum grill shall be provided as shown on TD drawing no CESZ/2015/TD/2014 but in no case weight of grill / grill frame shall be less than 3.13 Kg per sqm and 0.297 kg/m respectively.

10. FLOORING & FLOOR FINISHES

10.1. GENERAL

10.1.1. Refer Clause No. 13.2, 13.4, 13.5, 13.8, 13.14, 13.21, 13.23, 13.24, 13.25, 13.27, 13.32, 13.38, 13.39, 13.40, 13.46 and 13.47 of MES Schedule Part-I are to be adopted for Materials and Workmanship of laying Floor/ Skirting/Dado.

10.1.2. Polythene film shall conform to IS-2508, 150 micron thick of 01.0 Grade. If full width of polythene film is not available, the same shall be laid with an overlap of 150mm pasted with the resin solution.

10.1.3. Tiles (Non Skid/Glazed/PVC) shall be laid all as specified in Clause No. 13.39, 13.40 and 13.44 of MES Schedule Part-I.

10.1.4. Polished Kota stone shall be of machine cut polished and thickness shall be all as mentioned in Schedule 'A' (BOQ). The stone slab shall be hard, even, sound, durable, tough, and regular in shape of shade as approved.

10.2. TYPES OF FLOORING - Various types of flooring of buildings indicated in Drawings/Schedule of Finishes shall be provided with following specifications: -

10.3. PCC FINISH /PCC CHEQUERED FLOOR

10.3.1. PCC FLOOR:

10.3.2. The concrete mix shall be nominal mix. PCC floor shall be provided all as shown on drawings.

10.3.3. PCC floor shall be finished as specified in Schedule of finishes and in case no finish is specified it shall be considered finished smooth and even using extra cement. Where floors to be laid directly on RCC slab shall be given a coat of cement slurry @ 3 Kg/Sqm before floor is laid as specified in clause No 13.27 of MES Schedule Part I except to portion of sunken slab.

10.3.4. The earth wherever specified shall be approved earth as approved by GE.

10.4. PCC HARDSTAINDING

10.4.1. PCC hardstanding shall be carried out using Vacuum Dewatering process and Power Trowelling Method with a Skim Floater as specified in Clause No 13.49 of MES Schedule Part-I.

10.5. JOINT IN PCC HARDSTANDING /APRON / PAVED AREA:-

10.5.1. Irrespective of whatever shown or mentioned on drawings or schedule of finishes the expansion /contraction (dummy) joint shall be provided in all PCC hard-standing /apron / paved area as mention as below.

10.5.2. Refer Clause no 13.18 of MES Schedule, Part-I and IRC 15.

10.5.3. **Expansion Joint** :- The expansion joint shall be provided along perimeter (full length & full width) of hardstanding /apron / paved area and longitudinally every 20 meter(max) and transversely 9 meter (Max) interval. The joints shall be provided upto full thickens of hardstanding /apron / paved area and filled with 20 mm wide pre-formed bituminous filler and top 30mm thick primed and sealed with polysulphide sealant as specified in MES Schedule Part-I**.**

PARTICULAR SPECIFICATIONS (CONTD…..)

10.5.4. **Contraction(Dummy) joint** :- The contraction(dummy) joint shall be provided at 4.5 meter interval (max) in both direction. The joints shall be provided 10 mm wide and depth equal to $^1/_3$ of the thickness of PCC hardstanding /apron / paved area and filled with polysulphide sealant as specified in MES Schedule Part-I**.**

10.6. **PCC TILES/ PCC CHEQUERED TILE FLOOR**

10.6.1. Refer Clause no 13.17 of MES Schedule, Part-I. The PCC chequered Tiles shall conform to IS: 13801.

10.6.2. PCC Chequered tiles floor wherever shown in Schedule of finishes drawings shall be provided with PCC chequered tiles (factory made) shall be of size 200mm x 200mm x 20mm. Tiles shall be laid over 20mm thick screed bed in Cement Mortar 1:4, set, jointed and pointed in neat cement slurry over sub base floor/RCC slab/any other location as specified.

10.7. **PRECAST CEMENT CONCRETE INTERLOCKING PAVING BLOCKS**

10.7.1. Interlocking PCC tiles flooring wherever shown / indicated in Main Drawings / Schedule of Finishes drawings shall be provided as per Schedule 'A' BOQ.

10.7.2. Refer Clause no 13.18 of MES Schedule, Part-I(2009).

10.7.3. The precast cement concrete interlocking blocks shall be glazed top with colour and shall be confirming to IS: 15658.

10.8. **WEAR RESISTANT FLOOR TOPPING**

10.8.1. **METALLIC FLOOR HARDNER FLOORING**: Metallic floor hardener flooring wherever indicated in Drawings/Schedule of Finishes shall be provided 10mm thick Wear-proof topping as specified (incorporating an approved metallic floor hardner), laid on concrete floors and pavings, and finished even and smooth and metallic floor hardener and cement shall be thoroughly mixed with proportion 1:2 by volume.

10.8.2. **NON METALLIC FLOOR HARDNER FLOORING:** Non metallic floor hardner flooring wherever indicated in Drawings/Schedule of Finishes shall be provided @5 Kg/Sqm (minimum) quartz based floor non metallic hardner or as per manufacturer's instructions.

10.9. **SKIRTING/DADO:-**Skirting/dado of various types as shown on Drawings and Schedule of Finishes shall be provided as specified here-in-after.

10.9.1. **CEMENT SKIRTING/DADO**: - 5mm thick Setting Coat in Cement Mortar 1:3 over 10mm thick Cement Screed in Cement Mortar 1:6 finished even and smooth using extra cement. Edges of skirting and dado to be chamfered.

10.9.2. **PLAIN CEMENT CONCRETE TILES SKIRTING/DADO**

10.9.2.1. Refer Clause No.13.16 and 13.39 of MES Schedule Part-I(2009).

10.9.2.2. Cement Concrete Tiles (Factory Made) of size 250 x 250 x 22mm shall be laid over 10mm thick screed bed in Cement Mortar 1:3, set, pointed and pointed in neat cement slurry grouting and polishing.

10.10. **STAIR CASE (TREAD, RISER AND LANDING)/STEPS**

10.10.1. Stair case (Tread, Riser and Landing)/Steps as shown on Drawing and Schedule of Finishes shall be provided with the following Specifications:-

10.11. **PLINTH PROTECTION WITH/WITHOUT DRAIN:-**Unless otherwise specified in drawing/here-in-before, plinth protection shall be 75 mm thick PCC M-10(Nominal Mix)(using 20 mm graded stone aggregate) over 75 mm thick (consolidated thickness) hard core over rammed earth. Plinth protection shall be laid in 1:40 slope. The concrete shall be laid at a stretch not exceeding 3 Metre in length. Top surface of concrete shall be finished even and smooth without using extra cement. Joints between adjoining slabs shall not exceed 10 mm in thickness and shall be filled with bitumen mastic i.e. bitumen and sand in proportion 1:2. Hard core for plinth protection shall be crushed/broken stone aggregate (at the discretion of contractor) conforming to IS-383, not exc. 63 mm gauge. Unless otherwise specified in drawings, the width of plinth protection shall be 750mm.

10.12. **PCC DRAIN/SAUCER DRAIN :-**PCC drain shall be in PCC M-10(Nominal Mix)(using 20 mm graded stone aggregate) and constructed wherever and to the extent as shown on drawing. The inside surface of the drain shall be finished smooth using Extra Cement. Unless otherwise specified in drawings, the width of plinth protection shall be 900mm.

10.13. CERAMIC / VITRIFIED TILES:

10.13.1. All type ceramic tiles shall be conforming to **IS : 15622 – 2017.**

10.13.2. Refer Clause no's 13.14, 13.15, 13.40 & 13.41 of MES Schedule, Part-I.

10.13.3. The size of tiles shall be provided as specified in Schedule 'A' (BOQ) but in no case size of tiles shall be less than 300 mm x 450 mm x 6 mm for ceramic wall tiles, 300 mm x 300 mm x 7 mm for non-skid ceramic floor tiles and 600 mm x 600 mm x 9-10mm for vitrified tiles.

10.13.4. The vitrified tiles shall be fully vitrified and fully polished.

10.13.5. The floor ceramic tiles shall be **matt finish** of best Quality Superior type.

10.13.6. The wall ceramic tiles shall be **'glossy' matt or semi-matt finish** of best quality Superior type.

10.13.7. The top surface of vitrified tiles shall be **double charged finishing**.

10.13.8. Texture and colour of all ceramic tiles shall be as approved by GE.

10.14. VITRIFIED PAVING OUTDOOR TILES :-

10.14.1. Vitrified paving tiles / Outdoor vitrified tiles shall be provided with complete all as specified in Schedule 'A' (BOQ). The tiles shall be antiskid, polka dot, vitrified, coloured tiles of size as specified in BOQ. The joints shall almost invisible. The tiles shall be laid all as specified in para 13.40.1 & 13.40.2 of MES Schedule Part I, set and jointed in grey cement and pointed in white cement mixed with pigment to match tile colour.

10.14.2. Vitrified paving outdoor tiles shall be conforming to **IS : 4457 – 2007.**

10.15. GLAZED CERAMIC TILES SKIRTING/DADO –

10.15.1. Refer Clause No. 13.14 of MES Schedule Part-I(2009). Irrespective of whatever shown on drawings/schedule of finishes, glazed ceramic tile skirting/dado shall be with coloured glazed ceramic tiles not less than 350mm x 450mm(except in toilet of size 300mm X 200mm) x 6 mm to 7mm laid over 10mm thick screed in Cement Mortar 1:3, set and jointed in neat cement slurry and pointed in coloured cement to match. The contractor may provide the tiles of thickness more than 5.5mm in case it is not available without any extra cost to the Govt. However, this will not be treated as any deviation.

10.15.2. Unless otherwise indicated in schedule of finishes/main drawings/TD drawing the height of skirting shall be 100mm high, height of dado for Bath room, Counter type WHB area & its side walls, Kitchen, Water cooler point & its side wall, Hand wash area, Servary, Pan wash & Meat preparation room shall be 2000mm and height of dado for WC/ remaining portion of toilet room shall be 1200mm.

11. PLASTERING & POINTING

11.1. MATERIALS

11.1.1. CEMENT – Cement shall be as specified for Cement Concrete Work here-in-before.

11.1.2. SAND – Refer Clause No. 14.5, 14.5.1 and 14.5.2 of MES Schedule Part-I.

11.1.3. INTEGRAL WATER PROOFING COMPOUND – Refer Clause No.4.8 of MES Schedule Part-I and as approved by GE.

11.1.4. WATER – Water shall be as specified for Cement Concrete Work here-in-before.

11.2. WORKMANSHIPS

11.2.1. GENERALLY

11.2.1.1. Refer to Clause No.14.16 of MES Schedule Part-I (as applicable) and as directed by the Engineer-in-Charge.

11.2.1.2. The thickness of plaster shall be finished thickness exclusive of dubbing out. Plastering and dubbing out shall however, be done in one operation.

11.2.1.3. SCAFFOLDING – Refer Clause No.14.12 of MES Schedule Part-I and all as directed by Engineer-in-Charge.

11.2.1.4. PREPARATION OF MORTAR FOR PLASTERING – Refer Clause No.14.13 of MES Schedule Part-I as applicable and all as directed by Engineer-in-Charge.

11.2.1.5. PREPARATION OF BACKGROUND FOR APPLICATION OF PLASTER – Refer Clause No.14.14 of MES Schedule Part-I as applicable and all as directed by Engineer-in-Charge.

11.2.1.6. **ONE COAT PLASTER WORK** – Refer Clause No.14.17 of MES Schedule Part-I and as directed by Engineer-in-Charge.

11.2.1.7. **TWO COATS PLASTER WORK** – Refer Clause No.14.18 of MES Schedule Part-I and as directed by Engineer-in-Charge.

11.2.1.8. **CURING** – Refer Clause No.14.24, 14.24.1 of MES Schedule Part-I and as directed by Engineer-in-Charge.

11.2.1.9. **WATER PROOFING PLASTER** – Refer Clause No.14.25 of MES Schedule Part-I and as directed by Engineer-in-Charge.

12. RCC ROOF

12.1. All RCC Roof Slabs where shown on drawings shall be having the slope as indicated in the drawings.

12.2. WATER PROOFING TREATMENT TO ALL RCC ROOFS

12.2.1. RCC Roof Slabs shall be laid to proper slope. The Water Proofing Treatment to RCC Roof shall be provided with the following Specification irrespective of whatever shown on Drawings or specified elsewhere in the Tender documents :-

12.2.2. FOR ACCESSIBLE ROOF (OPEN TERRACE) EXCEPT SLOPING RCC ROOF :-

(i) Roof surface shall be plastered with 15mm thick cement mortar(1:4) with water proofing compound as in screed bed over slab when Concrete is still green.

(ii) Curing of slab by ponding with water shall be done all as specified.

(iii) Cleaning of plastered roof surface and applying one coat of cold bituminous primer @ 0.25 Litres per Sqm conforming to IS: 3384-1986

(iv) A layer of APP based polymeric membrane, 3mm thick (minimum), weighing 3.5 kg/Sqm reinforced with polyester non woven fabric matt (weighing not less than 160 gms/Sqm) shall be applied over bituminous primer as specified above. Application of polymeric membrane shall be by torch application. The overlaps shall be minimum 100mm on the transverse direction and 75mm on longitudinal direction.

(v) 22mm thick plain cement concrete tiles (conforming to IS) of size 250mmx250mm using grey cement laid over a bedding layer of 15mm screed in Cement Mortar 1:6, jointing with cement slurry and joints shall be 4mm wide and sealed with 2 mm deep polysulphide compound.

12.2.3. SLOPING RCC ROOF:

12.2.3.1. Half round glazed concrete tile/ Magalore roof tiles over RCC sloping roofing wherever shown/ mentioned on main drawing/ Schedule of finishes drawings shall be provided half round glazed concrete tile tiles shall be laid over 10mm thick screed bed in Cement Mortar 1:3 over sloping RCC roof slab and colour of tiles shall be approved by GE .

12.2.4. FOR NON ACCESSIBLE ROOF

(i) All layer same as per accessible roof treatment except Sl No. (v) i.e. PCC Tiles.

(ii) Two coats of Bitumen based UV reflective and water proofing paint with aluminium finish over APP membrane and maximum coverage 8.00 Sqm per litre per coat. The Membrane and this paint shall of same manufacturer.

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NOTES ; ACCESSIBLE ROOF/ NON ACCESSIBLE ROOF

(a) For parapets membrane shall be taken to the height of PCC curved fillet as specified here-in-after and then inserted inside the parapet/PCC curved fillet in groove.

(b) Curved fillet in PCC (1:2:4) type BO of radius 75mm shall be provided at the junction of roof and parapet Wall/Mumty/Chimney/Other Vertical Surface and surfaces painted with hot paving bitumen of penetration value 30/40 conforming to IS:73 @ 1.20 kg/Sqm over a coat of bituminous primer, confirming to IS : 3384 applied @ 0.3 litre per Sqm.

(c) **ATACTIC POLYPROPYLENE POLYMER MODIFIED WITH NON - WOVEN POLYESTER MATT PREFABRICATED MEMBRANE(APP)** :-The APP modified membrane shall be 3mm thick APP (Atactic Polypropylene Polymer) modified prefabricated five layer, weight not less than 3.50 kg/Sqm water proofing membrane, black

finished reinforced with non woven polyester matt and weight not less than 160 gms/Sqm. APP modified membrane shall be as specified in para 11.44 (including sub paras) of MES Schedule Part I

(d) In no case damage / puncture of APP modified proofing membrane shall be permitted.

(e) The entire work of water proofing treatment to RCC roof slabs shall be carried out by engaging authorised applicator of the manufacturer. For this purpose, contractor shall submit the details of authorised applicator alongwith supporting documents to GE well in time and the work shall be executed only after written approval by GE, all materials used for water proofing treatment shall be of same manufacturer. However, the main contractor shall give guarantee for efficiency of water proofing treatment for a period of Ten Years as specified here-in-before.

(f) Random samples of Polymeric Water Proofing membrane, as selected by GE, shall be got tested from National Test House, SEMT Wing CME, Regional Research labs, IITs, National Institutes of Technology, NABL laboratories, Govt Engineering Collage or Any Govt laboratories where such facilities exists before incorporation in the work.

(g) In case the test result does not meet the specifications given above, the entire lot shall be rejected by the GE. In such a case, the contractor shall bring the fresh lot of desired quantity of APP modified Water Proofing Membrane and the same procedure of testing repeated till the time satisfactory test results are obtained. Nothing extra will be paid to contractor on this account.

(h) The work of fixing membrane shall be got executed through the authorized applicator of the manufacturer of the membrane.

12.2.5. WATER PROOFING TREATMENT TO SUNKEN SLABS

12.2.5.1. Unless otherwise shown on drawings, provide water proofing treatment to sunken slabs of upper floor toilets, bath, wash, WC as stated below :-

12.2.5.2. The slope of slab shall be 1:20 towards outer wall. The water proofing treatment of various layers from bottom to top on all surfaces shall be as under with provision of 25mm dia GI pipe(Medium Grade) spout projection 100mm beyond outer face of the wall of sunken floor: -

(a) First layer of slurry of cement @ 0.488 Kg/Sqm mixed with Acrylic based polymer modified cementitous composite compound @ 0.253 kg/ sqm. This layer shall be allowed to air cure for 4 hours.

(b) Second layer of slurry of cement @ 0.242 kg/sqm mixed with Acrylic based polymer modified cementitous composite compound @ 0.126 kg/ sqm. This layer shall be allowed to air cure for 4 hours followed with water curing for 48 hours.

(c) 200 micron thick polythene film pasted with the help of adhesives shall be provided over the treatment.

(d) PCC M-5 for filling in the sunken portion.

(e) Concrete for sunken slabs shall be mixed with WP compound as per manufacturers Instructions.

(f) Joint of Trap/WC/Soil pipes shall be properly sealed with sealing compound

(g) P/S traps and Nahani / Floor traps with long arm shall be provided as per latest instructions.

(h) Testing of all traps & joints and sunken slab shall be as per IS-1742 before filling of sunken slabs, including smoke tests.

12.3. TESTING OF WATER PROOFING TREATMENT

12.3.1. After the water proofing treatment is completed, leak proof tests shall be done by ponding. For this purpose, in roof, mud, mortar, fillet ponds shall be made longitudinally one meter a part over entire treated surface of roof to form pond of suitable size as directed by the Engineer-in-Charge. (These ponds shall be filled with potable water so that an average 50 mm (minimum 25 mm and maximum 75 mm) height of water is maintained during the test period.

12.3.2. Test shall be carried out continuously for a period of 48 hours. Any seepage notified shall be rectified by the contractor and making good the defective portion to entire satisfaction of the GE, who will pass this stage.

12.3.3. All mud fillet bends shall be removed and surface made clear and tidy after completion of satisfactory testing.

12.3.4. Satisfactory completion of test shall not absolve the contractor from his responsibility of rectification of defects, which may arise during Defect Liability Period.

12.3.5. PREVENTION AND CURE OF LEAKAGE, SEEPAGE, DAMPMESS

12.3.6. Preventation and cure of Leakage, Seepage, Dampness in the building shall be carried out as shown on Drawings.

13. WHITE & COLOUR WASHING, CEMENT BASE PAINT, DISTEMPERING, PLASTIC EMULSION PAINT, WEATHER PROOF PAINT

13.1. LIME :-Refer Clause No.15.2 of MES Schedule Part-I.

13.2. BLUE VITRIOL :-Refer Clause No.15.4 of MES Schedule Part-I.

13.3. WHITE WASHING / COLOUR WASH:- Refer Clause No 15.2,15.3,15.4&15.12 of SSR part I 2009 and Clause No 15.2,15.3,15.4&15.12 of SSR Part-I 2009.

13.4. CEMENT BASE PAINT

13.4.1. Refer Clause No.15.7 of MES Schedule Part I.

13.4.2. Cement base paint shall be of superior quality manufactured by standard firms given in APPENDIX ‘B' here-in-after.

13.5. SMART PAINTING METHOD FOR APPLICATION OF ACRYLIC EXTERIOR WEATHER RESISTANT PAINT(WEATHER PROOF PAINT AS SHOWN IN DRAWING AND AS SPECIFIED IN BOQ):-

13.5.1. External plastered surfaces of walls/RCC/cill/jamb/RCC facia of chajja/Roof/RCC wall/parapet/soffit of slab shall be applied weather coats paint as mentioned in Schedule of finishes.

13.5.2. PREPARATION OF SURFACE

Wall surface shall be dust free, even and smooth to the satisfaction of Engr-in-Charge and apply one coat of primer on even surface as per manufacturer’s instruction.

13.5.3. APPLICATION OF ACRYLIC EXTERIOR WEATHER RESISTANT PAINT

13.5.3.1. Prepared acrylic exterior weather resistant paint(weather proof paint) shall be applied with airless spray technology in 2 coat and capable of operating single spray gun with a suitable hose length. The spray machine should be reputed manufacturer with pro-guard technology and capability of tracking pressure as well as paint & primer usage. A uniformly finished surface without patches, acrylic exterior weather resistant paint(weather proof paint) drop etc shall be obtained.

13.5.3.2. The entire work of painting shall be carried out by engaging authorised applicator of the manufacturer. For this purpose, contractor shall submit the details of authorised applicator alongwith supporting documents to GE well in time and the work shall be executed only after written approval by GE, all materials used for painting shall be of same manufacturer. Guarantee for efficiency of painting shall be submit by the contractor as given by manufacture of minimum 4 years.

13.6. SMART PAINTING METHOD FOR APPLICATION OF DISTEMPER (AS SHOWN IN DRAWING AND AS SPECIFIED IN BOQ)

13.6.1. PREPARATION OF SURFACE

Wall surface shall be dust free, even and smooth to the satisfaction of Engr-in-Charge and apply one coat of primer on even surface as per manufacturer’s instruction.

13.6.2. APPLICATION OF DISTEMPER

13.6.2.1. Prepared distemper shall be applied with airless spray technology in 2 coat and capable of operating single spray gun with a suitable hose length. The spray machine should be reputed manufacturer with pro-guard technology and capability of tracking pressure as well as paint & primer usage. A uniformly finished surface without patches, roller marks distemper drop etc shall be obtained.

13.6.2.2. The entire work of painting shall be carried out by engaging authorised applicator of the manufacturer. For this purpose, contractor shall submit the details of authorised applicator alongwith supporting documents to GE well in time and the work shall be executed only after written approval by GE, all materials used for painting shall be of same manufacturer. Guarantee for efficiency of painting shall be submit by the contractor as given by manufacture.

13.6.2.3. Notes for Smart painting method specified at para 13.5 & 13.6 here-in- before:

(a) Sanding of surface to be done by mechanical sander with 1500 - 2700 rpm, with internal dust extraction system for clean environment certified by paint manufacturer.

(b) The contractor shall submit the MoU letter from paint manufacturer and also with their authorized applicator as specified in tender before commencement of work.

(c) The work pertaining to finishing services shall be executed from authorized applicator through manufacturer of which the materials .i.e. Paint, Distemper, Putty etc. is proposed to be used by the contractor and approved by Garrison Engineer.

(d) Contractor shall submit approved paint manufacturer warranty card at the time of submission of RAR & final bill.

(e) Authorised applicator of paint manufacturer shall have valid authorization letter from manufacturer (minimum regional HQs) issued not earlier than 30 days from start date of said finishing work. The authorization letter needs to also be valid for the complete period of execution.

(f) The accessories for applying the finishes shall be able to spray in hard to reach areas and penetration of finish on the surface so as to achieve uniform application and quality surface finish.

(g) Equipment shall be certified by approved manufacturer.

14. GLAZING

14.1. SHEET GLASS

14.1.1. Refer Clause No.16.2 and subsequent Clause No. thereof of MES Schedule Part-I.

14.2. FIGURED GLASS (PIN HEADED GLASS)

14.2.1. Refer Clause No.16.3 of MES Schedule Part-I.

14.3. SILICONE SEALANT :- Silicone sealant shall be **silicone polymer base sealant.**

14.4. PUTTY :- Refer Clause No.16.5 of MES Schedule Part-I.

14.5. PREPARATION OF GLAZING :- Refer Clause No.16.7.1, 16.7.2 to 16.7.4 of MES Schedule Part-I.

14.6. GLAZING IN STEEL SURROUNDS :- Refer Clause No.16.10 and subsequent para thereof of MES Schedule Part-I.

14.7. GLAZING IN WOOD SURROUNDS :- Refer Clause No.16.9 and subsequent para thereof MES Schedule Part I.

15. PAINTING

15.1. PAINTS AND ALLIED MATERIALS

15.1.1. Refer Clause No.17.2 of MES Schedule Part-I.

15.1.2. Paint shall be of first quality and manufactured by Standard firms given in APPENDIX ‘B' to tender documents here-in- after.

15.2. PAINTING WORK GENERALLY :- Refer Clause No.17.3 and subsequent thereof MES Schedule Part-I.

15.3. PAINTING TO WOOD WORK

15.3.1.1. Refer Clause No.17.6 and subsequent thereof MES Schedule Part-I.

15.4. PAINTING TO STEEL AND IRON WORK .

15.4.1.1. Refer Clause No.17.8 of MES Schedule Part-I.

15.4.1.2. Painting for **Pressed steel windows/ventilators** frame & shutter, paint shall be applied by spray gun all as specified in appendix B and other relevant clause of IS 1477 Part-II. Colour of paint shall be approved by GE.

15.5. FRENCH POLISH

15.5.1. Refer Clause No.17.7.4 and subsequent thereof MES Schedule part I

15.6. COAL TAR PAINTING :- Coal tar paint shall conforming to IS-290. Tarred surfaces except wood shall be sanded before use.

15.7. CREOSOTING:- Hidden wood work shall be given two coats of creosote oil.

16. SANITARY APPLIANCE

16.1. GENERAL

16.1.1. Refer Clause No.18.32.1 and 18.32.1.1 of MES Schedule Part-I.

16.1.2. These shall be first quality, white, vitreous china & ISI marked.

16.2. SQUATTING PAN (ORISSA PATTERN) :-

16.2.1. Squatting pans water closet shall conform to IS-2556 (Part-III) and their pattern shall be Orissa pattern size as mentioned in Schedule 'A' (BOQ) and shall be as per MES schedule Part-I.

16.3. EUROPEAN TYPE WC (PEDESTAL PATTERN):-

16.3.1. Wash down water closet shall conform to IS-2556 (Part-II) and their pattern shall be Pattern – 4 (Pedestal WC with independent cistern and concealed S-trap or P-Trap), white, 410 mm high (minimum).

16.3.2. Refer Clause No. 18.32.2 of MES Schedule Part-I.

16.4. UNIVERSAL WATER CLOSET (ANGLO-INDIAN TYPE) :

16.4.1. Universal Water Closet (Anglo-Indian Type) shall conform to IS-2556 (Part-15) and their pattern shall be Size shall be as specified in Schedule 'A' (BOQ) with S-trap or P-Trap, and shall be as per MES schedule Part-I.

16.5. PLASTIC WATER CLOSET SEATS AND COVERS

16.5.1. Refer Clause No. 18.36 of MES Schedule Part-I.

16.5.2. Plastic water closet seat and cover shall be ISI marked on ISI 2548 recessed bottom made out of moulded poly propylene.

16.6. LOW LEVEL FLUSHING CISTERN

16.6.1. Refer Clause No. 18.34 and 18.87 of MES Schedule Part-I.

16.6.2. The flushing cistern shall be of PVC "feather touch" with symphonic fittings ISI marked moulded from vergin poly propylene.

16.7. URINAL

16.7.1. Refer Clause No. 18.32.7 and 18.88.8 of MES Schedule Part I.

16.7.2. These shall be with grating and unions, waste pipe 32mm dia nominal bore GI medium grade of required length.

16.7.3. **PARTITION WALL IN URINAL**. Wherever partition wall in urinal shown on drawings, partition wall in urinal shall be of 18mm thick marble.

16.7.4. PVC FLUSHING CISTERN FOR URINAL

16.7.4.1 Refer Clause No. 18.34, 18.88.2 and 18.88.3 of MES Schedule Part-I.

16.7.4.2 Flushing cistern shall be PVC automatic flushing cistern.

16.7.4.3 The flushing cistern shall be of PVC "Feather Touch" operated bearing IS-7231, diaphragm bearing IS-13049 and detrim valve, float valve.

16.8. WASH BASIN

16.8.1. WHB shall be provided complete all as specified in Schedule 'A' (BOQ).

16.8.2. Refer Clause No. 18.32.5 and 18.89 of MES Schedule Part-I.

16.8.3. Wash hand basin (WHB) shall conform to IS-2556 (Part-IV), white.

16.9. PVC CONNECTOR

16.9.1. Refer Clause No. 18.5 of MES Schedule Part-I.

16.10. MIRROR

16.10.1. Refer Clause No. 18.37, 18.38, 18.91 and 18.92 of MES Schedule Part-I.

16.10.2. The mirrors shall be of indigenous of make as per APPENDIX 'B', copper plated at the back free from ripples, spats, bubbles, and other defects with edges slightly rounded off around.

16.11. TOWEL RAIL

16.11.1. Refer Clause No. 9.24 of MES Schedule Part-I.

16.12. GLASS SHELVES.:-Glass shelves shall be of 6.0mm thick plain glass (minimum).

16.13. UNPLASTICIZED POLYVINYL CHLORIDE (PVC-U) PIPE FOR RAIN WATER SOIL/WASTE/VENT PIPES & FITTINGS

16.13.1. Refer Clause No. 11.20, 18.67.2 and 18.67.2.1 of MES Schedule Part-I and IS 13592-2013.

16.13.2. COLOUR OF PIPE:- Surface colour of the pipes shall be dark shade of grey.

16.13.3. RAIN WATER PIPE:- Rain Water Pipe shall be unplasticized polyvinyl chloride (PVC-U) pipes, ISI marked.

16.13.4. SOIL/WASTE/VENT PIPES & FITTINGS :-

16.13.4.1. Soil/waste/vent pipe shall be unplasticized polyvinyl chloride (PVC-U) pipes. In case of non availability of any diameter (Nominal outside diameter) mentioned the next available diameter (Nominal outside diameter) as per IS 13592 shall be provided without any price adjustment.

16.13.5. The contractor shall submit the manufacturer's test Certificate in original along with the test sheet giving the results as per IS 13592 and duly signed by the manufacturer with each consignment.

16.13.6. TESTING: - After laying of PVC (SWR) pipes or any other type of pipes, smoke testing shall be done all as specified in Clause No 18.79.2 and water testing shall be done as specified in Clause No 18.79.3 of MES SSR Part-I (2009).

16.14. GULLY TRAPS

16.14.1. Gully traps shall be of unplasticized polyvinyl chloride (PVC-U) with `P' or `S' trap with square/round mouth conforming to IS-651. PVC Gratings shall be of heavy quality as approved by GE. Gully Traps shall be provided with two cast iron gratings of appropriate size.

16.15. NAHANI TRAPS/FLOOR TRAPS

16.15.1. Refer Clause No. 18.25 of MES Schedule Part-I and shall conform to IS 3989.

16.15.2. Nahani Trap/Floor Trap shall be long arm type so that the joint must come outside the external walls.

16.16. (PVC-U) COWLS

16.16.1. Refer Clause No. 18.24 of MES Schedule Part-I.

16.17. STAINLESS STEEL PEG SET :-Stainless steel peg set shall have 3 Pegs (if otherwise not specifically indicated/shown on drawing) on stainless steel plate. The steel plate shall be fixed on walls with rawl plug.

17. MISCELLANEOUS ITEMS

17.1 MILD STEEL RAILING:- Mild Steel Railing (other than detail shown on main drawing/specified here-in-before) shall be provide as per drawing No. CESZ/2015/TD/016.

17.2 STAINLESS STEEL RAILING:-

17.2.1 Stainless steel railing shall be provide as per main drawing or as per drawing No. CESZ/2015/TD/016.

17.2.2 The stainless steel railing shall be of Grade SS304 and shall be provided as per the design given in drawings at various locations given in drawings. The pipes shall be seamless with elegant glossy finish without any marks on surface. The railings shall be fixed with stainless steel accessories only providing adequate strength as per the purpose for which the railing is

being provided. The minimum weight per running meter for 20, 40 and 50mm pipes shall not be less than 1.25, 3.10 and 3.9 Kg respectively. The wall thickness of any component of the railing shall not be less than 1.65mm.

17.3 MILD STEEL CHROMIUM HAND RAIL :-

17.3.1 Whenever shown on drawings mild steel (MS) chromium hand rail shall be of 65 mm nominal bore, light grade with hard chromium plated conforming to IS-1986 and thickness of coating shall not be less than 20 micron and shall be procured from standard manufacturers as approved by GE.

17.3.2 The contractor shall be submit original manufacturer's test Certificate duly signed by the manufacturer before any payment from Department.

17.4 CI STEPS/RUNGS:- Wherever MS rungs/steps shown in drawings of Septic Tank, Manhole, etc. and/or other locations, the same shall be cast iron conforming to IS-5455. The CI rungs/steps shall of be pattern 2 as per IS having minimum weight 5.30 Kg/Each. The steps shall be treated with 2 coats of tar.

17.5 RCC SHELVES

17.5.1 RCC shelves shall be constructed all as shown on drawing and as specified here-in-below:-

(a) All RCC works shall be M-25 Design mix irrespective of what is shown on drawings.
(b) Reinforcement shall be as specified here-in-before.

17.6 CRUMPLE JOINT

17.6.1 Crumple joints shall be provided as per drawings.

17.6.2 Unless otherwise shown on drawing, the surface of the plaster/RCC to come in contact with the PVC element shall be cleaned and any damage/undulations shall be repaired by application of rich cements and mortar so that the surface is made smooth and monoplane. The right angle between the inner face of the joint and the exposed, face on which the flange will rest, shall also be checked for correctness within reasonable limit. By running a template along the joints the gap shall be ensured that generally the gap shall be about 2 to 4mm lesser in width than the outer diameter of the fixocap bulb, outer if necessary where the joint is wider than the fixocap outer diameter, plastering with rich mortar shall be resorted to, so that the fixocap, when pushed into the joints, is firmly in contact with the walls/RCC face. Where the external face of the masonry plaster/ RCC is wavy and irregular, the fixocap after insertation shall be bonded to the masonry/RCC by use of an adhesive known as Chemistik-75 which shall be applied to the masonry/RCC surface and the PVC face which will come into contact with the RCC/masonry. The application is done with a brush and actual fixing of the fixocap is carried out when the adhesive is semi dry (by contact with a finger forms threads when the finger is deflected). The adhesive shall be used on one side of fixocap, leaving the other side of the joint free for movement. The face on which the ahdesive is applied shall be pressed firmly against the wall/RCC element, by means of a batten strutted by diagonal balleys for about 4 hours and then removed.

17.7 NUMBERING OF BUILDING: Numbering of Buildings shall be done in all Buildings on the format as shown on Drawing No TD/N/76 Sheet No. 1/1 with following specification and cost of the same is deemed to be included in quoted Lump Sum of Building :-

(a)	Plaster	:	Raised plaster 10mm thick in Cement Mortar 1:3
(b)	Letters	:	Letter shall be painted in black colour and white ground

17.7.1 The Letters/Words/Figure to be written shall be intimated by the Engineer-in-Charge. The cost of the same is deemed to be included in Cost of Buildings.

17.8 MANHOLES

17.8.1 Manholes shall be built all as specified in Clause No. 18.78 of MES Schedule Part-I, shown on Drawings and specified in Schedule ‘A'.

17.8.2 Manholes and Drain Pipes shall be tested for Water Test all as specified in Clause No. 18.79.3 of MES Schedule Part-I.

17.8.3 CI Rungs/Steps shall be of Cast Iron as specified here-in-before. The Steps/Rungs shall be treated with Two Coats of Tar. The Cost of which is deemed to be included in the Unit Rate/Amount.

PARTICULAR SPECIFICATIONS (CONTD…..)

17.9 **CUP BOARD** :- Wherever Cup board has been shown in drawing shall be all as shown in main drawing and if details of cupboard are not shown in main drawing then shall be all as shown on Drawing No. CESZ/2015/TD-012 with the following Specifications :-

(a) Shutter, Partition , Shelves — 19/18mm thick pre laminated plywood exterior grade, pre-laminated both side with decorative choice lamination.

(b) CURTAIN ROD — Aluminium curtain rod outer dia 19/20mm and inner dia 16mm with powder coated paint thickness 40-60 microns and pipe shall be 1.6mm thick

17.10 **DISPLAY BOARD/NOTICE BOARD**: Wherever display board / notice board as shown on main architectural drawing, the display board / notice board shall be provided as per following specifications, main drawing and TD drawing no CESZ/2015/TD/024:-

(i) All round frame shall be of anodized extruded aluminium alloy hollow section designation and size of the frame as shown on TD drawing.

(ii) Galvanized steel sheet shall be 0.25 + 0.02 mm thick.

(iii) The board shall be soft board 9mm thickness.

(iv) The pinning surface shall be made from colored blazer cloth of color navy blue/green/brown as decided by GE all as per manufacturer specification.

(v) The pin up board and display board shall be of make WHITEMARK or equivalent as approved by GE.

(vi) Fixing of display board shall be all as shown on drawing and as directed.

(vii) Rest all as specified and shown on drawing.

17.11 **STAINLESS STEEL SINKS WITH DRAINAGE BOARD**:

17.12.1. Refer Para 18.100 of MES SSR Part I. The sink with draining board shall be type ‘C’ and overall size as specified in Schedule ‘A’ (BOQ). The sink with drainage board shall be provided with plug, washer and waste and fixed with a pair of MS cantilever bracket. The sink shall be provided with waste pipe of 32mm bore light grade GI pipe with fittings, length not less than 1m, duly connected to union. The surface of the sink shall be plastic coated to make it safe and scratch free.

17.13. **STAINLESS STEEL SINK WITHOUT DRAINAGE BOARD**:

17.13.1. Stainless steel sink shall be made of sheet thickness 1mm (min). The overall size shall be as specified in Schedule ‘A’ (BOQ). The sink shall be provided with plug, washer and waste and fixed with a pair of MS cantilever bracket. The sink shall be provided with waste pipe of 32mm bore light grade GI pipe with fittings, length not less than 1m, duly connected to union. The surface of the sink shall be plastic coated to make it safe and scratch free. All other details shall be mentioned on SSR Part I clause 18.100.

17.14. **STAINLESS STEEL PLATE RACK**:

17.14.1. Stainless steel plate rack shall be made of sheet thickness 0.80mm (min). The size of plate rack shall be as specified in Schedule ‘A’ (BOQ). The surface of the plate rack shall be plastic coated to make it safe and scratch free. All other details shall be mentioned on SSR Part I clause 18.100.

18. **TUBES AND FITTINGS**

18.1. GI Tube and Fittings shall be of Mild Steel Galvanised Medium Grade bearing IS-1239 mark. Fittings conforming to IS-1239 if not available readily may alternatively comply with the requirement of IS-1879 for malleable Cast Iron Fittings but these shall also be ISI marked. GI Pipe shall be laid in floor/ on surface/ concealed in Wall/Floor conforming to provision given in IS-2065 and IS-12183.

18.2. **CAST IRON SOIL, WASTE AND VENTILATING PIPE:-** Refer clause No. 18.23 of MES Schedule Part-I.

18.3. **CAST IRON COWLS:-** Refer clause No. 18.24 of MES Schedule Part-I.

18.4. **CAST NAHANI TRAPS:-** Refer clause No. 18.25 of MES Schedule Part-I.

PARTICULAR SPECIFICATIONS (CONTD…..)

18.5. **CAST FLOOR TRAPS:-** Refer clause No. 18.26 of MES Schedule Part-I.

18.6. **HDPE PIPES:-** Refer clause No. 18.6 of MES Schedule Part-I and shall be confirm to IS 16098-2013.

18.7. **BIB TAPS AND STOP VALVE :-** These shall be screw down type and shall bear ISI marking. Bib taps & Stop valve shall be polished bright.

18.8. **FANCY BIB TAPS AND STOP VALVES :-**They shall be cast copper alloy screwed down type and shall conform to IS-8931. They shall be approved by the GE and shall be nickel chromium plated as given in Schedule 'A' (BOQ).

18.9. **GATE VALVES AND GLOBE VALVES :-** These shall be standard quality Class I conforming to IS:778, Specifications and as approved by the GE. Valve shall be of Class I suitable for 1.0 MPa Working Pressure.

18.10. **BALL VALVE :-** Ball Valves for Water Tanks shall be high pressure type of brass with Polythene Ball conforming to IS-1703.

18.11. **SHOWER ROSE :-** Shower Rose shall be brass chromium plated as specified in Schedule 'A' (BOQ).

18.12. **BOTTLE TRAP :** Refer Clause No. 18.22.1 of MES Schedule Part-I.

18.13. **TOILET PAPER HOLDER :** Refer Clause No. 18.39 of MES Schedule Part-I.

18.14. **GLASS SHELF ASSEMBLY :** Refer Clause No. 18.37 of MES Schedule Part-I.

18.15. **WATER STORAGE TANK** :-

18.15.1. Refer Clause No. 18.60 on of MES Schedule Part-I.

18.15.2. HDPE water tanks shall be provided all as per manufacturer's instructions. Tank shall be of rotational moulded polyethylene water storage tanks in double layers otherwise specified in Schedule 'A' (BOQ) and cylindrical vertical type with manhole lids (closed top) with ISI marked and Make as given in APPENDIX 'B' here-in-after.

18.15.3. **JOINTING OF PIPES TO STORAGE TANKS :-** For jointing steel pipes to storage tanks, the end of the pipe shall be threaded, passed through a hole in the tank and secured by back nuts both inside and out side. The Pipe End shall be flush with the face of the inside back nut to obviate corrosion of the pipe threads.

19. **INTERNAL ELECTRIFICATIONS**

19.1. **GENERAL REQUIREMENTS**

19.1.1. **KNOWLEDGE OF WORK :-** The tenderers shall be deemed to have thorough working knowledge of the exact requirement of the job. In case of any doubt clarification may be sought from CESZ before tender is submitted. Tenderers are advised to visit the site and get themselves conversant with site conditions. No claim what so ever shall be entertained/accepted in case the tenderer is coming up with an excuse of having incomplete knowledge of work involved.

19.1.2. **WORK IN ACCORDANCE WITH SPECIFICATION :-** Work under this contract shall be carried out as per brief description given in Schedule 'A' (BOQ), Particular Specifications. Work shall also be in conformity with IE Rules Latest Edition, relevant IS specification and MES Schedule which are forming part of the contract. The entire work shall be carried out properly with due care and under the supervision of competent technical personnel.

19.2. **GENERAL**

19.2.1. Refer Clause No. 19.2 of MES Schedule Part-I and requirements of Clause No. shall be strictly followed.

19.2.2. Supply and Installation of Electrical Equipments and Accessories shall strictly comply with the provisions contained in the Latest Edition of India Standard Code of Practice and/or IS Specification as applicable, Works except where such Regulations and Rules are modified by these Particular Specifications.

19.2.3. All Electrical Work shall be carried out in conformity with the requirements of the Indian Electricity Act 1910, Indian Electricity Rules 1956 framed there under, Indian Standard Codes of Practice, Fire Insurance Act as applicable and also the relevant Regulations for Electric

Supply and Installations of all Electrical Equipments, Fittings and Accessories as amended from time-to-time.

19.2.4. The supply and installation of all Electrical Equipments and Accessories shall strictly comply with the provisions contained in the latest edition of Indian Standard code of Practice IS Specification as applicable to above mentioned work except where such regulations and rules are modified by these specifications.

19.2.5. **QUALIFIED ELECTRICAL SUPERVISORS / ENGINEERS :-** Refer Condition 25 of General Conditions of Contracts, IAFW–2249. All Electrical Works shall be carried out by properly Skilled and Licensed Electricians under the supervision of qualified Electrical Supervisors/ Engineers. The Contractor shall on demand shall produce such evidence of qualifications of his Workmen/ Supervisors / Engineers, either at the commencement of Work or at any time thereafter/ during currency of the Contract. The entire Work shall be High Class with the best Workmanship and to the entire satisfaction of Engineer-in-Charge/ GE.

19.3. INTERNAL WIRING

19.3.1. MATERIALS

19.3.1.1. All the Electrical Accessories incorporated in the work shall conform to the relevant IS and shall bear ISI mark. The steel conduit shall be of maximum 16 SWG thickness and rigid PVC conduit shall be suitable for heavy mechanical stresses as given in IS and will be ISI marked.

19.3.1.2. Work under this part will cover the point wiring and fixing of various light fittings in the building as per drawing. Work will be carried out as per provision of IS 732-1989. IE Rules as per MES Schedule and as per Sound Engineering Practices.

19.3.1.3. The general layout of light fittings is shown in drawing. The location of fitting may be altered by Engineer- in-Charge to suit the site requirement in which case the contractor shall have no claim for any extra amount on account of such changes in point wiring.

19.3.1.4. No price adjustment in rates shall be made on account of changes in location of fittings and accessories shown on drawings.

19.3.1.5. Recessed terminal boxes for housing switches, socket outlets, power plug, regulator etc. shall be provided flush with the walls.

19.3.1.6. The internal wiring shall be in concealed PVC conduits. The PVC conduits and its fittings shall conform to the requirements of IS-9537-1983 and IS-3480-1966. PVC conduit shall be suitable for heavy duty as asked in IS. The size of wiring cable shall be as stipulated in Schedule 'A' (BOQ). The PVC unsheathed cable should conform to IS-694-2010 and suitable for 650/1100 volt grade. All cables and switches in work shall have ISI mark.

19.3.1.7. It shall however be noted by the contractor that layout of the fittings shall be completed to match the casted structural members and cost of extension/supports considered unavoidable and essential by the tenderer shall deemed to be inclusive in the unit rate quoted for wiring.

19.3.1.8. The switch boards shall be located in such a manner that they are easily accessible and such that there is adequate working space around the switch boards. The tentative location shall be actually marked on the walls in chalk and written approval of Engineer-in-Charge shall be obtained before proceeding with the erection of the switch boards. The Engineer-in-Charge and the contractor shall associate themselves during the civil work so as to finalize the firm locations of switch boards, DBs, etc in consultation with departmental staff while executing the civil work.

19.3.1.9. All conductors used in point wiring/ sub-main wiring shall be FR-LSH (Flame retardant low smoke and halogen) Copper conductor conforming to IS 694-2013 and shall have ISI mark. Insulated wire used in point wiring will have colour code matching with phase i.e. wire starting from 'R' phase will have red colour and similarly with 'Y & B' phase will have yellow and blue respectively. In neutral, only black colour wire will be used. No joints in PVC sheathed Copper conductor for phase and neutral or in earth wire is permitted. All stranded wires will be crimped with suitable Copper lug before they are connected to MCB. Green colour wire shall be used in earth wire along with other wires.

19.3.1.10. The cable and the flexible cords shall be stranded Copper conductor. Same colour code suggested in above said para shall be followed.

PARTICULAR SPECIFICATIONS (CONTD…..)

19.3.1.11. The surface wiring for points outside the building shall run on internal face of wall as far as practicable. Only at terminal points it shall be taken out by puncturing the wall and connect with light fitting/switches.

19.3.1.12. The maximum number of point wiring of light/socket 5/6 Amp on an independent circuit from DB shall not exceed 8 Nos in any case and not more than two power points in an independent circuit from DB.

19.3.1.13. Each circuit will have its independent neutral and looping is only permitted with in the circuit only. For example if five circuits are going out from one distribution board then five neutral wire will also go along with phase from neutral bus-bar.

19.3.1.14. For concealed wiring, it is advisable not to lay conduit horizontally in wall but all horizontal running of conduit should be done in RCC slab. As far as possible, only vertical run of conduit should be laid in the wall.

19.3.1.15. Wherever more than two conduits are running together vertically concealed in wall, tenderer is advised to wrap the PVC conduits in wire mesh and then do plaster otherwise cement plaster may not stick to the PVC conduit and it is likely to come out. Cost of wire mesh used is deemed to be included in the quoted cost of wiring.

19.3.1.16. Wherever Tube Light Fitting has been shown suspended by Conduit from Soffit of Roof Slab, to fix Conduit, Contractor shall fix Mild Steel Flat of suitable size in Slab while casting the Roof Slab/Floor Slab.

19.3.1.17. PVC Conduit used in Point Wiring for flushed Florescent Fitting and Ceiling Fan fixed at False Ceiling Area, can be laid on False Ceiling.

19.3.1.18. Mild Steel Conduit used for suspension of Tube Light Fittings shall conform to IS-9537.

19.3.2. **PVC CONDUIT**:

19.3.2.1. PVC conduit to be used for point wiring/sub main wiring shall be 'Heavy Grade' and all as specified in Para 19.29 of MES Schedule Part-I.

19.3.2.2. PVC circular box and bend used for PVC conduit shall also be conforming to IS-3419.

19.3.2.3. PVC terminal box shall be ISI Marked.

19.3.3. **FLEXIBLE CABLE** Flexible cable shall be with tinned annealed stranded copper conductors of 1 Sqmm 3 runs (phase, neutral and earth) with PVC insulation.

19.3.4. **CEILING ROSES**: Ceiling rose shall be surface type and shall comply with IS-371 having three terminals plate and as specified in Para 19.32 of SSR 2009 Part – I.

19.3.5. **EARTH WIRE**

19.3.5.1. Copper PVC insulated wire conforming to IS-694 green in colour of size given in Schedule 'A' (BOQ) will be continuously laid along with other colour coded wire of the each circuit as asked in Schedule 'A' (BOQ).

19.3.5.2. The cables shall be with multistranded copper conductor PVC insulated unsheathed suitable for 1100 volts, for PVC conduit as specified. For point/sub main wiring color code of cables as per IE rule shall be maintained.

19.3.5.3. Earthing shall conform to IS - 3043. Min. distance between Two Earth Pits shall be kept 4.5 Metre.

19.3.6. **MCB AND DISTRIBUTION BOARD**

19.3.6.1. Extra length of cable shall be kept inside the distribution board for connection with MCB. All cable terminals with MCB shall be provided with suitable crimping lugs. All MCB and their distribution board shall comply with IS 8828-1978 and shall have minimum rupturing capacity as 10.0 KA.

19.3.6.2. All MCB, DBs shall be located at Load Centre (as per as possible) irrespective of whatever is shown on Drawings.

19.3.6.3. All MCB, DBs shall be same manufacturers.

19.3.7. **SWITCH BOXES:-**The Switch Box to accommodate Switches for Light, Fan and Socket shall be fabricated with 16 SWG sheet conforming to IS-5133. The box will have cover of minimum 3 mm thick Laminated Sheet of suitable shade approved by GE. The Switch Boxes shall be of adequate sizes to house all Switches/Regulator/ 3 Pin Plug. If space inside the box

is found to be insufficient due to number of Wires then Two Switch Boxes shall be provided side by side. Before fixing the box, outer and inner surface will be treated against corrosion.

19.3.8. **FIXING OF FITTINGS AND FANS:-**All tube light fittings if suspended will be fixed at 2.60 Metre from finished floor level or as directed by Engineer-in-Charge. All ceiling fans will be kept at 2.70 Metre from finished floor level or as directed by Engineer-in-Charge. Air circulator will be fixed on wall with proper MS bracket grouted properly either with MS bolt or split bolt. Cost of MS brackets is deemed to be included in the quoted cost of air circulator.

19.3.9. **ELECTRICAL TEST**

19.3.9.1. On completion of wiring, following tests shall be carried out conforming to IS 732.

(a) Insulation Test.

(b) Polarity Test

(c) Testing of Earth Continuity

19.3.10. **FLAME PROOF POINT WIRING, FITTINGS, LIGHT FITTING AND ACCESSORIES:-**

19.3.10.1. All components shall be conforming to IS-2148 &13346 and where applicable shall be ISI marked.

19.3.10.2. Flame proof light fittings shall be cast aluminum alloy LM6 housing with epoxy heat resistant paint.

19.3.10.3. All flame proof any type switches shall be cast aluminum alloy LM6 housing with epoxy heat resistant paint.

19.3.10.4. Manufacturer’s test certificate of flame proof light fitting shall be submitted by the contractor before payment from Department.

19.3.11. **LED LIGHTS/LUMINARIES**:

19.3.11.1. All LED lights/luminaries shall be meet the provision laid down in IS No-16101, IS No - 16102 Part I/II, IS No -16130 Part I, IS No -15885 Part II/ Sec 13, IS No -16104, IS No - 16105, IS No -16106, IS No - 16108: 2012 and LM 79 & LM 80.

19.3.11.2. All LED light products shall be replacement warranty of complete **LED luminaries fitting shall not less than three years from the date of completion** and Life expectancy of LED's minimum 50,000 burning hrs. The product will replace free of cost in the following cases :-

(i) Manufacturing defects.

(ii) Failure due to mechanical and electrical impact.

(iii) Drop in lumen (or lux at lm) below 90% of claimed values of lumen (or lux at 1m) of the LED/Luminaries.

19.3.11.3. The. above warranty shall not include natural disaster, sudden surge in voltage recorded by digital electronic meter or due to wrong handling of light fittings

19.3.11.4. Manufacturer’s test certificate of LED light fitting shall be submitted by the contractor before any payment.

19.3.11.5. Guarantee certificate of LED light fitting shall be submitted by the contractor before any payment. Contractor shall be responsible for replacing at his cost any parts which are rendered defective and unserviceable during the above period.

19.4. **INSULATION TEST**

19.4.1. The insulation resistance shall be measured by applying between earth and the whole system of conductor or any section thereof with all fuses in place and all switches closed, and except in earthed concentric wiring, all lamps in position or both poles of installation otherwise electrically connected together, a DC voltage of not less than twice the working voltage, provided that it does not exceed 500 volts for medium voltage circuits. Where the supply is derived from three-wire (AC or DC) or a poly-phase system, the neutral pole of which is connected to earth either direct or through added resistance the Working Voltage shall be deemed to be that which is maintained between the outer or Phase Conductor and the Neutral.

19.4.2. The insulation resistance in megohms of an installation measured as in Particular Specification here-in-above shall not be less than 50 divided by the number of points on the circuit, provided that the whole installation need not be required to have an insulation resistance greater than one megohm.

19.4.3. **POLARITY TEST:-** Each Switch of Light Fitting, Ceiling Fan, Exhaust Fan and other Fixtures shall be connected with phase and not with neutral. For this Each Point should be tested by opening Switch Boxes and recorded in the Test Sheet.

19.4.4. **EARTH CONTINUITY TEST**

(a) The Earth Resistance of Each Electrode shall be measured.

(b) The Earth Resistance of Earthing Grid shall be measured.

(c) All Electrode shall be connected to the Grid and the Earth Resistance of the entire Earthing System shall be measured.

(d) These Test shall preferably be done during the Summer Months.

19.4.5. **CABLES**:-Cable installation shall be checked as follows :-

(a) It shall be ensured that the Cables conform to the relevant Indian Standards. Tests shall also be done as laid down in IS:1255-1983. The insulation resistance before and after the Tests shall be checked.

(b) The Insulation Resistance between each Conductor and against Earth shall be measured. The Insulation Resistance varies with the type of insulation used and with the type of insulation used and with the length of Cable.

(c) The following empirical Rule gives reasonable guidance

$$\text{Insulation resistance in megohms} = \frac{10 \text{ x voltage in KV}}{\text{Length in KM}}$$

19.4.5.1. The Test Results shall be recorded and signed by Engineer-in-Charge and the Contractor.

19.4.5.2. Electrical gadget (tools/devices), material & labour required to carry out the above test shall be the responsibility of the contractor.

19.5. **ELECTRONIC BALLAST**

19.5.1. Manufacturer of Electronic Ballast should have tested its proto type at any of the Electronic Regional Test Laboratory, Deptt of Electronics.

19.5.2. This should meet the standard laid out in IS-1534-1977 and IS-13021 Part I & II 1991.

19.5.3. In case of failure of the lamps connected across the ballast, the circuit should be designed to switch OFF to prevent repeated attempts at reignition. The ballast should switch on automatically after replacement of failed lamps without the need or any manual resetting. There should be provision of fuse within the ballast to prevent any Short Circuit from affecting the Main Supply. Power Factor of the system when used with filter coil should be > 0.96.

19.5.4. Ballast will have filter coil either fixed inside or outside the ballast. Ballast will offer min 32 Watts of power to fluorescent tube and tube should offer same lumen output as with 36 Watts TLD. The ballast should ensure a low stroboscopic effect of less than 4% meeting the requirement of IS-13021 Part II.

19.6. **DISTRIBUTION / FEEDER PILLARS BOX**

19.6.1. Refer clauses 19.21 of MES Schedule Part-I. , IS 5039 and TD drawing

19.6.2. The feeder pillars box shall be factory made and CPRI approved. Feeder pillars box shall be obtained from the specified manufacturers and shall be epoxy powder coated factory made and tested as per IS. All feeder pillars boxes shall have extra space for installing one MCCB in future.

19.6.3. The contractor shall design & submit drawings of panel board which shall be approved by GE before bringing same to the site.

19.6.4. The cable chamber of the board shall be with adequate space for housing cable and connecting for safety of working and for avoiding accidental contacts with live parts by providing barriers.

19.6.5. In case nothing is mentioned in Schedule ‘A’ (BOQ) bus bar shall be copper.

19.6.6. In case mounting of panel board has not been mentioned in Schedule ‘A’ (BOQ) the same shall be floor mounted and the fixing arrangement shall be provided by the contractor and the unit rate quoted shall be deemed to inclusive for the same.

19.6.7. Cable entry will be from the bottom, detouchable gland plate to be provided for ease of punching hole for entry of cable.

19.7. **LT SWITCH GEAR PANEL:-**

19.7.1. Refer Para (including sub paras) 19.100 of MES Schedule Part-I and IS 2147.

19.7.2. The panel board shall be factory made and CPRI approved. Panel boards shall be obtained from the specified manufacturers and panels will be epoxy powder coated factory made and tested as per IS. All LT panels shall have extra space for installing one MCCB in future.

19.7.3. The contractor shall design & submit drawings of panel board which shall be approved by GE before bringing same to the site.

19.7.4. The cable chamber of the board shall be with adequate space for housing cable and connecting for safety of working and for avoiding accidental contacts with live parts by providing barriers.

19.7.5. In case nothing is mentioned in Schedule 'A' (BOQ) bus bar shall be copper.

19.7.6. In case mounting of panel board has not been mentioned in Schedule 'A' (BOQ) the same shall be floor mounted and the fixing arrangement shall be provided by the contractor and the unit rate quoted shall be deemed to inclusive for the same.

19.7.7. Cable entry will be from the bottom, detouchable gland plate to be provided for ease of punching hole for entry of cable.

20. FOR EXTERNAL WATER SUPPLY

20.1. GENERAL REQUIREMENT

20.1.1. All as specified for Internal Water Supply here-in-before.

20.2. EXTERNAL WATER SUPPLY

20.2.1. LAYING OF GI PIPES

20.2.1.1. GI Pipes shall be laid conforming to various provision of IS-2065-1983. All overhead pipe line shall be clamped to the supporting structure with Clamps of Mild Steel Flat of 50 x 3 mm size. Rate/Amount quoted for laying of Pipe Line deemed to include the Cost of Clamps, Nut, Bolt and Washer etc.

20.2.1.2. The trench shall be so dug that the pipe may be laid to the required alignment and at the required depth. When pipe line is crossing a road, a minimum cover of 1.0 mtr is required. Width of trench shall be kept to a minimum consistence with the working space required. At the bottom between the faces, it shall be such as to provide not less than 200 mm clearance on either side of the pipe.

20.2.1.3. Proper implements tools and facilities satisfactory to the Authority shall be provided and used for the safe and convenient prosecution of the work. All Pipes, Fittings, Valves and Hydrants shall be carefully lowered into the trench, piece by piece, by means of a derrick, ropes or other suitable tools or equipment, in such a manner as to prevent damage to water main materials and protective coatings and linings. Under no circumstances shall water main materials be dropped or dumped into the trench. Pipes over 300 mm diameter shall be handled and lowered into trenches with the help of chain pulley blocks. Tripod supports used for this purpose shall be regularly checked to prevent all risks of accidents.

20.2.1.4. The pipe and fittings shall be inspected for defects and be rung with a light hammer preferable while suspended to detect cracks. If doubt persists, further confirmation may be obtained by pouring a little paraffin on the inside of the pipe at the suspected spot; if a crack is present, the paraffin seeps through and shows on the outer surface.

20.2.1.5. All lumps, blisters and excess coating materials shall be removed from the socket and spigot end of each pipe and the outside of the spigot and the inside of the socket shall be wire-brushed and wiped clean and dry and free from oil and grease before the pipe is laid.

20.2.1.6. Every precaution shall be taken to prevent foreign material from entering the pipe while it is being placed in the line. If the pipe-laying team cannot put the pipe into the trench and in place without getting earth into it, a heavy, tightly woven canvas bag of suitable size shall be placed over each end and left there until the connection is to be made to the adjacent pipe. Before lowering the pipe into the trench during laying operations, no debris, tools, clothing or other material shall be placed in the pipe.

20.2.1.7. After placing a length of pipe in the trench, the spigot end shall be centered in the socket and the pipe forced home and aligned to gradient. The pipe shall be secured in place with approved backfill material tamped under it except at the socket. Pipe and fittings which do not allow a sufficient and uniform space for joints shall be removed and replaced with pipe and fittings of proper dimensions to ensure such uniform space. Precautions shall be taken to prevent dirt from entering the joint space.

20.2.1.8. At times when pipe laying is not in progress, the open ends of pipe shall be closed by a watertight plug or other means approved by the GE.

20.2.1.9. Joints of pipe in the trench which cannot be poured shall be caulked with jointing material to make them as watertight as possible.

20.3. DUCTILE IRON PIPE AND FITTINGS

20.3.1. Refer Clause No. 18.94 of MES Schedule Part-I.

20.3.2. DI Pipe shall be conforming to IS 8329-2000. In case nothing is mentioned in Sch 'A' class & type of DI pipe shall be provide K-7 and Spigot and socket type. The DI pipe shall be used for drinking water supply purpose only and accordingly **excise duty(ED)** shall be applicable as per GOI notification. When pipe is crossing a road, a minimum cover of 1.0 Metre is required.

20.3.3. **DUCTILE IRON PIPE AND FITTINGS:-**All Fittings and Appliances shall be inspected, before delivery at the Site to see whether they bear where appropriate, the certification mark of the Indian Standards Institution or the mark of the testing station of the Authority. All fittings shall be inspected and tested by the manufacturers at their factory and shall comply with the requirements of this code. These shall be tested hydraulically under a pressure equal to twice the maximum permissible working pressure or under such greater pressure as may be specified. The pipes and fittings shall be inspected on site before laying and shall be sounded to disclose cracks. Any defective item shall be cearly marked as rejected and forthwith removed from the site.

20.3.4. After laying and jointing, the main shall be slowly and carefully charged with water, so that all air is expelled from the main by providing a 25mm inlet with a stopcock allowed to stand full of water for few days if time permits, and then tested under pressure. The test pressure shall be the working pressure of Pump or the OH Tank as the case may be.

20.4. PITS FOR JOINTS

20.4.1. For Lead Joint additional excavation as specified in IS and SSR shall be provided.

20.5. SLUICE VALVE

20.5.1. All sizes of Sluice Valve shall conform to IS-14846. It shall have Gun Metal Non-Rising Spindle with Gun Metal Check Nut.

20.5.2. All cut off valve shall be of the right hand type. Valve Wheels should have an arrow head engraved or cast thereon showing direction for turning 'Open' and 'Close'.

20.6. **VALVE PIT:-** The Valve Pit shall be constructed all as shown on Drawing with the following Specifications:-

(a)	Excavation and Earth Work	:	In Soft/Loose Soil.
(b)	Foundation	:	PCC M-10(Nominal Mix)(using 40 mm graded stone aggregate)
(c)	Masonry Work	:	Brick masonry built in Cement Mortar 1:4
(d)	Floor	:	100 mm thick PCC 1:4:8 (using 20 mm graded stone aggregate) over 75 mm thick Hardcore
(e)	Plaster	:	20 mm thick in Cement Mortar 1:4 mixed with Water Proofing Compound as per Manufacturer Instructions
(f)	PCC Block/Benching		PCC M-15(Nominal Mix)(using 20 mm graded stone aggregate) for Block/ PCC M-15(Nominal Mix)(using 12.5 mm graded stone aggregate) for Benching.
(g	Mild Steel Cover with Angle Frame		All Steel shall conform to be 410-SW of IS-226
(h)	Painting		All Exposed Steel Surfaces shall be painted with Two Coats of Bituminous Paint. Embedded Portion of Steel shall be treated with Two Coats of Tar
(j)	Rest all as per Drawings		

20.7. RECORD DRAWINGS:

20.7.1. The contractor shall submit six copies of the following drawings duly signed by him and the Engineer-in-Charge on completion of the work.

20.7.2. Overall layout plan for external water supply scheme showing entire details of the installation complete with actual layout of various pipe lines (indicating clearly different sizes in different colours) various valves and all other information necessary for record, maintenance and operation, all as directed without any extra cost.

20.7.3. Specification for all other materials and workmanship shall be all as mentioned here-in-before.

21. EXTERNAL ELECTRIFICATION

21.1. STANDARD OF QUALITY & WORKSMANSHIP

21.1.1. **KNOWLEDGE OF WORK**:- The tenderers shall be deemed to have thorough working knowledge of the exact requirement of the job. In case of any doubt clarification may be sought from CESZ before tender is submitted. Tenderers are advised to visit the site and get themselves conversant with site conditions. No

claim what so ever shall be entertained/accepted in case the tenderer is coming up with an excuse of having incomplete knowledge of work involved.

21.1.2. **WORK IN ACCORDANCE WITH SPECIFICATION**:-Work under this contract shall be carried out as per brief description given in Schedule 'A', Particular Specifications and as shown on drawings. Work shall also be in conformity with IE Rules, relevant IS specification and MES Schedule which are forming part of the contract. The entire work shall be carried out properly with due care and under the supervision of competent technical personnel.

21.2. GENERAL

21.2.1. Refer Clause No. 19.2 of MES Schedule Part-I and requirements of Clause No. shall be strictly followed.

21.2.2. Supply and Installation of Electrical Equipments and Accessories shall strictly comply with the provisions contained in the Latest Edition of India Standard Code of Practice and/or IS Specification as applicable, Works except where such Regulations and Rules are modified by these Particular Specifications.

21.2.3. All Electrical Work shall be carried out in conformity with the requirements of the Indian Electricity Act 1910, Indian Electricity Rules 1956 framed there under, Indian Standard Codes of Practice, Fire Insurance Act as applicable and also the relevant Regulations for Electric Supply and Installations of all Electrical Equipments, Fittings and Accessories as amended from time-to-time.

21.2.4. The supply and installation of all Electrical Equipments and Accessories shall strictly comply with the provisions contained in the latest edition of Indian Standard code of Practice IS Specification as applicable to above mentioned work except where such regulations and rules are modified by these specifications.

21.2.5. **QUALIFIED ELECTRICAL SUPERVISORS/ENGINEERS**:-Refer Condition 25 of General Conditions of Contracts, IAFW–2249. All Electrical Works shall be carried out by properly Skilled and Licensed Electricians under the supervision of qualified Electrical Supervisors/ Engineers. The Contractor shall on demand shall produce such evidence of qualifications of his Workmen/ Supervisors/Engineers, either at the commencement of Work or at any time thereafter/ during currency of the Contract. The entire Work shall be High Class with the best Workmanship and to the entire satisfaction of Engineer-in-Charge/ GE.

21.2.6. **LAYOUT OF EQUIPMENT**:-Layout of various Equipments, Cables, Poles, etc., shown on various Drawings attached with this Tender are tentative. Exact layout shall be as directed by the Engineer in Charge to suit the site requirements. No adjustment in the Quoted Rate/Amount shall be made on account of such varied layout.

21.2.7. **TECHNICAL LITERATURE**:-Technical literature of Gen Set, various Pumps, Starter containing Operation and and Maintenance Instructions, Spare Parts, Catalogue shall be submitted to the department.

21.2.8. Six copies of the above said drawings and literature shall be supplied in calico bound book form. Drawings should not be submitted on Ammonia print but it shall be a Xerox copy on white paper. Front and back cover shall be of laminated hard material or any other superior material approved by GE. Any cost incurred on this account is deemed to be included in the quoted rate for the Gen set. It should also indicate part list of all major items like pumps, motor, Gen set and list of tools and spares.

21.2.9. **SUBMISSION OF COMPLETION OF E\M PORTION**:-On physical completion of work, contractor is to submit the following drawings and other relevant literature as enumerated below to the deptt. This is a pre-requisite before Completion Certificate is issued to the Contractor.

21.2.10. DRAWINGS

21.2.10.1. Schematic Diagram of Power Wiring including cable sizes.

21.2.10.2. LT Cable Route/ HT/ LT over head line route from tapping point to termination.

21.2.10.3. Power Distribution Net Work.

21.2.11. After erection at site and before being connected to the power supply, all items of electrical equipment, all circuits shall be fully tested to prove correct connection, defects in installation pointed out/noticed shall be rectified immediately by repairing or replacing the defective part of the equipment at no extra cost. All instruments appliances, oil, grease and other materials etc, required for carrying out the test shall be provided by the contractor at his own cost. The insulation and other safety measures will be inspected and checked by electrical inspector before commissioning. The electrical inspector shall be nominated by the Accepting Officer.

21.2.12. If owing to storage or other cause, the electrical insulation resistance has deteriorated, the equipment shall be thoroughly dried out or replaced or other steps taken to restore proper insulation resistance before connection it to the power supply.

21.2.13. After installation of the equipment and laying and jointing of cables, necessary tests including relay co-ordination as per relevant ISS shall be carried out by the contractor to the satisfaction of GE.

21.2.14. After connection the power supply, the whole of the installation shall be tested to demonstrate its ability to operate satisfactorily.

21.2.15. The installation shall be deemed to be complete only after satisfactory completion of all tests. Approval by the Engineer-in-Charge for materials, workmanship etc, during manufacture of items at site will not relieve the contractor from his obligations to comply with all the requirements of the contract. The contractor shall supply schematic diagrams (Three copies) of the equipment installed by him on completion of the work.

21.2.16. One month before expiry of the defects liability period, the contractor shall arrange an experienced Engineer together with any additional labour required to attend at the site to open up the equipment to check the various connections clearances etc. The detail checks shall be done by the contractor in the presence of the GE before the end of the defects liability period.

21.3. EXCAVATION, EARTH WORK, CONCRETE, FORM WORK, STEEL WORK, ETC.

21.3.1. Refer respective Clause No. of MES Schedule Part-I and as specified here-in-before.

21.4. CABLE WORK

21.4.1. **GENERAL:-** Refer clauses 19.19 & 19.73 to 19.96 of MES Schedule Part-I. Where the cables are to be laid in the trenches, 80 mm thick cushion of clean dry sand shall be provided below the bottom of cable. The cable shall be covered by a further layer of clean dry sand above the bottom of the Cable, to the height as mentioned in MES Schedule. The Cable Cover shall be laid dry and side by side to protect the cable from Mechanical Injury, the cables shall be looped at ends for furthers joints. The above additional length of cable as directed by Engineer-in-Charge shall also be measured and paid for under the respective items of cable in Schedule ’A'. Laying Bricks at places where Cables are laid, shall be done as directed by the Engineer-in-Charge shall also be Measured and Paid for under the respective Items of in Schedule ‘A'. Cable gland shall be provided at Cable Termination in the DBS and Feeder Pillar Box, the cost of which is deemed to be included in the Rate of Cable.

21.4.2. All Jointing Materials such as Tape, Cables Compound and Cable Boxes, etc. for Straight Joints shall be provided by Contractor without Extra Cost to Government.

21.4.3. The cables being of Short Length for various Connections, there shall be no Straight Through Joints. The cable shall be terminated in Switch Fuse Units through suitable Pin Type Lugs.

21.4.4. The cost of cutting through Roads/Drains/Path for laying cable under the Road/Drain/Path and Making Good shall be deemed to be included in the Unit Rate/Amount quoted against GI Pipes.

21.4.5. **SAND CUSHIONING** :The sand shall be local sand as approved by GE. For laying of Cable in trenches, provision of cushioning shall be carried out all as specified in Clause No. 19.75 and 19.77 of MES Schedule Part-I.

21.4.6. LAYING OF CABLES

21.4.6.1. The LT underground cables shall be of 1100 volts, grade PVC armoured, heavy duty underground cable, conforming to IS-1554.

21.4.6.2. The cable should be laid in such a way that the minimum axial distance of cable from top surface shall be 45Cm + radius of cable for LT cable end and 75Cm + radius of cable for HT cable.

21.4.6.3. LT cable shall bear ISI certification mark.

21.4.6.4. Both ends of pipe protecting the cable shall be sealed with wooden bushes cut into two halves of appropriate size.

21.4.6.5. The cable shall be protected by means of 23 cm(punned down to 18cm) of Sand Cushion on top and under the cable as described in MES Schedule Part-I. On top of Sand Cushion, Pre-cast Cover shall be provided as protection.

21.4.6.6. Earth Work, Sand Cushioning and Cable Protection shall be Measured and Paid separately.

21.4.6.7. Cable path indicators shall be provided at a distance of 30 Metre and Every Turning Point and Cost of the same is deemed to be included in Unit Rates/Amount.

21.4.6.8. The scope of work will include proper laying of Cable in trenches/in pipes/on wall, fixing Identification Tags, crimping with Crimping Lugs, fixing Compression Glands at Termination Point in Panel/Marshalling Box. The above shall be deemed to be part of the installation work and the Contractor shall not be eligible for any Extra Charges other than those specifically mentioned in Schedule ‘A’.

21.4.6.9. LT Cable shall be 3½ Core aluminum conductor PVC insulated inner sheathed armoured and PVC sheathed conforming to IS-1554 suitable for 1100 Volt grade. LT Cables shall be laid on Cable Trays in Pipes /on stanchion as directed by Engineer-in-Charge. Cable shall be laid as per

provision of IS-1255- 1983. The cable drum and the cables shall be handled in the correct manner so as to ensure safety for personal and obviating kinks or other damage to the cable insulation, armouring or cable sheath.

21.4.6.10. All LT Cables shall be tested at manufacturer works as laid down in relevant Indian Standard Specifications including Physical Tests, Insulation Resistant Test and Voltage Tests. The Test Certificate shall be submitted to the Engineer-in-Charge. Termination of cable shall be carried out by means of crimping socket of aluminium. The glands shall be of brass double compression type and shall be provided at the termination. Armouring of cable shall be bonded to the earthing system at the termination through the Compression Glands. Joints in cable length is not permitted unless length of the cable between two terminating Ends is more than 400 Metre.

21.4.7. **TEST:-** After laying the HT and LT Cables, each core of HT/LT cables will be subjected to High Voltage Tests before commissioning. High Voltage Value for HT and LT Cable shall be as given in MES Schedule Part-I vide Clause No. 19.94. Each core of cable shall be subjected to high voltage for a duration of 15 Minutes. The results will be recorded and submitted to Engineer-in-Charge. Testing shall be done in the presence of Department Representative. Contractor is to make arrangements for High Voltage Test of cable at no Extra Cost to the Department. Apart from this Test, IR Value Test will also be conducted as given in IS-1255 and in MES Schedule Part I. The Contractor shall note that these Tests shall be carried out before connecting Cable to Switches.

21.5. DANGER NOTICE PLATES

21.5.1. Danger Notice Plate shall be provided on all Electrical Equipment like Transformer, HT/LT Switch Gear, Bus Duct, etc. The Danger Notice Plate shall conform to IS-2551-1963 250 x 250 mm and 200 x 150 mm size Danger Notice Plate for 11 KV and 433/380 Volts respectively shall be provided as directed by Engineer-in-Charge.

21.5.2. The Plates shall be made from 1.6 mm thick Mild Steel Sheet, Vitreous Enamel White. The Letter, Figures and Conventional Skull and Bones in Signal Red Colour shall be on Front Side. The Danger Notice Board shall be fixed either with 25 x 3 mm GI Clamps or Rivet/Nut Bolts.

21.6. LIGHTNING PROTECTION

21.6.1. Arrangement for lightning protection shall be made on Locations as directed by Engineer-in-Charge. The arrangement shall conform to IS-2309. On Ridge of Building aluminium Strip of 25 x 3.00 mm shall be fixed with J Bolts on both side of Ridge as shown in Drawing. Both side, aluminium Strip shall be in contact with each other through Air Termination Point. If Air Termination Point is separately fixed and not giving contact to both the aluminium strip in that case at every 10 Metre both the aluminium Strip shall be connected separately by a piece of same aluminium Strip so that Lightning have an easy and Shortest Passage to ground.

21.6.2. Air Termination Point shall be fixed as shown in Drawing. Aluminium Down Conductors shall run along CGI Sheet and on Wall fixed by means of Brass Saddles. Aluminium strip down conductor shall be terminated at Test Point. GI Strip coming from earth pit and from another test point shall meet Copper Down Conductor at Test Point. All Test Point shall be interconnected through Copper Strip as shown in Drawing.

21.6.3. All the Roof Conductors and Down Conductors (i.e aluminium Strip 25 x .15mm) and Air Termination Copper Rod (12 mm dia & 300mm long) shall be mounted on suitable size of Porcelain Insulator. Due care be taken to ensure that any portion of the Lightning Protection Air Termination Rod and the Strip are not in contact with the Building. The cost of porcelain insulator is deemed to be included in the Quoted Rate/ Amount of the concerned ite

21.7. DISTRIBUTION / FEEDER PILLARS BOX

21.7.1. Refer clauses 19.21 of MES Schedule Part-I. , IS 5039 and TD drawing

21.7.2. The feeder pillars box shall be factory made and CPRI approved. Feeder pillars box shall be obtained from the specified manufacturers and shall be epoxy powder coated factory made and tested as per IS. All feeder pillars boxes shall have extra space for installing one MCCB in future.

21.7.3. The contractor shall design & submit drawings of panel board before bringing same to the site.

21.7.4. The cable chamber of the board shall be with adequate space for housing cable and connecting for safety of working and for avoiding accidental contacts with live parts by providing barriers.

21.7.5. In case nothing is mentioned in Sch ‘A’ bus bar shall be copper.

21.7.6. In case mounting of panel board has not been mentioned in Sch ‘A’ the same shall be floor mounted and the fixing arrangement shall be provided by the contractor and the unit rate quoted shall be deemed to inclusive for the same.

21.8. LT SWITCH GEAR PANEL:-

21.8.1. Refer Para (including sub paras) 19.100 of MES Schedule Part-I and IS 2147.

21.8.2. The panel board shall be factory made IP-65 and CPRI approved. Panel boards shall be obtained from the specified manufacturers and panels will be epoxy powder coated factory made and tested as per IS. All LT panels shall have extra space for installing one MCCB in future.

21.8.3. The contractor shall design & submit drawings of panel board before bringing same to the site.

21.8.4. The cable chamber of the board shall be with adequate space for housing cable and connecting for safety of working and for avoiding accidental contacts with live parts by providing barriers.

21.8.5. In case nothing is mentioned in Sch 'A' bus bar shall be copper.

21.8.6. In case mounting of panel board has not been mentioned in Sch 'A' the same shall be floor mounted and the fixing arrangement shall be provided by the contractor and the unit rate quoted shall be deemed to inclusive for the same.

21.9. MOULDED CASE CIRCUIT BREAKERS (MCCBs):-

21.9.1. Moulded case circuit breakers (MCCBs) shall be suitable for operational voltage of 415V AC, 50 Hz, 3 phase, 4 wire system for a rated current and ultimate breaking capacity and it shall conform to IS : 13947 (Part 3, 1993) and IEC-60947 (Part 2).

21.9.2. MCCBs shall be of compact and elegant design suitable for reversible load and line terminations without affecting its performance. MCCBs shall be suitable for fixing flush on the panels and shall be provided with handle operating mechanism including Rotary Handle vari-depth type.

21.9.3. The insulating case and cover of MCCBs shall be made of high resistant and flame retardant thermosetting insulating materials. The switching mechanism shall be quick make, quick break and trip free. The position of the operating knob/handle shall clearly indicate ON, OFF and TRIP position.

21.9.4. Each pole shall be provided with a pair of contacts, which shall open at a high speed over a large distance under short circuit faults. The special designed arc chutes of insulating materials shall be provided to contain the arc by providing effective arc quenching device.

21.9.5. The tripping mechanism shall be Electronic release or thermal magnetic release for protection for over load and short circuit as per the details given below. –

Capacity Of MCCBs	Ultimate breaking Capacity	Overload release	Short circuit release
Upto 160 Amps	16 KA	Adjustable Thermal release (0.8 to 1.0 In)	Fixed magnetic type suitable for 10.0 In.
250 to 320 Amps	25 KA	Adjustable Thermal release 0.8 to 1.0 In	Adjustable magnetic release (5 to 10.0 In)
400 to 500 Amps	36 KA	Electronic release with over load Zone of adjustment 0.4 to 1.0 In.(Adjust current) with time.	Electronic release with Short circuit Zone of adjustment 1.5 to 10.0 Ir. (Adjusted current) with time.
630 Amps	36 KA	Electronic release with over load Zone of adjustment 0.4 to 1.0 In.(Adjust current) with time.	Electronic release with Short circuit Zone of adjustment 1.5 to 10.0 Ir. (Adjusted current) with time.

21.9.6. Under voltage trip, mechanical interlocks etc shall be provided as per standard practice and ISS. Under voltage trip shall be designed to operate when the control voltage drops below a tripping threshold i.e. 20% to 70% of rated voltage and shall be suitable for operation on 230V/415V AC.

21.9.7. The terminals shall be suitable for both copper terminations.

21.9.8. In case number of poles of MCCB are not specified in Sch A then 4 pole shall be provided.

21.10. EARTHING

21.10.1. Earthing shall be carried out by Earth Electrodes as shown in Electrical Plate No. 3 of MES Schedule Part-I and as specifed in IS-3043 including upto date amendments therto. The Earth Electrodes shall be placed all as shown on Drawings and 2.0 Metre away from the Main Structure. The surplus soil obtained from the excavation shall be removed to distance not exc. 50 Metre. The Concrete for Chamber shall be PCC 1:3:6 (using 20 mm graded stone aggregate) and exposed surfaces need to be finished fair. Funnel shall be prepared out 3.15mm thick PGI Sheet by welding and shall be leak proof. It shall be provided with wire gauge with 1.18mm

average width of aperture and 0.45mm average dia of wire soldered to Funnel Cover. Dust shall be used for Medium Grade 20mm with 12mmdia holes drilled staggered @ 300mm C/C through out Pipe. The bottom of Pipe shall be layed at an angle of 45º. Two numbers Earthing shall be interconnected with 32 x 6mm into GI Strip. GI Pipe for Earthing shall not be issued under Schedule 'B'.

21.10.2. The following Tests shall be carried out to the entire satisfaction of the Engineer-in-Charge :-

(a) Earth Resistance Test : Resistance of each Earthing shall be measured individually and shall not exceed 0.5 Ohm in ordinary soil and 3 Ohm in Rocky situation.

(b) In case the Resistance is more than as specified, the Contrctor shall increase the depth of the Bore hole, quantity of Charcoal, Salt and length of Electrodes to bring the Resistance of Earth to the Value specified above, whithout any Extra Cost to the Government.

21.11. GUARANTEE/ PRODUCTION OF TEST CERTIFICATE

21.11.1. After completion of work the contractor shall guarantee that materials and workmanship of the equipment, power system and connected items offered and installed by him are new and best quality in every respect and he will make good the defects, damages which are not attributed to normal wear and tear or misuse for which decision of the GE is final and binding. Contractor shall be responsible for repairing and or replacing at his cost any parts which are rendered defective and unserviceable during the defect liability period.

21.12. Manufacturer's test certificate in respect of **HT/LT Cables, Geyser, Water Cooler, Air Cooler, Refrigerator, Transformer, DG Set, Solar Water Heating System and** other equipment as directed by GE are to be submitted to the GE by the contractor before any payment from Department.

21.12.1. **INSPECTIONS AND TESTING**: -All equipments such as HT/LT cables (more then 500merte each type) and transformer shall be tested in the manufacturers workshop according to tests specified in the relevant Indian Standard Specification and elaborated in these specifications. The equipment shall bear the IS testing marks where manufactured accordingly. The HT/LT cables, DG set and transformer shall be inspected in presence of the rap of the Accepting Officer in factory/ manufacturers workshop premises before dispatch of the material. The contractors shall be facilitating for inspection/testing in factory/ manufacturers workshop premises. The contractor shall give one month notice to the Accepting Officer indicating readiness of equipments for testing. The cost of the representative's 1st visit to the factory shall be borne by the Department.

21.13. TUBULAR STEEL POLES SWAGED:-

Tubular steel poles shall be swaged type as specified in clauses 19.3 of MES Schedule Part-I & shall conform to IS 2713 (Parts 1 to 3) 1980 specification for Tubular poles for overhead power lines (Second Revision). The tubes for making poles shall be out of steel conforming to grade YSt 240 (with minimum tensile strength of 410 Mpa) except that Manual metal arc welding process may also be used to manufacture tubes and Cold bend test need not be conducted. Swaged poles shall be made of seamless or welded tubes of suitable lengths swaged and joined together. No circumstantial joints shall be permitted in the individual tube length of the poles. If welded tubes are used, they shall have longitudinal weld seam only, and the longitudinal welds shall be staggered at each swaged joints. Pole shall be well finished, clean and free from harmful surface defects. Ends of the poles shall be cut square. The poles shall be straight, smooth and cylindrical.

21.13.1. The dimensions of poles shall be as per clause 19.3.3 of SSR Part – I, 2009.

21.13.2. Workmanship shall be as per clause No. 19.50 of MES SSR Part – I, 2009.

21.13.3. Painting of pole shall be as per clause No. 19.71 of MES SSR Part – I, 2009.

21.13.4. The contractor shall procure steel tubular poles from the manufactures listed at Appendix-B. The contractor shall produce manufacturer test certificate before in-corporation in the work.

21.14. ALUMINIUM CONDUCTORS GALVANISED STEEL REINFORCED (ACSR):- 24.7.1.

Aluminium Conductors Galvanised Steel Reinforced (ACSR) shall be as specified in clause No. 19.10.2 of SSR Part – I, 2009 shall be made of seven or more aluminium and galvanised steel wires built up in concentric layers. The centre wire or wires are of galvanised steel and outer layer or layers of aluminium and conform to to IS 398-1976 Specification for aluminium conductors for overhead transmission purposes, Part 2, 1976 Aluminium conductors galvanised steel reinforced as indicated.

21.14.1. Workmanship shall be as per clause No. 19.56 to 19.65 of MES SSR Part – I, 2009.

21.15. **LT/HT CABLE ROOT INDICATOR** shall be as per clause No. 19.88.2 of MES SSR Part – I, 2009.

21.16. **TRANSFORMER**

21.16.1. Refer Para 19.14, 19.14.1 to 19.14.3 & 19.97 (including sub clause) of MES Schedule Part-I.

21.16.2. The transformers supplied shall comply with energy conservations norms given in Energy Conservation Building Code (ECBC).

21.16.3. The transformer shall be step down type suitable for 3 phase, 50 cycle/sec application and shall be indoor type/outdoor type, naturally cooled, oil immersed, core type.

21.16.4. **VOLTAGES:** - The primary voltage and secondary voltages shall be as given in Schedule A.

21.16.5. **VECTOR GROUP: -** The vector group shall be DYN 11 with primary winding connected in delta and secondary in star.

21.16.6. **WINDINGS: -** The windings primary and secondary shall be of copper. The windings round/strips shall conform to IS-7404 (Part I & II).

21.16.7. **CORE: -** The core materials shall be in accordance with IS and treated in such a way that the no load losses are minimized. The laminations shall be stress free and shall be of cold rolled steel grain.

21.16.8. **TAPPING: -** The transformer shall be provided with ON load externally manual operated tap changing gear on HT side with tapping ± 2.5%, ± 5% and ±7.5% to provide and maintain in normal voltages on LT side constantly by compensating the variation in the system.

21.16.9. **TAP SWITCH: -** The switch shall be provided with handle externally, with indicating disc for denoting the tap position in which it is set. The switch shall be provided with looking device to look the switch in set position.

21.16.10. **TANK: -** The tank shall be of robust design and with adequate mechanical stiffness of pressed steel sheets radiator shall be used for sufficient cooling and heat dissipation to limit the temperature rise within tolerance limits specified in IS-2026.

21.16.11. **TRANSFORMER OIL: -** The transformer shall be provided with first fill of transformer oil and oil shall conform to IS-335-1983.

21.16.11.1. Transformer shall be provided with built in protection for winding, temperature alarm and tripping with visual indication. Cable end box on HT side shall be suitable for Epoxy joint for XLPE cable with suitable size cable entry gland for upto 400 mm 3 core cable. LT side terminal box shall be suitable for entry of four aluminium strips 50x5mm and suitably separated duly insulated at the entry position of terminal box to avoid short circuiting.

21.16.11.2. Terminal marking, temperature rise, insulation levels and dielectric strength shall be in accordance with IS-2026 (Part-II, III & IV) 1977.

21.16.12. Fittings and accessories of transformers shall be conforming to IS- and SSR part -I.

21.16.13. Erection on installations:- Installation of transformer shall be carried as explained in clause Nos 19.97.1, 19.97.2, 19.97.3, 19.97.4 and 19.97.6 on Srl Page No 19.24 of SSR Part-I.

(a) Measurement of winding resistance
(b) Measurement of voltage rating and check for voltage vector relationship.
(c) Measurement of impedance voltage/short circuit (Principal tapping) and load less
(d) Measurement of no load loss and current.
(e) Measurement of insulation resistance.

21.16.14. Type test:-

(a) Impulse voltage withstand test
(b) Temperature rise tests.

A certificate for type tests shall be produced and submitted to GE.

21.16.15. **TRAINING: -** The contractor will impart training at least for one week within one month of the completion of whole work to the persons authorized by the Department. The training will cover the following aspects –

(a) Starting/stopping procedure
(b) Maintenance of fuel, lubricating system, starting control panels and Alternators
(c) Dismantling and assembling of certain components like fuel, filter elements, lubricating oil filter, changing of oil air cleaner etc.

21.17. **GENERATING SET**

21.17.1. **GENERAL**

PARTICULAR SPECIFICATIONS (CONTD…..)

21.17.1.1. The contractor shall be responsible for supplying, installing, testing and commissioning of diesel engine driven generating set with all its accessories, all the equipment's and fittings as indicated in the respective item of schedule' A' according to sound and "Latest Engineering Practice" and tested as specified herein after to the entire satisfaction of the Engineer-in-Charge.

21.17.1.2. On completion of complete work, the contractor shall submit in triplicate to the Engineer-in-Charge layout plan showing actual position of the Generating set with its accessories, switch gears, other equipment cable runs earthing pits, earthings leads and all other information that will be necessary for record, maintenance and for inspection. Scale of the layout plan will be as accepted by Engineer-in-Charge.

21.17.2. **SCOPE OF WORK:-**The work shall comprise of the following, all as indicated in schedule' A' and as specified herein after and as directed by the Engineer-in-Charge.

(a) Supply, installing, commissioning and testing of Radiator water cooled diesel engine driven Generating set with all accessories, controls, control panel and LT control panel than control panel to cable from Alternator to change over switch.

(b) Civil work connected with installation and commissioning Generating set, control panel, fuel tank foundation with anti-vibration pads, sand cushioning, fixing clamps and distribution pipes of radiator of engine complete.

(c) Painting of Generating set, control panel, fuel tank, switch gears, piping controls with two coats of synthetic enamel paint over a primer coat. The proper colors code for Generating set shall be as under:-

(i)	Generating set	As per manufacture's color and as approved by GE
(ii)	Base Plate	Black
(iii)	Solt Guard	Red

(d) Training of MES staff for one week after the test run of plant for operation and maintenance of plant

(e) Any other item not specifically mentioned but required for efficient functioning of plant

(f) The diesel engine driven Generating set shall be capable of giving net output as specified in schedule' A'.

(g) DHPP/HSD oil required for testing of DG set will be arranged by the contractor at his own cost. Contractor's quoted rate is also deemed to be inclusive of all type(s) of lubricating oil, required under the contract to execute the work up to full satisfaction of Engineer-in-Charge, no claim whatsoever will be entertained on this account at a later stage.

21.17.3. **ALTERNATOR**

The Alternator shall be of adequate KW to give the specified KVA output at 0.8 power factor AC three phase, four wire, 415 volts, 50 Hz, 1500 RPM brushless type self excited self regulated, self ventilated, screen protected, drip proof type enclosures with one end bearing with damper winding in the pole faces statically and dynamically balanced as per IS-4722. The tenderer shall supply complete specifications and technical literature.

(a) The Alternator shall be designed and built to withstand tropical conditions.

(b) The alternator should be with brushless excitation and shall be cooled by Alternator circulating air only.

(c)It shall have voltage regulation of range (+) 2.5% at NO LOAD to FULL LOAD at 0.8 PF. The voltage regulation shall be automatic.

(d) The Alternator shall be able to withstand over speeding of up to 20% of normal speed. Automatic speed control shall be provided with governor.

(e)In case of short circuiting, the Alternator shall be able to withstand short circuiting current up .to 5 times of the normal current.

21.17.4. **DIESEL ENGINE**

The Diesel engine shall be of adequate capacity, four cylinder to drive independently directly coupled through flexible coupling, alternators of specified capacities, industrial type as per BS-649 or IS-l0000 with 10% over load provision for one hour in every twelve hours consecutive running under STP conditions.

21.17.5. **ACOUSTIC ENCLOSURE (For noise level limit of less than 75dBA at 1.0 m distance)**:-

Acoustic enclosure shall be sound proof complete with fittings and fixtures to meet the requirements of Indian norms of noise pollution all as specified and directed as per "Environmental (protection) rule 1986" as laid down by Min of Environmental and Forest read

with GSR 371 (E) dated 17 Feb 2002 and its amendments". ENCLOSURES SHOULD MEET THE CPCB NORMS.

21.17.6. SAFETY PROTECTION :-

Following safety protections shall be provided :-

(a) High water temperature protection device.
(b) Low lub. oil pressure
(c) High enclosure temp.
(d) Emergency stop push button outside the enclosure

21.17.7. INSTALLATION OF GENERATING SET:-

The installation/erection work shall cover supply of all items mentioned in schedule 'A', in particular specification, commissioning and functionally testing the same. This shall cover the following additional works and rate quoted in schedule' A' shall be deemed include cost for same :-

(a) Generating set shall be mounted on a common base/bed plate of robust construction having suitable arrangements to fix with requisite number of bolts with nut, anti vibration pad fixed to base plate with bolts and nuts as per manufacturer's recommendation.
(b) Provision of clamp to hold the exhaust pipe including grouting and connection.

21.17.8. PAMPHILET LITERATURE AND DRAWINGS

21.17.8.1. CATALOGUE: - The contractor shall supply (from manufacturers) the following literature in English/Hindi

(a)	Manufacturers instruction/maintenance	Two sets each
(b)	Spare parts catalogue	Two sets each
(c)	Electric shock treatment charts in English/Bengali	Two sets each
(d)	Operation and maintenance instruction duly framed	Two sets each

21.17.8.2. TECHNICAL LITERATURE/PAMPHILETS :- The contractor shall supply in quadruplicate technical literature and pamphlets of Generating set, Control panel and Instrument panel.

21.17.9. GUARANTEE AND TEST CERTIFICATES :- The contractor shall stand guarantee for the equipments supplied under the contract for one year from the actual date of handing over of DG set.

21.17.9.1. The contractor shall produce test certificates from manufactures for the following equipments/instructions in accordance with relevant IS :-

(a) Diesel Engine
(b) Alternator

21.17.9.2. TESTING:- After execution of work/job as per Schedule 'A' and Particular Specifications with the satisfaction of GE, following test shall be carried out by 'Technical Board' ordered by the Accepting Officer in presence of contractor. GE/AGE (E/M) shall form part of the Board. These tests shall be carried out for 12 hours continuously as per IS 10000(Part I to XII) of 1980. The completion certificate will be issued after the tests have been carried out satisfactorily confirmed by the Technical Board'. The fuel for all tests shall be supplied by the contractor and the rate quoted shall be inclusive of the same.

(A) **INSTALLATION TEST**: - The test result shall not be lower than the minimum as specified in IE/IES rules.
(B) **LOAD TEST**: - Artificial load will be arranged by the contractor i.e. solt, water both with barrier.
(C) **LOAD TEST**: - The test will be conducted as under -

(i)	First hour	: 25 % of full load
(ii)	Second hour	: 50 % of full load
(iii)	Third hour	: 75 % of full load
(iv)	Fourth hour	: Full load
(v)	Fifth hour	: At 10 % over load
(vi)	Sixth to twelfth hour	: 100 % load

(vii) If the result of testing is not found satisfactory, the contractor will at own expense rectify/replace the defective equipment/material or any part thereof as directed by Engineer-in-Charge. After rectification the set shall be retested. The fuel and lubricating oil shall be provided by the contractor. The decision of the Accepting Officer shall be final and binding.

21.17.9.3. TRAINING: - The contractor will impart training at least for one week within one month of the completion of whole work to the persons authorized by the Department. The training will cover the following aspects –

(a) Starting/stopping procedure
(b) Maintenance of fuel, lubricating system, starting control panels and Alternators
(c) Dismantling and assembling of certain components like fuel, filter elements, lubricating oil filter, changing of oil air cleaner etc.

22. BOREWELLS:-

22.1. SITE CONDITION OF BORE WELL

The tenderer are advised to see the proposed site of bore well in order to assess the work Involved for setting of drilling rig/equipment before quoting the rates. Nothing shall be entertained on this account after receipt of tender. The location of bore well shown on site plan is tentative, slight variation in site at the time of execution may take place. The department shall entertain no extra claim in this account. The contractor rate is deemed to be inclusive of all these site factors.

22.1.1. METHOD OF DRILLING

Combination of DTH & reverse Rotary rig shall be adopted in boring of the bore well. The contractor from his or her own expense and sources shall arrange all equipment required for boring. Development of bore well shall be done as per IS 11189-1985 (Latest addition) & Boring/testing of bore well shall be done as per IS 2800-1 (1991) (Reaffirmed 2001).

22.1.2. BORING

A well boring rotary rig will do boring by reverse rotary circulation method from ground level to the depth of suitable strata, (depth depending on the availability of potable water bearing strata) within approx **100 Mtr** depth. The boring shall be vertical to ensure proper functioning of a submersible pump in the bore well. It is the responsibility of the contractor to select suitable final strata for the tube well, to ensure maximum yield of water, which shall not be less than that as specified in Schedule 'A'. After selection of final strata by the contractor the places where strainers will be inserted will be got approved from the GE and the strainers will be placed in position accordingly. During the period for approval of sample, the contractor has to continue, the washing process of Borehole to keep the borehole clear. No extra payment shall be entertained on this account.

The contractor will ensure that the pipes are sunk perfectly in plumb, while boring and sinking of the tube wells. The verticality will be checked by 'Plumb bob method' in the presence of the Engineer-in-Charge.

The anticipated depth of boring of the tube well at site shall be **100 Meter** below ground level. However, the depth of boring may be varied maximum of **50 Meter** (plus or minus) as ordered by GE in writing. If GE decides that suitable water bearing strata is not available in the aforesaid trial bore the contractor shall fill the abandoned bore well if any without any extra cost to the government or the boring cannot be continued further on account of the type of strata met with, he may order the contractor to bore at other site(s) selected by him until suitable water bearing strata is met with. Provided the total number of trial bores per tube well in CA (including the first bore) do not exceed **three**. The payment shall be made to the contractor for abandoned trial bore under respective item of Sch 'A'. The decision of the GE shall be final and binding as to whether or not the boring can be continued further in the strata met with.

22.1.3. COLLECTION OF GEOLOGICAL STRATA

Samples of drill cuttings from different strata should be collected at every **2 (TWO)** meters depth drilled or closer intervals, if change in the strata is met with. The sample should be dried and stored in neat cloth or polythene bags, which should be, leveled clearly indicating the depth ranges of the strata. The samples should be kept in an open type wooden box. The samples so collected for the successful bore well will be handed over to the department. The samples should be carefully examined and analyzed and a strata chart in proper form will be prepared indicating the probable strata which are likely to give required yield of water. While boring is in progress each strata (water bearing zone) is to be tested for iron content. Strata having iron-content more than 5% shall not be tapped for water and blanking pipe should be inserted. It is the responsibility of the contractor to collect the sample and get it tested for iron content. Record of testing of water for iron content shall be kept in record duly signed by contractor and Engineer-in-Charge. Testing charges shall be borne by the contractor. The rate quoted is deemed to be inclusive of this aspect.

22.1.4. MATERIALS AND WORKMANSHIP

Unless otherwise mentioned & specified in Schedule ‘A’ and particular specification, the materials incorporated in the work shall conform to relevant IS as mentioned in IS-2800 (Part I).

The MS pipe and fittings as specified in Schedule ‘A’ items shall be free from any manufacturing defects and the materials supplied by the contractor shall be approved by the GE before incorporation in the work.

22.1.5. PEA GRAVEL

Pea size gravels for packing surrounding the bore well shall be hard, rounded particles reasonably uniform in diameter and size 3mm to 5mm. The gravel shall be free from the impurities such as shale, mica, feldspar, sand clay, dirt, loam any other organic materials. Gravel shall be properly

sieved and washed with spray jets of water to remove the impurities. In case no other method of measurement of pee gravel is defined in Sch "A" then Gravel(s) in stack(s) shall be measured before incorporating in the work. Payment shall be made for the quantity utilized after deducting the quantity left over. Gravel packing around the pipe assembly shall be done by means of inverted cones. Stacks for gravel shall be made at fairly leveled surfaces and measurement shall be recorded in the measurement books by the Engineer-in-Charge and signature from both parties shall be obtained after the entry.

22.1.6. GRAVEL FILLING UP

After the tube well assembly has been placed in position the annular spat between the housing/blanking/casing pipe/ strainer and bore hole shall be packed with gravel. The gravel shall be packed upto 15cm depth above the top of strainer pipe and above gravel/clayed soil mixed with sand (obtained from the boring) shall be properly packed to ground level all as directed by Engineer-in-Charge.

22.1.7. DESIGN & LOWERING OF PIPE ASSEMBLY

From the data collected about the nature of soil met with the length of the housing pipe, the GE shall determine casing pipe and strainer to be incorporated. The depth of bore holes, length of casing pipe and strainers may vary from those given in Sch 'A' depending upon ground data collected as stated above and the contractor shall have no claim on account of variation in quantities of pipes.

22.1.8. SIZE OF BORING

The size of boring shall be such as that at least 10 cm clearance is kept around the outside surface of the housing pipe to facilitate easy gravel packing to the well. The housing pipe must be extended up to at least 60cm above the ground level.

22.1.9. HOUSING & CASING PIPE

All pipes used under this contract shall be new and shall be ISI marked with plain end. Housing and casing pipe shall be mild steel quality of approved make, jointed by suitable size collar with butt-welding. No extra payment will be made for welding as mentioned here in before over and above the collar joint. The strainers shall be MS slotted pipe of approved quality. MS pipe shall be automatic fusion welded pipe or electric resistance welded pipe or high frequency Induction welded pipe at the discretion of the contractor. Grade of steel for the pipe shall be Fe-410. Length of each pipe shall not be less that 6.0 meters.

22.1.10. PLUMBING AND ALIGNMENT

Bore well must be of proper alignment. The contractor will ensure that pipes are sunk perfectly in plumb while boring and sinking of tube wells.The housing pipe must be true to line within permissible limit of deviation of 10 cm per 30m depths from its top to the bottom of the pipe. Using plumb bob or a plunger, which shall be circular, well shall test the verticality of the bore and of size 6mm smaller in diameter than the internal diameter of the housing pipe and result found shall be recorded in the Performa attached as Appendix 'A'. GE will accept the boring when it satisfied the deviation limit as per IS 2800 (having deviation of 10cm per 30 meter in one direction and in one plane only). The verticality will be checked in presence of GE.

22.1.11. PAINTINGS

MS pipe shall be painted both inside and outside with two coats of anti corrosive black bituminous paint suitable for drinking purpose over a coat of zinc oxide primer of approved make before the pipes are lowered in the borehole.

22.1.12. WELL CAP, BAIL PLUG & REDUCER

Well cap, Bail Plug & Reducers are required for bore well shall be new and shall be provided by the contractor to suit the requirement. These fittings shall be got approved by the GE before incorporation in the work.

22.1.13. DEVELOPMENT OF TRIAL BORE

After the bore well has been gravel packed, the well shall be developed with Air compressor or over pumping system. As the development proceeds gravel shall be added whenever necessary.

The well shall be developed in the presence of GE either by surging including washing and agitating or by pumping and back washing with an airlift. The over pumping shall be 25% higher discharge than the expected discharge from the tube well. The following procedures shall be strictly adhered to during development:-

(a) Air surging by air compressor shall be intermittent and continuous. Air shall be sent for ten minutes at a stretch followed by an equal period of shut down.

(b) Development shall be carried out with air compressor of suitable capacity and diameter of pipe. The pipe of compressor shall be kept in the middle of each water-bearing stratum and each stratum shall be developed separately till the water becomes sand free. Development shall be started from the top most stratum and proceed down to the bottom most. After each water bearing stratum has been developed individually as mentioned above there will be a cumulative development of all water bearing strata. This procedure shall be continued till the following: -

(i) The well ceases to absorb further gravel, which shall be added continuously throughout the developing if level is going down.

(ii) Depression ceases to improve.

(iii) Discharge ceases to improve.

(iv) Sand content in the water shall not exceed 20 parts in millions parts of water by volume after 20 minutes of starting the pump.

(c) The tube well shall be developed to minimum discharge capacity as specified in Sch "A".

22.1.14. HANDING OVER OF THE TUBE WELL

The following information shall be furnished by the drilling agency on completion of the tube well.

(a) Total depth of tube well drilled.

(b) Strata chart of the tube well indicating different type of soil formations met with at different depths and indicating the depth of each type of soil formation.

(c) Samples of strata collected, neatly packed and correctly marked in sample bays. (d) Position of every joint in well assembly.

(e) Method used for development.

(f) Total hours of development done.

(g) Developed discharge in L.P.S.

(h) Discharge is totally sand free or presence of sand particles is there.

(i) P.P.M and turbidity after development.

(j) Pumping water level at developed discharge, and

(k) Static water level.

22.2. TESTING:-

The following test as per IS-2800 of latest edition shall be carried out in the bore well-constructed and results noted in proper form:-

(a) Alignment (vertically) tests

To ensure the verticality of the tube well verticality test shall be carried out by plumbing method at the contractor's own arrangement to the entire satisfaction of the GE as per IS specifications.

(b) Step draw down including yield test

The contractor shall have to lower down his own pump and same shall be run for 24 hours and discharging recorded either through a monometer orifice meter or through "V" Notch to be provided for measurement of draw down and static levels directly with steel tape. Test shall be carried out in presence of Engineer-in-Charge and test sheet shall be submitted in triplicate to the Deptt by the contractor duly signed by the contractor and Engineer-in-Charge. The contractor shall supply the necessary pumping set, apparatus, instrument and labour (required for the test without any extra charge to the Deptt). The performance as per IS 2800 shall be followed for carrying the test.

(c) If the water level and discharge are found to be fluctuating, development shall be carried out for some more hours until discharge becomes steady and sand contents are within permissible limit. After completion of tube well the yield test shall be carried out by installing a test pump in tube well temporarily and pumping out as specified in IS 2800 (Part II).

22.2.1. SAMPLE OF WATER AND TESTING FOR POTABILITY

(a) As the development proceeds, whenever clear and sand free water is obtained at any strata this will be collected by contractor in four sample bottles. The bottles shall be surgical, sterilized of size 250ml capacity or nearest higher capacity. The bottles shall also be provided with glass stopper preferably having an over tapping rim. The sterilized bottles shall not be opened except at the time of sampling. They shall be sealed, labeled and initialed by both the contractor and Engineer-in-Charge.

(b) Any Govt. approved/recognized Laboratory shall be approached for testing the potability of the water. It is the responsibility of the contractor to send the samples to any Govt. approved/recognized Laboratory for testing at his own arrangements and obtain test result. Testing charges shall be borne by the contractor. The test result shall be kept in record duly signed by the contractor, Engineer-in-Charge and GE.

22.2.2. BORE HOLE CHART

The contractor shall maintain bore hole chart for each tube well prepared on tracing cloth without any extra cost to the Govt. The chart shall furnish the following information:-

(a) Description and depth of strata encountered.

(b) Actual layer opposite of which strainers have been placed.

(c) Water bearing strata below ground level and draw down level.

(d) Rate of progress of boring.

(e) Type and make of strainer.

(f) Position of joints in strainer/blank pipes.

(g) Yield, duration of test, depression, level interference between adjacent levels, prevalence of sand etc.

(h) The quantity of gravel initially poured during packing around the tube well as specified here in before and quantity of gravel poured during development of well.

(j) Full particulars of final test.

Four copies of bore hole chart will be handed over to the GE along with the original tracing duly signed by both the contractor and Engineer-in-Charge.

22.2.3. CHAIN PULLEY BLOCK

Chain pulley block shall conform to relevant IS. Test certificate for pulley block, lifting chain hook to be given along with the material by the contractor.

22.2.4. PAINTING

Black **MS ERW** pipes shall be painted externally with one coat of red oxide and final coat of approved anti corrosive black bitumastic paint suitable for drinking water on both inside and outside face of the pipes, specially before sinking.

22.3. GUARANTEE FOR TUBE WELLS

The contractor shall lodge guarantee in writing for satisfactory performance of the tube well and the yield shall not be less than as specified in Sch “A” (BOQ) for a period of twelve months from the date of handing over each tube well to the department. During this period if the specified yield is found to dwindle down, any work such as washing /developing of tube well and chemical treatment to remove the scales etc from body of the tube well in order to restore the yield shall be carried out by the contractor without any extra cost to the Govt.

22.4. ABANDONMENT OF BORE WELL

If the bore well construction fails due to the non-availability of suitable water strata met with even after boring maximum depth as specified here-in-before due to some other reasons for which the contractor is not responsible, the payment for boring only shall be paid to the contractor under **Schedule 'A'** according to depth of boring actually done. However contractor shall have no claim for payment/compensation for any portion of works of bore well carried out by him if the bore well fails entirely due to his negligence or fault.
The decision of the GE with regards to the type of failure of the bore well and whether or not the cause of failure accrued to the contractor will be final and binding**.**

22.5. SLUICE VALVE

(i) All sizes of sluice valve shall conform to IS-14846. It shall have brass spindle with brass check nut.

(ii) All cut off valve shall be of the right hand type. Valve wheels should have an arrow head engraved or cast thereon showing direction for turning 'open' and 'close'.

Cast iron pipes shall be centrifugally cast, pressure pipe, flanged end including CI screws and drilled flanges conforming to IS-1536. Cast Iron fitting shall be suitable for pressure pipe and fitting shall be provided as specified and as shown on drawing. Exposed surfaces of all CI pies and fittings shall be painted with 2 coats of synthetic enamelled paint over a coat of red oxide primer. The colour code for various pipe lines shall be followed as directed by GE.

All CI pipe and fitting shall be embedded in the concrete at the time of casting of concrete. Flanges of pipe and fittings shall be jointed using 3mm thick rubber gasket of approved quality with suitable bolts, nuts and washers of appropriate size.

22.6. CAST IRON OUTLET PIPE

Cast iron outlet pipe with sump in primary section building shall be provided all as shown on drawing. The cast iron pipe shall be 250 mm bore with flanged joints, centrifugally cast class ‘A’. The CI Pipe shall be laid with gratings form sump and taken across the building upto 2.5m from the outer face of external wall of the building.

22.7. SAMPLE OF WATER

(a) As the development proceeds, the sample of water shall be drawn by the contractor.

(b) In final stage of development when clean sand free water is obtained, clear water so obtained shall be collected in sample bottle to be provided by the contractor. Chemical and bacteriological test of water shall be done by the contractor from Public Health Dept/National Test House/Recognized private agency and test result shall be submitted by GE within 15 days.

23. PCC DRAIN

PCC Drain shall be constructed all as directed by the Engineer-in-Charge. The Inside Surface of Drain shall be finished Even and Smooth with using Extra Cement which shall be measured separately under relevant Item(s) of Schedule ‘A’.

24. SEWERAGE PIPE:-

24.1. **SGSW PIPES** – Unless otherwise mentioned in Schedule ‘A', Salt Glazed Stone Ware Pipes and Fittings shall be of Grade ‘A' conforming to IS 651 and shall be of Approved make. Jointing shall be done in Cement Mortar 1:1 as specified in Clause No. 18.70 and 18.70.1 of MES Schedule Part I.

24.2. **HDPE PIPE FOR SEWERAGE SYSTEM** – High density polyethylene (HDPE) for sewerage shall be conform to IS 14333. The contractor shall submit the manufacturer's test Certificate in original along with the test sheet giving the results of each physical test as applicable and the chemical composition text and authenticated copy thereof, fully signed by the manufacturer with each consignment.

24.3. **SPUN YARN** – Refer to Clause No. 18.10 of MES Schedule Part-I.

24.4. **CONCRETE PIPES –** Refer Clause No. 18.29 MES Schedule Part-I. Unless otherwise mentioned in Schedule ‘A', Reinforced Concrete Pipes for Drains and Culverts shall be non-pressure type Class NP2 and shall conform to IS- 458-2003. The Ends of Pipes shall be suitable for Butt End Joints. The Butt Ends shall be prepared for for Collar Joint with Grooves. The Pipe Joints shall be capable of withstanding the same pressure as the Pipe. Pipes shall be procured from manufacturers as approved by GE.

24.5. WORKMANSHIP

24.6. **GENERAL** – Refer relevant Clauses of MES Schedule Part I and all specified in Particular Specifications here in before for various Trades.

24.7. **EARTH WORK** – Refer relevant Clauses of MES Schedule Part-I as applicable. Earthwork and Excavation involved in this Contract shall be in Soft / Loose Soil to the depth and gradient as required at Site and ordered by Engineer-in-Charge. Excavation for sewer trenches shall be in straight lines and to the correct depths and gradients.

PARTICULAR SPECIFICATIONS (CONTD…..)

24.8. **CONCRETE BEDDING** – Refer to Clause No. 18.68.2 and 18.68.3 of MES Schedule Part-I and all as directed by Engineer-in-Charge.

24.9. **CONCRETE HAUNCHING** – Refer to Clause No. 18.68.4 of MES Schedule Part-I and all as directed by Engineer-in-Charge.

24.10. **CONCRETE SURROUND OR ENCASING** – Refer to Clause No. 18.68.5 of MES Schedule Part-I and all as directed by Engineer-in-Charge.

24.11. **LAYING OF HDPE PIPES & FITTINGS** – Salt Glazed Stoneware Pipes shall be laid all as specified in Clause No. 18.69.1 to 18.69.3 of MES Schedule Part-I and all as directed by Engineer-in-Charge

24.12. **JOINING OF HDPE PIPES & FITTINGS** – Jointing of Pipes and Fittings shall be carried out all as specified in Clause No. 18.70 and 18.70.1 of MES Schedule Part-I and all as directed by Engineer-in-Charge. Jointing shall be carried out using Spun Yarn and Cement Slurry and filling the joint with Cement Mortar 1:1.

24.13. **TESTING** – After laying of Salt Glazed Stoneware Pipes Testing shall be done all as specified in Clause No.18.79.3 of MES Schedule Part-I.

24.14. **MANHOLES**

24.14.1. Manholes shall be built all as specified in Clause No. 18.78 of MES Schedule Part I, shown on Drawings and specified in Schedule ‘A'.

24.14.2. Manholes and Drain Pipes shall be tested for Water Test all as specified in Clause No. 18.79.3 of MES Schedule Part I.

24.14.3. CI Rungs/Steps shall be of Cast Iron as specified here-in-before. The Steps/Rungs shall be treated with Two Coats of Tar. The Cost of which is deemed to be included in the Unit Rate/Amount

24.15. **SOAK WELL AND SEPTIC TANK**

Soak Well shall be provided all as per details shown on Drawings No. CESZ/DIN/2024/AR-512, Sheet No. 1/1 and specified here in below : -

(a)	Excavation and Earth Work	:	In Soft/Loose Soil. Surplus spoil shall be removed to a distance not exc. 50 Metre and deposited as directed by Engineer-in- Charge.
(b)	Lean Concrete in Foundation	:	PCC 1:4:8 (using 40 mm graded stone aggregate)
(c)	All RCC Work	:	M-25 (Design Mix) using 20 mm graded coarse aggregate
(d)	Masonry Work	:	Brick masonry built in Cement Mortar 1:6
(e)	Plastering (External)	:	15 mm thick in Cement Mortar 1:4 upto 15 cm below ground level
(f)	Inlet Pipe	:	150 mm dia Cast Iron (Sand Cast) Inlet Pipe including Tee. All Joints shall be with Lead Joint.
(g)	Vent Pipe	:	110 mm dia PVC Vent Pipe with PVC Slotted Cowl and as specified in Clause No. 18.27 of MES Schedule Part-I.
(h)	Brick Bat Filling	:	Loose filling with 150mm to 80mm graded broken stone aggregate for filling the bottom layer and 80 mm to 40 mm graded broken stone aggregate for the top layer all as specified in Clause No.3.28 of MES Schedule Part-I.
(j)	Other Details	:	All as shown on Drawings to be follow

25. **ROAD WORKS**

25.1. **GENERAL**

25.1.1. Refer Clause No. 20-A.12, 20-A.16, 20-A.19 of MES Schedule Part-I.

25.1.2. Irrespective of whatever mentioned in these Tender documents, Carpeting, bituminous macadam of road shall be done using **Hot Mix Plant** and WMM, Carpeting, bituminous macadam shall be laid using with **Mechanical Paver** only.

PARTICULAR SPECIFICATIONS (CONTD…..)

25.2. MATERIALS

25.2.1. Materials like Cement, Aggregate, Sand for Mortar, Hardcore, Water, etc. required for Cement Concrete, etc. shall be as specified here-in-before.

25.3. BITUMEN:

25.3.1. Refer Para 20.A.11 of MES Schedule Part I. Bitumen shall be of paving bitumen of grade VG-10 for priming/tack coat and VG30 for other road work. The contractor shall arrange to procure bitumen on his own arrangement.

25.3.2. Paving bitumen shall conform to IS-73(Latest edition): Specification for paving Bitumen and shall be of Grade VG-10 for priming/tack coat and VG30 for other road work meeting all other requirement of IS.

25.3.3. The paving bitumen shall be procured directly from main producers like **IOC, Bharat Petroleum and HP**.

25.3.4. The manufacturer is to carryout inspections and testing of bitumen with the relevant BIS provisions. The contractor shall submit the manufacturer's test certificate in original alongwith the test giving the results of each as applicable or authenticated copy thereof, duly signed by the manufacturer with each consignment.

25.3.5. The GE shall organise independent testing of random samples of paving bitumen drawn from the various lots which shall be got tested by the contractor under his own arrangements and the cost of both materials and labour required for sampling, testing, packing, transportation of samples to the testing places and testing charges thereof, and for subsequent identification of cement sampled, shall be borne by the contractor.

25.4. COMPACTION :-Where rolling by Power Roller is not possible, Compaction shall be done by Rammers as directed by Engineer-in-Charge.

25.5. FORMATIONS:-Compaction of Formation Surface in cutting shall be done by 8 to 12 Tonne Power Roller (for Road) or 0.8 to 1.5 Tonne Animal/Manually Operated Hard Roller (for Path).

25.6. SOLING :-Refer Clause No. 20-A.8 and 20-A.2.1 of MES Schedule Part-I. Soling shall be done with broken boulders.

25.7. WET MIX MACADAM (WMM) BASE COARSE:-

25.7.1 Refer Clause No. 20-B.2.5 of MES Schedule Part-I for Coarse Aggregate, Screening and Binding Materials.

25.7.2 Refer Clause No. 20-B.5 of MES Schedule Part-I for Workmanship.

25.8. BITUMINOUS PREMIX ASPHALTIC DENSE CONCRETE(BPADC) :-

25.8.1. Refer Clause No. 20.B.2.5.4, 20.B.2.11 and 20.B.4 sub para 20.B.4 of MES Schedule Part-I and as per IRC 111. BPADC shall be done with broken stone.

25.8.2. Mix design criteria shall be specified in Clause No. 20.B.2.11 of MES Schedule Part-I. Design Mix shall be got designed from National Test House, SEMT Wing CME, Regional Research labs, IITs, National Institutes of Technology, Govt Engineering Collage or Any Govt laboratories

25.8.3. Quality Control of BPADC strictly compiled as per Clause No. 20.B.4.9 of MES Schedule Part-I.

25.9. BITUMINOUS CARPET

25.9.1. Refer Clause No. 20-A.5 of MES Schedule Part-I for Aggregate. Aggregate shall be of broken hard stone.

25.9.2. Refer Clause No. 20-A.23 and 20-A.27 of MES Schedule Part-I for Workmanship.

25.10. PREMIXED SEAL COAT :-Refer Clause No. 20-A.28, 20-A.28.1(b), 20-A.28.2, 20-A.28.3, 20-A.28.6 and 20-A.28.8 of MES Schedule Part-I.

25.11. EDGING TO ROADS : Edging to Road shall be as specified in Schedule 'A' and shall be laid as per Clause No. 5.54 of MES Schedule Part-I

25.12. PAINTING TO ROADS :

25.12.1. The contractor shall inform GE within a weeks of the commencement of the work as per work order the name of the manufacturer and brands of paint he want to use in the works and submit samples there of and obtain prior written approval of GE before their use in the work and containers of capacity not exceeding 4 litres.

25.12.2. Contractor shall when so required by the GE produce certificates from the Manufacturer or their representatives to establish that the brands of paints purchases by the contractor form them satisfy the quality of paint as specified here-in-before.

25.12.3. Also refer Clause No. 20-A.32 of MES SSR Part I, 2009.

26. COMPOUND WALL/ SECURITY FENCING / GATE

26.1. Refer relevant Clauses of of MES Schedule Part I and all specified in Particular Specifications here in before for various Trades.

26.2. **BARBED WIRE,** Galvanised steel barbed wire for fencing shall confirm to IS 278-2001, Specification for galvanized steel barded wire for fencing. The galvanized barbed wire shall be manufactured from galvanized mild steel wire conforming to IS 280-2006, Galvanised coating of steel wire shall conform to the requirements as laid down for medium coated wire in IS 4826, Hot dipped galvanized coatings on round steel wires. The barbed wire shall consist of two line wire 2.24 mm nominal dia, one or both containing 2 mm dia barbs at 75 mm centre to centre and weighing 97 to 106 Kg per Km as specified in clause 10.32 of MES Schedule Part-I.

26.3. CHAIN LINK FENCE

26.3.1. Chain link fence shall be made of 3.15mm dia galvanized wire and 75 mm (+/-4) mesh Size, Line Wire Dia (mm): 4.00, No. of wires upto & including 2m width: 2, No. of wires above 2m width: 3. The mesh wire and line wire of the fabric shall be manufactured from galvanized steel wire conforming to IS 280:2006 having zinc coating of type heavy as laid down in IS 4826, Hot dipped galvanized coatings on round steel wires or IS 12753, Electrogalvanised coatings on round steel wires. Tensile strength of wires shall be within 400 to 500 MPa. Dia of wire and the length of side of the mesh of chain link fence and the dia of line wire shall be as indicated. The wires shall be free from scale, irregularities, imperfections, flaws, sand splits and other defects. Zinc coating shall be smooth, even and bright. Chain link in fencing shall be fixed to the fencing posts as indicated or directed.

26.3.2. Galvanised Steel Chain Link Fence Fabric with both ends of the fabric knuckled.

26.3.3. Contractors shall be produce original purchase invoice and manufacturer test certificate from wire manufacturer before any payment.

27. AMN STORAGE ACCN AND WHERE APPLICABLE:-

27.1. PCC flooring shall be provide as shown in drawing. The top surface shall be treated with sodium silicate as per Clause 13.9 and 13.35 of MES schedule-2009, Para 1. Top surface of ramp shall be finished and making impressions of expanded metal, whilst the concrete is green.

27.2. Refer Para 13.32 & 13.49 and sub paras of 13.32 & 13.49 of MES Schedule Part-I and specifications of concrete hereinbefore.

27.3. PCC floor in igloo's and where applicable shall be casted by vacuum dewatering method as given in Para 13.49 of MES Schedule, Part-I. No adjustment shall be made on this account while pricing deviation order.

27.4. Admixture(super plasticiser) shall be used as specified hereinbefore.

27.5. Irrespective of whatever shown on drawings/schedule of finishes copper strip shall be 50mm x 3mm thick of copper strip in igloo's and where applicable and other details shall be provide as shown in drawing and panel size 3.00m (Maximum) x 4.00 m (Maximum) as shown on drawing. Copper strip shall be connected to earth lead. Top edge of both copper strips shall be at same level and there should not be any undulation at top level. Two Nos copper strip shall be soldered at interval of 0.60m

27.6. The groove for copper strip of suitable size shall be formed in PCC floor with using mechanical equipment(diamond cutter) within 6 to 36 hours of placing of concrete(depending on weather conditions) and when the concrete has sufficient hardened. The slab edges adjacent to the joint shall be formed truly vertical. The copper strip shall be insert in groove and filled with joint filler.

27.7. WEAR RESISTANT FLOOR TOPPING(NON METALLIC FLOORING)

Non metallic quartz based floor hardener shall be applied as per manufacturer's instructions over PCC flooring of magazine shed and platforms minimum @5 Kg/Sqm.

28. FIRE FIGHTING

28.1. **GENERAL**:-The work shall be carried out as per Section 27 of MES SSR Part-1 2009. These particular specifications are intended to amplify the provisions in SSR Part-1 and shall be read in conjunction with the same. For any discrepancies between the provisions in SSR Part-1 and these particular specifications or Schedule 'A', the more stringent shall apply.

28.2. **SCOPE OF WORK**:- The scope of work shall cover supply, fabrication, installation, testing and commissioning of the fire hydrant system covering the following and complete all as specified in Schedule 'A' :-

28.2.1. The contractor shall get the Fire Fighting System executed by authorized agency holding license from any state Govt / Local Fire Authority to carry out Fire Fighting System Works.

28.3. STANDARDS :-

28.3.1. The fire hydrant installation shall conform to and meet with the requirements set out by the following :-

28.3.2. As relevant IS Code of practice for the safety of buildings (General) fire fighting equipment and its maintenance.

28.3.3. As relevant IS Code of practice for installation of internal fire hydrant in multi-storeyed building.

28.3.4. Compliance with the local fire brigade and the fire enforcing authorities as specifically laid down by them.

28.4. WORKMANSHIP:-The workmanship shall be best of its kind and shall conform to the specifications, as below or as per Indian Standard Specifications in every respect or latest trade practices and shall be subject to approval of GE. All materials and/or Workmanship which in the opinion of GE is defective or unsuitable shall be removed immediately from the site and shall be substituted with proper materials and/or workmanship forthwith.

28.5. MATERIALS:-

28.5.1. All materials and equipment shall conform to the relevant Indian Standards and shall be of the approved make and design. Makes shall be strictly in conformity with list of approved manufacturers as per Appendix – 'B'. As and when required by GE, the contractor shall arrange for testing of the materials and/or portions of works at his own cost to prove their soundness and efficiency. If after tests any materials, work or portions of work are found defective or unsound by GE, the contractor shall remove the defective material from the site, pull down and re-execute the works at his own cost to the satisfaction of GE. To prove that the materials used are as specified the contractor shall furnish original purchase vouchers on demand.

28.5.2. Where manufacturer has furnished specific instructions, relating to the material and equipment used in this project, covering points not specifically mentioned in these documents, such instructions shall be followed in all cases.

28.6. ASSOCIATED CIVIL WORKS:- The civil works associated with Fire Protection installation equipments such as foundations shall be provided as per manufacturer's instructions. Nothing extra shall be paid on this account and the rates quoted for respective items shall be inclusive of such civil works.

28.7. BYE-LAWS AND REGULATIONS:-

28.7.1. Refer SSR Part-1 clause 27.2.2 in this respect. The installation shall be in conformity with the Byelaws, Regulations and Standards of the local authorities concerned, in so far as these become applicable to the installation. But if these Specifications and Drawings call for a higher standard of materials and/or workmanship than those required by any of the above regulations and standards, then these Specifications and Drawings shall take precedence over the said regulations and standards. However, if the Drawings and specifications require something which violates the Bye-laws and Regulations, then the Bye-laws and Regulations shall govern the requirement of this installation.

28.8. FEES AND PERMITS:-

28.8.1. The contractor shall obtain all permits/ licenses and pay for any and all fees required for the inspection, approval and commissioning of their installation. No extra payment shall be made on this account.

28.8.2. As and when notified in writing or instructed by the GE the Contractor shall submit shop drawing and attend all tests and inspections carried out by Local Fire Authorities, Water Authority and other Statutory Authorities, and shall forthwith execute free of charge any rectification work ordered by the GE as a result of such tests and inspections where these indicate non-compliance with Statutory Regulations. Some of these tests may take place after the issue of Completion of the work and the Contractor shall make all allowances in this respect.

28.8.3. The Contractor shall be responsible for the submission of all necessary forms and shop drawings to the Statutory Authorities which shall conform in layout to the latest architectural plans submitted to and kept by these Authorities.

28.8.4. The submission shall comply with the requirements set forth in the current Codes of Practice and circular letters of the Statutory Authorities. The shop drawings to be submitted shall be forwarded to GE for checking before submission.

28.8.5. The Contractor shall notify GE at least seven days in advance of his application for local Authority tests and inspections. On receipt of a confirmed date for test and inspection the Contractor shall inform GE without delay.

28.9. DRAWINGS:-

28.9.1. The Fire Protection Drawings listed under List of drawings are diagrammatic only and indicate arrangement of various systems and the extent of work covered in the contract. These Drawings indicate the points of supply and of termination of services and broadly suggest the routes to be followed. Under no circumstances shall dimensions be scaled from these Drawings. The architectural/interiors drawings and details shall be examined for exact location of sprinklers, hydrants, equipments and water supply / drainage piping etc.

28.9.2. The contractor shall follow the tender drawings in preparation of his shop drawings, and for subsequent installation work. He shall check the drawings of other trades to verify spaces in which his work will be installed.

28.9.3. Maximum headroom and space shall be maintained at all points. Where headroom appears inadequate, the contractor shall notify GE before proceeding with the installation. In case installation is carried out without notifying, the work shall be rejected and contractor shall rectify the same at his own cost.

28.9.4. The contractor shall examine all architectural, structural, plumbing, electrical and other services drawings and check the as-built works before starting the work, report to GE any discrepancies and obtain clarification. Any changes found essential to coordinate installation of fire fighting with other services and trades, shall be made with prior approval of GE.

28.9.5. Contractor shall periodically submit completion drawings as and when work in all respects is completed in a particular area. These drawings shall be submitted in the form of two sets of CD's and four portfolios (300 x 450 mm) each containing complete set of drawings on approved scale indicating the work as installed. These drawings shall clearly indicate complete plant room layouts, piping layouts, location of wiring and sequencing of automatic controls, location of all concealed piping, valves, controls, wiring and other services. Each portfolio shall also contain consolidated control diagrams and technical literature on all controls. The contractor shall frame under glass, in the plant room, one set of these consolidated control diagrams.

28.10. ACCESSIBILITY:-The Contractor shall verify the sufficiency of the size of the shaft openings, clearances in cavity walls and suspended ceilings for proper installation of his piping and other ancillaries. His failure to communicate insufficiency of any of the above, shall constitute his acceptance of sufficiency of the same. The Contractor shall locate all equipment which must be serviced, operated or maintained in fully accessible positions. The exact location and size of all access panels, required for each concealed valve or other devices requiring attendance, shall be finalized and communicated in sufficient time, to be provided in the normal course of work. Failing this, the Contractor shall make all the necessary repairs and changes at his own expense. Access panel shall be standardized for each piece of equipment / device / accessory and shall be clearly nomenclature/marked.

28.11. FIRE DETECTION & FIRE ALARM SYSTEM :-

28.11.1. Refer SSR Part-I clause 27.4, 27.5, 27.6 & 27.7 in this respect.

28.11.2. SCOPE OF WORK : This scope of work covers the supply, installation, testing and commissioning and getting all necessary approvals from competent authorities of fire and electrical safety departments for the complete fire detection and alarm system which comprises of following and complete all as specified in Schedule 'A' :-

28.11.2.1. Smoke and heat detectors

28.11.2.2. Manual call points

28.11.2.3. Electronic hooters

28.11.2.4. Response indicators

28.11.2.5. Zonal panels with mimic panels

28.11.2.6. Main Control and indicating panel, mimic panel with battery

28.11.2.7. Wiring / conduiting /cabling from Zonal / local control panel to detector heads and response indicators and from zonal panel to the main control panel.

28.11.2.8. All other allied and necessary equipment and accessories to complete the system up to the satisfaction of Engineer-in- Charge and for proper functioning of the entire system.

28.11.2.9. In general the complete installation of fire alarm and detection system shall strictly confirm to the IS: 2189, and other relevant IS specifications and amendments if any made till completion of the work.

28.11.3. EQUIPMENTS

28.11.3.1. Multisensor (Smoke & Heat) detectors (Ionization and optical Type)

Operating Voltage range	-	12 V/24 V DC
Ionization source	-	Less than one micro C maximum Americium 241
Detection (ionization type)	-	Unipolar type dual ionization chamber
Detection (Optical type)	-	Photo optic sensing chamber
Visual	-	Twin L.E.D. (Stand by & Alarm)
Operating Temperature	-	0oC to 49oC
Construction	-	Flame retardant

28.11.3.2. The detector head should be cylindrical incorporating two Ionization chambers and double source mounted on printed circuit board. One chamber shall be sealed from the environment and other open to the atmosphere. A LED lamp should be incorporated which should normally flicker (Six flashes per minute) indicating alterness and should turn steady when a fire is sensed enabling immediate identification of the detector.

28.11.3.3. Detector should be capable of sensing fire in the smoldering or incipient stage. The detector shall be equipped with a smoke response delay of approximately 5 second to prevent false alarm due to transient smoke. Smoke detectors shall be sensitive to products of combustion of all materials like wood, paper, ammonia processing paper, cloth, PVC, Bakelite, nylon, foam, rubber, acrylic, thermocole, photo film, leather etc. The detector shall be fixed to the base by a twist lock action. The enclosure shall be of off white textured high impact plastic. Units shall be suitable for mounting directly to a 75mm/100 mm round box.

28.11.3.4. Optical type smoke detector shall consists of an infra red LED and a high speed light sensing photo diode within a sensing chamber operating on photo electric principal and be capable of detecting visible products of combustion like PVC.

28.11.3.5. The detectors sensitivity shall remain constant and not vary with change in ambient temperature, humidity, pressure or voltage upto + or - 10% The performance of the detectors shall not be effected by air currents upto10 Mtr/Sec. Detectors shall be suitably protected against dust accumulation and increase of moisture etc

28.11.3.6. It should be possible to functionally test the detector as well as assess its actual sensitivity without having to remove it. Insect screen to protect the detector shall be provided.

28.11.3.7. The sensitivity of all types of detectors shall be set / adjusted by the supplier to suit the site conditions.

28.11.4. Heat Detectors

28.11.4.1. The heat detector shall be plug-in-type with a cylindrical head, housing a thermostat as temperature sensing element. The detector shall be designed to operate when the temperature at the detectors exceeds a predetermined value 57oC to 60oC (135oF). They shall be suitable for operation on rate of rise (15oF/min.) and fixed temperature setting.

28.11.4.2. Electrical contacts and orders moving parts of the detector shall be enclosed in such a manner that will afford protection against moisture, dust, insects and other foreign matter. All make and break contacts shall be silver or its alloy of equivalent characteristics. Any adjustment made in the factory shall be securely sealed and any adjusting screws shall be of appropriate type with reliable means of locking. The means of adjustment should be rendered inaccessible to prevent tempering when the detector is mounted metal. It should have the provision of connecting a response indicator. The detectors shall be suitable for 2 wire connections with minimum current consumption.

28.11.4.3. On detection, the detector shall transmit the signal to the control panel and the alarm should latch. The detector should have no fusible element but should be resettable automatically. Resetting of the system shall be done from the control panel.

28.11.4.4. The detectors shall meet the performance requirements as per IS : 2175- 1988.

28.11.5. Manual Call Point

28.11.5.1. Manual call points shall be of wall mounting type. The housing shall be dust and moisture proof sheet steel enclosure of 1.5 mm thickness properly sealed with rubber lining. The glass

surface should be minimum 30 sqcm in area and glass thickness should not exceed 2 mm. Once the glass is broken the alarm should sound on the floor as well as on the control and indicating equipment and light should glow to indicate its operation. The alarm should maintained by the control and indicating equipment even if someone press the button subsequently.

28.11.5.2. A breakable glass cover fixed in such a way that the actuating push button is kept depressed as long as the glass is intact and released when the glass is broken.

28.11.5.3. A small steel hammer shall be attached to the assembly with a steel chain to facilitate breaking of glass. The manual call points shall be suitable for surface or fully / partly recessed mounting. 5mm high red painting/printed word “IN CASE OF FIRE BREAK GLASS” shall be provided on the front face.

28.11.6. Electronic Hooter

28.11.6.1. The hooter/sounder should work in conjunction with Public Address System. It should be wall/ceiling mounting with speaker to cover. The required audible range is 25 meters. The enclosure should be M.S. sheet of 1.5 mm thickness duly painted and marked "FIRE ALARM". All hooters shall produce a similar sound and shall maintain the same during their operation.

28.11.7. Response Indicators

28.11.7.1. Lamp assembly consisting of lamp holder and lamp shall be suitable for mounting on walls, partition etc. The housing shall be suitable for surface/flush mounting of 1.5mm thick sheet steel and the facia 3mm acrylic mounted outside a cabin, and directly connected to the detector inside the cabin. In normal circumstances the lamp should flicker but in the event the detector inside the cabin senses a fire, the lamp should glow.

28.11.8. Control & Indicating Panel

28.11.8.1. The control & indicating panel shall be of modular construction type and consisting of all solid state circuitry enclosed in 16 gauge (minimum) M.S. dust and vermin proof enclosure. The panel provides power to audio alarms for fire and also feeds supply to public address system.

28.11.8.2. The supply unit inside the control & indicating shall consist of transformers, bridge rectifier, and filter condenser card and regulator cards. Secondary of the step down 230/240 volts transformer shall feed a silicon diode bridge rectifier to get D.C. output, which is filtered by filter condenser assembly to minimize ripple on secondary side. Unregulated D.C. shall finally be regulated to 24 volt D.C. by regulator card containing integral circuits.

28.11.8.3. Panel should be made in a module of different zones. Each zone will have audible and visual indications and will monitor the circuit conditions.

28.11.8.4. The system shall be so designed that in case of failure of A.C. main supply it shall automatically change over to battery supply.

28.11.8.5. All control shall be clearly labeled to indicate the mode of operation and their function.

28.11.8.6. The control and indicating equipment should have electronic/electrical relays conforming to relevant Indian Standards voltmeter and ammeter on main DC power circuit. Following switches, push buttons and indications shall be provided on panel facia :-

28.11.8.6.1. Mains “ON” switch with indication

28.11.8.6.2. System “ON” indication

28.11.8.6.3. Mains Failure indication

28.11.8.6.4. Battery low indication

28.11.8.6.5. Lamp test push button

28.11.8.6.6. Fire indication

28.11.8.6.7. Fault indication

28.11.8.6.8. Fire test push button per zone

28.11.8.6.9. Zone isolate switch per zone

28.11.8.6.10. Standby on indication

28.11.8.6.11. System reset push button

28.11.8.6.12. Alarm cancel push button

28.11.8.6.13. Trickle boost toggle switch

28.11.8.6.14. Open/short circuit fault.

28.11.8.7. The panel should have a sounder with distinct audible sound for fire and fault alarm.

28.11.8.8. There should be the facility to isolate zonal/sector control panel from the rest of the system. There should be a visual indicator at the main panel if any zone at the zonal / sector panel is isolated.

28.11.8.9. Control panel shall be complete with screw type terminal block and cable glands. Two potential free contacts under fire condition shall be provided for purchaser's use.

28.11.8.10. Necessary end of line resistors shall be included in the control & indicating panel for the system. It shall be possible for the system to monitor the removal and pilferage of any of the detectors. The panel shall be complete with all internal wiring labels, and duly painted with two coats of red oxide primer and one coat of spray paint as required. The panels should preferably be powder coated in PO red colour.

28.11.8.11. The control & indicating panel shall also incorporate public address system with selector switches for each zone and all call facility to address all the speaker/hooters simultaneously. Arrangement should be so made, so that PA system of the zone giving fire indication automatically get activated. There should be a switch to changeover from sounder to announcement.

28.11.8.12. Local control panel should have talk back facility. One potential free contact shall also be provided on each local control panel for future use. The master station for the talk back system shall be located at the main control panel. The master station shall have the overriding facility to choose and select the particular talk back station which is apparently more important. The reset facility should be at the master station.

28.11.8.13. All panels should be provided with proper earthing of copper plate.

28.11.8.14. Mimic panel shall also be provided with main control and indicating panel and zonal control panels as per standard requirements and design showing building drawings indicating all fire fighting and alarm equipment, escape routes etc.

28.11.8.15. Main control & indicating panel and zonal control panels shall fully satisfy all their requirements laid down in clause 6 and 7 of IS : 2189 amended up to date.

28.11.9. WIRING

28.11.9.1. Wiring for fire alarm system in general shall comply with IS-15908(2011), IS : 2189 and IS : 732 - amended up to date. The detectors shall be wired upto the main junction boxes by ISI marked 1.5 sq mm PVC insulated stranded copper conductor wires of 1100 volts grade in ISI marked 16 SWG MS conduits of 25 mm and 20 mm dia. as required and desired on site unless otherwise specified in Schedule ‘A’. Crimping type of lugs are also included in the rates of laying wires and its connections.

28.11.9.2. PVC insulated copper conductor wires shall confirm to IS : 694 and PVC insulated copper conductor armoured cables should confirm to IS : 1554 (Part-1).

28.12. TESTING AND COMMISSIONING

28.12.1. GENERAL

28.12.1.1. The contractor shall be responsible for testing and commissioning the entire services installation described in these specifications and will demonstrate the operation of the system up to the entire satisfaction of the GE.

28.12.1.2. After the installation is complete, the contractor shall conduct operating and commissioning tests complete all as per SSR Clause No. 27.9 of SSR Part-I, 2009 and as per the provisions of various codes of Practice, fire protection manual of TAC and National Building Code. The equipment will be demonstrated to operate in accordance with the requirements of the specification. The system installation, testing and commissioning shall be as per Local approvals and requirements.

28.12.1.3. Prior to commissioning, the Contractor shall provide a certificate of installation and cable test certificates of the system to the Engineer-in-Charge.

28.12.2. ON SITE TESTING

28.12.2.1. The contractor shall provide on site all the necessary instruments, plants, equipment, materials, water, electricity and labour necessary for carrying out the specified tests. All tests shall be carried out as required to meet the construction programme and the contractor shall be include for all necessary isolation and other works as may be required for testing the whole or parts of the installation. The contractor shall also be responsible for re-testing, if necessary, until satisfactory test results achieved.

28.12.2.2. The fire alarm system shall be completely programmed in accordance with Fire Department requirement and a specialist from the manufacturer shall attend and demonstrate the complete

system. A company trained representative shall personally supervise the complete installation and final testing of the system. All tests shall be carried out in the presence of the Engineer-in-Charge. Upon the completion of the acceptance tests, the representatives shall instruct operatives in the proper operation, maintenance programming, configuration, and testing of the system. The contractor shall provide equipment and / or software which are necessary to allow field modification of the programming and configuration.

28.12.3. COMMISSIONING AND ACCEPTANCE TESTS FOR FIRE ALARM SYSTEM

28.12.3.1. The commissioning and acceptance tests shall be apart from the standard or routine tests prescribed and normally conducted by the manufacturer/contractor and will be irrespective of the fact whether the same are covered by such tests or not.

28.12.3.2. Each zone shall be tested by a test fire or by a heat source such as hair dryer or shielded heat lamp below any one detector selected arbitrarity and the time required for detection shall be noted.

28.12.3.3. Each sounder circuit shall be energized separately and the sound level reading taken to check for conformity with the minimum standards.

28.12.3.4. Open circuit and removal of detector for each detection circuit shall be tested.

28.12.3.5. Short circuit for each detection circuit will be tested.

28.12.3.6. Mains failure performance.

28.12.3.7. Open circuit of each sounder circuit to be tested.

28.12.3.8. Short circuit of each sounder circuit to be tested.

28.12.3.9. The results of the above tests either by faults warning or fire alarm shall be recorded in the log books which will be signed both by the contractor and the employer.

28.13. COMPLETION DRAWINGS

28.13.1. The contractor shall submit, after completion Six copies of completion drawings and literature in calico bound book form with duly signed by him and the Engineer-in-Charge alongwith soft copy in Auto CAD-2008 or higher version, giving the following information :-

28.13.1.1. Position of all detectors, hooters, manual call points, response indicators, exit signs etc.

28.13.1.2. Runs of all conducting and wirings on all floors.

28.13.1.3. Position of all zonal control panels and main control panels etc.

28.14. OPERATION & MAINTENANCE MANUALS

28.14.1. The contractor shall have to hand over all operation and maintenance manuals of the plant and equipment supplied and installed by him. Only manufacturer's catalogues, wiring diagram and installation drawings relevant to particular items of equipment concerned shall be submitted. General catalogues will not be acceptable.

28.15. ON SITE TRAINING:-Upon completion of all work and all tests, the Contractor shall furnish necessary operators, labour and helpers for operating the entire installation for a period of fifteen (15) working days of ten (10) hours each, to enable the MES staff to get aquainted with the operation of the system. During this period, the contractor shall train the MES personnel in the operation, adjustment and maintenance of all equipment installed**.**

29. JUNCTION / COMBINER BOX

29.1. All junction/ combiner boxes including the module junction box, string junction box, array junction box and main junction box should be equipped with appropriate functionality, safety (including fuses, grounding, etc.), string monitoring capabilities, and protection.

29.2. The terminals will be connected to copper bus-bar arrangement of proper sizes to be provided. The junction boxes will have suitable cable entry points fitted with cable glands of appropriate sizes for both incoming and outgoing cables. Suitable markings shall be provided on the bus-bars for easy identification and cable ferrules will be fitted at the cable termination points for identification.

29.3. The junction boxes shall be dust, vermin, and waterproof and made of thermoplastic/ metallic in compliance with IEC 62208, which should be sunlight/ UV resistive as well as fire retardant & must have minimum protection to IP 65(Outdoor)/ IP 54(indoor) and Protection Class II.

30. TURFING:- The readymade grass turfing locally available shall be provided on the top and sides of earthen filling of NAT and VIFT or wherever directed by Engr-in-Charge by sodding. The height of earth filling as shown in drawings is without grass turfing and the turfing shall be provided above this height. The blocks of readymade grass turfs (with minimum 5cm of soil covering the grass roots) shall be obtained from any source outside defence land and shall be

transported carefully and planted on the earth filling in traverses. If found necessary, pegs or nails shall be used to hold down the grass sods in the initial stage. The turfing shall be properly maintained as directed by GE.

30.1. A tenderer shall himself ascertain the level of sub-soil water at site of work to allow in his tender for any bailing or pumping out water etc, required as per site conditions all as specified in clause 3.17 of MES Schedule (Part I). In the event of deviation, no adjustment shall be made for cost of bailing, pumping and de-watering arising out of the reasons as specified herein before similarly no adjustment in lump sum will be made if no sub-soil water is met with at site.

31. LIST OF BUILDINGS WHERE ATT AND WATER PROOFING TREATMENT TO BE CARRIED OUT:

Sr..No	Name of building	No. of BLDGS
1	2	3
1	AMN STORAGE ACCN URP HD 1.1, CHAMBER 5 X 2MT. (NEC) (URP NO.1, 2 & 4 WITH TRAVERSE	3
2	AMN STORAGE ACCN URP HD 1.1, CHAMBER 6 X 2MT. (NEC) (URP NO. 3 WITH TRAVERSE	1
3	AMN STORAGE ACCN (W P SHED) HD 1.2, W/O TRAVERSE(ESH-5)	1
4	AMN STORAGE ACCN (STORAGE) HD 1.3, W/O TRAVERSE (ESH-6)	1
5	AMN STORAGE ACCN (STORAGE) HD 1.4, W/O TRAVERSE (ESH-7)	1
6	RETURNED AMMUNITION AND EMPTIES AREA ESH NO.8	1
7	RECEIPT AND ISSUE SHED/COUNTER	2
8	AMN POINT MAIN OFFICE	1
9	GARRAGES FOR ‘B’ VEH	1
10	FIRE STATION	1
11	FUEL STORE	1
12	GUARD HOUSE	1
13	MAIN GATE WITH INFRASTRUCTURE AT IN & OUT GATE	2
14	PUMP HOUSE CUM CHLORINATION PLANT	1
15	PUMP HOUSE	1
16	SENTRY ROOM	3
17	GARAGE FOR PARKING OF 33 CYCLES AND 02 SCOOTERS	1
18	SHED FOR GEN SET	2
19	SUB STATION BLDG	1
20	GARAGE FOR 2 X TFF	1
21	UNIT STORAGE ACCN	1
22	GUARD ROOM WITH WATCH TOWER	3
23	WATCH TOWER	3
24	COOK HOUSE WITH DINING HALL	1
25	GUARD ROOM	3
26	DRYING AND IRONING ROOM	1
27	SINGLE LIVING ACCN FOR JCO’S (JCO-01, DSC-01)	1
28	LIVING ACCN FOR 2 HAV AND 28 OR’S	1
29	SCOOTER SHED FOR 2 HAV AND 28OR’S	1

PARTICULAR SPECIFICATIONS (CONTD…..)

32. **LIST OF DRAWING**

Ser No	Description	Drawings No	Sheet No	Date	
				Original	Last revision
1	2	3	4	5	6
	LIST OF DRAWING				
1.	LIST OF DRAWINGS	CESZ/DIN/2024/LD-112	1/2	26.09.22	-
2.	LIST OF DRAWINGS	CESZ/DIN/2024/LD-112	2/2	26.09.22	-
	EXTERNAL E/M & B/R SERVICES				
3.	SITE PLAN (AREA DRAINAGE, SEWAGE PLAN)	CESZ/DIN/2024/LP-112	1/3	26.09.22	-
4.	SITE PLAN INCLUDING ROADS, RETAINING WALL, COMPOUND WALL AND FENCING	CESZ/DIN/2024/LP-112	2/3	26.09.22	-
5.	FIRE FIGHTING, EXT WATER SUPPLY & EXT ELECTRIC SUPPLY LAYOUT	CESZ/DIN/2024/LP-112	3/3	26.09.22	-
	AMN STORAGE ACCN URP HD 1.1, CHAMBER 5 X 2MT. (NEC) (URP NO.1, 2 & 4 WITH TRAVERSE				
6.	PLAN AND ELEVATION	CESZ/DIN/2024/AR-473	1/4	26.09.22	-
7.	SECTIONS AND ELEVATION	CESZ/DIN/2024/AR-473	2/4	26.09.22	-
8.	ROOF PLAN, CROSS SECTION AND ELEVATION	CESZ/DIN/2024/AR-473	3/4	26.09.22	-
9.	ELEVATION, SECTIONS AND STEEL DOOR DETAILS	CESZ/DIN/2024/AR-473	4/4	26.09.22	-
10.	STRUCTURAL FOUNDATION PLAN, PLAN OF ROOF LEVEL	CESZ/DIN/2024/ST-473	1/5	11.07.22	21.05.24
11.	TYPICAL FLOOR PLAN & PART SECTION AT A-A	CESZ/DIN/2024/ST-473	2/5	11.07.22	21.05.24
12.	SECTION B-B, C-C AND D-D	CESZ/DIN/2024/ST-473	3/5	11.07.22	21.05.24
13.	STRUCTURAL DETAILS OF RETAINING WALL AND TRAVERSE	CESZ/DIN/2024/ST-473	4/5	11.07.22	-
14.	STEEL DOOR(2500 X 3000) ESH 1 TO 4	CESZ/DIN/2024/ST-473	5/5	11.07.22	21.05.24
	AMN STORAGE ACCN URP HD 1.1, CHAMBER 6 X 2MT. (NEC) (URP NO. 3 WITH TRAVERSE				
15.	PLAN AND ELEVATION	CESZ/DIN/2024/AR-474	1/4	26.09.22	-
16.	SECTIONS AND ELEVATION	CESZ/DIN/2024/AR-474	2/4	26.09.22	-
17.	ROOF PLAN, ELEVATION AND CROSS SECTION	CESZ/DIN/2024/AR-474	3/4	26.09.22	-
18.	ELEVATION, SECTIONS AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-474	4/4	26.09.22	-
19.	TYPICAL PLAN OF WALL & SECTION (SHOWING REIN DETAILS)	CESZ/DIN/2024/ST-474	1/1	11.07.22	21.05.24
	AMN STORAGE ACCN (W P SHED) HD 1.2, W/O TRAVERSE(ESH-5)				
20.	PLAN ELEVATION AND ROOF PLAN	CESZ/DIN/2024/AR-475	1/3	26.09.22	-
21.	SECTIONS	CESZ/DIN/2024/AR-475	2/3	26.09.22	-

PARTICULAR SPECIFICATIONS (CONTD…..)

22.	ELEVATION AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-475	3/3	26.09.22	-
23.	STRUCTURAL LAYOUT OF FOUNDATION, COMBINED FOOTING, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-475	1/1	11.07.22	-
	AMN STORAGE ACCN (STORAGE) HD 1.3, W/O TRAVERSE (ESH-6)				
24.	PLAN AND ELEVATION	CESZ/DIN/2024/AR-476	1/3	26.09.22	-
25.	ELEVATION AND SECTIONS	CESZ/DIN/2024/AR-476	2/3	26.09.22	-
26.	ROOF PLAN, STEEL DOOR AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-476	3/3	26.09.22	-
27.	STRUCTURAL LAYOUT OF FOUNDATION, COMBINED FOOTING, PLINTH BEAM, FLOOR BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-476	1/1	11.07.22	21.05.24
	AMN STORAGE ACCN (STORAGE) HD 1.4, W/O TRAVERSE (ESH-7)				
28.	PLAN AND ELEVATION	CESZ/DIN/2024/AR-477	1/3	26.09.22	-
29.	ELEVATION AND SECTIONS	CESZ/DIN/2024/AR-477	2/3	26.09.22	-
30.	ROOF PLAN AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-477	3/3	26.09.22	-
31.	STRUCTURAL LAYOUT OF FOUNDATION, COMBINED FOOTING, PLINTH BEAM, FLOOR BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-477	1/1	11.07.22	21.05.24
	RETURNED AMMUNITION AND EMPTIES AREA ESH NO.8				
32.	PLAN, ELEVATION AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-478	1/3	26.09.22	-
33.	SECTIONS	CESZ/DIN/2024/AR-478	2/3	26.09.22	-
34.	ROOF PLAN, ELEVATIONS AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-478	3/3	26.09.22	-
35.	STRUCTURAL LAYOUT OF FOUNDATION, COMBINED FOOTING, PLINTH BEAM, FLOOR BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-478	1/1	11.07.22	21.05.24
	RECEIPT AND ISSUE SHED/COUNTER				
36.	PLAN, ELEVATIONS AND ROOF PLAN	CESZ/DIN/2024/AR-479	1/2	26.09.22	-
37.	SECTION AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-479	2/2	26.09.22	-
38.	STRUCTURAL LAYOUT OF FOUNDATION, COMBINED FOOTING, PLINTH BEAM, FLOOR BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-479	1/1	11.07.22	-
	AMN POINT MAIN OFFICE				
39.	PLAN, ELEVATIONS AND ROOF PLAN	CESZ/DIN/2024/AR-480	1/2	26.09.22	-
40.	SECTION AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-480	2/2	26.09.22	-
41.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-480	1/1	11.07.22	21.05.24
	COOK HOUSE WITH DINING HALL				
42.	ENLARGED PLAN OF COOK HOUSE AND DINING HALL	CESZ/DIN/2024/AR-481	1/5	26.09.22	-
43.	ROOF PLAN AND SECTION	CESZ/DIN/2024/AR-481	2/5	26.09.22	-

PARTICULAR SPECIFICATIONS (CONTD…..)

44.	ELEVATIONS	CESZ/DIN/2024/AR-481	3/5	26.09.22	-
45.	DETAILS OF SUMP KITC. WASTE, WORKING PLATFORM, CUPBOARD WITH THREE TIER SHELVES AND PREP. TABLE	CESZ/DIN/2024/AR-481	4/5	26.09.22	-
46.	MISC DETAILS	CESZ/DIN/2024/AR-481	5/5	26.09.22	-
47.	STRUCTURAL LAYOUT OF PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-481	1/1	26.09.22	21.05.24
	DRYING AND IRONING ROOM				
48.	PLAN, ELEVATIONS, ROOF PLAN	CESZ/DIN/2024/AR-482	1/2	26.09.22	-
	FUEL STORE				
49.	PLAN, ELEVATIONS, SECTION, ROOF PLAN	CESZ/DIN/2024/AR-482	2/2	26.09.22	-
	FUEL STORE AND IRONING ROOM				
50.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-482	1/1	11.07.22	
	GUARD ROOM				
51.	PLAN, ELEVATIONS AND ROOF PLAN	CESZ/DIN/2024/AR-483	1/2	26.09.22	-
52.	SECTIONS, TOILET DET. & MISC DET.	CESZ/DIN/2024/AR-483	2/2	26.09.22	-
53.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-483	1/1	26.09.22	21.05.24
	FIRE STATION				
54.	PLAN, ROOF PLAN	CESZ/DIN/2024/AR-484	1/3	26.09.22	-
55.	ELEVATION, SECTION	CESZ/DIN/2024/AR-484	2/3	26.09.22	-
56.	TOILET SECTION AND MISC DETAIL	CESZ/DIN/2024/AR-484	3/3	26.09.22	-
57.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-484	1/1	26.09.22	-
	UNIT STORAGE ACCN				
58.	PLAN, ROOF PLAN AND ELEVATIONS	CESZ/DIN/2024/AR-485	1/2	26.09.22	-
59.	SECTIONS & MISC DET.	CESZ/DIN/2024/AR-485	2/2	26.09.22	-
60.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-485	1/1	11.07.22	-
	GARAGE FOR 2 X TFF				
61.	PLAN, ELEVATION, ROOF	CESZ/DIN/2024/AR-486	1/1	26.09.22	-
62.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-486	1/1	11.07.22	-
	GARRAGES FOR 'B' VEH				
63.	PLAN, ELEVATION, ROOF	CESZ/DIN/2024/AR-487	1/2	26.09.22	-
64.	SECTIONS AND OTHER MISC DETAILS.	CESZ/DIN/2024/AR-487	2/2	26.09.22	-
65.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-487	1/1	11.07.22	-
	GUARD HOUSE & ANCILLARIES				
66.	PLAN, ROOF PLAN	CESZ/DIN/2024/AR-488	1/4	26.09.22	-
67.	ELEVATIONS	CESZ/DIN/2024/AR-488	2/4	26.09.22	-

PARTICULAR SPECIFICATIONS (CONTD…..)

68.	TOILET DETAILS, SECTION & MISC. DETAILS	CESZ/DIN/2024/AR-488	3/4	26.09.22	-
69.	STEEL DOOR	CESZ/DIN/2024/AR-488	4/4	26.09.22	-
70.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-488	1/1	26.09.22	-
	SENTRY ROOM, GARAGE FOR PARKING OF 33 CYCLES AND 02 SCOOTERS				
71.	1. PLAN, ELEVATIONS, SECTIONS, ROOF PLAN AND OTHER MISC DETAILS OF SENTRY ROOM 2. PLAN, ELEVATIONS, SECTION, ROOF PLAN AND OTHER MISC DETAILS OF GARAGE FOR PARKING OF 33 CYCLES & 02 SCOOTERS	CESZ/DIN/2024/AR-489	1/1	26.09.22	-
72.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-489	1/1	11.07.22	-
	SHED FOR GEN SET				
73.	PLAN, ELEVATIONS, SECTION, ROOF PLAN AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-490	1/2	26.09.22	-
	SUB STATION BLDG				
74.	PLAN, ELEVATIONS, SECTION, ROOF PLAN	CESZ/DIN/2024/AR-490	2/2	26.09.22	-
	SUB STATION AND SHED FOR GEN SET				
75.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-490	1/1	11.07.22	-
	LIVING ACCN FOR 5 OR'S & 28(DSC) (2 HAV AND 28 OR'S)				
76.	GROUND FLOOR PLAN, FIRST FLOOR PLAN AND FRONT ELEVATION	CESZ/DIN/2024/AR-491	1/4	26.09.22	-
77.	ROOF PLAN, FRONT & REAR ELEVATION, SIDES ELEVATION AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-491	2/4	26.09.22	-
78.	DETAILS OF STAIRCASE, SECTIONS AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-491	3/4	26.09.22	-
79.	DETAILS OF TOILET-1 (GROUND FLOOR) DETAILS OF TOILET No.2 (FIRST FLOOR)	CESZ/DIN/2024/AR-491	4/4	26.09.22	-
80.	STRUCTURAL DETAILS OF FLOOR BEAM, ROOF BEAM & SLAB AND TYPICAL DETAILS	CESZ/DIN/2024/ST-491	1/2	26.09.22	-
81.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, FLOOR BEAM & SLABS, FOUNDATION, PLINTH BEAM AND COMBINED FOOTING DETAILS	CESZ/DIN/2024/ST-491	2/2	26.09.22	21.05.24
	MAIN GATE WITH INFRASTRUCTURE AT IN & OUT GATE				
82.	PLAN, ROOF PLAN	CESZ/DIN/2024/AR-492	1/4	26.09.22	-
83.	ELEVATIONS	CESZ/DIN/2024/AR-492	2/4	26.09.22	-
84.	RIGHT SIDE ELEVATION, TOILET DETAIL	CESZ/DIN/2024/AR-492	3/4	26.09.22	-
85.	SECTION A-A, MISC DET.	CESZ/DIN/2024/AR-492	4/4	26.09.22	-
86.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-492	1/1	26.09.22	-
	WATCH TOWER				
87.	GROUND FLOOR PLAN, FIRST FLOOR, SECOND FLOOR PLAN,	CESZ/DIN/2024/AR-493	1/1	26.09.22	-

PARTICULAR SPECIFICATIONS (CONTD…..)

	ELEVATIONS, SECTION AND ROOF PLAN				
88.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, FLOOR BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-493	1/1	11.07.22	-
	PUMP HOUSE				
89.	PLAN, ELEVATIONS AND ROOF PLAN	CESZ/DIN/2024/AR-494	1/4	26.09.22	-
90.	SECTION AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-494	2/4	26.09.22	-
	PUMP HOUSE CUM CHLORINATION PLANT				
91.	PLAN, ELEVATIONS, ROOF PLAN	CESZ/DIN/2024/AR-494	3/4	26.09.22	-
92.	SECTION & MISC DET.	CESZ/DIN/2024/AR-494	4/4	26.09.22	-
	PUMP HOUSE & PUMP HOUSE CUM CHLORINATION PLANT				
93.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-494	1/1	11.07.22	-
	SINGLE LIVING ACCN FOR JCO'S (JCO-01, DSC-01)				
94.	GROUND FLOOR PLAN & ROOF PLAN	CESZ/DIN/2024/AR-495	1/4	26.09.22	-
95.	SECTION AND MISC DETAILS	CESZ/DIN/2024/AR-495	2/4	26.09.22	-
96.	ELEVATIONS	CESZ/DIN/2024/AR-495	3/4	26.09.22	-
97.	DETAIL OF TOILET SECTION AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-495	4/4	26.09.22	-
98.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-495	1/1	26.09.22	-
	GUARD ROOM WITH WATCH TOWER				
99.	GROUND FLOOR PLAN, FIRST FLOOR, SECOND FLOOR PLAN AND ELEVATION	CESZ/DIN/2024/AR-496	1/3	26.09.22	-
100.	ELEVATIONS AND SECTIONS	CESZ/DIN/2024/AR-496	2/3	26.09.22	-
101.	DETAILS OF STAIRCASE, DETAILS OF TOILET AND OTHER MISC DETAILS	CESZ/DIN/2024/AR-496	3/3	26.09.22	-
102.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, FLOOR BEAM, ROOF BEAM & SLABS AND TYPICAL DETAILS	CESZ/DIN/2024/ST-496	1/2	26.09.22	-
103.	STRUCTURAL LAYOUT OF STAIRCASE AND DETAILED SECTIONS	CESZ/DIN/2024/ST-496	2/2	11.07.22	-
	LIVING FOR 5 OR'S & 28(DSC) (2 HAV AND 28OR'S) SCOOTER SHED				
104.	PLAN, FRONT ELEVATION, SECTION AND ROOF PLAN	CESZ/DIN/2024/AR-497	1/1	26.09.22	-
105.	STRUCTURAL LAYOUT OF FOUNDATION, PLINTH BEAM, ROOF BEAM & SLABS	CESZ/DIN/2024/ST-497	1/1	26.09.22	-
	OTHER STRUCTURAL DRAWING				
106.	STATIC WATER TANK 225 KL	CESZ/DIN/2024/ST-498	1/1	11.07.22	21.05.24
107.	DETAIL OF COMPOUND WALL & MAIN GATE 6M WIDE	CESZ/DIN/2024/ST-499	1/1	26.09.22	-
108.	SCHEDULE OF RCC COLUMNS, RCC BEAMS, RCC SLABS AND RCC FOOTINGS FOR EXPLOSIVE STORAGE SHEDS AND RECEIPT AND ISSUE SHED	CESZ/DIN/2024/ST-500	1/1	11.07.22	21.05.24
109.	GENERAL NOTES AND MISC DETAILS	CESZ/DIN/2024/ST-501	1/2	11.07.22	-

PARTICULAR SPECIFICATIONS (CONTD…..)

110.	MISCELLANEOUS DETAILS	CESZ/DIN/2024/ST-501	2/2	11.07.22	-
111.	RCC UG SUMP 225 KL CAPACITY	CESZ/DIN/2024/ST-502	1/1	11.07.22	21.05.24
112.	RETAINING WALL TYPE 1, 2, 3 & 4	CESZ/DIN/2024/ST-503	1/1	11.07.22	-
113.	SCHEDULE OF RCC COLUMNS, RCC BEAMS, RCC SLABS AND RCC FOOTINGS FOR NON EXPLOSIVE STORAGE AND OTHER BUILDING	CESZ/DIN/2024/ST-504	1/1	11.07.22	-
114.	STEEL DOOR DETAILS (2000 X 2500) AND (3000 X 3500)	CESZ/DIN/2024/ST-505	1/1	11.07.22	-
115.	BOX CULVERT 1M X 1M, 2M X 2M & 3M X 3M	CESZ/DIN/2024/ST-506	1/1	11.07.22	-
	TYPICAL MISCELLANEOUS ARCHITECTURAL DRAWING				
116.	TYPICAL DETAILS OF PANELED DOORS, PVC DOOR AND FIXED GLAZING WITH MISC DETAILS	CESZ/DIN/2024/AR-507	1/1	26.09.22	-
117.	PLINTH PROTECTION, RAMP, RCC CHAJJA, BRICK STEPS, HARD STANDING, SPACE FOR WATER COOLER, DETAILS OF CUP BOARD, 3 TIER SHELVES, DETAIL OF STEEL WINDOWS AND VENTILATORS	CESZ/DIN/2024/AR-508	1/1	26.09.22	-
118.	DETAILS OF ALARM POST AND DHOBI GHAT	CESZ/DIN/2024/AR-509	1/1	26.09.22	-
119.	DETAILS OF BARBED WIRE FENCING AND CHAIN LINK FENCING	CESZ/DIN/2024/AR-510	1/1	26.09.22	-
120.	DETAILS OF CABLE DUCT & INSPECTION PIT	CESZ/DIN/2024/AR-511	1/1	26.09.22	-
121.	SEPTIC TANK AND SOAK PIT (10 USERS AND 50 USERS)	CESZ/DIN/2024/AR-512	1/1	26.09.22	-
122.	TY. DETAILS OF FLAG POST, FIRE POINT, SIGN BOARD AND INCINRETOR	CESZ/DIN/2024/AR-513	1/1	26.09.22	-
123.	DETAILS OF ROAD AND FOOTPATH SECTION	CESZ/DIN/2024/AR-514	1/1	26.09.22	-
	SCHEDULE OF FINISHES				
124.	BUILDING NOS. 1 TO BUILDING NOS. 4	CESZ/DIN/2024/AR-516	1/6	26.09.22	-
125.	BUILDING NOS. 5 TO BUILDING NOS. 7	CESZ/DIN/2024/AR-516	2/6	26.09.22	-
126.	SCHEDULE OF FINISHES	CESZ/DIN/2024/AR-516	3/6	26.09.22	-
127.	SCHEDULE OF FINISHES	CESZ/DIN/2024/AR-516	4/6	26.09.22	-
128.	SCHEDULE OF FINISHES	CESZ/DIN/2024/AR-516	5/6	26.09.22	-
129.	SCHEDULE OF FINISHES	CESZ/DIN/2024/AR-516	6/6	26.09.22	-
	ELECTRIFICATION				
130.	INTERNAL ELECTRIFICATION & LIGHTNING PROTECTION	CESZ/DIN/2024/EM-517	1/15	26.09.22	-
131.	INTERNAL ELECTRIFICATION & LIGHTNING PROTECTION	CESZ/DIN/2024/EM-517	2/15	26.09.22	-
132.	INTERNAL ELECTRIFICATION & LIGHTNING PROTECTION	CESZ/DIN/2024/EM-517	3/15	26.09.22	-
133.	INTERNAL ELECTRIFICATION & LIGHTNING PROTECTION	CESZ/DIN/2024/EM-517	4/15	26.09.22	-
134.	INTERNAL ELECTRIFICATION	CESZ/DIN/2024/EM-517	5/15	26.09.22	-

PARTICULAR SPECIFICATIONS (CONTD…..)

135.	INTERNAL ELECTRIFICATION	CESZ/DIN/2024/EM-517	6/15	26.09.22	-
136.	INTERNAL ELECTRIFICATION	CESZ/DIN/2024/EM-517	7/15	26.09.22	-
137.	INTERNAL ELECTRIFICATION	CESZ/DIN/2024/EM-517	8/15	26.09.22	-
138.	INTERNAL ELECTRIFICATION	CESZ/DIN/2024/EM-517	9/15	26.09.22	-
139.	SCHIMATIC DIAGRAM OF HT & LT DISTRIBUTION	CESZ/DIN/2024/EM-517	10/15	26.09.22	-
140.	INTERNAL ELECTRIFICATION & LIGHTNING PROTECTION	CESZ/DIN/2024/EM-517	11/15	26.09.22	-
141.	FIRE DETECTION SYSTEM	CESZ/DIN/2024/EM-517	12/15	26.09.22	-
142.	FIRE DETECTION SYSTEM	CESZ/DIN/2024/EM-517	13/15	26.09.22	-
143.	FIRE DETECTION SYSTEM	CESZ/DIN/2024/EM-517	14/15	26.09.22	-
144.	SCHIMATIC LAYOUT OF FIRE ALARM CONTROL SYSTEM	CESZ/DIN/2024/EM-517	15/15	26.09.22	-
	TD DRAWINGS				
145.	GENERAL NOTES AND ARCHITECTURAL NORMS FOR VARIOUS FITTING/ FIXTURE	CESZ//2015/TD-001	1/2	05.01.15	-
146.	GENERAL NOTES AND ARCHITECTURAL NORMS FOR VARIOUS FITTING/ FIXTURE	CESZ//2015/TD-001	2/2	05.01.15	-
147.	TUBULAR STEEL POLE FOR STREET LIGHT	TD/EM/05	1/1	20.02.89	-
148.	TYPICAL DETAILS OF VALVE PIT	TD/EM/06	1/2	08.12.99	-
149.	TYPICAL DETAILS OF VALVE PIT	TD/EM/06	2/2	08.12.99	-
150.	METHOD OF EARTHLING	TD/EM/11	1/1	22.01.04	-
151.	PLUMBING (SINGLE STACK SYSTEM)	CESZ/TD/31	1/2	14.10.97	-
152.	PLUMBING (SINGLE STACK SYSTEM)	CESZ/TD/31	2/2	14.10.97	-
153.	DETAIL OF INDIAN TYPE WC, SINGLE STACK SYSTEM	CESZ/TD/38	1/2	14.10.97	-
154.	DETAILS OF NAHANI TRAP & PIPE CONNECTIONS	CESZ/TD/38	2/2	14.10.97	-
155.	DETAILS OF WATER PROOFING TREATMENT TO BATH/EWC	CESZ/TD/39	1/1	14.10.97	20.11.99
156.	TYPICAL DETAIL OF GULLY TRAP AND MAN HOLE	CESZ/TD/40	1/1	14.10.97	-
157.	TYPICAL DETAIL OF FIXING HDPE WATER STORAGE TANK OVER RCC ROOF SLAB	CESZ/TD/41	1/1	14.10.97	-
158.	DETAILS OF FIXING EXHAUST FAN	CESZ/TD/42	1/1	14.10.97	-
159.	FIXING DETAILS OF FAN HOOK	CESZ/TD/97	1/1	14.07.00	-
160.	TYPICAL MASONRY DETAILS	CESZ/STD/3	1/5	12.05.98	-

PARTICULAR SPECIFICATIONS (CONTD…..)

161.	TYPICAL MASONRY DETAILS	CESZ/STD/3	2/5	12.05.98	-
162.	TYPICAL MASONRY DETAILS	CESZ/STD/3	3/5	12.05.98	-
163.	TYPICAL MASONRY DETAILS	CESZ/STD/3	4/5	12.05.98	-
164.	TYPICAL MASONRY DETAILS	CESZ/STD/3	5/5	12.05.98	-
165.	FLUSH DOORS WITH PRESSSED STEEL FRAME.	CESZ//2015/TD-003	1/2	05.01.15	16.05.15
166.	FLUSH DOORS WITH PRESSSED STEEL FRAME.	CESZ//2015/TD-003	2/2	05.01.15	16.05.15
167.	DETAILS OF MOULDED PVC DOORS	CESZ//2015/TD-008	1/3	05.01.15	-
168.	DETAILS OF MOULDED PVC DOORS	CESZ//2015/TD-008	2/3	05.01.15	-
169.	DETAILS OF MOULDED PVC DOORS	CESZ//2015/TD-008	3/3	05.01.15	-
170.	MISC TYPICAL DETAILS-1	CESZ//2015/TD-010	1/3	05.01.15	27.04.15
171.	MISC TYPICAL DETAILS-1	CESZ//2015/TD-010	2/3	05.01.15	-
172.	MISC TYPICAL DETAILS-1	CESZ//2015/TD-010	3/3	05.01.15	27.04.15
173.	TYPICAL DETAILS OF PREVENTION & CURE & LEAKAGE I SEEPAGE & DAMPNESS	CESZ//2015/TD-011	1/2	05.01.15	-
174.	TYPICAL DETAILS OF PREVENTION & CURE & LEAKAGE I SEEPAGE & DAMPNESS	CESZ//2015/TD-011	2/2	05.01.15	05.10.15
175.	TYPICAL DETAILS OF DEAPERY ROD	CESZ//2015/TD-013	1/1	05.01.15	-
176.	FUNCTIONAL HEIGHTS OF TOILET/LAVATORY FITTINGS	CESZ/TD/18	1/1	14.10.97	14.09.99
177.	CABINET BELOW COOKING PLATFORM & MEAT SAFE BELOW PLATFORM (OTM ACCN)	CESZ/TD/22	1/1	14.10.97	14.09.99
178.	LT PANEL STAGE (OUTDOOR)	TD/EM/14	1/1	28.12.04	-
179.	CABLE ROUTE INDICATOR	TD/EM/17	1/1	30.07.05	-
180.	E/M NOTES & LEGENDS	TD/EM/07	1/1	24.08.2000	28.01.05
181.	NUMBERING OF BUILDING	TD/N/76	1/1	31.03.73	21.05.73

1. Under column 3 of List of Drawings (TD drawings covered in booklet), reference to drawings No. "CESZ/TD" and "CESZ/STD" have been made and may not be upload alongwith tender documents. Contractor are deemed to be in possession of all said standard drawings amended upto date as issued by "Chief Engineer Shillong Zone", Shillong / Engineer – in -Chief, New Delhi. These drawings may however be referred to in this HQ/CWE/GE office during working hours. No claim whatsoever shall be admissible to the contractors on account of any misunderstanding if any on this account.
2. In case of any discrepancy in Drawings No, Sheet No, Original date and Revision Date in between the “LIST OF DRAWINGS” as stated above and mentioned in architectural Drawings, the details as shown in “LIST OF DRAWINGS” above shall take precedence over the Drawings. In this regards,

PARTICULAR SPECIFICATIONS (CONTD…..)

decision of Accepting Officer shall be final and binding.

3. In case of varying provisions between main drawing and typical drawing (TD), the main drawing(s) shall take precedence over TD drawings.
4. In case of varying provisions between 2015, 2016 & 2017 series (CESZ/2015/TD, CESZ/2016/TD, CESZ/2017/TD or CESZ/STD) TD drawing and older version typical drawing, the provision catered in 2015/2016/2017 series (CESZ/2015/TD, CESZ/2016/TD, CESZ/2017/TD or CESZ/STD) TD drawing (s) shall take precedence.

SIGNATURE OF CONTRACTOR **DY DIR/AAD(CONTRACTS)**

FOR ACCEPTING OFFICER

APPENDIX 'A'

FORMAT FOR QUALITY CONTROL PLAN

(To be submitted by contractor within 30 days of commencement of work)

PART I

1. Contract Agreement Reference No :

2. CPM network prepared and approved by GE :

3. Resource scheduling done based on CPM :

4. Site Laboratory (With Equipments) Set Up As Per Contract Agreement (CA) :

5. Concrete mix design submitted and approved :

6. Preliminary works completed to standard engineering practice :

7. Arrangements for water made (including testing of water) :

8. Arrangement for electric supply made :

9. Materials

Sl No	Item	Source as per CA	Contractor's plan of sourcing	Refer to testing clause	Agency for testing	Responsibility for testing

10. List of all T & P, make and numbers that the contractor would deploy at site of work
11. Name of person nominated by contractor for exercising quality control :

12. Qualifications/Experience of person at Serial No.11 above :

13. Names of supervisors with their qualifications/ experience employed by contractor :

Contractor Dy Dir/AAD(Contracts)

APPENDIX 'A'

14. Confirmation that contract requirements relating to quality of all materials and quality standards of workmanship and finishes and acceptance criteria are explained and understood by all

15. Confirmation that requirement of tests to be conducted on materials before approval of sample and during execution, test on workmanship, tests before acceptance including the testing procedure, sampling techniques, frequency and agencies responsible for testing are understood and shall be complied with.

16. Method to be adopted for maintaining records of test result

17. Certificate that contractor shall maintain a log of all materials received at site as per the following format.

Sl No	Date	Material	Quantity received	Source	Whether as per approved sample or not	Tests carried out by supplier	Tests to be carried out before incorporation

18. General Remarks by contractor of his plan of actions to ensure that quality standards

Date : (Signature of Contractor)

Contractor Dy Dir/AAD(Contracts)

1	2		3

LIST OF PRODUCTS FOR WHICH SPECIFIC MAKES ARE TO BE INCORPORATED IN WORKS

GENERAL

Materials to be incorporated in the Work shall be any one of the following MAKE as approved by the GE in writing before incorporated in the Work. These shall strictly comply with latest appropriate Indian Standard Specification or where IS Specification is not available these shall comply with the latest BS Specification.

Sl No.	MATERIALS		MAKE / NAME OF MANUFACTURER
1	2		3
A.	BUILDING MATERIALS		
1.	ATT CHEMICALS	–	ISI Marked with valid CML NO of BIS
2.	PLASTICISER	–	M/S HINDCON CHEMICALS LTD./M/S CHOKSEY CHEMICALS PVT. LTD./ BERGER PAINTS INDIA LTD./ M/S IWL INDIA LTD./ TIKI TAR DANOSA (INDIA) PVT. LTD/ MARUTI BITUMEN PVT. LTD/ ASIAN PAINTS PVT LTD./M/S SIKA INDIA PVT LTD./ BOSTIK INDIA PVT. LTD/ M/S STP LTD.
3.	WATER PROOFING COMPOUND	–	M/S HINDCON CHEMICALS LTD./M/S CHOKSEY CHEMICALS PVT. LTD./ BERGER PAINTS INDIA LTD./ M/S IWL INDIA LTD./ TIKI TAR DANOSA (INDIA) PVT. LTD/ MARUTI BITUMEN PVT. LTD/ ASIAN PAINTS PVT LTD./M/S SIKA INDIA PVT LTD./ BOSTIK INDIA PVT. LTD/ M/S STP LTD.
4.	FACTORY MADE 2ND CLASS HARD WOOD PANELED DOOR SHUTTER/SKELETON SHUTTER/FLUSH DOOR SHUTTER (Sample of Item as per Scales of Testing given in IS:1003 shall be subject to Testing at reputed Test House/Govt. Test House as approved by GE at Contractor's Cost and after getting satisfactory Test Result Payment shall be released)	–	M/S DOORKING INDUSTRIES 27 Gr block, Sector –V, Salt Lake, Kolkata Phone : 033-22511334
		–	M/S PREMIER WOOD CRAFTS (P) LTD. Jessore Road,PO Ganganagar, Dist-24 Parganas (WB)
		–	M/s Jain Wood Industries (Brand: JAYNA) A-228, Vardhaman Grand Plaza, Mangalam Place, sector-3, Rohini, New delhi-85 Tele: 9313694205 / 8860055818
		–	M/S PIONEER TIMBER, CHANDIGARH, M/S JAIN DOORS, HARYANA, M/S A-1 TEAK PRODUCTS INDORE
5.	PVC DOORS & FRAMES, WINDOWS, PARTITIONS, FALSE CEILINGS & WALL CLADDINGS	–	M/S RAJSHRI PLASTIWOOD/EVEREST INDUSTRIES LTD/ M/S (ACCUCEL) ACCURA POLYTECH PVT LTD/ M/S SINTEX INDIA LTD. / DUROPLAST/UNIFIX PLAST PVT. LTD./ KUMAR ARCH TECH PVT. LTD/ CHANDNI INDUSTRIES/YAMUNA INTERIORS PVT. LTD. (BRAND: BLACK COBRA)
6.	FRP DOOR	–	KRAFTO DOOR/ M/S Fibre ways technology(254 A/1, Gr floor, TK House, Malviya Nagar Delhi-17) , M/S Syntax Ind. Ltd, M/S Deluxe Poly Fab Pvt Ltd, M/S S M Fibre, M/s Jain Wood Industries (Brand: JAYNA)/ M/S RAJSHRI PLASTIWOOD/ DUROPLAST
7.	MILD STEEL BUILDER'S HARD-WARES VIZ BARREL TOWER BOLT, SLIDING DOOR BOLT, BUTT HINGE, HANDLE, DOOR CLOSER, ETC.	–	HARDWIN/ EVERLITE/ GODREJ/ HARRISON/ UNIVERSAL OR ANY MAKE BEARING IS CERTIFICATION MARK
8.	AUMINIUM BUILDER'S HARD-WARES VIZ BARREL TOWER BOLT, SLIDING DOOR BOLT, HANDLE, DOOR CLOSER, ETC.	–	ARGENT/ HARDWIN/ EVERLITE/ GODREJ/ OR ANY MAKE BEARING IS CERTIFICATION MARK
9.	ALUMINIUM DOOR & WINDOW SECTION, ALUMINIUM SNAP GRID or any other ALUMINIUM SECTION	–	HINDALCO/ JINDAL / HALCO(GRP-II) / BHAGWATI SAI METAL ALLOYS/ BENGAL ROLLING SHUTTERS & ENGINEERING WORKS/ CHANDNI INDUSTRIES/
10.	FACTORY MADE STEEL WINDOWS/ VENTILATORS (IS 1038-1980) &FACTORY MADE PRESSED STEEL DOOR FRAMES (ISI -4531) AND SHALL BE FLASH BUTT WELDED (Sample of Item as per Scales of Testing	–	M/S MADHU INDUSTRIES 63, N.S Palya, Banner Ghatla Road, Bangalore - 560 076 Phone : 080-6688907/6685195/ 6682263, Fax : 080-6688907
		–	M/S MULTIWYN INDUSTRIAL CORP, Camac Street (B Block, 2nd Floor), Kolkata - 700 016 Phone : 033-2299543/2297525/2298904, Fax : 033-2290748

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	given in IS:1038 shall be subjected to Testing at reputed Lab/Govt. Test House as approved by GE at Contractor's Cost and after getting satisfactory Test Result Payment shall be released)	–	M/S TRISUL INDUSTRIES 284-A, Chappel street Meerut Cant (UP)-250001, Phone: 0121-4026578, Mo-955777020
		–	M/S CHANDNI INDUSTRIES J-142, PATEL NAGAR 1ST GHAZIBAD (UP)- 201001
11.	PRE-COATED FACTORY MADE STEEL WINDOWS/VENTILATORS (IS-1038-1980) & (Sample of item as per Scales of Testing given in IS:1038 shall be subjected to Testing at reputed Lab/Govt. Test House as approved by GE at Contractor's Cost and after getting satisfactory Test Result Payment shall be released)	–	(i) TATA Steel, Mumbai, (ii) Ashwani & Sons, 4-66, Rajnagar, Ghaziyabad(UP) (iii) JINDAL, 2-32, Gurgaon, Haryana-122001 (iv) Ashish Industries, 1337, Manapur Industrial Area, Meerut Road(UP), Pin-201003, Ph-9891542779 (v) NCL Alltek&Seccolor Ltd, Reg. & Admin Ofc:1, Ganga enclave, Petbasheerabad,Hyderabad-500067
12.	ROLLER SHUTTERS, COILING DOOR, ROLLER DOOR, OR SECTIONAL OVERHEAD DOORS	–	BENGAL ROLLING SHUTTERS & ENGINEERING WORKS/ M/S DURA ROOF PVT LTD. (BRAND : DURA ULTIMA) OR ANY MAKE BEARING IS CERTIFICATION MARK
13.	PRE-ENGINEERING /GALVALUME SHEET as per IS 15965	–	DURASHINE/LYSAGHT BY M/S TATA BLUE SCOPE/ M/S DURA ROOF PVT LTD.
		–	SARAF ROOFING (BRAND: TUFFROOF)/ SBM ROOFING (BRAND: TAJ ROOFING)/ M/S INDIAN EXPORTS ENTERPRISERS /DURA ROOF BRAND DURA PREMIUM
		–	COLOURON PLUS BY JSW / DYNA ROOF/SHALINI ROOFING(BRAND : SPECTRUM)/ JINDAL(INDIA) LTD./ STELLAR BUILDTECH PVT. LTD.
14.	CGI/PGI STEEL SHEET	–	JINDAL(INDIA) LTD./ SAIL / TATA
15.	Wire mesh for wire crate	–	ISI MARKED WITH VALID CML NO OF BIS
16.	Prelaminated playwood/ Plywood Grade BWR type BB /decorative plywood	–	GREEN PLY /KIT PLY / CENTURY PLY./ ARCHID PLY /M/S JAIN WOOD INDUSTRIES(BRAND ; JAYANA)
17.	Plastic Laminated Sheet	–	GREEN LAM / ARCHID LAM / CENTURY LAM/ KIT LAM
18.	Veneered Particle Boards	–	ECOBOARD/ NOVAPAN/ CENTURY PLY / ARCHID PLY / GREEN PLY/ SHIRDI INDUSTRIAL LTD (ASIS)
19.	Prelaminated Particle Board/ Prelaminated MDF	–	GREEN LAM / ARCHID LAM/CENTURY LAM / KIT LAM
20.	Adhesive	–	M/S ASIAN PAINTS LIMITEDM/S HINDCON CHEMICALS LTD./M/S CHOKSEY CHEMICALS PVT. LTD./ BERGER PAINTS INDIA LTD./ M/S IWL INDIA LTD./ PIDILITE INDUSTRIES LTD./ FOSROC/ TIKI TAR/ STP LTD./ SOMANY CERAMICS LTD
21.	APP MODIFIED WATER PROOFING MEMBRANES	–	HYPERPLAS POLYSTER (M/S IWL INDIA LTD.)/ SUPERTHERMOLAY(M/S STP LIMITED) / BERGER (PROSHIELD MEMBRANE)/ M/S ASIAN PAINTS LIMITED/ M/S BENGAL BITUMEN (BRAND BENGAL AQUAFLUX, BENGAL AQUAFLEX ALUMINIUM AND BENGAL SUPERMAT)/M/S TORCHTAR MEMBRANES & BITUMEN PRODUCT PVT LTD/M/S TIKI TAR DANOSA (INDIA) PVT LTD/M/S SHIVAM TAR PRODUCTS/M/S IWL INDIA LIMITED
22.	BITUMEN PRIMER, BITUMEN, ETC FOR ROOF TREATMENT ONLY	–	MAKE BEAR IS CERTIFICATION MARK
23.	GYPSUM BOARDS	–	ARMSTRONG/ SAINT GOBAIN (GYPROC)/USG BORAL
24.	TERRAZZO TILES (IS-1237:1980)	–	ISI MARKED
25.	CHEQUERED TILES (IS-13801)	–	EAST WEST TILES/ ULTRA/ BANSAL/ NITCO/ ANJALI/ MULTIWYN OR CONFORMING TO IS (TILES SHALL NOT BE MADE AT SITE. ITEMS SHALL BE GOT TESTED AT NTH AT CONTRACTOR'S COST. PAYMENT SHALL BE RELEASED ONLY AFTER GETTING SATISFACTORY TEST RESULTS)
26.	CERAMIC TILES(ANTISKID/NON SKID/ GLAZED TILES), OUTDOOR	–	FIRST QUALITY OF JOHNSON TILES, SOMANY, NITCO, KAJARIA, RAK CERAMICS INDIA PVT LTD/ ORIENT

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	VITRIFIED PAVING TILES		BELL LTD. (BRAND: ORIENTBELL), M/s SUNSHINE TILES (BRAND: SUNHEARRT), VARMORA/ ASIAN GRANITO PVT. LTD
27.	ACID RESISTANCE CERAMIC TILES(INDUSTRIAL TYPE) AS CONFORM TO IS 4457 WITH CHEMICAL RESISTANT MORTAR	–	JHONSON, SOMANY, REGENCY, KAJARIA, NITCO
28.	VITRIFIED TILES	–	JOHNSON TILES, SOMANY, KAJARIA, NITCO ,RAK CERAMICS INDIA PVT LTD, ORIENT BELL LTD. (BRAND: ORIENTBELL), M/S APARNA ENTERPRISES LTD. (APARNA VITERO), M/s SUNSHINE TILES (BRAND: SUNHEARRT), ASIAN GRANITO PVT. LTD
29.	SHEET GLASS	–	MODIGUARD / SAINT GOBAIN GLASS(SSG)/ ATUL/ GOLD PLUS
30.	PRECAST INTERLOCKING PAVER BLOCK	–	ANJALI TILES, ULTRA TILES PVT. LTD., TERRA FIRMA, BANSAL, WONDER CRETE, A-ONE TILES, MEHTAB, EAST WEST TILES
31.	CEMENT BASE PAINT	–	SHALIMAR (MAHACEMKOTE) / SNOWCEM PLUS / DUROCEM / ROBBIACEM (J&N) / ACROCEM ULTRA / BERGER/ ASIAN PAINTS/ CRIMSON PAINTS PVT. LTD.
32.	DRY DISTEMPER / OIL BOUND DISTEMPER / ACRYCLIC DISTEMPER	–	ASIAN PAINTS, BERGER, DULUX of ICI, NERROLAC, CRIMSON PAINTS PVT. LTD.
33.	ACRYCLIC EMULSION PAINT, PLASTIC EMULSION PAINT	–	ASIAN PAINTS (ROYAL), BERGER (SILK), ICI(DULUX VELVET TOUCH)/ BAKELITE COATINGS AND PAINTS (P) LTD(HERITAGE DECORATIVE).
34.	ACRYCLIC BASED POLYMER MODIFIED CEMENTITIOUS WATER PROOFING	–	FORSOC, MYK LATICRETE, HI TECH, PIDILITE, SIKA, HINDCON CHEMICALS LTD/ BAKELITE COATINGS AND PAINTS (P) LTD(HERITAGE DECORATIVE).
35.	WEATHER PROOF PAINT	–	ASIAN PAINT, DULUX, BERGER, NEROLAC. BAKELITE COATINGS AND PAINTS (P) LTD(HERITAGE DECORATIVE), CRIMSON PAINTS PVT. LTD.
36.	WHITE CEMENT/ WALL PUTTY	–	ASIAN PAINTS, BERGER, J & N, BIRLA WHITE, J K WALL PUTTY, SHALIMAR, BAKELITE COATINGS AND PAINTS (P) LTD.
37.	SYNTHETIC PAINT (FIRST QUALITY)	–	ASIAN PAINTS, BERGER, ICI(DULUX), NERROLAC
38.	HEAT RESISTANT PAINT	–	ASIAN PAINTS, BERGER, ICI(DULUX), NERROLAC
39.	FRENCH POLISH	–	ASIAN PAINTS, BERGER, NERROLAC, J & N, JOTUN INDIA PVT. LTD.
40.	MIRROR /GLASS	–	ATUL, GOLD PLUS, KOHINOOR, MODI, SAINTGOBAIN, SWASTIK, PRAYAG, MODIGUARD, CERA,
41.	PVC CISTERN (ISI MARKED) & PLASTIC SEAT COVER (SOILD HEAVEY DUTY)	–	CERA/ COMMANDER/ HINDWARE/ JOHNSON PEDDAR/ PARRYWARE/ SPEED FLO (BLUE STAR POLYMERS)/ MS PLSTO MOULD PVT. LTD/ JAQUAR/ PRAYAG/ JAINKO/ ASIAN PAINTS PVT. LTD.
42.	STAINLESS STEEL PLATE RACK/ S S SINK	–	CERA/ PARRYWARE/ BLUE STAR /PRAYAG/ JAINKO/ ANJALI/ KOHLER/ SHAKTI
43.	BATH ROOM CHROMIUM PLATED / CAST COPPER ALLOY FANCY BIB TAPS, STOP VALVES & PILLAR TAPS (IS:8931-1993) IS-8934-1978)	–	JAQUAR / MARC/ PARRYWARE/ HINDWARE/ RAK CERAMICS INDIA PVT LTD M/S ASIAN PAINTS LTD.(BRAND: APROYALE, BATHSENCE, ESSESS)/ KOHLER/ / SHAKTI / GROHE/ CERA/ PRAYAG/ STATUS SANITECH PVT. LTD./ M/S DHAWAN SANITARY UDYOG (BRAND: PRIMA (AURA & SWAN COLLECTION ONLY))
44.	HDPE & LLDPE TANK, PVC OVER HEAD TANK (MULTI LAYERED CONSTRUCTION AS PER IS-12701:1996)	–	ROTAX/ SINTEX/ POLYCON/ M/S R C PLASTO TANKS & PIPES PVT (BRAND : PLASTO)/ JINDAL/ ASTRAL/ FINOLEX/ SUPREME
45.	NON ASBESTOS FIBER CEMENT BOARD PLAIN/CORRUGATED	–	EVEREST INDUSTRIES LTD./ ARMSTRONG/ HIL LTD./ RAMCO
46.	UPVC DOOR, WINDOW, FRAMES AND VENTILATOR		M/S RAJSHRI PLASTIWOOD/CHANDNI INDUSTRIES/ M/S (ACCUCEL) ACCURA POLYTECH PVT LTD/ M/S SINTEX INDIA LTD. /
47.	STAINLESS STEEL TOWEL RAIL, TOWEL RACK, TOWEL RING, SOAP DISH, TOILET PAPER HOLDER ETC.		KOHLER/ GROHE/ JAQUAR/ BLUE STAR & SILVER SHINE, CERA, HINDWARE

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	WATER SUPPLY ITEMS		
48.	GI/ MS/ ERW PIPE (ANY GRADE)	–	BANSAL/ BST/ JINDAL/ NEZONE TUBES LTD (NEZONE) (NTL)/ PRAKASH/ TATA/ SWASTIK/ ZENITH/SURYA ROSHANI LTD.
49.	GI/ MS/ ERW FITTINGS	–	BANSAL/ BST/ JINDAL/ NEZONE (NTL)/ PRAKASH/ TATA/ SWASTIK/ ZENITH/SURYA ROSHANI LTD. OR MAKE BEARING IS CERTIFICATION MARK
50.	PP-R PIPES & FITTINGS	–	PRINCE/ SAVOIR-FAIRE MANUFACTURING CO. PVT LTD. (SFMC)/ VECTUS/ /M/S KANHA PLASTICS PVT LTD/ SUPREME/ FINOLEX/ KISAN/FUSION INDUSTRIES LTD. (BRAND: FUSION)
51.	CAST COPPER BIB COCK/STOP VALVE	–	LEADER/ JAQUAR/ M/S ASIAN PAINTS LTD.(BRAND: APROYALE, BATHSENCE, ESSESS)/ PRAYAG
52.	BRASS BALL (FLOAT) VALVE	–	LEADER/ KIRLOSKAR/ ZOLOTO/ CERA/ PRAYAG
53.	GUN METAL GATE VALVE	–	KIRLOSKAR/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ LEADER/ZOLOTO/ BIR
54.	CAST IRON (VERTICALLY CAST)PIPE (CENTRIFUGALLY CAST) AND FITTINGS	–	ELECTRO STEEL/ KESORAM/ /JAYASWAL NECO INDUSTRIES LTD / SINGHAL IRON FOUNDRY PVT LTD(BRAND : SKF)/ M/S BINAYUDYAG PVT. LTD./ RAJ PATTERN MAKERS & FOUNDERS PVT LTD(BRAND: RPMF)
55.	Vitreous China, WC/HWB/ Urinal/Soap Dish/ Urinal Partition Wall/ Laboratory sink with ISI Marked	–	HINDUSTAN SANITARY WARE (Hindware) (Regular) / PARRYWARE (SUPERFINE) / CERA (Regular)/ JAQUAR/ RAK CERAMICS INDIA PVT LTD/ M/S ASIAN PAINTS LTD.(BRAND: APROYALE, BATHSENCE, ESSESS)/ PRAYAG/ KOHLER/ JOHNSON PEDDER
56.	DUCTILE IRON PIPE	–	ELECTROTHERM INDIA /ELECTROSTEEL CASTINGS LIMITED / JINDAL SAW LIMITED / TATA METALIKS / LANCO
57.	CI /DUCTILE IRON SPECIALS	–	ELECTROSTEEL/ELECTRO-THERM/ KESORAM/ LANCO/ TISCO
58.	HDPE /PVC PIPES & FITTINGS	–	JAIN IRRIGATION /ANANT/ TUSKER/ ORI-PLAST/ SUPREME/ PRINCE/ RELLIANCE / KISAN/ FINOLEX/ KANHA PLASTICS PVT LTD (BRAND : KPT) / MS PLASTO MOULD PVT LTD. / M/S R C PLASTO TANKS & PIPES PVT (BRAND : PLASTO)/ SFMC/ ASTRAL POLYTECHNIC LTD / KIRAN INTRA TECH PVT. LTD./ M/S PREMIER IRRIGATION ADRITEC PRIVATE LTD
59.	UPVC / RWP/CPVC PIPES AND FITTINGS	–	JAIN IRRIGATION / KISAN/ SUPREME/ PRINCE / ORI-PLAST/ MS PLASTO MOULD PVT LTD. / M/S R C PLASTO TANKS & PIPES PVT (BRAND : PLASTO)/ FINOLEX/ SFMC/ ASTRAL POLYTECHNIC LTD/FUSION INDUSTRIES LTD. (BRAND: FUSION)
60.	RCC DRAIN PIPES	–	EVEREST, HIMALAYA, INDIAN HUME PIPE ANY MAKE BEARING IS CERTIFICATION MARK
61.	SLUICE VALVE/ REFLUX VALVE	–	KIRLOSKAR(KBL)/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ LEADER/ ZOLOTO/ VENUS
62.	AIR RELEASE VALVE/ FOOT VALVE/ NON-RETURN VALVE	–	KIRLOSKAR(KBL)/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ LEADER ZOLOTO/ VENUS
63.	BLEACHING DOZER	–	AQUAPURA/ AVON/ ION EXCHANGE/ MAIC-INDIA
64.	GRANULAR CHLORINATION	–	BRIOZING/ DECLIBAC (ALTOCH "GP")/ POWER & WATTS
65.	LIQUID CHLORINE	–	DECLIBAC/ HYDRIOL/ RMCO
66.	CHLORONOME PLANT /CHLORINATOR	–	AQUA/ JESCO/ PEARL FILTERS/ ADVANCE 2000
67.	LIQUID CHLORINE DOZING	–	DECLIBAC/ MILTON ROY/ RMCO/ TECHNOMAC
68.	WATER METER	–	CAPSTON / DASHMESH / KIRLOSKAR
69.	CENTRIFUGAL PUMP / NON CLOGGING PUMP	–	CROMPTON GREAVES/ KIRLOSKAR/ KSB/ MATHER & PLATT
70.	SUBMERSIBLE PUMP	–	KIRLOSKAR / KSB/ CROMPTON GREAVES/ MATHER & PLATT
71.	SOLAR WATER HEATER SYSTEM	-	M/S V-GUARD SOLAR WATER HEATER, M/S TATA BP SOLAR INDIA PVT LTD
72.	VERTICAL TURBINE	–	JOHNSON/ KIRLOSKAR/ KSB/ WORTHINGSTON

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	ELECTRICAL ITEMS		
73.	POINT WIRING CABLES	–	FINOLEX/HAVELLS INDIA LTD/RR KABLE/ KEI / M/S HPL ELECTRIC & POWER LTD / M/S DYNAMIC CABLES LIMITED / M/S POLYCAB INDIA LIMITED/ ULTRACAB (INDIA) LTD./ M/S BENTEC INDIA LTD. (BRAND: BENLO)/ GOLD MEDAL ELECTRICALS PVT. LTD. / ASIAN GALAXY PVT LTD./ M/S ADHUNIK SWITCHGEARS PVT. LTD./ RICHA CABLES/ HAGER ELECTRO PVT. LTD./ PANASONIC (ANCHOR)/ M/S NOVATEUR ELECTRICAL & DIGITAL SYSTEMS PVT LTD (INDOASIAN)/ MESCAB INDIA PVT. LTD.
74.	PIANO SWITCHES, SOCKETS etc	–	ISI MARKED PRODUCTS OF ANCHOR / LAURITZ KNUDSEN ELECTRICAL & AUTOMATION / HAVELLS / LEGRAND/ M/S POLYCAB INDIA LIMITED/ BCH ELECTRIC LTD. / M/S BENTEC INDIA LTD. (BRAND: BENLO) / GOLD MEDAL ELECTRICALS PVT. LTD. / M/S ADHUNIK SWITCHGEARS PVT. LTD./ HAGER ELECTRO PVT. LTD./ M/S HPL ELECTRIC & POWER LTD/ BCH ELECTRIC LTD.
75.	PVC CASING CAPPING	–	FINOLEX / PRESTO PLAST/ PLAZA/KALINGA / M/S AKG INDUSTRIES / M/S RICHA CABLES PVT LTD
76.	PVC CONDUITS	–	FINOLEX / PRESTO PLAST/ PLAZA/KALINGA / M/S AKG INDUSTRIES / M/S POLYCAB INDIA LIMITED / GOLD MEDAL ELECTRICALS PVT. LTD./ / ANCHOR
77.	STEEL CONDUITS	–	TATA/JINDAL//SWASTIK/PRAKASH
78.	MODULAR SWITCHES/ FIXTURES	–	CRABTREE/ LEGRAND/ SCHNEIDER/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ ANCHOR / M/S NOVATEUR ELECTRICAL & DIGITAL SYSTEMS PVT LTD (INDOASIAN) /M/S HAGER ELECTRO PVT LTD / M/S C&S ELECTRIC LTD /M/S POLYCAB INDIA LIMITED /M/S HPL ELECTRIC & POWER LTD / M/S BENTEC INDIA LTD. (BRAND: BENLO) /GOLD MEDAL ELECTRICALS PVT. LTD. / M/S ADHUNIK SWITCHGEARS PVT. LTD.
79.	MCBs, RCCBs, DBs & ISOLATOR	–	LEGRAND/ SCHNEIDER (MERLIN GERIN)/ SIEMENS/ABB / M/S NOVATEUR ELECTRICAL & DIGITAL SYSTEMS PVT LTD (INDOASIAN) /LAURITZ KNUDSEN ELECTRICAL & AUTOMATION /M/S HAVELLS INDIA LTD / M/S PANASONIC LIFE SOLUTIONS INDIA PVT LTD / M/S HAGER ELECTRO PVT LTD /M/S C&S ELECTRIC LTD /M/S POLYCAB INDIA LIMITED/ M/S HPL ELECTRIC & POWER LTD /M/S BENTEC INDIA LTD. (BRAND: BENLO) / GOLD MEDAL ELECTRICALS PVT. LTD. / M/S ADHUNIK SWITCHGEARS PVT. LTD./ M/S SHALABH (INDIA) LTD.
80.	MCCBs	–	ABB/ LEGRAND/ M/S NOVATEUR ELECTRICAL & DIGITAL SYSTEMS PVT LTD(INDOASIAN) / LAURITZ KNUDSEN ELECTRICAL & AUTOMATION / M/S HAVELLS INDIA LTD / M/S C&S ELECTRIC LTD/ M/S HPL ELECTRIC & POWER LTD / M/S BENTEC INDIA LTD. (BRAND: BENLO) / GOLD MEDAL ELECTRICALS PVT. LTD. / M/S ADHUNIK SWITCHGEARS PVT. LTD./ M/S HAGER ELECTRO PVT LTD/ SCHNEIDER/ SIEMENS
81.	SWITCH FUSE	–	LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ SCHNEIDER (MERLIN GERIN)/SIEMENS/ABB / M/S NOVATEUR ELECTRICAL & DIGITAL SYSTEMS PVT LTD(INDOASIAN) / M/S HAVELLS INDIA LTD / M/S C&S ELECTRIC LTD/ M/S HAGER ELECTRO PVT LTD/ M/S HPL ELECTRIC & POWER LTD / LEGRAND/ M/S SHALABH (INDIA) LTD. /
82.	ALL TYPE FLUORESCENT FITTING/ HPMV / HPSV / CFL etc	–	BAJAJ/ CROMPTON GREAVES/ HAVELLS INDIA LTD / PHILIPS/ WIPRO/ M/S POLYCAB INDIA LIMITED
83.	ALL TYPE LED LIGHT FITTINGS	–	BAJAJ/ PHILIPS/ WIPRO/ SYSKA LED/ HAVELLS INDIA LTD / JAQUAR / M/S SIMOCO TELECOM / M/S HPL ELECTRIC & POWER LTD / M/S CROMPTON GREAVES CONSUMER ELECTRICALS LTD / M/S HALONIX TECHNOLOGIES PVT LTD / M/S CENTURY LED LTD (BRAND : MAGIC) / M/S C&S ELECTRIC / M/S PANASONIC LIFE SOLUTIONS INDIA PVT LTD (BRAND : PANASONIC) / M/S DHANSHREE ELECTRONICS LTD (BRAND : RASHMI) / M/S POLYCAB INDIA LIMITED/ SHRI RAMSWAROOP MULTI ENGINEERING SOLUTION PVT. LTD. (BRAND: NAMASTE LED) / M/S BENTEC

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			INDIA LTD. (BRAND: BENLO) / GOLD MEDAL ELECTRICALS PVT. LTD. / M/S ADHUNIK SWITCHGEARS PVT. LTD./ M/S SIGNIFY INNOVATIONS INDIA LTD./ M/s SURYA ROSHNI LTD/ M/S ASCENTECH LIGHTENING SOLUTIONS (BRAND: ASCENTECH)
84.	FLAME PROOF FITTINGS	–	BALIGA / SUDHIR / CROMPTON GREAVES/ BAJAJ/ ALTAS/ SHYAM
85.	HIGH MAST GI POLE/ OCTAGONAL POLE	–	BAJAJ/ CROMPTON GREAVES/ M/S SIGNIFY INNOVATIONS INDIA LTD. /M/s SURYA ROSHNI LTD
86.	LED SOLAR STREET LIGHT FITTING	–	TATA POWER SOLAR /BAJAJ/PHILIPS/ CROMPTON GREAVES /M/S HALONIX TECHNOLOGIES PVT LTD / M/S DHANSHREE ELECTRONICS LTD (BRAND : RASHMI)/ HAVELLS INDIA LTD
87.	EXHAUST FANS/CEILING FAN/ WALL MOUNTING FAN/ AIR CIRCULATION FAN	–	CROMPTON GREAVES / KHAITAN / BAJAJ, HAVELLS INDIA LTD / M/S ORIENT ELECTRIC LIMITED / M/S POLYCAB INDIA LIMITED / GOLD MEDAL ELECTRICALS PVT. LTD. / M/S ADHUNIK SWITCHGEARS PVT. LTD./ USHA/ V-GUARD/ M/s SURYA ROSHNI LTD
88.	HEATER WATER STORAGE TYPE ELECTRIC GEYSER	–	RECOLD / BAJAJ / HAVELLS INDIA LTD/ CROMPTOM GREAVES/ VENUS / M/S V-GUARD INDUSTRIES LTD (UPTO 35 LTRS)/ M/S ADHUNIK SWITCHGEARS PVT. LTD./ CERA/ JAQUAR AND COMPANY PVT. LTD.
89.	STREET LIGHT TIME SWITCHES	–	GE CONTROLS / LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ LEGRAND / ABB/ SIEMENS/ SCHNIEDER/ M/S HPL ELECTRIC & POWER LTD /
90.	STEEL POLES (ISI MARKED)	–	BANSAL POLES / CALCUTTA POLES / JINDAL POLES / NATIONAL TUBING COMPANY/ M/S SKIPPER LIMITED (BRAND : SKIPPER)/ BOMBAY TUBES AND STORE/ CROMPTON GREAVES CONSUMER ELECTRICAL LTD./ M/S SIGNIFY INNOVATIONS INDIA LTD./ M/S HAGER ELECTRO PVT LTD
91.	HT INSULATORS	–	BHEL/ JAYSHREE/ RASTRIYA/ JAYPURIYA BROTHERS/ PACTIL
92.	GOD WITH ISOLATOR	–	PACTIL/ JAIPURIA/BHEL/ CROMPTON GREAVES
93.	LIGHTNING ARRESTOR	–	ELPRO / OBLUM/ RASHTRIYA/ BHEL/ ABB/ JAYPURIYA BROTHER
94.	CHANGE OVER SWITCHES	–	LAURITZ KNUDSEN ELECTRICAL & AUTOMATION / SCHNEIDER /SIEMENS /ABB/ M/S NOVATEUR ELECTRICAL & DIGITAL SYSTEMS PVT LTD(INDOASIAN) / M/S HAVELLS INDIA LTD / M/S C&S ELECTRIC LTD/ M/S HAGER ELECTRO PVT LTD/ M/S HPL ELECTRIC & POWER LTD / M/S SHALABH (INDIA) LTD.
95.	STARTERS	–	M/S BCH ELECTRIC / LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ SIEMENS / M/S C&S ELECTRIC LTD/ ABB/ SCHNIEDER
96.	LT XLPE/ PVC CABLES (UG) (ARMOURED & UNARMOURED)	–	KEC (RPG)/ GLOSTER/ FINOLEX/ HAVELLS INDIA LTD / M/S RR KABEL LTD / M/S DYNAMIC CABLES LIMITED / M/S KEI INDUSTRIES LTD/ M/S POLYCAB INDIA LIMITED / ULTRACAB (INDIA) LTD./ ASIAN GALAXY PVT LTD. / M/S ADHUNIK SWITCHGEARS PVT. LTD./ M/S SHALABH (INDIA) LTD. / M/S HPL ELECTRIC & POWER LTD/ M/S BENTEC INDIA LTD. (BRAND: BENLO)/ RHINO M/S RICHA CABLES PVT LTD/ MESCAB INDIA PVT. LTD.
97.	HT XLPE/ PVC CABLES (UG) (ARMOURED & UNARMOURED)	–	KEC (RPG)/ GLOSTER/ FINOLEX/ HAVELLS INDIA LTD / M/S RR KABEL LTD / M/S DYNAMIC CABLES LIMITED / M/S KEI INDUSTRIES LTD/ M/S POLYCAB INDIA LIMITED / ULTRACAB (INDIA) LTD./ / ASIAN GALAXY PVT LTD. / M/S ADHUNIK SWITCHGEARS PVT. LTD./ M/S SHALABH (INDIA) LTD. / M/S HPL ELECTRIC & POWER LTD/ M/S BENTEC INDIA LTD. (BRAND: BENLO)/ RHINO
98.	HT XPLE CABLE JOINTS	–	DENSON/ M-SEAL/ RAYCHEM/3M / HAVELLS INDIA LTD/ VISHWA ELECTRICAL PVT. LTD. (BRAND: MS-

1	2		3
			SEAL)/ YMUNA POWER & INFRASTRUCTURE LTD.
99.	LT ACB	–	ABB/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ SCHNEIDER (MERLIN GREIN)/ SIEMENS / M/S HAVELLS INDIA LTD / M/S BCH ELECTRIC/ M/S HAGER ELECTRO PVT LTD/ M/S NOVATEUR ELECTRICAL & DIGITAL SYSTEMS PVT LTD(INDOASIAN) / LEGRAND
100.	HT VCB/OCB, 33KV & 11KV (INCLUDING PANEL)	–	INDIAN TRANSFORMERS AND ELECTRICALS PVT. LTD., M/S ABB INDIA LTD, SIEMENS, BHEL, SCHNIEDER, LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ YMUNA POWER & INFRASTRUCTURE LTD./ PASCAL SWITCHCARE INDIA LTD./ STELMAC LTD./ C G POWER & INDUSTRIAL SOLUTION LIMITED. (BRAND: CGPISL)/ M/S C-SEC TECHNOLOGIES PRIVATE LTD (MAKE: C-SEC) (MAKE OF ACCESSORIES LIKE CT's, PT's, ETC. SHALL BE AS APPROVED BY MANUFACTURER OF OCB/VCB PANEL)
101.	33 KV CTs / PTs	–	AE / CROMPTON GREAVES/ ABB/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ SCHNEIDER / HAGER/ ANY MAKE HAVING IP-65, CPRI AND ERDA APPROVED
102.	CURRENT TRANSFORMER & POTENTIAL TRANSFORMER 11 KV RANGE	–	KAPPA/ PRAGATI/ ABB/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ SCHNEIDER / HAGER/
103.	RELAY (ELECTRO MECHANICAL	–	ABB / GE CONTROLS / LAURITZ KNUDSEN ELECTRICAL & AUTOMATION / BCH ELECTRIC LTD. / YMUNA POWER & INFRASTRUCTURE LTD./ M/S HAGER ELECTRO PVT LTD/ C G POWER & INDUSTRIAL SOLUTION LIMITED.(BRAND: CGPISL)
104.	ENERGY METER AND PANEL METER	–	M/S HPL ELECTRIC & POWER LTD / M/S BENTEC INDIA LTD. (BRAND: BENLO)/ M/S ADHUNIK SWITCHGEARS PVT. LTD./ HAGER ELECTRO PVT. LTD/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ M/S HAVELLS INDIA LTD/ SCHNEIDER / SIEMENS/SECURE METER LTD
105.	INDICATOR LAMP	–	HAVELLS INDIA LTD / HAGER ELECTRO PVT. LTD/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION
106.	SELECTOR SWITCH	–	HAVELLS INDIA LTD / HPL LTD/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ SIEMENS/ HAGER ELECTRO PVT. LTD
107.	AMMETER, VOLTMETER, ETC.	–	AE/ HAVELLS INDIA LTD / LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ LEGRAND/ MECO/ HAGER ELECTRO PVT. LTD./ BCH ELECTRIC LTD/ SCHNEIDER
108.	CONTACTORS	–	BCH ELECTRIC LTD./ GE CONTROLS/ HAVELLS INDIA LTD / LAURITZ KNUDSEN ELECTRICAL & AUTOMATION / SIEMENS/ HAGER ELECTRO PVT. LTD
109.	LT PANEL BOARD/ LT FEEDER PILLAR BOX	–	INDIAN TRANSFORMERS AND ELECTRICALS PVT. LTD./ BCH ELECTRIC LTD./ M/S ADHUNIK SWITCHGEARS PVT. LTD. / SHALABH (INDIA) INDUSTRIES LTD./ ELECTRO ALLIED PRODUCTS (BRAND : EAP) ANY MAKE HAVING CPRI APPROVED
110.	CAPACITOR BANK/ APFC PANEL	–	CROMPTON GREAVESS/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION/ SIEMENS/ABB/ M/S BCH ELECTRIC / M/S C&S ELECTRIC LTD/ SHALABH (INDIA) INDUSTRIES LTD./ M/S KANYAKA PARAMESHWARI ENGINEERING LTD/ HAVELLS INDIA LTD/ M/S NOVATEUR ELECTRICAL & DIGITAL SYSTEMS PVT LTD(INDOASIAN) C G POWER & INDUSTRIAL SOLUTION LIMITED.(BRAND: CGPISL)
111.	DISTRIBUTION AND POWER TRANSFORMERS FROM 10 KVA TO 20 MVA 11/33/66/132 KV	–	RTS POWER CORPORATION LTD, M/S KANYAKA PARAMESHWARI ENGINEERING LTD, M/S ESENNAR TRANSFORMERS (P) LTD (UPTO 5 MVA OF 11 KV AND UP TO 5 MVA ABOVE 11 KV AND UP TO 33 KV), INDIAN TRANSFORMERS AND ELECTRICALS PVT. LTD (UPTO 10 MVA)., M/S KOTSONS PVT LTD(UP TO 2500 KVA 33 KV CLASS), M/S JAYBEE INDUSTRIES (UPTO 630 KVA), M/S VOLTAMP ELECTRICALS PVT LTD (1 MVA, 11/0.433 KV) / CROMPTON GREAVESS

1	2		3
112.	ISOLATION TRANSFORMER	–	INSTRUMENTS & EQUIPMENT CO./ M/S FUJI ELECTRIC CONSUL NEOWATT PVT. LTD., M/S VINTEK ELECTRONICS (VOLINA) (1KVA TO 1050 KVA) , M/S LIVE LINE ELECTRONICS
113.	ENGINE FOR DG SET	–	M/S STERLING GENERATORS PVT LTD, STAMFORD M/S JACKSON & COMPANY, KIRLOSKAR OIL ENGINE, CUMMINS, ASHOK LEYLAND, CROMTON GREAVES / M/S MAHINDRA & MAHINDRA LTD. (MAKE: MAHINDRA/MAHINDRA POWEROL)
114.	ALTERNATOR FOR DG SET	–	CROMPTON GREAVES/ KIRLOSKAR ELECTRIC/ STAMFORD/BHEL / M/S STERLING GENERATORS PVT LTD
115.	BATTERY CHARGERS	–	AMARA RAJA / STATCON / M/S INSTRUMENTS & EQUIPMENT CO/ ESCROP/ SU-KAM/ MICROTECH/ EXIDE
116.	BATTERY MAINTENANCE FREE	–	AMARA RAJA/ AMCO/ EXIDE/ OKAYA/ AMARON/ STANDARD
117.	ELECTRIC MOTORS	–	ABB/ CROMPTON GREAVES/ KIRLOSKAR/ SIEMENS/ LAURITZ KNUDSEN ELECTRICAL & AUTOMATION// HAVELLES
118.	VOLTAGE STABILIZER	–	BRENTFORD / VINTEK ELECTRONICS / INDIAN TRANSFORMERS AND ELECTRICALS PVT. LTD./ INSTRUMENTS & EQUIPMENT CO./ M/S FUJI ELECTRIC CONSUL NEOWATT PVT. LTD.
119.	BUS BAR TRUNKING	–	LAURITZ KNUDSEN ELECTRICAL & AUTOMATION, CONTROL & SWITCH GEAR, SCHNEIDER, SIEMEN, GE INDIA, SHALABH (INDIA) INDUSTRIES LTD.
120.	ACSR, AAAC, AAC CONDUCTORS	–	M/S DYNAMIC CABLES LIMITED, M/S GUPTA POWER INFRASTRUCTURE LTD, HAVELLS INDIA LTD/ CROMPTOM GREAVES, JAYPURIA BROTHERS
121.	UPS 1 KVA TO 30 KVA	–	M/S FUJI ELECTRIC CONSUL NEOWATT PVT. LTD., V-GUARD, TATA-LIEBERT, LUMINOUS, SUKAM, APLAB
122.	PRECAST CABLE COVER FOR HT/LT	–	ANJALI TILES/ ULTRA TILES PVT LTD/ TERRA FIRMA/ BANSAL /WONDER CRETE/ MEHTAB
123.	SCROLL TYPE WATER COOLED CHILLER	–	BLUESTAR/VOLTAS/CARRIER /DAIKIN/ KIRLOSKAR/ HITACHI
124.	FRP COOLING TOWER	–	ADVANCE/MIHIR/DELTA/PAHARPUR/POLO
125.	AIR HANDING/HANDLING UNIT/ COOLING COIL	–	EDGETECH/NATIONAL/ BLUE STAR/ VOLTAS/ ZECO/ LLOYD/ CARRIER/ ACCEL/ DAIKIN/ HOTCHI
126.	MONOBLOCK CENTRIFUGAL PUMP SET	–	KIRLOSKAR/CG/MATHER & PLATT
127.	AIR DIFFUSERS	–	RUSKIN TITUS/AIR FLOW/CARYAIRE/SYSTEMS AIR /AIRMECH
128.	INSULATION (FOR CHILLED WATER PIPE	–	KFLEX/ ARMACELL/ AEROFLEX/ STAR SHIELD TECHNOLOGIES
129.	BUTTERFLY VALVE	–	SANT/ KIRLOSKAR/ ZOLOTO
130.	DYNAMIC BALANCING VALVE	–	ADVANCE/ LEADER/ AUDCO/ ZOLOTO
131.	THERMOMETER	–	H-GURU/TSI/GENERAL/WIKA
132.	PRESSURE GAUGE	–	H-GURU/TSI/GENERAL/WIKA
133.	FLAME PROOF MCB DB	–	SHYAM SWITCHGEAR/SUDHIR
134.	DISC/PIN TYPE PORCELAIN INSULATORS	–	JAYASHREE BARODA, JAYPURIYA BROTHERS, PACTIL
135.	FIRE FIGHTING EQUIPMENTS, ACCESSORIES	–	M/S AN EXFLAME FIRE PROTECTION PVT LTD, MINIMAX, FIREX, SAFEX, DUNLOP, COSMOS, JAYSHREE, NEW AGE, INDIAN RAYONS, M/S SAFEGUARD, TYCO, CEASEFIRE, SUPEREX, AGNI SUKAN, M/S TEXTEL TECHNOLOGY PVT. LTD.
136.	FIRE PUMPS & PANEL	–	CROMPTON GREAVES, KIRLOSKAR, KSB, JYOTI, BEACON, MATHER & PLATT
137.	11/22/33 KV RMU	–	M/S LUCY ELECTRIC INDIA PVT. LTD., BHEL, ABB, MEGEWIN SWITCHGEAR PVT LTD, SIEMENS STELMEC LTD, ANDREW YULE, SCHNIEDER, LAURITZ KNUDSEN ELECTRICAL & AUTOMATION, C G POWER & INDUSTRIAL SOLUTION LIMITED.(BRAND: CGPISL) M/S C-SEC TECHNOLOGIES PRIVATE LTD (MAKE: C-

1	2		3
			SEC)
138.	JUNCTION BOX	–	M/S HENSEL ELECTRIC INDIA PVT LTD, SINTEX INDUSTRIES LTD, C&S ELECTRIC LTD
139.	CONDENSER & CHILLER PUMPS	–	CROMPTON GREAVES, KIRLOSKAR, ABB, KSB, MATHER & PLATT
140.	CABLE TRAY	–	YMUNA POWER & INFRASTRUCTURE LTD, SHALABH (INDIA) INDUSTRIES LTD., SUMIP, ERCON
141.	EARTHING BACKFILL COMPOUND	–	ASHLOK SAFE EARTHING ELECTRODE LTD., SHIWANGI ENGG WORKS LUCKNOW, JAYPURIA BROTHERS
	MISCELLANEOUS		
142.	SEALING COMPOUND & JOINT FILLER	–	M/S HINDCON CHEMICALS LTD./M/S CHOKSEY CHEMICALS PVT. LTD./ BERGER PAINTS INDIA LTD./ M/S IWL INDIA LTD./ TIKI TAR DANOSA (INDIA) PVT. LTD/ MARUTI BITUMEN PVT. LTD/ ASIAN PAINTS PVT LTD./M/S SIKA INDIA PVT LTD./ BOSTIK INDIA PVT. LTD/ M/S STP LTD.
143.	HDPE-DWC PIPES & FITTINGS	–	M/S KIRAN INFRA TECH / SUPREME / PRINCE / FINOLEX
144.	METTALIC FLOOR HARDNER	–	IRONITE / STILLONITE / HARDONATE
145.	ROAD SAFETY ANY TYPE FITTING or ROAD FURNITURE	–	TATA BP/ DARK EYE/3M/ SHAKTI
146.	FENCING RELATED ITEMS (CHAIN LINK, CONCERTINA COIL, WELDMESH, GABIONS, HONEYCOMB, RAZOR WIRE, BARBED WIRE, TREE GAURD), SECURITY GATE, ELECTRIC FENCE	–	M/S SHANKAR WELDMESH PVT. LTD./ SK WELDEDMESH PVT LTD. (GRP-II)/ A-1 FENCE PRODUCT COMPANY PVT. LTD./ TATA STEEL GLOBAL WIRES
147.	ROAD MARKING PAINT	–	ASIAN/APCOMARK/STP/SUNRISE/FAITH/AUTOMARK/SUNNY/ SHALIMAR/ BERGER/ JENSON & NICHOLSON PAINTS/ JOTUN INDIA PVT. LTD.
148.	REFRIGERATOR/ WATER COOLER/ DEEP FREEZER/SPLIT & WINDOW& CASSETTE TYPE AC UNIT	–	LG /SAMSUNG/ VEDIOCON / GODREJ/ VOLTAS/ WHIRLPOOL/ BLUE STAR
149.	WOODEN FLOORING	–	ACTION TESA/ ARMSTRON & BRUCE/ PERGO/ M/S GREEN PLY INDUSTRIES LTD./ VISTA/ HARO
150.	DRAPERY ROD/CURTAIN ROD	–	VISTA LEVOLOR/ MAC-DECOR/ DECO WINDOW/ JAYESH METAL CORPN (JMC)
151.	HDPE/LLDPE SEPTIC TANK	–	SINTEX/ SUPRIME/ UNITED SANITATION DEVICES/ RELIANCE POLYMERS

Note:-

1. The makes have been given tradewise only for the purpose of easily locating the items and otherwise makes given are applicable to the entire contract.

2. Wherever word "equivalent make" has been specified the option of equivalent make will only be considered after taking approval from Accepting Officer and only where products of brand/ manufacturer specified are not available in market.

SIGNATURE OF THE CONTRACTOR **DY DIR/AAD(CONTRACTS)**
DATE: **FOR ACCEPTING OFFICER**

MATERIALS OF LOCAL ORIGIN

APPENDIX 'C' (contd…)

S/No	Station	Brick size & strength	Fine sand	Coarse sand	Stone aggregate
1. **CWE Shillong Area**					
(a)	Shillong	Not Applicable	Umtangar/ Mowpat/ Happy Valley/Laitkore peak	Umtangar/ Mowpat/ Happy Valley/Laitkore peak	Mylliem
(b)	Umroi	Not Applicable	-do-	-do-	-do-
(c)	Narangi & Guwahait	Old size bricks (23x11.5x7.5cm of strength 75 Kg/Sqcm	Sonapur	Sonapur	Patharquary/ Narangi
2. **CWE Tezpur Area**					
(a)	Tezpur/Chardaur/ Missa/Missamari	Locally available of size 23x11.5x7.5cm of strength 75 Kg/Sqcm	Mara Bharali	Mara Bharali	Bhalukpong/ Shijusa
(b)	Bomdila/Rupa/ Tenga	-	Locally available from river bed	Locally available from river bed	Shingle from river bed
3. **CWE Dinjan Area**					
(a)	Dinjan	Old size brick of size 23x11.5x7.5 cm of strength 75 Kg/Sqcm	Guinjan	Guinjan	Namrup
(b)	Lekhapani	-do-	Jagun/Margharita	Jagun/Marghrita	Jagun/Marghrita
(c)	Jorhat/Mariani/ Bhalukmara	Old size brick of size 22x11x7.5 cm of strength 75 kg/sqcm	Bhogdoi river/ Kanighat	Bhogdoi river/ Kanighat	Bhihu-bari
4. **HQ 137 WE Area**					
(a)	Dimapur/ Rangapahar	Locally available of size 22x11x7.5 cm of strength 75 kg/sqcm	Dhansiri river	Deopnai	Diphu river
(b)	Zakhama	Old size of brick of size 23x11.5x7.5 cm of strength 75 kg/sqcm			
(c)	Leimakhong	Old size of brick of size 25x 10x7.0 cm of strength 75 kg/sqcm	Imphal	Imphal	Imphal
(d)	Agartala/Tripura	Locally available of size 25x12.5x7.5 cm and of strength 75 kg/sqcm	Locally available	Locally available	Silchar
(e)	Masimpur/Silchar	Old size bricks of size 23x11.5x7.5 cm of strength 75 kg/sqcm.	Madhura/Jatinga	Madhura/Jatinga	Madhura/Jatinga

Note – 1. The above sources are given for guidance only. This does not absolve the contractor from his responsibility of incorporating various materials in the work as per specifications mention in contract. Contractor has to ascertain the availability of material at the source given above. In case of non availability/scarcity of material mentioned at the source above the contractor has to arrange the materials from other sources without any extra cost to the Government. No claim whatsoever shall be entertained by the department on this account at later stage.

SIGNATURE OF CONTRACTOR
DATED

Jt/Dy DIRECTOR/AAD (CONTRACTS)
FOR ACCEPTING OFFICER

APPENDIX 'D' (contd…)

CEMENT SUPPLY & ACCEPTANCE REGISTER

1. CA NO & Name of work :
2. Control No* :
3. Name of Manufacturer/Brand/Gde of Cement (A) Manufacturer _____ (B) Brand _____ (C) Grade :
4. Qty of cement & Lot No/Week No (in bags) : (A) (Qty _____) (B) Lot No/Week No _____
5. Manufacturer's test certificates no _____
6. Random Test details (a) Physical test report from _____ vide their letter No _____ (Name of approved Lab/Engg College)

   (b) Chemical test report from _____ vide their letter No _____ (Name of approved Lab/Engg College)
7. Details of Physical & Chemical properties :

	Specific Surface Area (m^2 per Kg	Soundness by Le Chatellar	Soundness by Auto Clave	Initial Setting Time (Minutes)	Final Setting Time (Minutes)	Compressive strengths			Temp during testing ^0C	Standard consistency (%)	Lime Saturation (Ratio)	Aclumina Iron Ratio	Insoluble ratio (%)	Magnesium (%)	Sulphuric Anhydride (%)	Loss on ignition (%)	Alkalies (%)	Chlorides (%)
						03 days	07 days	28 days										
As per relevant IS																		
As per manufacturer's test certificate																		
As per random test certificate																		

Remarks with Signature : Accepted/Rejected

Contractor Junior Engineer Engineer-in-Charge Garrison Engineer

Remarks of BOO/Inspecting Officer/CWE

* To be allotted Serially by GE consignment wise

MATERIAL AND THEIR TESTS **APPENDIX 'E'**

RECOVERY RATES OF TESTING CHARGES

Sl No	Material	Test	Method of testing	**Frequency of Tests**	**Level of test**	**Rate per test Rs. Ps**	Remarks
1	2	3	4	**5**	**6**	**7**	8
1.	Bricks	(i) Compressive strength	IS-3495 (Part -II)	As per IS-5454 as given under:- Lot size Sample Permissible No of defective bricks	`A'	180/-	Checks for visual and dimensional characteristics shall also be carried out as per IS-5454.
		(ii) Water absorption	-do-	-do-	'A'	150/-	-do-
		Efflorescence	-do- (Part-I)	1001 5 0 to10000 0 10001 10 0 to 35000 35001 15 1 to 50000	'A'	180/-	
2.	Coarse aggregate	(i) Sieve analysis	IS-2386 (Part-I)	One test for every 15CUM of aggregates for part thereof brought to site.	'A'	120/-	
		(ii) Flakiness Index	-do-	-do-	'A'	90/-	
		(iii) Estimation of deleterious material	IS-2386 (Part-I)	One test for every 100CM of aggregate or part thereof	'A'	120/-	
		(iv) Organic impurities	-do-	One test per source of supply	'C'	120/-	
		(v) Moisture content	-do- (Part-II)	Regularly as required.	'A'	120/-	
		(vi) Specific gravity	-do-	One test for each source of supply	'B'	120/-	
3.	Fine aggregate	(i) Sieve analysis	IS-2386 (Part-I)	One test for every 15CUM of FA or part when brought to site.	'A'	180/-	
		(ii) Test for clay, silt and impurities	-do- (Part-I)	-do-	'A'	90/-	
		(iii) Specific gravity	-do- (Part-II)	One for each source of supply	'A'	180/-	
		(iv) Moisture content	-do- (Part-II)	Regularly as required subject to 2 test/day when being used.	'A'	180/-	
		(v) Test for organic impurities	-do- (Part-II)	One test for each source of supply.	'C'	180/-	

Legend
A-site lab.
B- Zonal Laboratory/ **Command Test Laboratory**
C- National Test House, SEMT Wing CME, Regional Research labs, IITs, National Institutes of Technology, , any Govt Engineering Collage or NABL Labs approved any Govt laboratories, Command Test Laboratory(if such facilities exists)

Contd..

1	2	3	4	5	6	7	8
4.	RCC/PCC concrete (nominal Mix & Design Mix)	(i) Slump test for IS-119 compacting factor test or vee-bee time.	IS-119	The minimum frequency of sampling of concrete of each grade shall be as under:- Qty of concrete in the work (cubic metre) — No of samples	'A'	180/-	Random sampling shall be carried out to cover all mixing units.
		Compressive strength		1-5 — 1 6-15 — 2 16-30 — 3 31-50 — 4 51 and above — 4+ 1 addl for each 50Cum or part thereof.	'A'	120/-	Refer IS 456-2000 clause 14 for frequency of sampling
5.	(a) PCC block for walling (Hollow block)	(i) Compressive strength	IS-2156-1984	08 block out of 14	'A'	60/-	Sample:- 14 blocks from consignment of every 5000 blocks or part thereof.
		(ii) Water absorption	-do-	03 blocks out of 14	'B'	120/-	
		(iii) Density	-do-	03 block out of 14	'B'	90/-	
	(b) PCC solid block for walling	**(i)** Compressive strength	IS-2185	12 block out of 18	'A'	60/-	18 blocks shall be taken from every lot/ consignment of 1000 blocks or part thereof These blocks to be checked for dimension and weight
		(ii) Water absorption	-do-	03 blocks out of 18 block	'B'	120/-	
		(iii) Density	-do-	03 blocks out of 18 block	'A'	120/-	
6.	Cement flooring tiles/terrazzo tiles	(i) Water absorption	IS-1237	06 tiles out of 18	'B'	180/-	Samples:- 18 tiles from each source of supply selected at Random.
		(ii) Wet transverse strength	-do-	-do-	'B'	144/-	
		(iii) Resistance to wear	-do-	-do-	'C'	540/-	

1	2	3	4	5	6	7	8
7.	Burnt clay roofing tiles (hand made) As per IS-2690 (Part-II) **Length** 150mm to 200mm **Width** 100mm to 200mm **Thickness** 35 mm to 50mm	(i) Water absorption	IS-3495 (Part-II)	6 tiles out of 12	'B'	216/-	Samples:- 12 tiles from each source of supply selected at Random.
		(ii) Compressive Strength	-do- (Part-II)	-do-	'A'	180/-	
8.	Mangalore pattern roofing tiles	(i) water absorption	IS-654 (	6 tiles out of 32	'B'	180/-	Samples:- 32 tile from each consignment of 3000 tiles or part thereof. These tiles shall be checked for dimension.
		(ii) Breaking load	-do-	-do-	'B'	120/-	
9.	Timber	(i) Specific gravity and weight	IS-1708-1960	Minimum 3 samples from a lot of 4 Cum or 250 pieces of seasoned timber.	'B'	120/-	
		(ii) Moisture content	-do-		'A'	120/-	
10.	Water for construction purposes	(i) Test for acidity	IS-4564 & 3015	Once at the stage of approval of source of water.	'B'	240/-	Also refer clause 4.3 of IS-456 and its subsequent sub clauses regarding suitability of water.
		(ii) Test for alkalinity	-do-		'B'	240/-	
		(iii) Test for solid content.	-do-		'C'	300/-	

1	2	3	4	5	6	7	8
11.	Welding of steel work	Visual inspection test.	IS-822-1970 clause 7.1	100% by visual inspection	Work site	360/-	Specialised tests their method and frequency to be decided on consideration of their importance by the Accepting Officer.
12.	Timber panelled and glazed door/window & shutters (including Factory mad shutters)	(a) Dimensions, sizes, workmanship and finish	IS-1003-1977 (Part-I)	Frequency of sampling from each lot shall be as under:- Lot size / Sample size 26 to 50 / 5 51 to 100 / 8 101 to 150 / 13 151 to 300 / 20 301 to 500 / 32 501 to 1000 / 50 1001 and above / 80			
		b) Strength test (i) Slamming	IS-1003	From each lot 5% of the factory made shutters shall be manufacturer tested for strength test		-	-
		(ii) Impact indentation	-do-	-do-			
		(iii) Shock resistance	-do-	-do-			
		(iv) Edge loading	-do-	-do-			
13.	Plywood (IS:3003)	(a) Moisture content	IS-1734-(Part-I)	Six test pieces cut from each of the boards selected as per table shall be subjected to tests.	'C'	240/-	Sampling shall be as per IS-7835-1975 tables
14.	Wood particle board (medium density)IS:3097	(a) Density	IS-2360 (Part-III)	Three test specimens from each sample size 150mm x 75mm)	'A'	60/-	Sampling shall be as per IS-3487-83 with moisture metre.
		(b) Moisture content	-do-	-do-	'A' & 'B'	60/-	

1	2	3	4	5	6	7	8
		(c) Water absorption	-do- (Part-16)	-do-(size 300 mm x 300mm)	'A'	60/-	
		(d) Swelling due to surface absorption	-do- (Part-17)	-do- (size 125 mm x 100 mm)	'A'	60/-	
		(e) Swelling in water	-do-	-do- (size 200mm x 100 mm)	'A'	60/-	
		(f) Modulus of rupture	-do- (Part-4)	Three test specimens as per IS –2380-77	'B'	90/-	
		(g) Screw withdrawal strength	-do- (Part-4)	-do- as per IS 2385.	'C'	120/-	
15.	Cement	(a) Setting time	IS:4031-63 RE-AFFIRMET – 1980	Once for each consignment or as and when required.	'B'	180/-	
		(b) Soundness	-do-	-do-	'B'	120/-	
		(c) Compressive Strength	-do-	-do-	'B'	360/-	
		(d) Fineness	-do-	-do-	'B'	120/-	
16.	Solid PVC door	(a) Dimensions and squareness	IS-4020	Frequency & Sample sizes shall be as per IS provisions	A	180/-	
		(b) General Flatness test	-do-	-do-	A	180/-	
		(c) Local planeness test	-do-	-do-	A	180/-	
		(d) Impact identification test (Steel ball impact test)	-do-	-do-	C	-	
		(e) Edge loading test	-do-	-do-	C	-	
		(f) Shock resistance test	-do-	-do-	C	-	
		(g) Buckling test	-do-	-do-	C	-	
		(h) Slamming test	-do-	-do-	C	-	
		(j) Misuse test	-do-	-do-	C	-	
		(k) Screw withdrawal resistance test	-do-	-do-	C	-	
17.	Bitumen use for road pavement work	**(a)** Penetration value	IS 73	1 test per batch of bitumen supplied in bulk or drums	B & C		
		(b) Softening points	**IS 73**	-do-	B & C		
		(c) Elongation	**IS 73**	-do-	B & C		
		(d) Wax Content	IS 1209	-do-	B & C		
		(e) Flash point/ fire point	**IS 1209**	-do-	B & C		
		(f) Ductility	**IS 1209**	-do-	B & C		

1	2	3	4	5	6	7	8
18.	All type Glazed tiles	(i) Water absorption	IS-777	06 tiles out of 18	'B'	180/-	
		(ii) Wet transverse strength	-do-	-do-	'B'	144/-	
		(iii) Resistance to wear	-do-	-do-	'C'	540/-	

SIGNATURE OF CONTRACTOR
DATED:

Jt/Dy Dir/AAD (CONTRACT)
FOR ACCEPTING OFFICER

STEEL SUPPLY & ACCEPTANCE REGISTER **APPENDIX 'F'(cont...)**

1. CA NO & Name of Work
2. Contract No
3. Name of Manufacturer's T.C No
4. Manufacturer
5. Random Test details (a) Physical test report from _____ vide their letter No _____
(Name of NABL approved Lab/ Govt Engg College)

(b) Chemical test report from _____ vide letter No_____
(Name of NABL approved Lab/ Govt Engg College)

6. Types of steel, Dia & Qty

(a) type : TMT/CRS
(b) Dia_____ mm
(c) Actual Wt _____ MT
(d) Conversion Wt _____ MT

		Chemical test						Mechanical test						
	Carbon %	Sulphur %	Phosphorous %	Sulphur % + Phosphorous %	Manganese %	Silicon %	Corrosion Resistant element	Wt per metre	Stress (N/mm^2) 0.2% proof stress	Tensile strength (N/mm^2)	Elongation%	Bend Test	Rebend Test	Remarks
As per IS 1786-2008														
As per Manufacturer's test certificate														
As per independent test														

Remarks with Signature Accepted/Rejected

Contractor Junior Engineer Engineer-in-Charge Garrison Engineer

Remarks of BOO/Inspecting Officer/CWE

CONTROL TESTS AND THEIR MINIMUM FREQUENCY FOR SUB-BASES AND BASES (EXCLUDING BITUMEN BOUND BASES)

GENERAL:-Various quality control tests and their frequency for flexible and rigid pavement must be carried out for each layer as per Ministry of Road Transport & Highways (MoRTH) specifications for Roads & Bridges (Fifth Revision).

SI No.	TYPE OF CONSTRUCTION	TEST	FREQUENCY (MINIMUM)
1	2	3	4
1.	GRANULAR	(i) Gradation	One test per 400 cum
		(ii) Atterberg limits	One test per 400 cum
		(iii) Moisture content prior to compaction	One test per 400 cum
		(iv) Density of compacted layer	One test per 1000 sqm
		(v) Deleterious constituents	As required
		(vi) CBR	As required
2.	Lime/ Cement Stabilised Soil Sub-base	(i) Quality of lime/ cement	One test for each consignment subject to a minimum of one test per 5 tonnes
		(ii) Lime/Cement content	Regularly, through procedural checks
		(iii) Degree of pulverization	Periodically as considered necessary
		(iv) CBR or Unconfined Compressive Strength test on a set of 3 specimens	As required
		(v) Moisture content prior to compaction	One set of two tests per 500 sqm
		(vi) Density of compacted layer	One set of two tests per 500 sqm
		(vii) Deleterious constituents	As required
3.	Water Bound Macadam	(i) Aggregate Impact Value	One test per 1000 cum of aggregate
		(ii) Grading of aggregate	One test per 250 cum
		(iii) Combined Flakiness and Elongation Indices	One test per 500 cum of aggregate
		(iv) Atterberg limits of binding material	One test per 50 cum of binding material
		(v) Atterberg limits of screenings	One test per 100 of aggregate
4.	Wet Mix Macadam	(i) Aggregate Impact Value	One test per 1000 cum of aggregate
		(ii) Grading of aggregate	One test per 200 cum of aggregate
		(iii) Combined Flakiness and Elongation Indices	One test per 500 cum of aggregate
		(iv) Atterberg limits of portion of aggregate passing 425 micron sieve	One test per 200 cum of aggregate
			One set of three tests per 1000 sqm

CONTROL TESTS FOR BITUMINOUS WORKS AND THEIR MINIMUM FREQUENCY

Sl No.	TYPE OF CONSTRUCTION	TEST	FREQUENCY (MINIMUM)
1.	Prime Coat/ Tack Coat/ Fog Spray	(i) Quality of binder	Number of samples per lot and tests as per IS:73, IS:217 and IS:8887 as applicable
		(ii) Binder temperature for application	At regular close intervals
		(iii) Rate of spread of Binder	Three tests per day
2.	Seal Coat/ Surface Dressing	(i) Quality of Binder	Same as mentioned under Serial No. 1
		(ii) Aggregate Impact Value or Los Angeles Abrasion Value	One test per 200 cum of each source and whenever there is change in the quality of aggregate
		(iii) Combined Flakiness and Elongation Indices	One test per 100 cum of aggregate for each source and whenever there is change in the quality of aggregate
		(iv) Stripping value of aggregates (Immersion Tray Test)	One test of each source and whenever there is change in the quality of aggregate
		(v) Water absorption of aggregate	-do-
		(vi) Water sensitivity of mix	-do-
		(vii) Grading of aggregate	Two tests per day
		(viii) Soundness (Magnesium Sulphate/ Sodium Sulphate)	One test for each source and whenever there is change in the quality of aggregate
		(ix) Polished stone value (not applicable for SAM/SAMI)	-do-
		(x) Temperature of binder in boiler, aggregate in dryer and mix at the time of laying and compaction	At regular intervals
		(xi) Rate of spread of materials	Same as mentioned under Serial No. 1
		(xii) Percentage of fractured faces (When gravel is used)	One test per 100 cum of aggregate
3.	Open-graded Premix Surfacing/ Close-graded Premix Surfacing	(i) Quality of binder	Same as mentioned under Serial No. 1
		(ii) Aggregate Impact Value or Los Angeles Abrasion Value	Same as mentioned under Serial No. 2
		(iii) Combined Flakiness and Elongation Indices	Same as mentioned under Serial No. 2
		(iv) Stripping value	Same as mentioned under Serial No. 2
		(v) Water absorption of aggregates	Same as mentioned under Serial No. 2
		(vi) Water Sensitivity of mix	Same as mentioned under Serial No. 2
		(vii) Grading of aggregates	Same as mentioned under Serial No. 2
		(viii) Soundness(Magnesium Sulphate and Sodium Sulphate)	Same as mentioned under Serial No. 2
		(ix) Polished stone value	At regular interval
		(x) Temperature of binder at application	Two tests per day per plant
		(xi) Binder content	Same as mentioned under Serial No. 2

		(xii) Percentage of fractured faces	
4.	Bituminous Macadam	(i) Quality of binder	Same as mentioned under Serial No. 1
		(ii) Aggregate Impact Value or Los Angeles Abrasion Value	Same as mentioned under Serial No. 2
		(iii) Combined Flakiness and Elongation Indices	One test per 350 cu.m for each source Same as mentioned under Serial No. 2
		(iv) Stripping value	Same as mentioned under Serial No. 2
		(v) Water absorption of aggregates	Same as mentioned under Serial No. 2 Same as mentioned under Serial No. 2
		(vi) Water Sensitivity of mix	Same as mentioned under Serial No. 2
		(vii) Grading of aggregates	
		(viii) Soundness (Magnesium Sulphate/ Sodium Sulphate)	Same as mentioned under Serial No. 2
		(ix) Percentage of fractured faces	
		(x) Binder content	Same as mentioned under Serial No. 3
		(xi) Control of temperature of binder and aggregate for mix and of the mix at the time of laying and rolling	Same as mentioned under Serial No. 2
		(xii) Density of Comp layer	One test per 700 sq.m area
		(xiii) Rate of spread of Mixed Material	At regular intervals
5.	Dense Bituminous Macadam/ Bituminous Concrete	(i) Quality of binder	Number of samples per lot and tests as per IS:73 or IRC:SP:53, IS:15462
		(ii) Aggregate Impact Value/ Los Angeles Abrasion Value	One test per 350 cu.m of aggregate for each source and whenever there is change in the quality of aggregate
		(iii) Flakiness and Elongation Indices	One test per 350 cu.m of aggregate for each source and whenever there is change in the quality of aggregate
		(iv) Soundness test (Sodium or Magnesium Sulphate test)	One test for each source and whenever there is change in the quality of aggregate
		(v) Water absorption of aggregates	One test for each source and whenever there is change in the quality of aggregate
		(vi) Sand equivalent test	One test for each source and whenever there is change in the quality of aggregate
		(vii) Plasticity Index	One test for each source and whenever there is change in the quality of aggregate
		(viii) Polished stone value	One test for each source and whenever there is change in the quality of aggregate
		(ix) Percentage of fractured face (x) Mix grading	One test for each source and whenever there is change in the quality of aggregate One test for each source and whenever there is change in the quality of aggregate
		(xi) Stability and voids analysis of mix including theoretical maximum specific gravity of loose mix	One test per 350 cu.m of aggregate when crushed gravel is used
		(xii) Moisture Susceptibility of mix (AASHTO T283)	One set for individual constituent and mixed aggregate from dryer for each 400 tonnes of mix subject to minimum of two tests per day per plant
		(xiii) Temperature of binder in boiler, aggregate in dryer and mix at	Three tests for stability, flow value,

		the time of laying and compaction	density and void contents for each 400 tonnes of mix subject to minimum of two tests per day per plant
		(xiv) Binder content	One test for each mix type whenever there is change in the quality or source of coarse or fine aggregate
		(xv) Rate of spread of mix material	At regular intervals
		(xvi) Density of Compacted layer	One set for each 400 tonnes of mix subject to minimum of two tests per day per plant
			After every 5th truck load
			One test per 700 sq.m area
6.	Sand Asphalt Base course	(i) Quality of binder	Same as mentioned under Serial No. 2
		(ii) Aggregate Impact Value or Los Angeles Abrasion Value	Same as mentioned under Serial No. 2
		(iii) Sand equivalent test	Same as mentioned under Serial No. 2
		(iv) Plasticity Index	Same as mentioned under Serial No. 5
		(v) Mix grading & binder content	Same as mentioned under Serial Nos. 2 and 3
		(vi) Stability of Mix	Same as mentioned under Serial No. 5
		(vii) Control of temperature of binder in boiler, aggregate in the dryer and mix at the time of laying and rolling	Same as mentioned under Serial No. 2
		(viii) Thickness of layer	Same as mentioned under Serial No. 5
		(ix) Density of Compacted layer	Same as mentioned under Serial No. 5
7.	Slurry seal and Micro surfacing	(i) Quality of Aggregate Sand Equivalent Value Water Absorption Soundness Test	One per source/ site
		(Sodium/ Magnesium Sulphate Test)	
		(ii) Quality of Emulsion	One per lot of 20 t as per IS:8887
		(iii) Aggregate Moisture	Two per day
		(iv) Aggregate Gradation	Two per day at site
		(v) Binder Content	Two per lane per Km
		(vi) Calibration of Machine	Once per project
		(vii) Quantity of Slurry (By weight of aggregate)	Daily (Travel time of Machine)
8.	Stone Matrix Asphalt	(i) Quality of binder	Number of samples per lot and tests as per IS:73 or 'RC:SP:53, IS:15462
		(ii) Aggregate Impact Value/ Los Angeles Abrasion Value	One test per 100 cu.m of aggregate
		(iii) Flakiness and Elongation Indices	One test per 100 cu.m of aggregate
		(iv) Soundness Test (Sodium and Magnesium Sulphate Test)	One test for each method for each source and whenever there is change in the quality of aggregate
		(v) Water absorption of	One test for each source and

		aggregate	whenever there is change in the quality of aggregate
		(vi) Sand equivalent test	One test for each source
		(vii) Plasticity Index	One test for each source
		(viii) Polished stone value	One test for each source
		(ix) Percent of fractured faces	One test per 50 cu.m of aggregate when crushed gravel is used
		(x) Mix grading	One set for individual constituent and mixed aggregate from dryer for each 400 tonnes of mix subject to minimum of two tests per day per plant
		(xi) Air voids and VMA analysis of mix including theoretical maximum specific gravity of loose mix	Three tests per day
		(xii) Moisture Susceptibility of mix (AASHTO T 283)	One test for each mix type whenever there is change in the quality or source of coarse or fine aggregate
		(xiii) Temperature of binder in boiler, aggregate in dryer and mix at the time of laying and compaction	At regular intervals
		(xiv) Binder content	One set for each 400 tonnes of mix subject to minimum of two tests per day per plant
		(xv) Rate of spread of mix material	After every 5th truck load
		(xvi) Density of compacted layer	One test per 250 sqm area
9.	Mastic asphalt	(i) Quality of binder	Same as mentioned under Serial No. 5
		(ii) Aggregate Impact Value and Los Angeles Abrasion Value	Same as mentioned under Serial No. 5
		(iii) Combined Flakiness and Elongation Indices	Same as mentioned under Serial No. 5
		(iv) Stripping value	Same as mentioned under Serial No. 2
		(v) Water Sensitivity of mix	Same as mentioned under Serial No. 5
		(vi) Grading of aggregates	Two tests per day per plant on the individual constituent and mixed aggregates from the dryer
		(vii) Water absorption of aggregates	Same as mentioned under Serial No. 5
		(viii) Soundness (Magnesium Sulphate/ Sodium Sulphate)	Same as mentioned under Serial No. 5
		(ix) Percentage of fractured faces	Same as mentioned under Serial No. 5
		(x) Binder content and aggregate grading	Same as mentioned under Serial No. 3
		(xi) Control of temperature of binder and aggregate for	At regular close intervals

		mixing and of the mix at the time of laying and rolling	
		(xii) Rate of Spread of Mixed Material	Regular control through check of layer thickness
		(xiii) Hardness number	Minimum two tests per day
10.	Recycled Material Grading of aggregate	-	Two tests per day
11.	Cold Mixes	-	All tests as per S. No.5
12.	Quality of Modified Binder	-	Number of samples per lot and tests as per IS:15462.
13.	Geotextiles	-	The requirements of Section 700 shall apply

Note:-Daily, weekly, monthly reports on test results shall be prepared indicating the location of sampling and testing, deviation from the specified values for materials and works and remedial action taken in respect of removal of defective work shall be prepared by the Contractor. The test record shall be certified by the Engineer that these tests were done in his presence and testing carried as per prescribed methodology.

SIGNATURE OF CONTRACTOR **DY DIR/ AAD (CONTRACTS)**
DATED: **FOR ACCEPTING OFFICER**